MARKETSANDMARKETS™

ADVANCED ANALYTICS MARKET

GLOBAL FORECAST TO 2026

BY COMPONENT (SOLUTIONS AND SERVICES),
BUSINESS FUNCTION (SALES & MARKETING, OPERATIONS),
TYPE (BIG DATA ANALYTICS, RISK ANALYTICS),
DEPLOYMENT MODE (ON-PREMISES AND CLOUD), VERTICAL, AND REGION

Report Code: TC 2361

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
AIR	Advanced Interaction Recorder
ALMO	Arms' Length Management Organization
APAC	Asia Pacific
ASR	Automated Speech Recognition
BFSI	Banking, Financial Services, and Insurance
BMGF	Bill and Melinda Gates Foundation
BU	Business Units
C1CX	ConvergeOne Cloud Experience
CAD	Computer-Aided Design
CAGR	Compound Annual Growth Rate
CC	Contact Center
CCPA	California Consumer Privacy Act
CEO	Chief Executive Officer
CIO	Chief Information Officer
COAI	Cellular Operators Association of India
COO	Chief Operating Officer
CPM	Collaboration Performance Management
CPO	Chief Product Officer
CRA	Call Recording Assurance
CRM	Customer Relationship Management
CSAT	Customer Satisfaction
CSO	Chief Strategy Officer
CTI	Computer Telephony Integration
CTO	Chief Technology Officer
CX	Customer Experience
DEM	Digital Experience Monitoring
DERA	Division of Economic and Risk Analysis
DNC	Do Not Call

EDGAR	Electronic Data Gathering, Analysis, and Retrieval
EFM	Enterprise Fraud and Misuse Management
EHR	Electronic Health Record
EICC	Enghouse Interactive Communications Center
ESI	Electronically Stored Information
EU	European Union
FAQ	Frequently Asked Question
FRA	Federal Railroad Administration
FSO	Financial Service Organization
FY	Financial Year
GCC	Gulf Corporations Countries
GCP	Google Cloud Platform
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
IBM	International Business Machines
ICT	Information and Communications Technology
IoT	Internet of Things
ISMS	Information Security Management System
ISO	International Standards Organization
ISV	Independent Software Vendors
IT	Information Technology
IVR	Interactive Voice Response
LFPDPPP	Law on Protection of Personal Data Held by Individuals
LTE	Long-term Evolution
MD	Managing Director
MEA	Middle East and Africa
MIC	Ministry of Internal Affairs and Communications
ML	Machine Learning
MMS	Multimedia Messaging Service
MNO	Multinational Organization
MPA	Mitel Performance Analytics
MSP	Managed Service Provider

MTT	Meaning-Text Theory
NCH	Nottingham City Homes
NEC	Nippon Electric Company
NER	Named Entity Recognition
NGN	Next-Generation Network
NHS	National Health Service
NLP	Natural Language Processing
NLU	Natural Language Understanding
NPS	Net Promoter Score
NRECA	National Rural Electric Cooperative Association
NTR	NICE Trading Recording
OCR	Optical Character Recognition
OS	Operating System
OTC	Over-The-Counter
PaaS	Platform-as-a-Service
PBX	Private Branch Exchange
PCI DSS	Payment Card Industry Data Security Standard
PDPA	Personal Data Protection Act
PDPL	Personal Data Protection Law
PLC	Product Life Cycle
PoS	Point-of-Sale
PPE	Personal Protective Equipment
PSAP	Public Safety Answering Point
R&D	Research and Development
RoA	Return on Asset
RoI	Return on Investment
SaaS	Software-as-a-Service
SBC	Session Border Controller
SEC	Securities and Exchange Commission
SfB	Skype for Business
SI	System Integrator
SLA	Service Level Agreement

SME	Small and Medium-sized Enterprise
SMS	Security Management System
SOC 2	Service Organizational Control 2
TaaS	Testing as a Service
TCO	Total Cost of Ownership
UAE	United Arab Emirates
UC	Unified Communications
UCaaS	Unified Communication as a Service
UK	United Kingdom
US	United States
USD	United States Dollar
USPTO	United States Patent and Trademark Office
VAR	Value-added Resellers
VoIP	Voice over Internet Protocol
VoLTE	Voice-over-Long-Term Evolution
VP	Vice President
WebRTC	Web Real-Time Communication
WFO	Workforce Engagement and Optimization
WHO	World Health Organization
Y-o-Y	Year-on-Year

1 INTRODUCTION

1.1 INTRODUCTION TO COVID-19

COVID-19 is an infectious disease caused by the novel coronavirus. Largely unknown before its outbreak in Wuhan, China, in December 2019, COVID-19 moved from a regional crisis to a global pandemic in just a few weeks. The World Health Organization (WHO) declared COVID-19 as a pandemic on March 11, 2020.

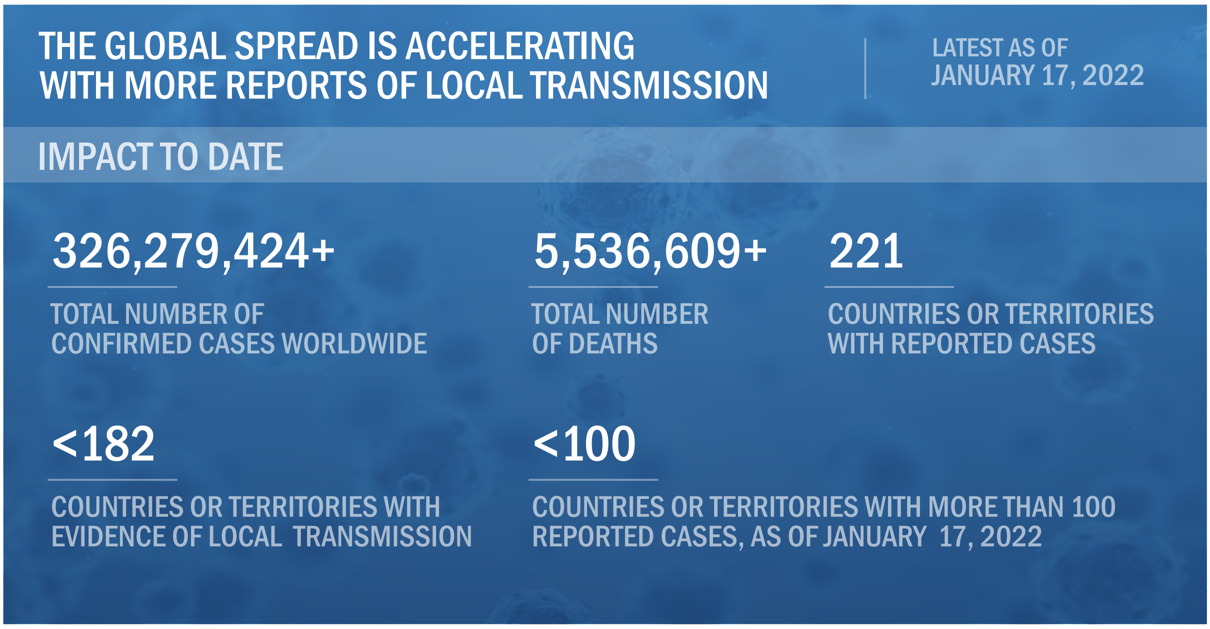
As the virus is spreading rapidly at a global level, countries globally are trying to prevent further contagion by adopting measures such as social distancing, contact tracing, self-quarantining, surveillance, communication, and testing. So far, China, South Korea, Japan, and Singapore have managed to flatten the curve, while the US, Italy, Spain, Germany, France, and Iran have imposed drastic measures to slow down the spread and control fatalities.

Government agencies have announced special financial aid packages to develop preventive and curative drugs, the purchase of critical care medical devices, and the fast-track approval of diagnostic tests. Various organizations across the globe, such as the Bill and Melinda Gates Foundation (BMGF) and Wellcome Trust, are also coming together to accelerate and strengthen the efforts to fight the COVID-19 pandemic.

1.2 COVID-19 HEALTH ASSESSMENT

It is an unprecedented crisis creating an enormous strain on healthcare providers across the globe. According to the WHO, within 67 days, the first 100,000 confirmed cases were reported worldwide. It took the first 11 days to reach 200,000 confirmed cases. In the next four days, the number reached 300,000 confirmed cases, while in another 12 days, the count reached one million confirmed cases.

FIGURE 1 COVID-19: GLOBAL PROPAGATION

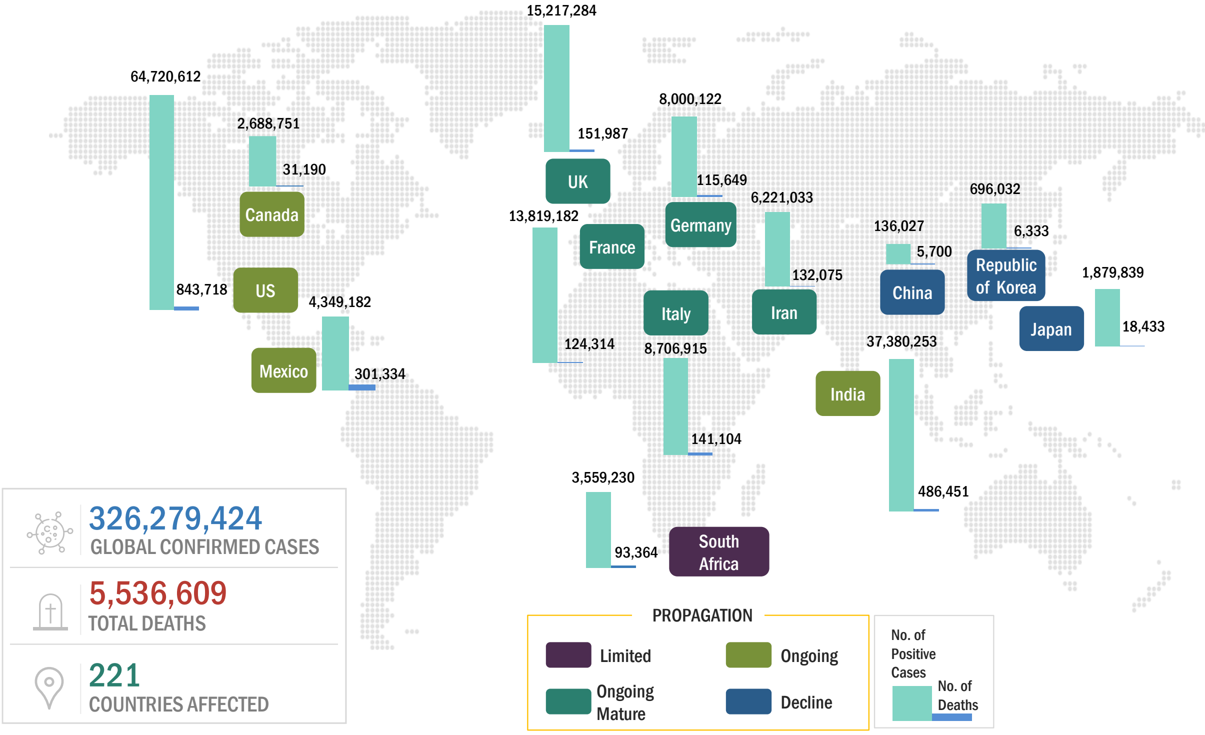


Source: WHO and ECDC

As of January 17th, 2022, the virus has spread to more than 221 countries, with over 326,279,424 reported cases and close to 5,536,609 deaths.

The actual number of infected people is probably higher than the official total, as some mild and asymptomatic cases are not tested and counted.

FIGURE 2 **COVID-19 PROPAGATION: SELECT COUNTRIES**



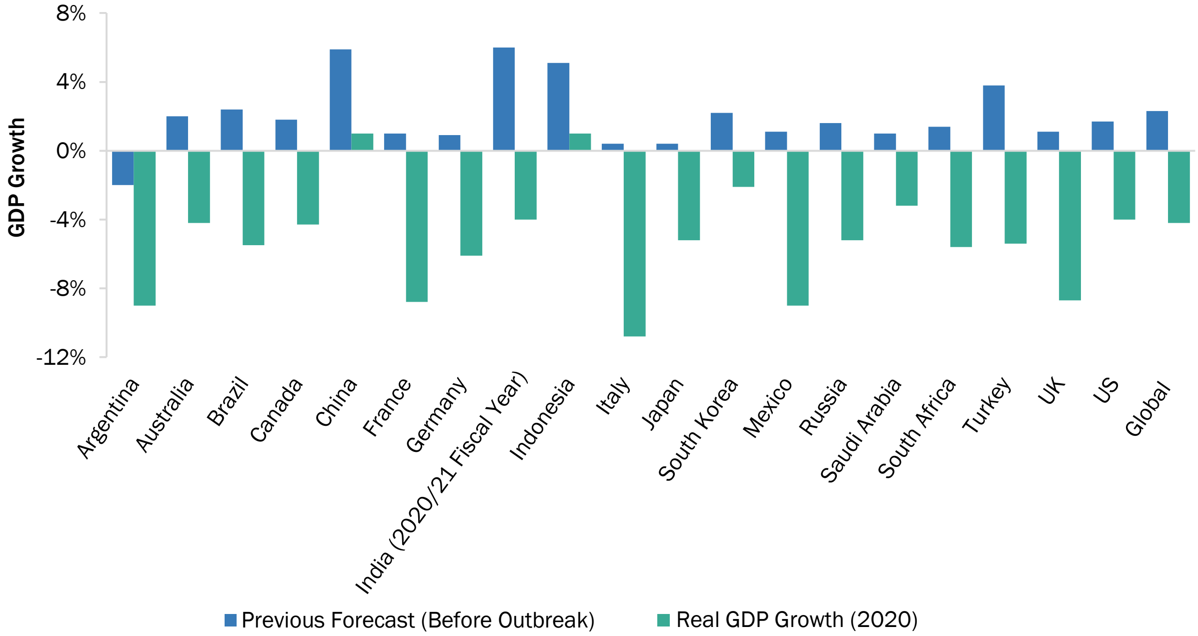
Source: WHO Coronavirus Disease 2019; data as of January 17th, 2022

Disease progression patterns are similar in most countries; however, flattening of the curve and disease containment largely depend on various measures taken by each country. China and South Korea, for example, have managed to control the spread of the disease by implementing severe countermeasures, such as restricting travel; imposing city-wide quarantines, curfews, and lockdowns; building specialty hospitals; increasing medical aid; and ramping up testing. Most other countries are trying to emulate these measures to contain the spread of the disease.

1.3 COVID-19 ECONOMIC ASSESSMENT

The COVID-19 outbreak has not only affected the healthcare sector but also impacted the global economy. It has had a significant economic impact on various financial and industrial sectors, such as energy, oil and gas, transportation and logistics, manufacturing, and aviation. It is predicted that the global economy will go into recession due to the loss of trillions of dollars. With the increasing number of countries imposing and extending lockdowns, economic activities are declining, which would impact the global economy.

FIGURE 3 REVISED GROSS DOMESTIC PRODUCT FORECASTS FOR SELECT G20 COUNTRIES IN 2020



Source: The Economist Intelligent Unit

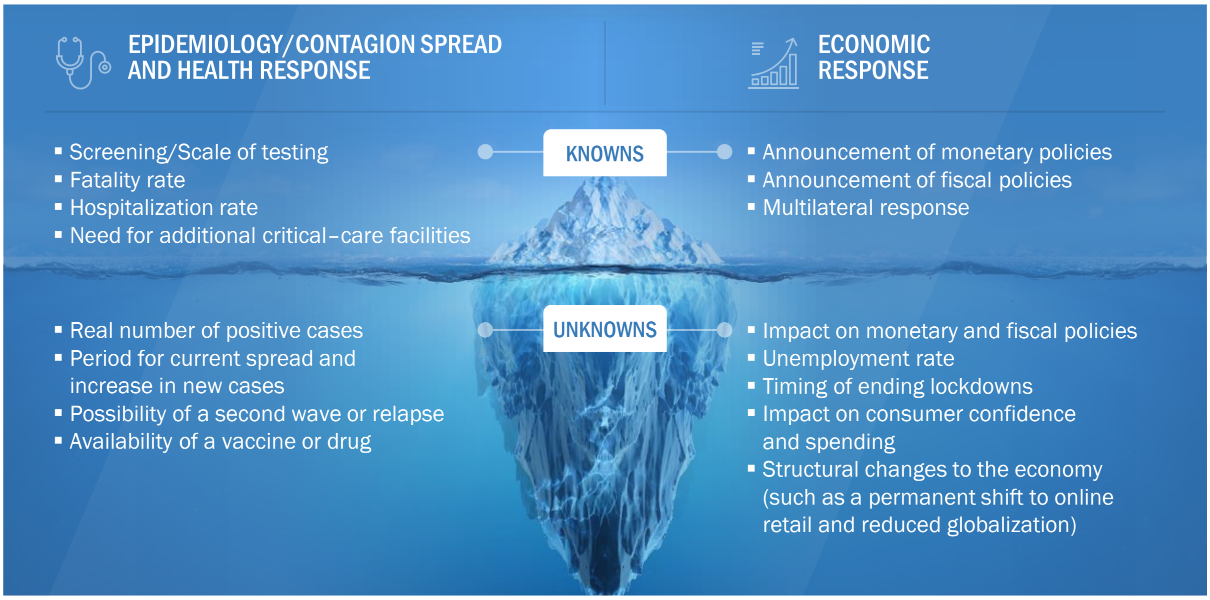
With the combined effects of the pandemic and the slump in global oil prices, investments are expected to contract sharply this year, especially in the energy sector. Export growth is also expected to reduce drastically. A recovery is expected in the second half of the year, provided there is no emergence of another wave of the pandemic, and countries manage to control the spread quickly and resume routine.

China has managed to contain the spread and is currently on track to resume routine. It is expected to recover in the second half of the year. However, the Eurozone, one of the hardest-hit regions, is expected to undergo a recession throughout 2021. Within the Eurozone, Germany, which is a manufacturing hub, would recover at a much slower pace than the other countries. The US economy is also expected to see a slump, with unemployment set to rise; recovery will be slow and will only be seen in the second half of the year.

1.3.1 COVID-19 ECONOMIC IMPACT—SCENARIO ASSESSMENT

Recovery from this crisis largely depends on various knowns and unknowns, which most countries will have to deal with and respond to. The speed and strength of the economic recovery will depend on the health and economic responses and the agility and speed of implementing various measures.

FIGURE 4 CRITERIA IMPACTING GLOBAL ECONOMY



In the recent past, the global economy became substantially more interconnected. The adverse consequences of various steps related to containment are evident from global supply chain disruptions, weaker demand for imported products and services, and the increase in the unemployment rate. Risk aversion has increased in the financial market, with all-time low-interest rates and sharp declines in equity and commodity prices. Consumer and business confidence have also reduced significantly.

This has magnified over time, and the spillover is seen in other countries as well. The latest forecasts by multiple sources predict a four percent decrease in the global Gross Domestic Product (GDP) in 2020 and a slow recovery, making the recession caused by this pandemic worse than the 2009 global recession. Based on these forecasts, the global economy is expected to recover by 2022; however, for some countries, including the UK, Germany, India, and Italy, recovery is predicted to take longer.

In this fast-changing environment, the full impact of the pandemic on the global economy may not entirely be known. MarketsandMarkets uses three scenario-based approaches (epidemiology, health response, and economic response) to assess the economic impact and recovery period at the global level. Countries and regions are expected to have different impacts and recovery periods.

FIGURE 5 SCENARIOS IN TERMS OF RECOVERY OF GLOBAL ECONOMY

SCENARIO	OPTIMISTIC	REALISTIC	PESSIMISTIC
EPIDEMIOLOGY/CONTAGION SPREAD AND HEALTH RESPONSE	FATALITY RATE 1.5–2% HOSPITALIZATION RATE: LESS THAN 15% ADDITIONAL ICU BEDS AND LESS THAN 2.5 - 5% ADDITIONAL VENTILATORS 15%	FATALITY RATE 2–3.5% HOSPITALIZATION RATE: 15–20% ADDITIONAL ICU BEDS AND 5% ADDITIONAL VENTILATORS 15–20%	FATALITY RATE MORE THAN 3.5% HOSPITALIZATION RATE: MORE THAN 20% ADDITIONAL ICU BEDS AND 15% ADDITIONAL VENTILATORS 25%
ECONOMIC RESPONSE	FISCAL STIMULUS TO THE TUNE OF USD 7.5–8 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF USD 100 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING TO THE TUNE OF USD 100 BILLION GLOBALLY	FISCAL STIMULUS TO THE TUNE OF USD 5–7.5 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF USD 75–100 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING TO THE TUNE OF USD 50–75 BILLION GLOBALLY	FISCAL STIMULUS TO THE TUNE OF LESS THAN USD 5 TRILLION GLOBALLY EMERGENCY FUNDING TO SUPPORT EFFORTS TO FIGHT COVID-19 TO THE TUNE OF LESS THAN USD 75 BILLION GLOBALLY MULTILATERAL AGENCIES FINANCING TO THE TUNE OF LESS THAN USD 50 BILLION GLOBALLY
RECOVERY SCENARIO	▪ Milder recession in 2020 Q1 and Q2; stronger recovery in 2020 Q3 and after ▪ US: Recovery by 2020 Q3 ▪ EU5: Recovery by 2020 Q3 ▪ China: Recovery by 2020 Q2 ▪ India: Recovery by 2020 Q3	▪ Recession in 2020 Q1 and Q2; partial bounce back in 2020 Q3, then slow growth ▪ US: Recovery by 2020 Q4 ▪ EU5: Recovery by 2020 Q4 ▪ China: Recovery by 2020 Q2 ▪ India: Recovery by 2020 Q3	▪ Deeper recession in 2020 Q1, Q2, and Q3; modest rebound in 2020 Q4 ▪ US: Recovery by 2021 Q1 ▪ EU5: Recovery by 2021 Q1 ▪ China: Recovery by 2020 Q3 ▪ India: Recovery by 2020 Q4

Source: UNCTAD, Moody's, FAS, OECD, and MarketsandMarkets Analysis

With more than four million confirmed cases of COVID-19, businesses are coping with lost revenues and disrupted supply chains as factory shutdowns and quarantine measures are spreading across the globe, restricting movement and business activities. The economic impact is mounting, and the top economies of the world are expected to enter a global recession.

To provide a perspective on various scenarios, we assume the factors that would impact the economic recovery from the global recession.

In a realistic scenario, we assume that possible health and economic responses would help economies bounce back partially in 2021 Q4. In Q1, Q2, and Q3 of 2021, countries have ramped up testing capacities and critical healthcare services and increased the availability of financial aid. This will help contain the spread of the virus. The economic recession will last until 2022 Q2; a slow recovery will be seen from 2022 Q3. This is expected to be a 'U-shaped' economic recovery, with a gradual improvement in the economy.

In an optimistic scenario, the economy is expected to recover much faster and continue from 2021 Q4. A quick return to the routine will be seen after the infection curve flattens toward the end of 2021 Q3 due to time-bound and efficient measures related to health and the economy. In this scenario, most economies would experience a mild recession, and the growth would accelerate in 2022. This is expected to be a 'V-shaped' economic recovery, with a strong and fast improvement in the economy.

In a pessimistic scenario, we assume that all health and economic measures would neither help control the spread of the virus nor lead to economic recovery. Most economies would experience unprecedented economic losses, along with social and political turmoil. In this scenario, most economies will face a recession till 2021 Q3, and moderate recovery will begin in 2021 Q4. This is expected to be an 'L-shaped' economic recovery, with countries taking longer to return to their pre-crisis levels.

Similar scenario-based approaches are considered in this report to assess the impact of COVID-19 and provide market forecasts.

1.4 OBJECTIVES OF THE STUDY

- To define, describe, and predict the advanced analytics market by component (solutions and services), type, business function, organization size, deployment mode, vertical, and region
- To provide detailed information related to major factors (drivers, restraints, opportunities, and industry-specific challenges) influencing the market growth
- To analyze opportunities in the market and provide details of the competitive landscape for stakeholders and market leaders
- To forecast the market size of segments for five main regions: North America, Europe, Asia Pacific (APAC), Middle East and Africa (MEA), and Latin America
- To profile key players and comprehensively analyze their market rankings and core competencies
- To analyze competitive developments, such as partnerships, new product launches, and mergers and acquisitions, in the advanced analytics market
- To analyze the impact of the COVID-19 pandemic on the advanced analytics market

1.5 MARKET DEFINITION

According to TIBCO Software, advanced analytics employs predictive modeling, statistical methods, Machine Learning (ML), and process automation techniques beyond the capacities of traditional Business Intelligence (BI) tools to analyze data or business information. It leverages data science that involves mature methods of analysis to project future trends and anticipate the likelihood of potential events. Advanced analytics solutions provide the ability to forecast future trends or outcomes for a deeper understanding of the business and offer a wider set of capabilities to deal with challenges that traditional rearview BI cannot, enabling stronger strategic decision-making for the future.

As per Oracle, advanced analytics describes the analysis of data using complex techniques to forecast trends and predict events. It uses state-of-the-art tools, such as ML and AI, along with various statistical analyses and algorithms, to examine large data sets. The goal is to identify patterns and discover deeper insights that go beyond traditional BI. Descriptive, diagnostic, predictive and prescriptive analytics, as well as data mining and other advanced, high-level data science methods, all fall under the advanced analytics umbrella.

According to RapidMiner, advanced analytics is about optimizing, correlating, and predicting the next best action or the next most likely action. Advanced analytics utilizes a wide variety of techniques drawn from statistical methods algorithms, as well as computer science to identify relationships and forecast trends.

1.5.1 INCLUSIONS AND EXCLUSIONS

CATEGORY	INCLUSION	EXCLUSION
Advanced Analytics Market	▪ The Business-to-Business (B2B) approach is considered for revenue estimation. ▪ Vendors offering solutions and services are included in this study.	The Business-to-Consumer (B2C) approach is not considered for calculating revenue.
Component	The report includes solutions and services offered on-premises and cloud.	The hardware related to advanced analytics is not included in the scope of the report.
Solution	The report includes advanced analytics solution. The solution includes software tools and platform. The software covers Application Programming Interfaces (APIs) and Software Development Kits (SDKs) specific to advanced analytics.	▪ The report does not include advanced analytics solutions offered by Managed Service Providers (MSPs). ▪ Open-source platforms have not been considered; however, open-source platforms that are integrated with supporting technologies and imply cost toward developing solutions have been considered.
Service	▪ The report includes managed and professional services. ▪ Professional services include consulting, deployment and integration, and support and maintenance services.	▪ Consulting services mentioned in the report are pre-deployment services. Post-deployment services are not included in the current scope. ▪ Services such as IT outsourcing, application development, and Business Process Outsourcing (BPO) are not included in the study.
Vertical	Verticals covered in the report include BFSI, retail and consumer goods, healthcare and life sciences, energy and utilities, IT and telecom, government and defense, transportation and logistics, and other verticals (education, travel and hospitality, and media and entertainment).	Verticals other than those mentioned in the report are not included in the scope of the report.

Source: MarketsandMarkets Analysis

1.6 MARKET SCOPE

The study provides an in-depth analysis of the advanced analytics market based on contemporary market trends and developments, and its potential growth from 2021 to 2026. It includes detailed market trends, competitive landscape, market size, forecasts, and analysis of key advanced analytics solution providers.

1.6.1 MARKET SEGMENTATION

ADVANCED ANALYTICS MARKET

BY COMPONENT

- Solution
- Services
- Professional Services
  - Training and Consulting
  - Support and Maintenance
  - System Integration and Implementation
- Managed Services

BY ORGANIZATION SIZE

- SMEs
- Large Enterprises

BY DEPLOYMENT MODE

- On-premises
- Cloud

BY VERTICAL

- BFSI
- Government and Defense
- IT and Telecom
- Retail and Consumer Goods
- Travel and Hospitality
- Healthcare and Life Sciences
- Transportation and Logistics
- Media and Entertainment
- Other Verticals

BY BUSINESS FUNCTION

- Finance
- Sales and Marketing
- HR
- Operations
- Supply Chain

BY TYPE

- Big Data Analytics
- Customer Analytics
- Risk Analytics
- Predictive Analytics
- Prescriptive Analytics
- Statistical Analytics
- Other Types

BY REGION

- North America
- Europe
- APAC
- MEA
- Latin America

Other types include augmented analytics, multimedia analytics, and social and video analytics

Other verticals include education, travel and hospitality, and energy and utilities

Source: MarketsandMarkets Analysis

1.6.2 REGIONS COVERED

NORTH AMERICA

- US
- Canada

EUROPE

- UK
- Germany
- France
- Rest of Europe*

APAC

- China
- India
- Japan
- Rest of APAC**

MEA

- KSA
- UAE
- South Africa
- Rest of MEA***

LATIN AMERICA

- Brazil
- Mexico
- Rest of Latin America****

**Rest of Europe includes Italy, Sweden, and Denmark*

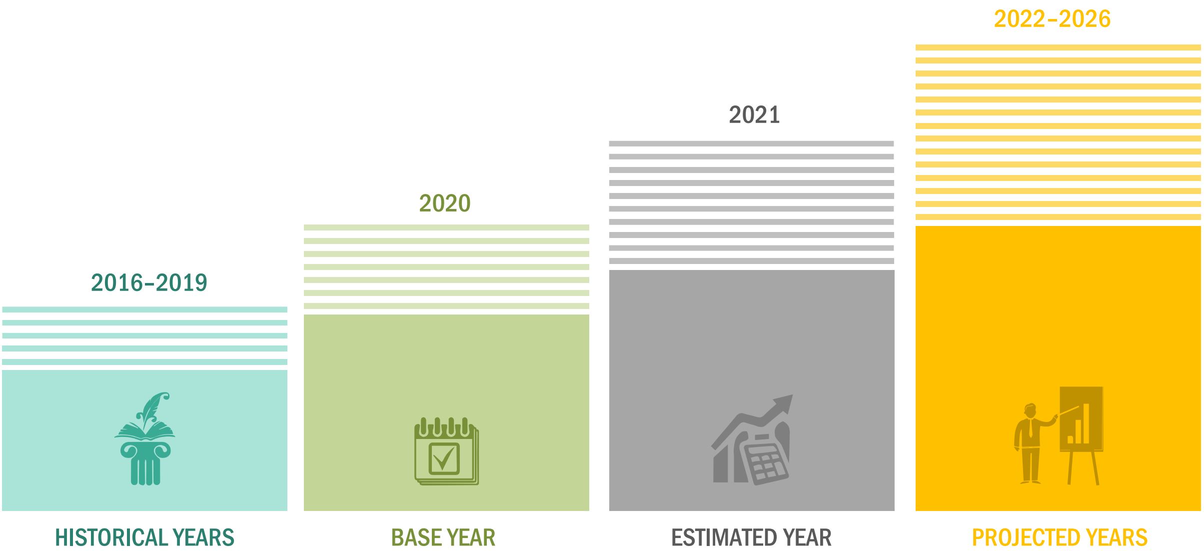
***Rest of APAC includes ANZ, South Korea, Singapore, and Malaysia*

****Rest of MEA includes Israel, Qatar, Turkey, and Kuwait*

*****Rest of Latin America includes Argentina, Colombia, Peru, and Chile*

Source: MarketsandMarkets Analysis

1.6.3 YEARS CONSIDERED FOR THE STUDY



Note: The forecast period is 2021–2026

Source: MarketsandMarkets Analysis

1.7 CURRENCY CONSIDERED

The base currency used in the report is the United States Dollar (USD), with the market size indicated in USD million/billion.

- The revenue figures were sourced from the annual reports of companies.
- The average annual currency conversion rate was used for companies that reported their revenues in the annual reports in currencies other than the USD.

TABLE 1 UNITED STATES DOLLAR EXCHANGE RATE, 2018–2020

Currency	2018	2019	2020
Euro	0.848	0.893	0.877

Source: The US Internal Revenue Service (IRS)

1.8 STAKEHOLDERS

- Advanced analytics solution vendors
- Managed service providers
- Support and maintenance service providers
- System Integrators (SIs)/migration service providers
- Distributors and Value-added Resellers (VARs)
- Independent Software Vendors (ISV)
- Cloud Service Providers
- Research organizations
- Analytics solutions vendors
- Marketing analytics executives
- Predictive analytics providers
- Third-party providers
- Technology providers

1.9 SUMMARY OF CHANGES

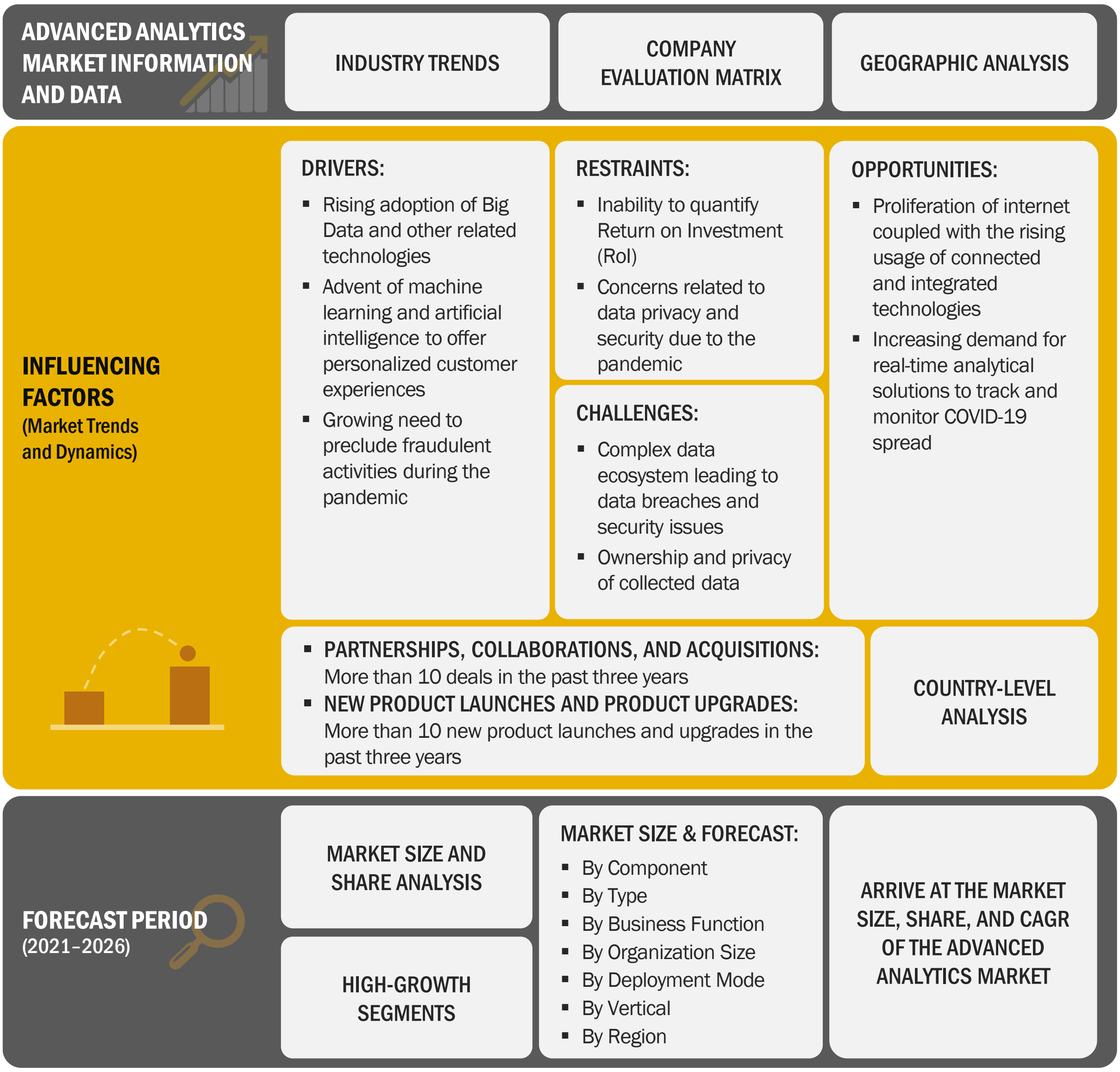
- The study is segmented by component (solution and services), type, business function, organization size, deployment mode, vertical, and region. The business function segment has been newly added and service and type segment have been updated as per growth and development witnessed by the advanced analytics vendors in recent years.
- The new research study includes the impact of COVID-19 on the overall advanced analytics market as well as on segments and regions.
- The new study also comprises operational drivers for adoption in each segment and region.
- The new market study includes the breakup of the organization size segment into Small and Medium-sized Enterprises (SMEs) and large enterprises.
- The new research study features 31 players as compared to 12 in the previous version.
- Updated financial information/product portfolio of players: The new edition of the report provides updated financial information in the context of the advanced analytics market till 2020–2021 for each listed company in graphical representation.
- The new research study includes the updated market developments of profiled players, including those from 2019 to 2021.
- The new study includes quantitative data for historical years (2016–2020), base year (2020), and forecast years (2021–2026).
- The new study also includes quantitative data of adjacent markets.

2 RESEARCH METHODOLOGY

2.1 RESEARCH DATA

The research study for the advanced analytics market involved extensive secondary sources, directories, and several journals, including International Journal of Data Analytics (IJDA), Journal of Big Data Analytics in Transportation, and International Journal of Business Analytics (IJBAN). Primary sources were industry experts from the core and related industries, preferred advanced analytics providers, third-party service providers, consulting service providers, end users, and other commercial enterprises. In-depth interviews were conducted with various primary respondents, including key industry participants and subject matter experts, to obtain and verify critical qualitative and quantitative information, and assess the market's prospects. The following figure highlights the market research methodology applied to make the advanced analytics market report:

FIGURE 6 ADVANCED ANALYTICS MARKET: RESEARCH DESIGN



Source: MarketsandMarkets Analysis

2.1.1 SECONDARY DATA

The market size of companies offering advanced analytics solutions and services was arrived at based on secondary data available through paid and unpaid sources. It was also arrived at by analyzing the product portfolios of major companies and rating the companies based on their performance and quality.

In the secondary research process, various sources were referred to, for identifying and collecting information for this study. Secondary sources included annual reports, press releases, and investor presentations of companies; white papers, journals, and certified publications; and articles from recognized authors, directories, and databases. The data was also collected from other secondary sources, such as journals, government websites, blogs, and vendors' websites. Advanced analytics spending of various countries was extracted from the respective sources. Secondary research was used to obtain key information related to the industry's value chain and supply chain to identify key players based on solutions, services, market classification, and segmentation according to offerings of major players, industry trends related to solutions, services, deployment modes, functionalities, applications, verticals, and regions, and key developments from both market-and technology-oriented perspectives.

2.1.2 PRIMARY DATA

In the primary research process, various primary sources from both supply and demand sides were interviewed to obtain qualitative and quantitative information on the market. The primary sources from the supply side included various industry experts, including Chief Experience Officers (CXOs); Vice Presidents (VPs); directors from business development, marketing, and advanced analytics expertise; related key executives from advanced analytics solution vendors, SIs, professional service providers, and industry associations; and key opinion leaders.

Primary interviews were conducted to gather insights, such as market statistics, revenue data collected from solutions and services, market breakups, market size estimations, market forecasts, and data triangulation. Primary research also helped in understanding various trends related to technologies, applications, deployments, and regions. Stakeholders from the demand side, such as Chief Information Officers (CIOs), Chief Technology Officers (CTOs), Chief Strategy Officers (CSOs), and end users using advanced analytics solutions, were interviewed to understand the buyer's perspective on suppliers, products, service providers, and their current usage of advanced analytics solutions and services, which would impact the overall advanced analytics market.

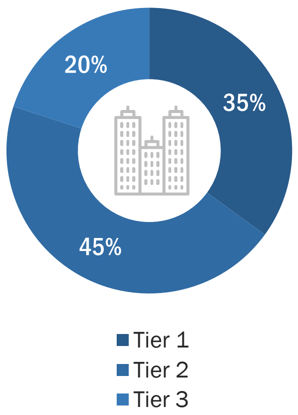
TABLE 2 PRIMARY INTERVIEWS

ADVANCED ANALYTICS SOLUTION AND SERVICE PROVIDERS	
SAS Institute	Provana
Cypher Analytica	Kwantics
VoiceBase	CallMiner

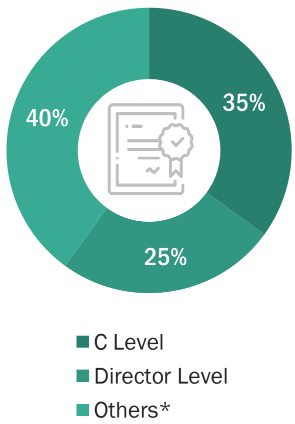
Source: MarketsandMarkets Analysis

2.1.2.1 Breakup of primary profiles

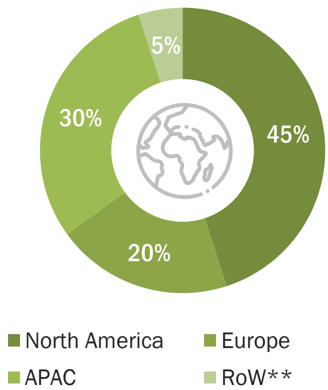
BY COMPANY



BY DESIGNATION



BY REGION



**Others include sales managers, marketing managers, and product managers*

Note: Tier 1 companies' revenue is more than USD 10 billion; Tier 2 companies 'revenue ranges in between USD 1 and 10 billion; and Tier 3 companies' revenue ranges in between USD 500 million and USD 1 billion

Source: Industry Experts

2.1.2.2 Key industry insights

Advance Analytics is the autonomous or semi-autonomous examination of data or content using sophisticated techniques and tools, typically beyond those of traditional business intelligence (BI), to discover deeper insights, make predictions, or generate recommendations.

- Product Head
- Leading Advanced Analytics Service Provider

Advanced analytics specifically focuses more on forecasting, using the data it gathers to find trends that can be used to determine what might happen in the future.

- CEO
- Leading Advanced Analytics Vendor

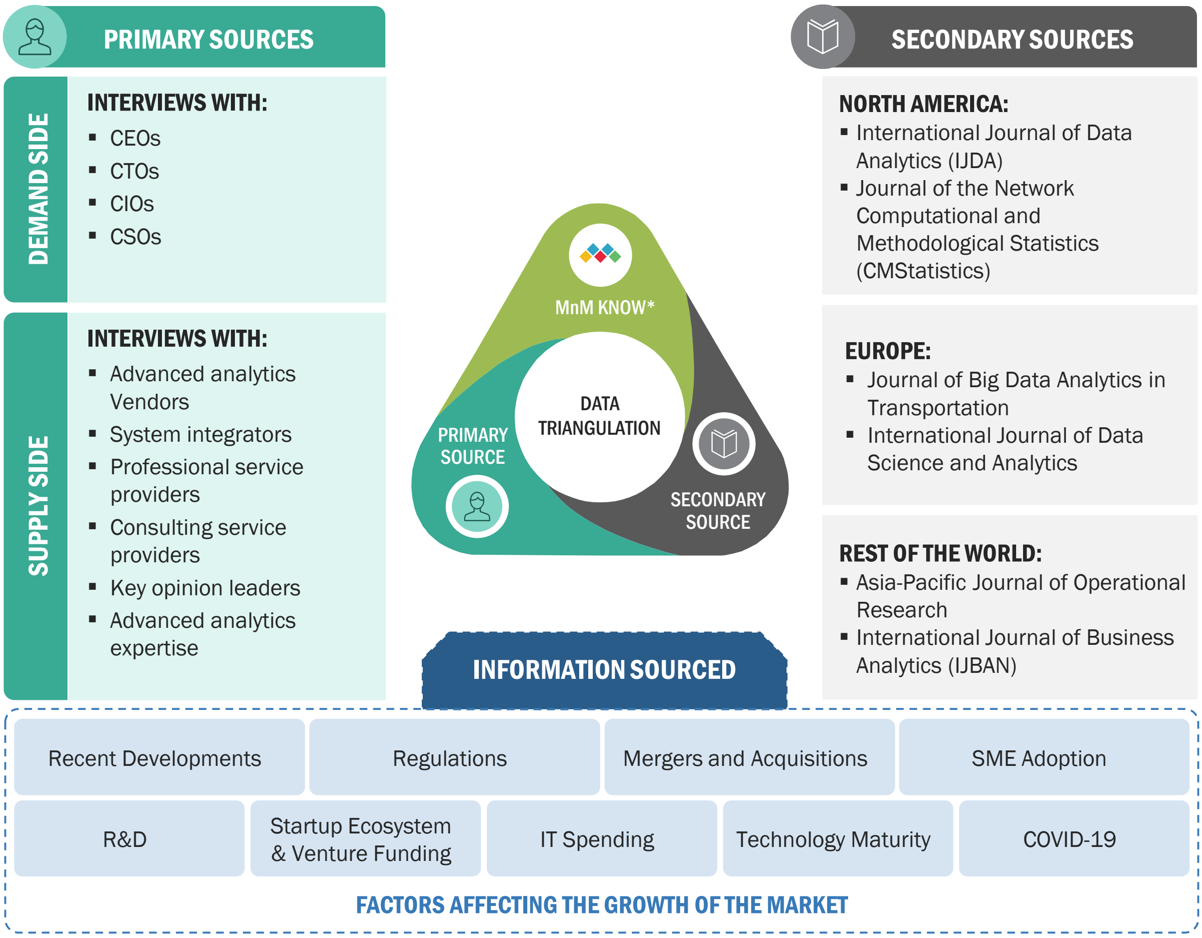
However, even smaller companies can leverage advanced analytics to enhance decision making and boost competitive advantage now as well as for years to come. After all, advanced analytics use cases and applications are incredibly vast.

- Cofounder
- Leading Advanced Analytics Solution Provider

Source: Industry Experts

2.2 MARKET BREAKUP AND DATA TRIANGULATION

FIGURE 7 DATA TRIANGULATION



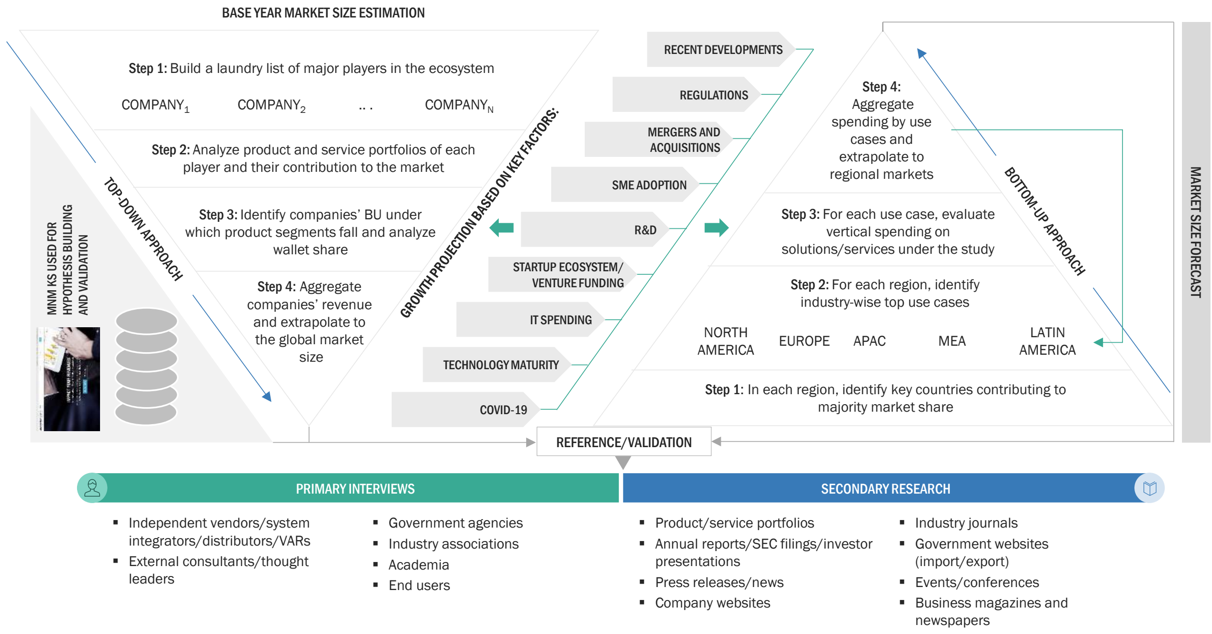
*MnM KNOW stands for MarketsandMarkets' 'Knowledge Asset Management' framework. In this context, it stands for the existing market research knowledge repository of over 5,000 granular markets, our flagship competitive intelligence and market research platform 'Knowledge Store,' subject matter experts, and independent consultants. MnM KNOW acts as an independent source that helps us validate information gathered from primary and secondary sources. Research studies referred from MarketsandMarkets' repository included the speech analytics market and predictive analytics market.

Source: MarketsandMarkets Analysis

2.3 MARKET SIZE ESTIMATION

Multiple approaches were adopted for estimating and forecasting the advanced analytics market. The first approach involves estimating the market size by summation of companies' revenue generated through the sale of solutions and services.

FIGURE 8 ADVANCED ANALYTICS MARKET: TOP-DOWN AND BOTTOM-UP APPROACHES



Source: MarketsandMarkets Analysis

2.3.1 TOP-DOWN APPROACH

In the top-down approach, an exhaustive list of all vendors offering solutions and services in the advanced analytics market was prepared. The revenue contribution of the market vendors was estimated through annual reports, press releases, funding, investor presentations, paid databases, and primary interviews. Each vendor's offerings were evaluated based on the breadth of solutions and services, deployment modes, applications, and verticals. The aggregate of all the companies' revenue was extrapolated to reach the overall market size. Each subsegment was studied and analyzed for its global market size and regional penetration. The markets were triangulated through both primary and secondary research. The primary procedure included extensive interviews for key insights from industry leaders, such as CIOs, CEOs, VPs, directors, and marketing executives. The market numbers were further triangulated with the existing MarketsandMarkets' repository for validation.

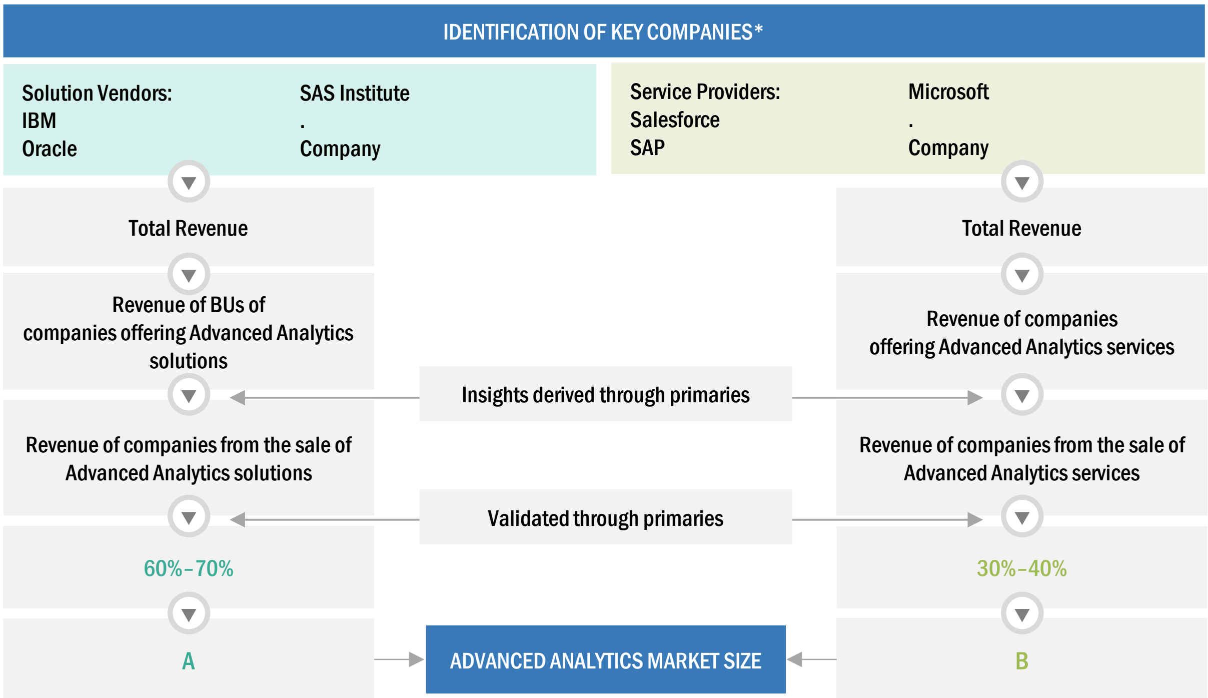
The list of vendors considered for estimating the market size is not limited to the vendors profiled in the report. However, MarketsandMarkets prepared a laundry list of vendors offering advanced analytics solutions and services, and mapped their products related to the advanced analytics market to identify major vendors operating in the market.

2.3.2 BOTTOM-UP APPROACH

In the bottom-up approach, the adoption rate of advanced analytics solutions and services among different end users in key countries with respect to their regions contributing the most to the market share was identified. For cross-validation, the adoption of advanced analytics solutions and services among industries, along with different use cases with respect to their regions, was identified and extrapolated. Weightage was given to use cases identified in different regions for the market size calculation.

Based on the market numbers, the regional split was determined by primary and secondary sources. The procedure included the analysis of the advanced analytics market's regional penetration. Based on secondary research, the regional spending on Information and Communications Technology (ICT), socio-economic analysis of each country, strategic vendor analysis of major advanced analytics providers, and organic and inorganic business development activities of regional and global players were estimated. With the data triangulation procedure and data validation through primaries, the exact values of the overall advanced analytics market size and segments' size were determined and confirmed using the study.

FIGURE 9 MARKET SIZE ESTIMATION METHODOLOGY - APPROACH 1 (SUPPLY-SIDE): REVENUE FROM SOLUTIONS/SERVICES OF THE ADVANCED ANALYTICS MARKET



**Key companies should contribute to more than 50% of the total market*

Source: MarketsandMarkets Analysis

FIGURE 10 MARKET SIZE ESTIMATION METHODOLOGY- APPROACH 2, BOTTOM-UP (SUPPLY-SIDE): COLLECTIVE REVENUE FROM ALL SOLUTIONS/SERVICES OF THE ADVANCED ANALYTICS MARKET

**Key companies should contribute to more than 50% of the total market*

Source: MarketsandMarkets Analysis

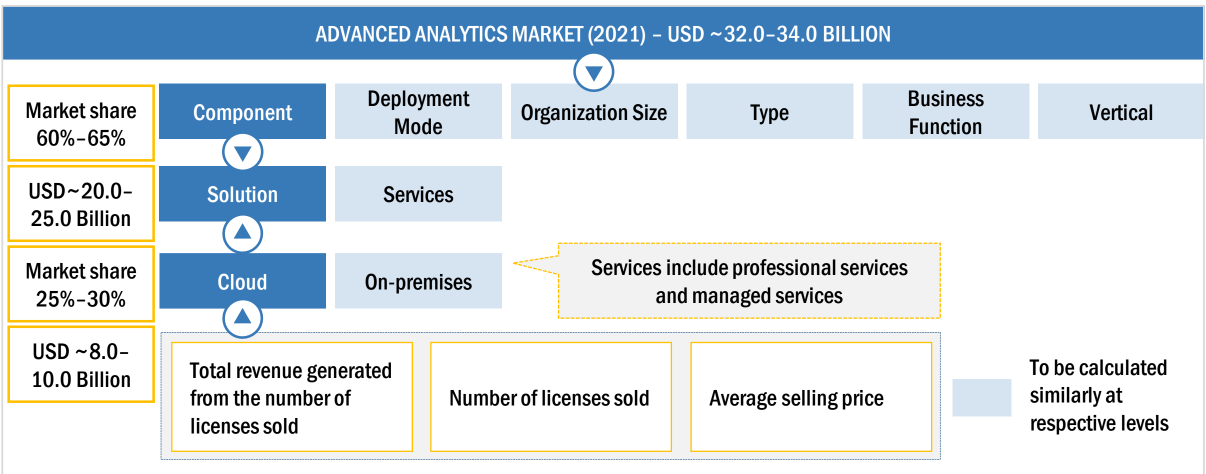
In this approach for market estimation, key advanced analytics solution and service vendors such as IBM (US), Oracle (US), SAS Institute (US), SAP (Germany), FICO (US), KNIME (Switzerland), Microsoft (US), Altair (US), RapidMiner (US), AWS (US), Salesforce (US), TIBCO Software (US), Alteryx (US), Teradata (US), Adobe (US), Absolutdata (US), Moody's Analytics (US), Qlik (US), Databricks (US), Dataiku (US), Kinetica (US), H2O.ai (US), Domino Data Lab (US), DataRobot (US), DataChat (US), Imply (US), Promethium (US), Siren (Ireland), Tellius (US), SOTA Solutions (Germany), and Vanti Analytics (Israel) were identified. These vendors contribute nearly 45%–55% to the global advanced analytics market. The market is fragmented due to the presence of several vendors. After confirming these companies through primary interviews with industry experts, their total revenue through annual reports, Securities and Exchange Commission (SEC) filings, and paid databases was estimated. The revenue pertaining to Business Units (BUs) that offer advanced analytics solutions was identified through similar sources. Then through primaries, the data of revenue generated from specific advanced analytics solutions were collected. The collective revenue of key companies that offer advanced analytics solutions comprises 40%–50% of the market, which was again confirmed through primary interviews with industry experts.

FIGURE 11 MARKET SIZE ESTIMATION METHODOLOGY-APPROACH 3, BOTTOM-UP (SUPPLY-SIDE): COLLECTIVE REVENUE FROM ALL SOLUTIONS/SERVICES OF THE ADVANCED ANALYTICS MARKET

STEP 1	STEP 2	STEP 3	STEP 4	STEP 5
Identification of 30-35 major Advanced analytics companies, such as IBM, Oracle, SAS Institute, SAP, FICO, KNIME, Microsoft, Altair, RapidMiner, AWS, Salesforce, TIBCO Software, Alteryx, Teradata, Adobe, Absolutdata Analytics, Moody's Analytics, Qlik, Databricks, and Dataiku	Estimation of top players' revenue from BUs offering advanced analytics solutions and services	Identification of the revenue contribution of advanced analytics solutions and services to the relevant BU/BUs	Calculation of the total market size of advanced analytics solutions and services using the combined share of top companies and their revenues	Revalidation of the total market size and market share estimations with primary respondents
KEY SOURCE OF INFORMATION				
Secondary sources such as company financials, press releases, paid databases (Factiva), and primary research	Secondary sources such as company financials, press releases, paid databases (Factiva), and primary research	Primary research (CEOs, VPs, marketing directors, and technology and innovation directors, among other stakeholders)	MarketsandMarkets calculations, company websites, and annual reports	Primary research (directors - product development, VPs – operations, warehouse managers, COOs, and CSOs)

Source: MarketsandMarkets Analysis

FIGURE 12 MARKET SIZE ESTIMATION METHODOLOGY-APPROACH 4, BOTTOM-UP (DEMAND-SIDE): SHARE OF ADVANCED ANALYTICS THROUGH OVERALL ADVANCED ANALYTICS SPENDING



Source: MarketsandMarkets Analysis

Top-down and bottom-up approaches were used to estimate and validate the size of the advanced analytics market and various other dependent subsegments. The research methodology used to estimate the market size included the following details:

- Key market players were not limited to IBM (US), Oracle (US), SAS Institute (US), SAP (Germany), FICO (US), KNIME (Switzerland), Microsoft (US), Altair (US), RapidMiner (US), AWS (US), Salesforce (US), TIBCO Software (US), Alteryx (US), Teradata (US), Adobe (US), Absolutdata (US), Moody's Analytics (US), Qlik (US), Databricks (US), Dataiku (US), Kinetica (US), H2O.ai (US), Domino Data Lab (US), DataRobot (US), DataChat (US), Imply (US), Promethium (US), Siren (Ireland), Tellius (US), SOTA Solutions (Germany), and Vanti Analytics (Israel).
- The market players were identified through extensive secondary research, and their revenue contribution in respective regions was determined through primary and secondary research.
- The entire procedure included studying annual and financial reports of top market players and extensive interviews for key insights from industry leaders, such as CEOs, VPs, directors, and marketing executives.
- All percentage splits and breakups were determined using secondary sources and verified through primary sources.

All the possible parameters that affect the market covered in the research study have been accounted for, viewed in extensive detail, verified through primary research, and analyzed to get the final quantitative and qualitative data. The data is consolidated and added with detailed inputs and analysis from MarketsandMarkets.

2.4 MARKET FORECAST

Key factors that have a direct or indirect impact on the market were analyzed to estimate the overall trend of the market during the forecast period. Each factor was analyzed on a scale of 1–3, 1 being the lowest and 3 being the highest. After analyzing each factor and its impact on the market, various forecasting techniques were used to arrive at the market size.

TABLE 3 FACTOR ANALYSIS

FACTOR	INFERENCE	IMPACT
Recent Developments	Major vendors account for 30%–40% of the developments in the market.	Medium
Regulations	There are various existing regulations related to advanced analytics which significantly affect the market. Hence, the factor of regulations has been considered as high for the advanced analytics market.	High
SME Adoption	Advanced analytics solution and services are witnessing increased adoption among Small and Medium-sized Enterprises (SMEs), followed by large enterprises across the globe.	Medium
R&D Spending	The players operating in the advanced analytics market are allocating a significant part of their revenues for R&D activities to innovate their solutions and meet the emerging demand from customers.	Medium
Startup Ecosystem/Venture Funding	Startups account for more than 40% of the advanced analytics market; hence, the growth of startups in the advanced analytics market is high. Startups offering advanced analytics solution have received significant investments and are coming up with innovative advanced analytics solutions.	High

IT Spending	The top three verticals in the advanced analytics market are contributing between 40% and 60% to the overall market. Various stakeholders across verticals in different regions are looking at adopting advanced analytics solutions to gain a competitive advantage in the global market.	Medium
Technology Maturity	Analytics and AI technology are nearing the maturity phase.	Medium
Government Initiatives/Projects	There are various regional and global initiatives related to advanced analytics in the market, but they indirectly affect the adoption of the advanced analytics market.	Medium
Merger and Acquisition (M&A)	In the last year, there were more than five acquisition-related developments in the advanced analytics market. Hence, the M&A strategy is considered to have a medium impact on the advanced analytics market.	Medium
COVID-19	The recent COVID-19 pandemic is projected to have a minimal impact on the growth of the advanced analytics market. The adoption of advanced analytics solutions has increased among the customer-facing verticals such as BFSI, travel and hospitality, healthcare and life sciences, and retail verticals. The companies are still leveraging advanced analytics solutions and services for enhanced customer experience during the COVID-19 pandemic.	Low

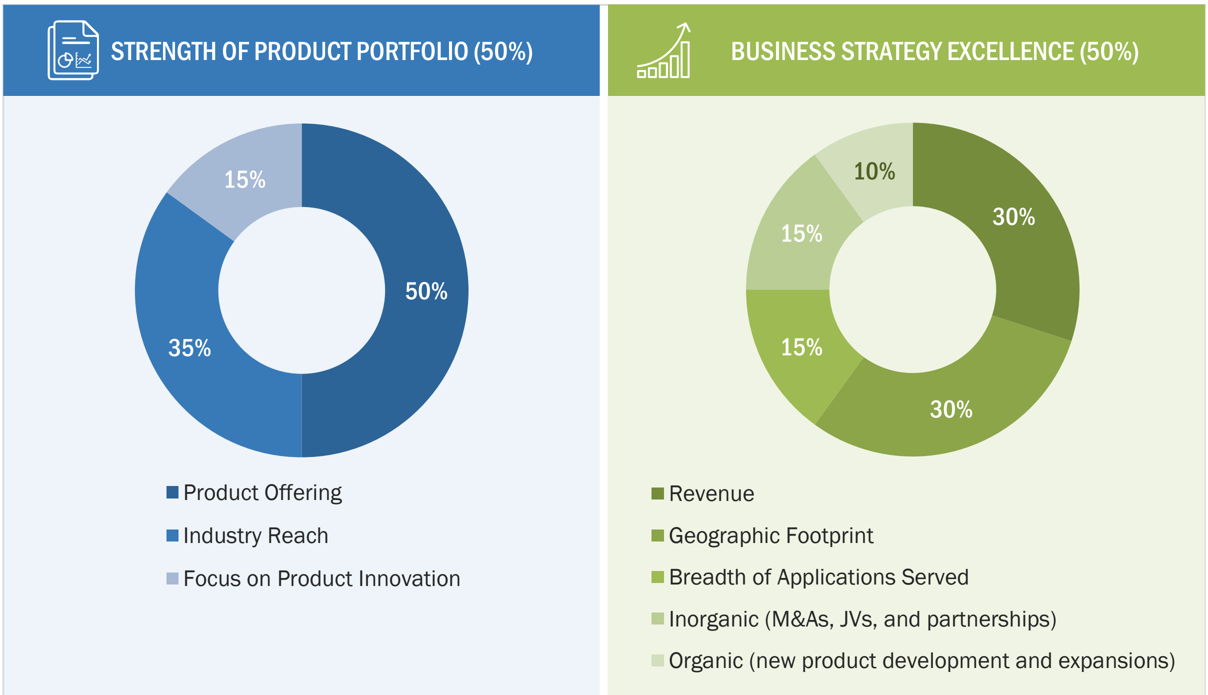
Source: MarketsandMarkets Analysis

2.5 COMPANY EVALUATION MATRIX METHODOLOGY

The company evaluation matrix reviews major players offering advanced analytics solution and services, and outlines findings and analysis based on their core competencies. Vendor evaluations are based on two broad categories: strength of product portfolio and business strategy excellence. Each category has various criteria based on which vendors are evaluated.

The figure given below depicts the criteria and weightage used for scoring the vendors in the advanced analytics market:

FIGURE 13 COMPANY EVALUATION MATRIX: CRITERIA WEIGHTAGE



Source: MarketsandMarkets Analysis

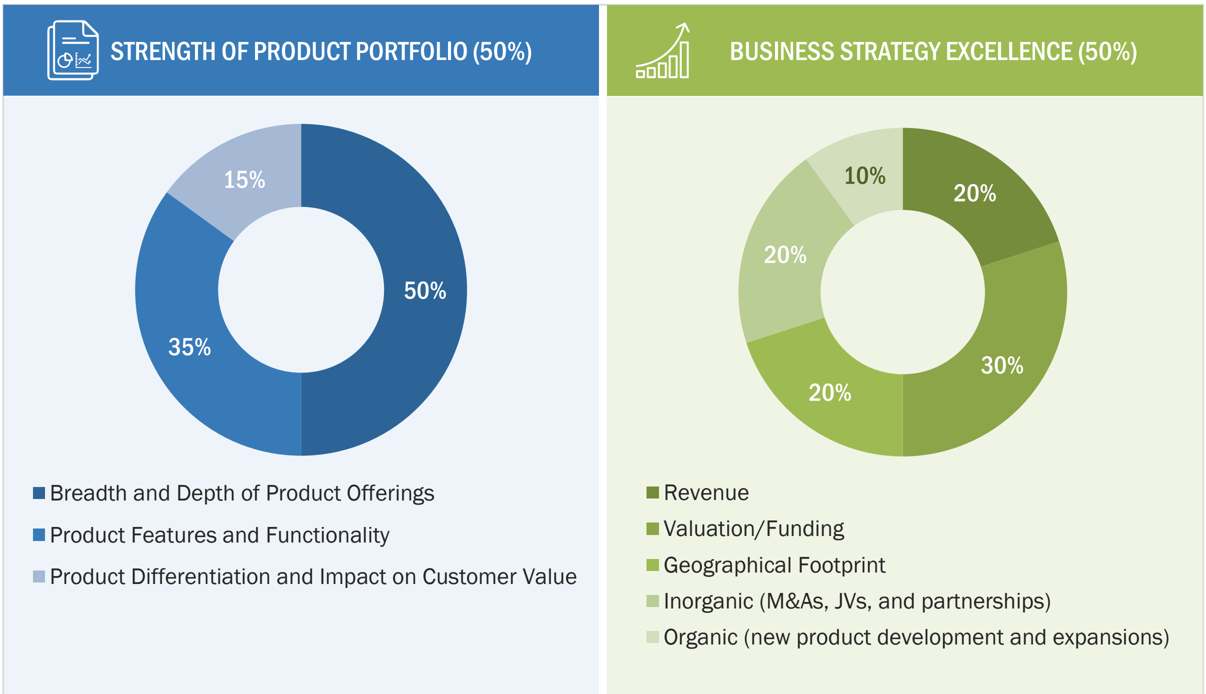
Extensive secondary and primary research was carried out to gather key information related to vendors' strength of product portfolio and business strategy excellence based on a questionnaire. A scale of 0–10 was formulated for each question, after which each criterion for every vendor was scored based on the collected information. After completing the data gathering and verification process, the scores and weightage for shortlisted vendors against each parameter were finalized. A comparison scorecard was prepared after evaluating all vendors, and each vendor was placed in the company evaluation matrix based on its strength of product portfolio and business strategy excellence scores.

2.6 STARTUP/SME EVALUATION MATRIX METHODOLOGY

The startup/SMEs evaluation matrix reviews startups/SMEs offering advanced analytics solutions and services. Ten startups/SMEs founded after 2012 were considered for evaluation in the startup/SME evaluation matrix.

The figure below depicts the criteria and weightage used for scoring startups/SMEs in the advanced analytics market.

FIGURE 14 STARTUP/SME EVALUATION MATRIX: CRITERIA WEIGHTAGE



Source: MarketsandMarkets Analysis

2.7 ASSUMPTIONS FOR THE STUDY

The following assumptions were taken into consideration to complete the overall market estimation of the advanced analytics market:

FACTOR	ASSUMPTION	IMPACT
Economy	According to an update released by the International Monetary Fund (IMF) in January 2021, the global economy grew 5.5% this year, a 0.3% increase from October's forecasts.	▪ A positive global economy is projected to continue until the end of 2026. ▪ In developed countries, a modest and uneven recovery is expected to continue with the gradual narrowing of output gaps.
Exchange Rate	MarketsandMarkets assumes that dollar fluctuations would not be severe enough to affect the forecast period to a significant extent.	A fall in the dollar value is projected to create revenue growth opportunities for US multinationals, whereas stability in the dollar value is projected to stabilize the price of imports and exports.
Financial Statistics	MarketsandMarkets extracts all the revenue and financial insights from company websites or annual reports.	It ensures the authenticity of the financials mentioned in the report.
Solution	Solution spending has increased over the last two years, and organizations invest in adopting solutions to achieve digital transformation.	Organizations are expected to increase their spending on IT and business services.
Service	▪ The growth of the services segment has been steady. Government austerity programs have included direct reductions in the level of IT services spending. Cannibalization from the cloud is still projected to be a drag on the overall growth. ▪ On the upside, it is expected that some downstream spending from solutions might happen, but lesser than previous cycles.	Solution spending would increase the adoption of services.
COVID-19	▪ MarketsandMarkets assumes a short-term effect on the technology sector, affecting the electronics value chain and initiating an inflationary risk on products. ▪ MarketsandMarkets assumes delays in scheduled new projects and reduced discretionary spending by enterprises.	The impact of COVID-19 is believed to be short-term and may have a minimal effect on the forecast period.

Market Estimation	▪ The pricing trend is assumed to vary over time. ▪ All the forecasts are made with the standard assumption that the accepted currency is USD. ▪ For the conversion of various currencies to USD, average historical exchange rates are used according to the year specified. For all the historical and current exchange rates required for calculations and currency conversions, the US Internal Revenue Service's website is used. ▪ All the forecasts are made under the standard assumption that the globally accepted currency USD remains constant during the next five years. ▪ Vendor-side analysis: The market size estimates of associated solutions and services are factored in from the vendor side by assuming an average of licensing and subscription-based models of leading and innovative vendors in the market. ▪ Demand/end-user analysis: End users operating in verticals across regions are analyzed in terms of market spending on advanced analytics solutions based on some of the key use cases. These factors for the advanced analytics industry per region are separately analyzed, and the average spending was extrapolated with an approximation based on assumed weightage. This factor is derived by averaging various market influencers, including recent developments, regulations, mergers and acquisitions, enterprise/SME adoption, start-up ecosystem, IT spending, technology propensity and maturity, use cases, and the estimated number of organizations per region.	NA

Source: MarketsandMarkets Analysis

2.8 LIMITATIONS OF THE STUDY

The following limitations were taken into consideration to complete the overall market engineering process of the advanced analytics market:

METHODOLOGY-RELATED LIMITATIONS

- The report mainly focuses on offerings of major vendors in the advanced analytics market. The market size estimation is done based on the approximation of the market ranking/contribution of these vendors, along with corresponding investments by various regulatory bodies in the market.
- The overall market size is arrived at using a holistic and systematic approach.

SCOPE-RELATED LIMITATIONS

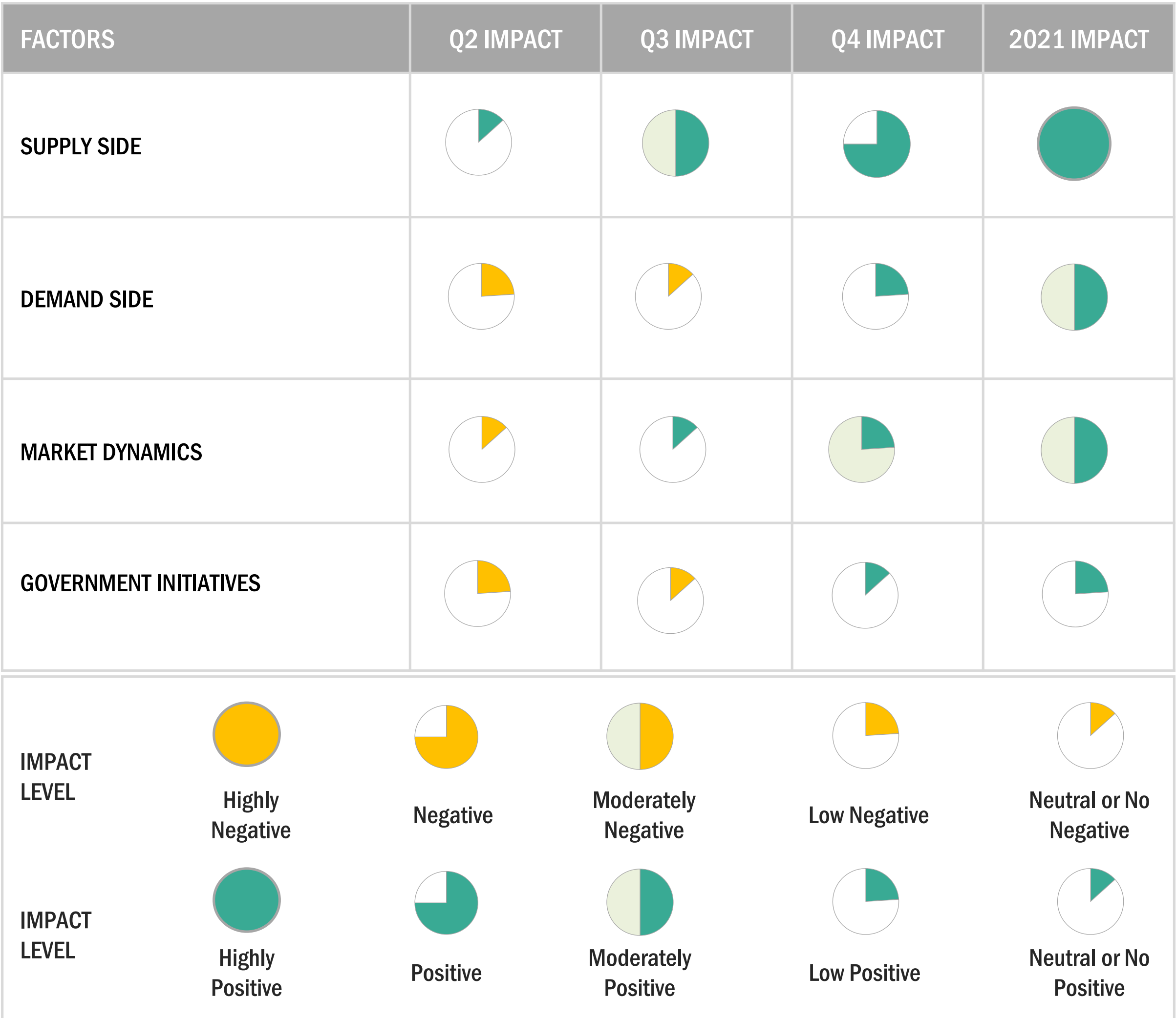
- The report is focused on the B2B business model; the B2C model is not considered in the report.
- The hardware component in the advanced analytics ecosystem is not considered while sighting the market.
- Advanced analytics solutions offered by MSPs are not considered for forecasting the market size.

Source: MarketsandMarkets Analysis

2.9 IMPLICATIONS OF COVID-19 ON THE ADVANCED ANALYTICS MARKET

The impact of COVID-19 on the advanced analytics market was evaluated globally. The pandemic led to changes in the entire business equation across organizations. Several factors were considered while computing the market size during the forecast period.

FIGURE 15 QUARTERLY IMPACT OF COVID-19 DURING 2020–2021



Note: COVID-19 related inputs have been included in various sections of the report to showcase the impact of COVID-19 on the advanced analytics market

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

2.9.1 IMPACT OF COVID-19 PANDEMIC:

SUPPLY SIDE

- New practices, such as work from home and social distancing, have led to the requirement of advanced analytics solutions and services, and the development of digital infrastructures for large-scale technology deployments.
- The COVID-19 pandemic in 2020 brought accelerating changes in consumer preferences and behaviors and put pressure on brands to keep pace and provide a personalized customer experience.
- Organizations that have already invested in cloud technology, advanced data solutions, and mobile employee technology before COVID-19 have been able to navigate the impact more quickly and confidently.
- Companies have started designing an end-to-end experience and touchpoint strategy post-COVID-19 to remain competitive and provide the best customer experience.

DEMAND SIDE

- Enterprises have witnessed a reduction in their operational spending and are now focusing more on business continuity and sustainability.
- Companies are focused on building real-time channel dashboards to continuously re-evaluate and re-calibrate channel investments to keep ahead of demand and drive profitable growth.
- Core verticals such as BFSI, retail, healthcare and life sciences, and telecom and IT prefer advanced technologies (cloud, AI, big data analytics, and IoT) for accelerating their digital transformation journey.
- Customer behavior is changing at a staggering pace, and digital adoption has become necessary for survival. When the pandemic eventually recedes, sales and service organizations will have to continue to accommodate new attitudes and behaviors to enhance customer experience.
- The pandemic has affected the overall analytics market, but companies are still leveraging advanced analytics solution to gain insights from structured as well as unstructured data.
- Governments, as well as private companies across verticals, are adopting advanced analytics solutions to keep citizens indoors as well as to track them.
- Healthcare providers are leveraging data from countries that were affected earlier by the pandemic for predicting requirements, such as hospital beds, masks, and ventilators. Business retailers are utilizing Point-Of-Sale (POS) data to help distributors in identifying and shipping items most important to their customers.

DYNAMICS

- The COVID-19 crisis has shown that internal data is no longer sufficient to inform and predict customer and employee needs. A combination of internal and external data will be required in the future for making informed business decisions.
- Digital channel usage throughout the pandemic has continued to spike due to which customer experience expectations will continue to rise.
- The pandemic has forced businesses to revive their business strategies, focus more on reducing dependency on an on-premises deployment, and resulted in a shift toward the adoption of cloud-based solutions.

- Organizations have started looking for ways to engage their employees in personalized and adaptive ways to improve workforce productivity and satisfaction.
- Government agencies, healthcare providers, non-profit organizations, and businesses in virtually every industry were quick to adopt the technology as an effective method of responding to large invasions of customer calls while many onshore and offshore call centers closed due to work-at-home policies and social distancing.
- Compliance with the stay-at-home orders was one of the primary behavioral drivers to help contain the spread of the disease as well as economic activity. As a result, agent-based modeling and simulation was one of the primary advanced analytics and AI techniques used to perform scenario analysis.

GOVERNMENT INITIATIVES

- A regular monitoring and remote detection system for individuals will assist in the fast-tracking of suspected COVID-19 cases. Using such systems will generate a huge amount of data, which will provide opportunities for applying big data analytics tools that are likely to improve the level of healthcare services. There is many open-source software such as the big data components for the Apache project, which are designed to operate in a cloud computing and distributed environment to assist in the development of big data-based solutions.
- During the COVID-19 pandemic, the healthcare vertical is under immense pressure to enhance and provide PPE, ventilators, prophylactic, and anti-viral drugs across the world. Healthcare organizations are using advanced technologies, such as analytics, AI, and machine learning to analyze the complex data around COVID-19 to monitor and reduce the impact of the virus.

3 EXECUTIVE SUMMARY

Advanced analytics describes the sophisticated analysis of data using complex techniques to forecast trends and predict events. It uses state-of-the-art tools, such as ML and AI technologies and complex statistical analyses and algorithms to examine the rising big data and identify patterns to discover deeper insights. Advanced analytics uses quantitative and qualitative methods to uncover relationships, trends, correlations, and outliers. Advanced analytics tools cover data mining, ML, cohort analysis, cluster analysis, retention analysis, complex event analysis, predictive analysis, regression analysis, sentiment analysis, and time series analysis.

With huge amounts of data being generated every day, businesses are looking for new ways to take advantage of all that data. It enables companies to optimize their operations and innovate to gain a competitive advantage. With better customer analysis, predictive analytics, and statistical modeling, advanced analytics is helping companies improve decision-making and keep pace with extremely competitive, quick-changing markets.

The major factors driving the growth of the advanced analytics market are the advent of ML and AI to offer personalized customer experiences and the growing need to preclude fraudulent activities during the pandemic. The proliferation of internet coupled with the rising usage of connected and integrated technologies and the increasing demand for real-time analytical solutions to track and monitor COVID-19 spread are the major factors adding value to the advanced analytics offerings, which is expected to provide opportunities for enterprises operating in various verticals in the advanced analytics market. Cloud adoption is said to have increased in recent times because vendors are making use of Software-as-a-Service (SaaS) to deliver cloud-based solutions. Business users are always on the lookout to ensure they are providing the most effective yet economical solutions. Cloud-based solutions provide the ability to outsource the operational IT work to another company. The complex data ecosystem leading to data breaches and security issues and lack of advanced analytical knowledge among the workforce may hinder the overall growth of the advanced analytics market.

The COVID-19 pandemic has accelerated many companies to instigate the use of advanced analytics and AI solutions and implement innovative strategies to helps engage customers through digital channels, manage fragile and complex supply chains, and support workers through disruption to their work and lives. Companies should focus on where they can obtain new insights rather than rely on lagging information. These can come from both new sources of data and using existing data in new ways. For instance, banks that traditionally use credit scores to analyze risk can instead turn to customer-account data, where they might identify gaps in deposits. The crisis has also exposed that one of the most advanced analytics techniques relies on principles of behaviors and patterns repeated periodically. But the patterns exposed are revealed through data and even suggested during identification, which changes the customer’s mind. Many firms that had started benefitting its client with data analytics usage are growing in all departments. The departments most directly affected by the virus see the most incredible growth in marketing, finance, and customer service use cases for small businesses.

The advanced analytics market is expected to witness a slowdown in 2020 due to the global lockdown, which is impacting global manufacturing, and supply chains and logistics. The manufacturing, transportation and logistics, and retail and consumer goods sectors have been most severely affected. The availability of essential items has also been impacted due to the lack of manpower to work on production lines, supply chains, and transportation, even though essential items are exempted from the lockdown. The situation is expected to come under control by early 2021. Analytics have also played a vital role in the determination of fraudulent activities with a system updated with fraudulent activities that help in determining security boundaries for an organization. For instance, IBM Cloud Pak for Security can help you uncover hidden threats and make more informed risk-based decisions. An economic model has also been represented by data scientists to forecast the various insights about customers' probable future behavior, suppliers, and staff is a top priority. These predicted behaviors have helped the business in deriving the consequential financial effect through continuous, rapid decision-making based on accurate, data-driven analytics, and simulations.

Based on regions, North America is estimated to have the largest market size in 2021, followed by Europe; this growth can be attributed to the technological upgradations in these regions. Being an early adopter of technologies, North America is an innovation hub and is expected to present strong opportunities for advanced analytics vendors to expand in this market. However, APAC is projected to grow at the highest Compound Annual Growth Rate (CAGR), due to the rapid rise in technology adoption across verticals for enhancing customers experience and productivity. The adoption of advanced analytics solutions and services is expected to rise in MEA and Latin America due to the growing need to engage customers throughout their entire journey and enhance customer experience.

The global advanced analytics market has been segmented by component, business function, type, deployment mode, organization size, vertical, and region. The component segment is further categorized into solution and services. The services segment is further divided into managed services and professional services. The professional services segment includes training and consulting, system integration and implementation, and support and maintenance. The business function segment is further categorized into sales and marketing, finance, operations, human resources (HR), and supply chain. The type segment includes big data analytics, customer analytics, predictive analytics, risk analytics, prescriptive analytics, statistical analytics, and other type (augmented analytics, multimedia analytics, and social and visual analytics). The deployment mode segment is categorized into on-premises and cloud. Based on organization size, the market is divided into large enterprises and Small and Medium-sized Enterprises (SMEs). The vertical segment includes IT and telecom, transportation and logistics, media and entertainment, retail and consumer goods, BFSI, healthcare and life sciences, government and defense, manufacturing, and other verticals (travel and hospitality, energy and utilities, and education). The market study has also been segmented into five regions: North America, Europe, APAC, MEA, and Latin America.

Some of the major players in the advanced analytics market include International Business Machines Corporation (IBM), Amazon Web Services, Inc. (AWS), SAP SE (SAP), SAS Institute Inc. (SAS Institute), Fair, Isaac and Company (FICO), KNIME AG (KNIME), Microsoft Corporation (Microsoft), Altair Engineering, Inc. (Altair Engineering), RapidMiner, Inc. (RapidMiner), Salesforce.com, inc. (Salesforce), TIBCO Software Inc. (TIBCO Software), Teradata Corporation (Teradata), Adobe Inc. (Adobe), ALTERYX, INC. (Alteryx), Teradata Corporation (Teradata), Adobe Inc. (Adobe), Absolutdata Holdings, Inc. (Absolutdata), Moody's Analytics, Inc. (Moody's Analytics), QlikTech International AB (Qlik), Databricks, Inc. (Databricks), Dataiku Inc (Dataiku), Kinetica DB Inc. (Kinetica), H2O.ai, Inc. (H2O.ai), Domino Data Lab, Inc. (Domino Data Lab), DataRobot, Inc. (DataRobot), DataChat, Imply Data, Inc. (Imply), Promethium, Inc. (Promethium), Siren Analytics (Siren), Tellius, Inc. (Tellius), SOTA ANALYTIC SOLUTIONS, LLC. (SOTA Analytics), and Vanti Analytics. Partnerships, new product developments, mergers and acquisitions, product enhancements, and acquisitions are some key strategies implemented by major vendors in the advanced analytics market.

The advanced analytics market is expected to grow from USD 33,771 million in 2021 to USD 89,823 million by 2026, at a CAGR of 21.6% during the forecast period

TABLE 4 GLOBAL ADVANCED ANALYTICS MARKET SIZE AND GROWTH RATE, 2016–2020 (USD MILLION, Y-O-Y%)

Advanced Analytics Market	2016	2017	2018	2019	2020	CAGR (2016–2020)
Market Size	8,400	12,776	18,431	25,360	29,063	36.4%
Y-o-Y		52.1%	44.3%	37.6%	14.6%	

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

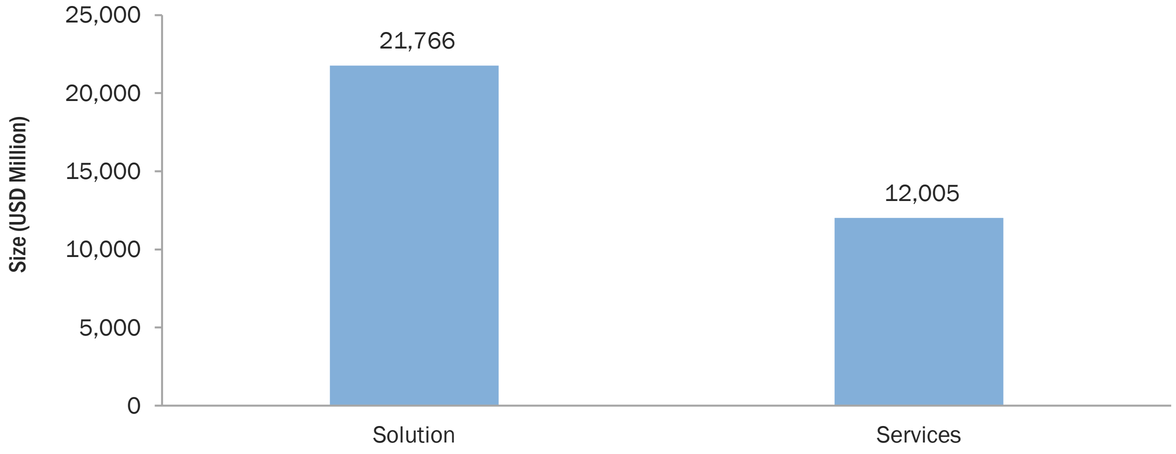
TABLE 5 GLOBAL ADVANCED ANALYTICS MARKET SIZE AND GROWTH RATE, 2021–2026 (USD MILLION, Y-O-Y%)

Advanced Analytics Market	2021-e	2022-p	2023-p	2024-p	2025-p	2026-p	CAGR (2021–2026)
Market Size	33,771	39,816	47,660	57,954	71,515	89,823	21.6%
Y-o-Y	16.2%	17.9%	19.7%	21.6%	23.4%	25.6%	

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

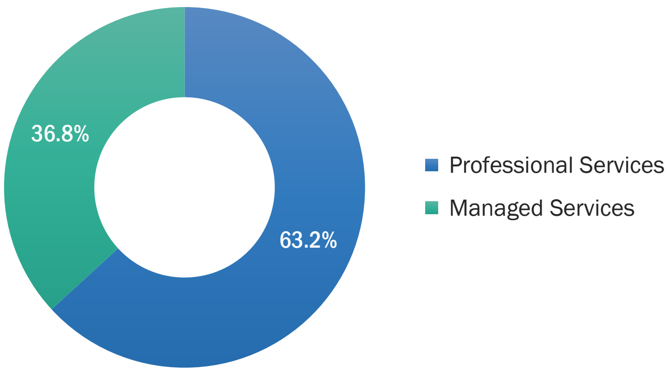
FIGURE 16 ADVANCED ANALYTICS MARKET SNAPSHOT, BY COMPONENT



Note: Numbers represent the market size for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

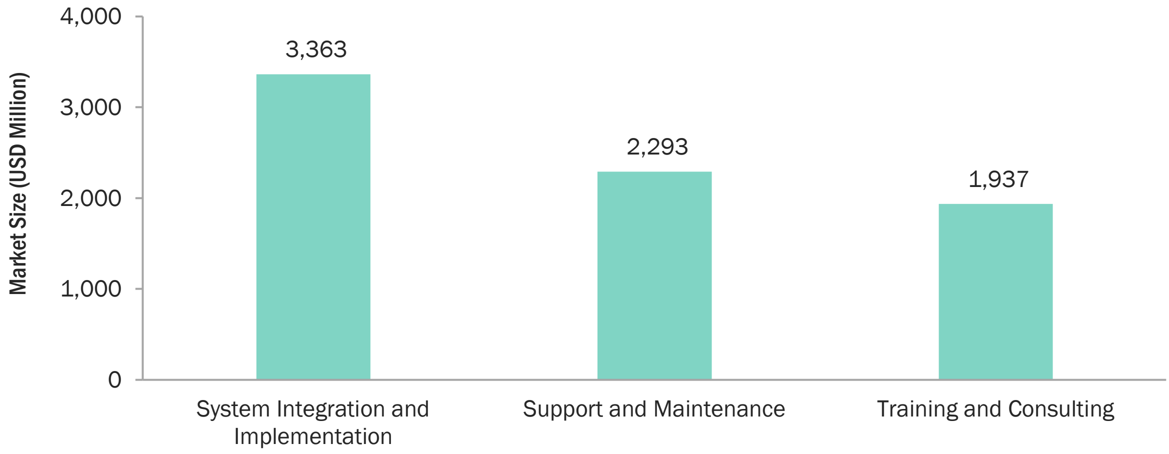
FIGURE 17 ADVANCED ANALYTICS MARKET SNAPSHOT, BY SERVICE



Note: Numbers represent the market share for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

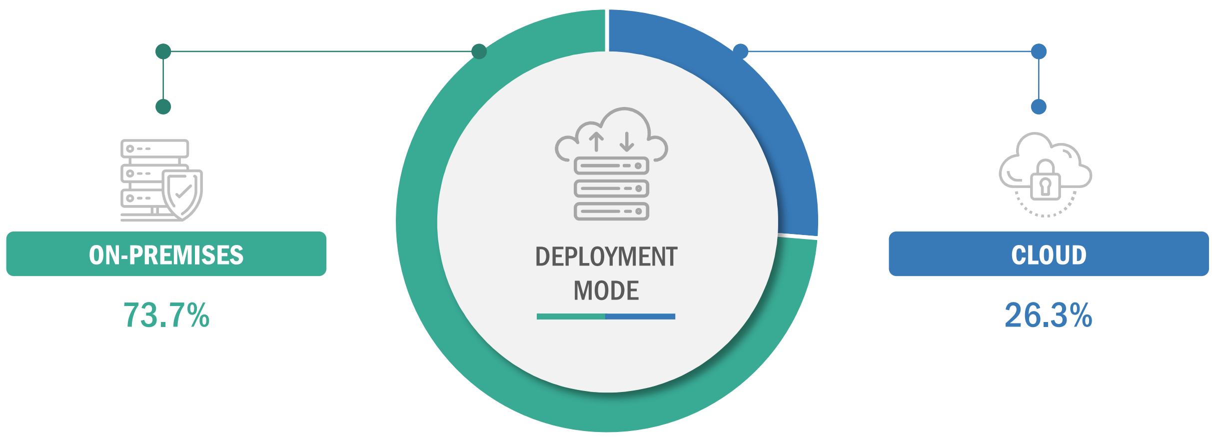
FIGURE 18 ADVANCED ANALYTICS MARKET SNAPSHOT, BY PROFESSIONAL SERVICE



Note: Numbers represent the market share for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

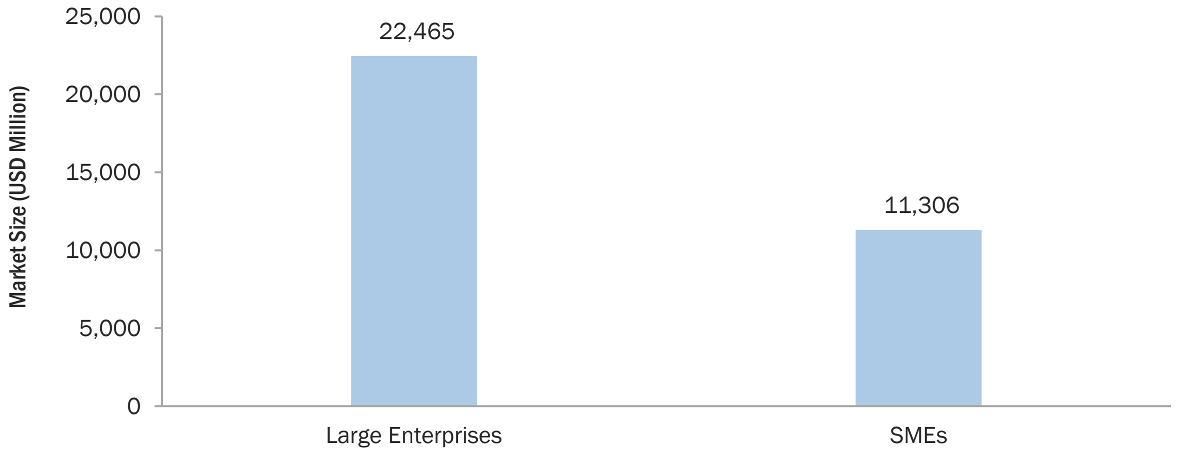
FIGURE 19 ADVANCED ANALYTICS MARKET SNAPSHOT, BY DEPLOYMENT MODE



Note: Numbers represent the market share for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

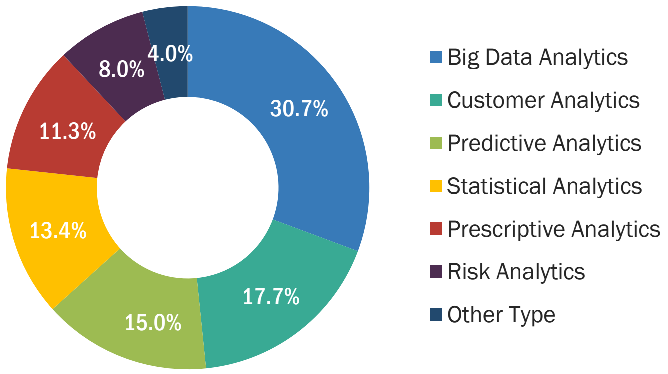
FIGURE 20 ADVANCED ANALYTICS MARKET SNAPSHOT, BY ORGANIZATION SIZE



Note: Numbers represent the market share for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 21 ADVANCED ANALYTICS MARKET SNAPSHOT, BY TYPE

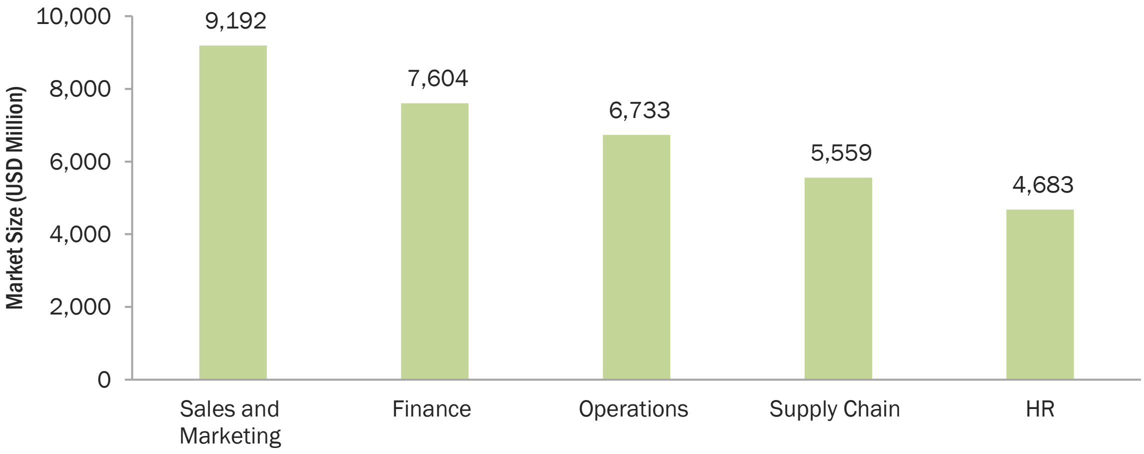


Note: Numbers represent the market size for 2021

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

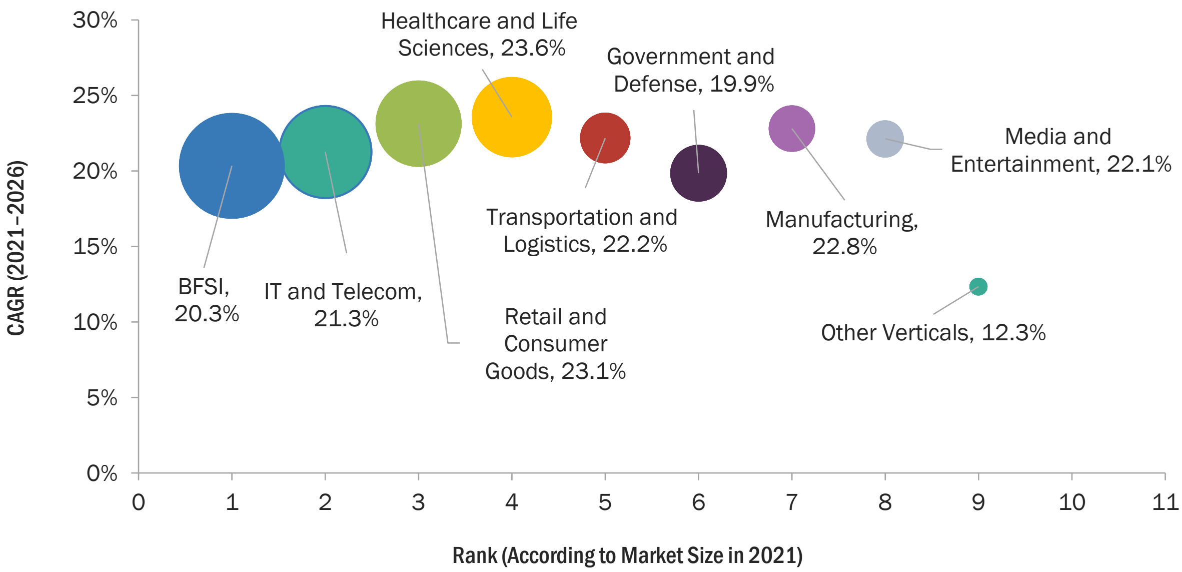
FIGURE 22 ADVANCED ANALYTICS MARKET SNAPSHOT, BY BUSINESS FUNCTION



Note: Numbers represent the market size for 2021

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 23 ADVANCED ANALYTICS MARKET SNAPSHOT, BY VERTICAL

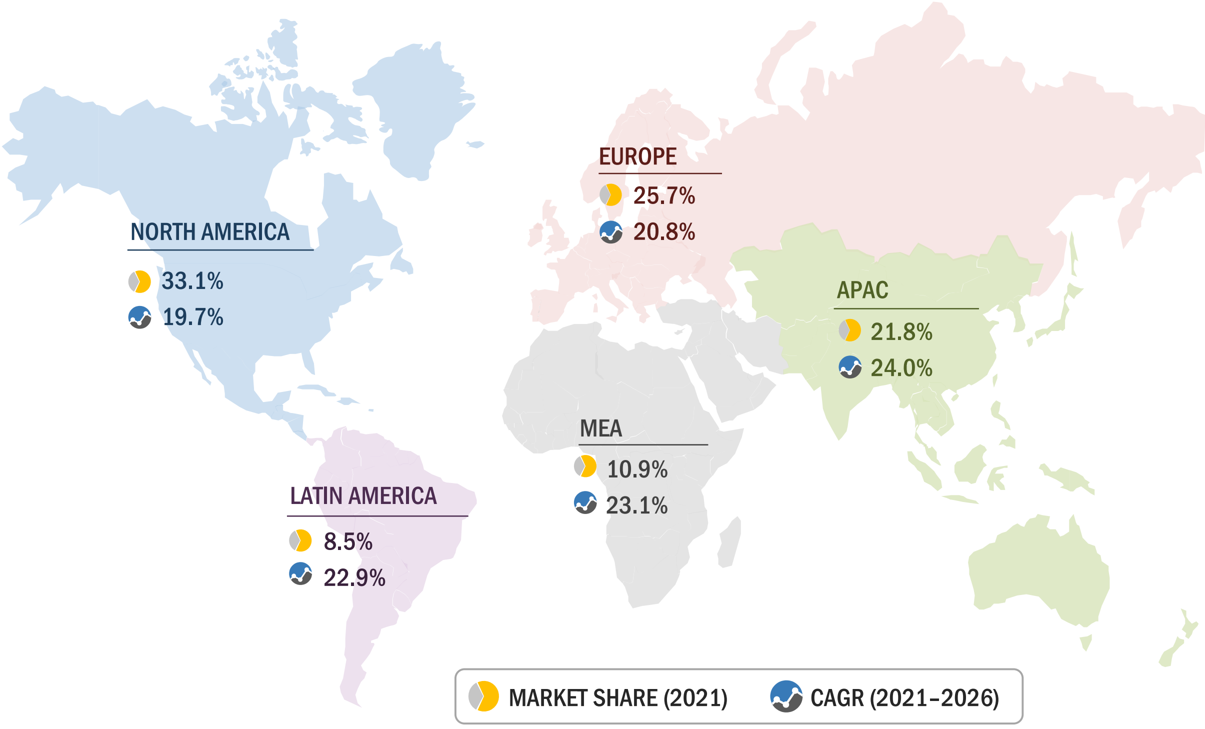


Note: Bubble size represents the market size in terms of value in 2021

Other verticals include education, energy and utilities, and travel and hospitality

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 24 ADVANCED ANALYTICS MARKET SNAPSHOT, BY REGION

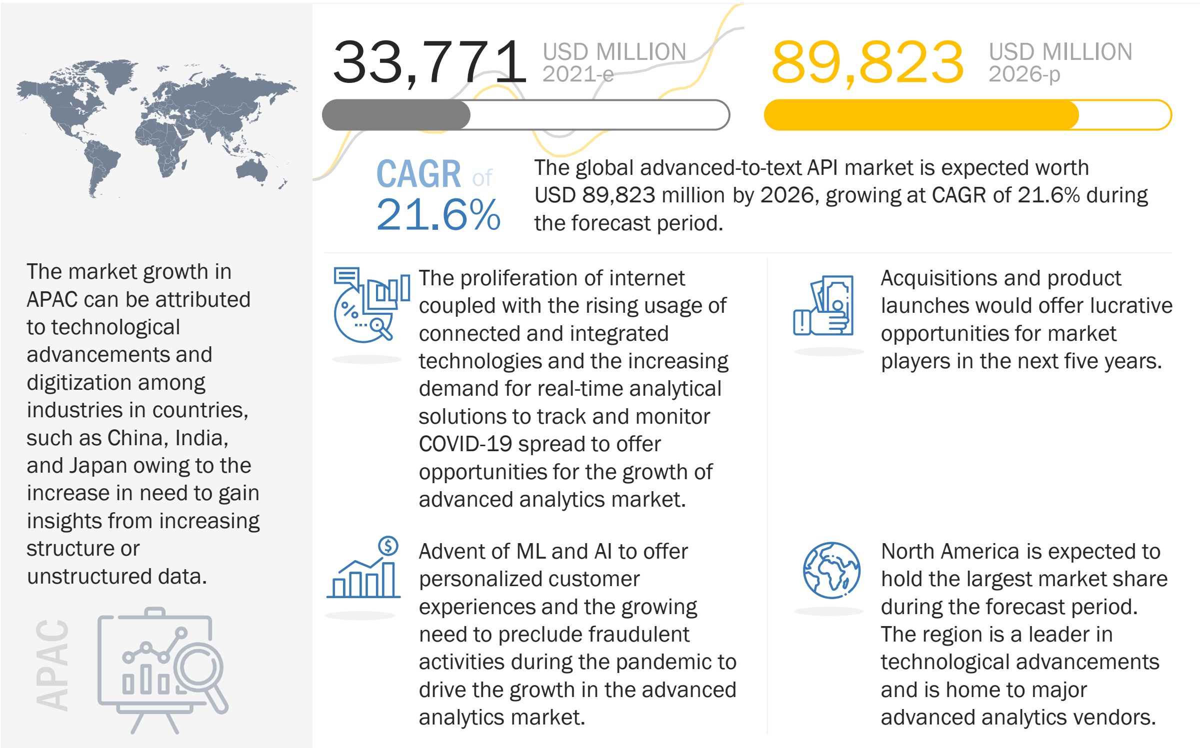


Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

4 PREMIUM INSIGHTS

4.1 ATTRACTIVE MARKET OPPORTUNITIES IN THE ADVANCED ANALYTICS MARKET

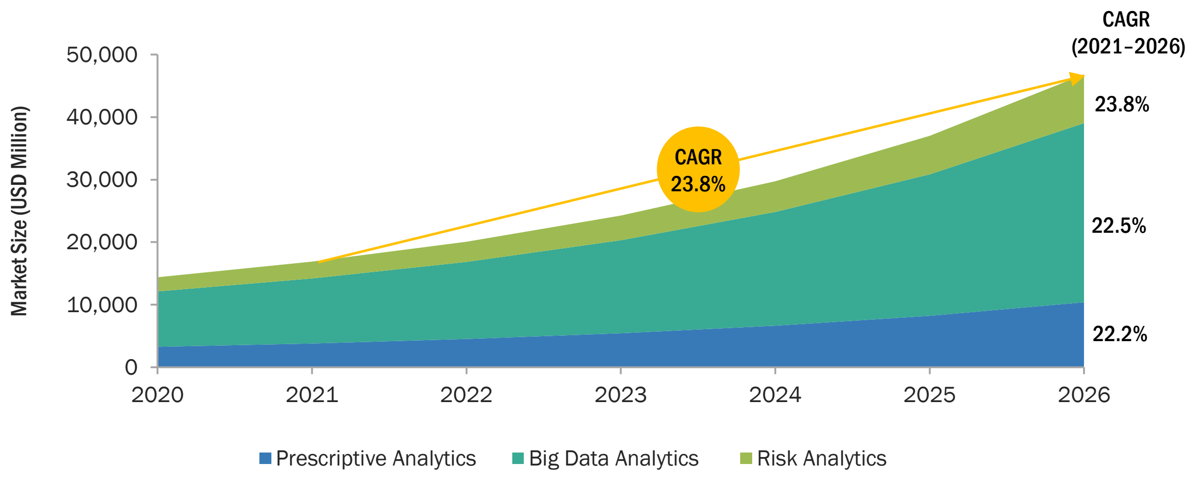
FIGURE 25 ADVENT OF MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE TO OFFER PERSONALIZED CUSTOMER EXPERIENCES TO DRIVE THE GROWTH OF THE ADVANCED ANALYTICS MARKET



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

4.2 ADVANCED ANALYTICS MARKET: TOP THREE TYPES

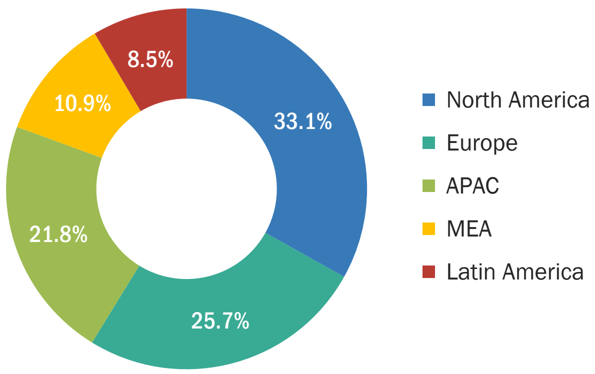
FIGURE 26 BIG DATA ANALYTICS SEGMENT TO GROW AT THE HIGHEST CAGR DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

4.3 ADVANCED ANALYTICS MARKET: BY REGION

FIGURE 27 NORTH AMERICA TO HOLD THE LARGEST MARKET SHARE IN 2021

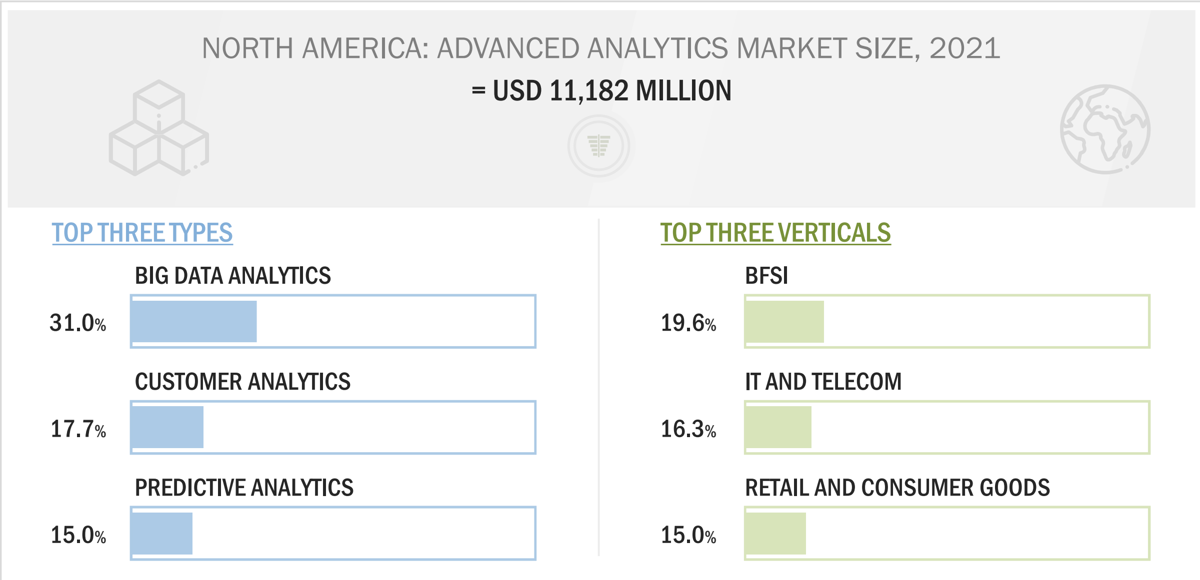


Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The advanced analytics market has a presence across all the major regions of the world. The advent of ML and AI to offer personalized customer experiences and the growing need to preclude fraudulent activities during the pandemic to boost the demand for the advanced analytics market. This is resulting in an increased number of pilots and deployments of advanced analytics solutions and services across the globe. In North America, advanced analytics is widely used for extracting meaningful insights from structure as well as unstructured data in real time. APAC is projected to grow at the highest CAGR during the forecast period. The major reasons for the high growth rate in APAC are the growing need to enhance customer experience. Countries in MEA and Latin America are catching pace in terms of advanced analytics adoption and are expected to achieve decent growth during the forecast period.

4.4 ADVANCED ANALYTICS MARKET IN NORTH AMERICA, TOP THREE TYPES AND VERTICALS

FIGURE 28 BIG DATA ANALYTICS TYPE AND BFSI VERTICAL TO ACCOUNT FOR THE LARGEST SHARES IN THE ADVANCED ANALYTICS MARKET IN 2021



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

5 MARKET OVERVIEW AND INDUSTRY TRENDS

5.1 INTRODUCTION

The market overview chapter offers detailed information about the factors responsible for the growth of the advanced analytics market. It comprises market dynamics, such as drivers, restraints, opportunities, and challenges. It further explains the impact of use cases and various industry standards on the market. Globally, companies are rapidly adopting advanced analytics solutions and services to gain a competitive edge in the market. Key players in the advanced analytics market are IBM (US), Oracle (US), SAS Institute (US), SAP (Germany), FICO (US), KNIME (Switzerland), Microsoft (US), Altair (US), RapidMiner (US), AWS (US), Salesforce (US), TIBCO Software (US), Alteryx (US), Teradata (US), Adobe (US), Absolutdata (US), Moody's Analytics (US), Qlik (US), Databricks (US), Dataiku (US), Kinetica (US), H2O.ai (US), Domino Data Lab (US), DataRobot (US), DataChat (US), Imply (US), Promethium (US), Siren (Ireland), Tellius (US), SOTA Solutions (Germany), and Vanti Analytics (Israel). These companies are continuously innovating their products and services to overcome large and complex data challenges compared to other advanced analytics solutions.

5.2 MARKET DYNAMICS

The research study indicates that the advanced analytics market is of utmost importance for every firm and technology vendor. The advanced analytics market is expected to grow at a significant rate during the forecast period due to various business drivers. Some factors driving the growth of the advanced analytics market are the rising adoption of big data and other related technologies, the advent of ML and AI to offer personalized customer experiences, and the growing need to preclude fraudulent activities during the pandemic. The drivers, restraints, opportunities, and challenges impacting the growth of the advanced analytics market are listed below:

FIGURE 29 DRIVERS, RESTRAINTS, OPPORTUNITIES, AND CHALLENGES: ADVANCED ANALYTICS MARKET

DRIVERS	▪ Rising adoption of big data and other related technologies ▪ Advent of machine learning and artificial intelligence to offer personalized customer experiences ▪ Growing need to preclude fraudulent activities during pandemic
RESTRAINTS	▪ Inability to quantify RoI ▪ Concerns regarding data privacy and security due to the pandemic
OPPORTUNITIES	▪ Proliferation of internet coupled with rising usage of connected and integrated technologies ▪ Increasing demand for real-time analytical solutions to track and monitor COVID-19 spread
CHALLENGES	▪ Ownership and privacy of collected data ▪ Complex data ecosystem leading to data breaches and security issues ▪ Lack of advanced analytical knowledge among the workforces

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

5.2.1 DRIVERS

5.2.1.1 Rising adoption of big data and other related technologies

The use of advanced technologies, such as cloud, IoT, big data and analytics, mobility, and social media, has led to innovation and transformation, thereby stimulating growth in the business ecosystem. Digital technologies have transformed the legacy approach to business into a modern approach. For example, online services are now firmly established in the banking and financial sector, resulting in the proliferation of online activities and websites. According to a blog published by Usabilla (Netherlands) in June 2020, the top 100 organizations are 50% more likely to address customer empowerment as a key parameter for business growth. The blog also stated that 92% of leaders had employed advanced digital transformation strategies to enhance their customers' experience.

Today, data is recognized as a new resource. It requires efficient and effective utilization to ensure a competitive market edge. Businesses are interested in deriving insights from collected data for making better, real-time, and fact-based decisions. This demand for in-depth knowledge has accelerated the adoption of big data and related technologies. The exponential growth in data volume is due to the expansion of businesses worldwide, which is driving the rise in data volumes and sources. The accumulation of big data in a single location has rapidly developed the evaluation capabilities of data science experts in every organization. Companies prefer to provide standalone solutions rather than combined solutions. This is eventually increasing the number of big data analytics start-ups, which are driving noteworthy innovations.

Big data has captured attention among businesses and consumers. It has not only become a major subject in technology and media, but has also made its way into many compliances, internal audit, and fraud risk management-related discussions. As an integral part of business operations in every industry, big data has influenced the adoption of advanced analytics worldwide. By leveraging the opportunities and challenges of big data analytics, enterprises can optimize vital business processes, functions, and objectives. Advanced analytics enables organizations to meet stakeholder demands, manage data volumes, manage risks, improve process controls, and boost administrative performance by turning information into intelligence. Advanced analytics solutions play a vital role in enabling an effective Intelligent Enterprise (IE) approach, which helps create a single view across the entire organization through the combination of standard reporting and data visualization.

5.2.1.2 Advent of machine learning and artificial intelligence to offer personalized customer experiences

DRIVERS	According to an IBM survey, 61% of company executives indicated that machine learning and AI are their company's most significant data initiatives in 2019. These leaders recognize that advanced analytics is transforming the way companies traditionally operate.

Source: www.experian.com

Advanced analytics is the key to understanding the data and extracting the crucial information needed to unlock customer insights. AI and ML are two emerging technologies with advanced analytics capabilities that can help companies achieve their business goals. Organizations are increasingly focusing on the re-engineering of businesses with the help of the immense power of the internet and digital technology. The adoption of digital operations assists advanced analytics solution providers to find newer ways to tackle the evolving challenges. Due to the digital transformation of retail and eCommerce businesses, operating costs have reduced and had positively impacted the advanced analytics market growth. The advent of AI and ML has enhanced organizations' ability to gather information, analyze it, and make quick decisions with minimum human intervention. Organizations are also adopting AI systems to speed up investment decisions. For instance, Fukoku Mutual Life Insurance has adopted AI to calculate payouts. AI systems analyze and suggest Governance, Risk Management, and Compliance (GRC)- related changes. Hence, they help in monitoring and collecting the appropriate quality of data to meet regulatory standards. The introduction and implementation of ML and AI have helped providers in understanding customer perceptions by analyzing the data gathered from multiple sources. Enterprises have shifted their focus to deliver a unified customer experience across all channels and provide offers in real time by analyzing customer buying patterns.

5.2.1.3 Growing need to preclude fraudulent activities during the pandemic

DRIVERS	A recent industry survey identified that fraud and compliance monitoring is increasingly the focus of senior management. This is due to the rising frequency and value of fines being issued for non-compliance. Firms typically spend 4% of their total revenue on compliance, but that could rise to 10% by 2022. Banks spend most on enhancing their transaction monitoring, and fraud and payment systems. Financial Institutions are actively looking for applying advanced analytics to augment resilience in their compliance function.

Source: Primary Insights

Advanced analytics provides businesses with a comprehensive understanding of the business, past, present, and future to better identify and manage risk. As per data from the Association of Certified Fraud Examiners' (ACFE) 2018 report, most typical organizations ran the risk of losing approximately 5% of their revenues due to fraud. Among sectors that suffer huge losses due to fraud is healthcare, where companies lose approximately USD 68 billion annually, which amounts to 3% of the total healthcare spending. Advanced analytics refers to the use of analytics software to identify trends, patterns, anomalies, and exceptions within data. Big data provides access to new sources of data as well as real-time events, which can be used as inputs for decision support system tools and models for fraud detection. By analyzing current indicators of fraudulent behavior and correlating them with occurrences of fraud, it becomes easier to identify fraud before it occurs or at an early stage in the cycle. Advanced analytics focuses on identifying anomalies to detect fraud. It includes detecting values that exceed standard deviation averages, besides an analysis of high and low values to detect abnormalities, which often indicate the likelihood of fraud. With the explosion of cyber and data breaches during the pandemic, verticals have created a ready supply of talented, cross-disciplinary resources agile by legacy organizational structures. Advanced analytics provides a unique and powerful means to transform fraud operations across key verticals.

5.2.2 RESTRAINTS

5.2.2.1 Inability to quantify RoI

RoI relies on gained value, which is often linked with customer satisfaction and process improvements, it becomes increasingly difficult to quantify, due to which advanced analytics might not be the top priority for management when budgets are set. Higher upfront investments make it further challenging to build a business case; the overall data processing cost consists of software, services, and software licensing costs. The efficacy of certain complex deployments also depends upon ongoing support, upgrades, and associated professional and managed services. Many organizations halt or delay their speech processing projects when they become aware of the overall resource requirements. Costly pricing structures have restricted the growth of advanced analytics deployments among SMEs. However, cloud-based deployments as well as flexible pricing and scaled-down versions of advanced analytics solutions are slowly changing the scenario.

5.2.2.2 Concerns related to data privacy and security due to the pandemic

One of the major concerns for users of advanced analytics is the privacy of data. The backbone of advanced analytics solutions, data, remains a critically important aspect that most organizations find difficult to manage. The inefficiency of managing exabytes and petabytes of data has increased the chances of security breaches and data losses. In today's competitive marketplace, marketing teams require real-time and secure data to deliver an outstanding customer experience. Organizations are gathering data through multiple touchpoints and measure them virtually. Such data is used in support and communication and may include a variety of data types. These data types comprise public information, big data, and small data collected from customers.

This data can include permissions, individual preferences, and updated contact information on products, services, and communication platforms. Hence, vendors need to ensure high-level data security to maintain customer trust. Recently, cybercriminals have gained access to widespread tools to obtain anything from passwords to secret questions and token-generated passwords. In such situations, marketing and IT teams need to work concurrently, providing each other with insights on when and how the data is gathered, processed, and used in operations. Such potent cyberattacks continue to hinder the widespread adoption of advanced analytics across data-intensive industrial sectors. However, this scenario is expected to gradually change throughout the forecast period as the benefits of technology adoption continue to outweigh the risks associated and with overall developments on the data security front.

5.2.3 OPPORTUNITIES

5.2.3.1 Proliferation of internet coupled with the rising usage of connected and integrated technologies

The proliferation of the internet and the availability of various means to access the internet have led to a massive increase in data volumes generated. This would help in the advancement and expansion of high-speed internet services. Globalization and economic growth are also playing major roles in driving greater data generation worldwide. With the rise in touchpoints and the need for collecting data to understand consumer behavior, every touch by a consumer has become an important data point that can be processed to reveal user behavior. With the exponential rise in individual and organizational data, businesses are now deploying data scientists and analysts for processing collected data. This is compelling firms to invest in advanced analytics. The rise in connected and integrated technologies has provided a platform to advanced analytics solution vendors for leveraging the development and unprecedented growth of the internet. The eCommerce sector has modified the traditional shopping behavior of customers. Dedicated email campaigns, online/social media advertising, and cognitive customer analysis are key enablers driving sales and increasing customer loyalty. With connected devices coming to the forefront, retailers are focusing on the real-time analysis of customers' shopping behavior and market basket analysis for analyzing consumers' perception, which can be used for building tailor-made offers to increase customer retention. With the rise in the global IoT analytics demand in the retail vertical, the market is expected to offer unprecedented growth opportunities to advanced analytics vendors

5.2.3.2 Increasing demand for real-time analytical solutions to track and monitor COVID-19 spread

During the ambiguous situation of the COVID-19 pandemic, the healthcare vertical has sought to use big data and advanced analytics tools to better understand the virus and its spread. Advanced analytics has helped researchers worldwide build advanced analytics models that can track COVID-19 surges in different countries. Researchers from Binghamton University and the State University of New York have developed several ML models to examine COVID-19 trends and patterns. Hospitals and health systems have leveraged predictive models to gain insights into COVID-19 risks, disease outcomes, and the virus' potential impact on resources. Communities across countries have started relaxing social distancing orders; hence, it has become crucial for health systems to plan for changes in healthcare demands and surges in COVID-19 cases. The team at CommonSpirit Health has built predictive models using de-identified cell phone data, public health information, and data from the CommonSpirit Health system's care sites. Using these models, organizations can gain insights into the surges and dips of COVID-19 infection rates, allowing hospitals to plan better and work toward resuming their full spectrum of necessary services. Recently, a team from Florida Atlantic University (FAU) has utilized ML technology to build a COVID-19 knowledge base and risk assessment dashboard, which aims at addressing the discrepancies that exist in the context of the pandemic. Hence, using these predictive models, researchers can help organizations stay ahead of possible increases in COVID-19 patients by continuing to collect data from around the world to make advanced analytics models more accurate.

5.2.4 CHALLENGES

5.2.4.1 Complex data ecosystem leading to data breaches and security issues

Data consolidation helps transform the gathered data into a single format, thus simplifying the decision-making process for companies. Data exchanges and data ecosystems deliver the tools to analyze the collected data at a centralized location and help extract and cross-check business-critical components. The development of data exchanges and data ecosystems varies based on the assumptions made on the value of the data for each customer segment. However, since centralized data is more prone to cyberattacks, data exchanges are a major concern for data-sensitive organizations in industry verticals such as Banking, Financial Services, and Insurance (BFSI) and healthcare, which deal with individual data. Such organizations need to take extreme care to ensure the safety of customers' personal information. Data breaches impact an organization's identity and security, and fool-proof security measures are a priority. In some countries, the concerned organizations are partially responsible for the damage triggered by the loss of confidential data. Enterprises must also ensure the security of data outsourced to a third-party BI vendor for analysis. However, with companies increasingly adopting data security solutions, consumer data security is not expected to be a major issue for long.

5.2.4.2 Ownership and privacy of collected data

The market competition is increasing every day due to the introduction and expansion of new technologies, such as IoT, big data, and ML. These technologies have significantly contributed to the number of endpoints collecting data, thereby resulting in an increase in dynamic data that keeps changing/updating every second. For every organization that predicts customer behaviors by analyzing the collected sets of data, maintaining data privacy is a huge security mandate. Concerns related to the data ownership and privacy of collected data pose potential risks to the identity of individuals. Therefore, organizations are bound to follow certain limitations in data collection types, quantities, and methods. These limitations are likely to vary from nation to nation across the globe. Therefore, enterprises must stay abreast with the changing rules and data-related regulations.

Today, in the technology era, the advent of the internet has brought cybercrimes into the picture. Hackers target data storage for financial gains and other reasons. The breach of data impacts organizations' identity and security. Some governments hold enterprises responsible for damages triggered by the loss of confidential data. Therefore, enterprises ought to implement the latest security measures to ensure the privacy of collected data. While outsourcing data analysis to a third-party BI vendor, organizations must ensure that the BI vendor appropriately secures firms' data.

5.2.4.3 Lack of advanced analytical knowledge among the workforces

The enormous value proposition harnessed around using advanced analytical computations stems from using predictive and prescriptive analytics. However, still, the usage of the technology is very little. Enterprises have still not been able to come to terms with the massive adoption of the technologies that are futuristic and to promise the next course of action that can be taken in the disruptive world. Prescriptive analytics, which promises enhanced decision-making and empowers business users to choose one among many viable options using an optimization approach, ML, and other tools in deciding a solution, still hovers around in the infancy stage. As per data scientists across the world, the cause of apprehension of adopting advanced technology is the low level of skillsets among the workforce that are not adequate to utilize these advanced pervasive technologies.

Enterprises breathing under the Volatility, Uncertainty, Complexity, and Ambiguity (VUCA) atmosphere need to adopt the advanced analytical technology by leaps and bounds to help stay afloat in the disruptive world. To combat current challenges of disruption, enterprises need to invest deeply in the training and certification courses of using advanced technology to remain competitive and have a distinct place in the market.

5.2.5 CUMULATIVE GROWTH ANALYSIS

The cumulative growth analysis is carried out considering the short-term and long-term impact of the current scenarios and dynamics, such as drivers, opportunities, restraints, and challenges, on the advanced analytics market. This analysis involves inputs from industry experts and multiple companies associated with the advanced analytics ecosystem.

MARKET DYNAMICS	SHORT-TERM IMPACT (2020–2021)	LONG-TERM IMPACT (BEYOND 2021)
Drivers and Opportunities	(+) 17.0% - 19.0%	(+) 22.5% - 23.5%
Restraints and Challenges	(-) 2% - 3%	(-) 2.5% - 1.5%
Projected Cumulative Growth	(+) 15.0% - 16.0%	(+) 20.0% - 22.0%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

5.3 INDUSTRY TRENDS

5.3.1 ADVANCED ANALYTICS TECHNIQUES

5.3.1.1 Data mining

Data mining involves the effective collection of data, storage of data, and the computer processing of the data. The various data mining techniques can also be instrumental in helping a business cut down on costs, generate more revenue, and improve customer relationships. Data mining is a technique that is relevant across industries including banking, retail, manufacturing, and research projects such as those of genetics, mathematics, and cybernetics.

5.3.1.2 Machine learning

ML uses computational methods to find patterns and inferences in data and automatically create statistical models to produce reliable results with minimal human intervention. This advanced analytics tool employs computational methods to identify patterns in data, which is then uses to create statistical models capable of producing reliable results without requiring much human intervention. ML is seen to be a key component of the AI subset of advanced analytics.

5.3.1.3 Cohort analysis

Cohort analysis is a technique that examines the behavior of a group of people to draw broadly applicable insights. It is a subset of behavioral analytics. It takes a selection of data from a larger data set over a period, and instead of looking at all the users as one single unit, it segregates them into smaller related groups based on different types of attributes for analysis.

5.3.1.4 Cluster analysis

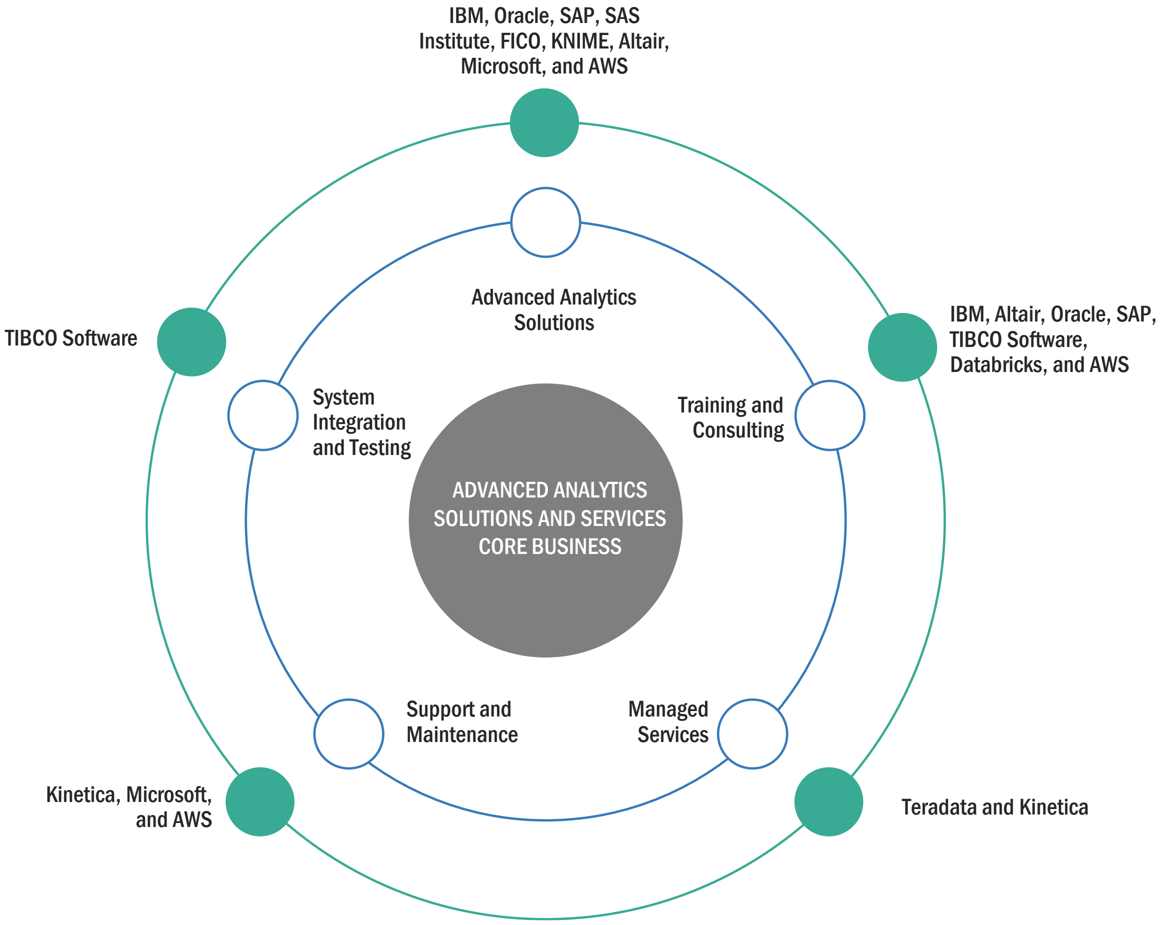
Cluster analysis is a way to identify similarities and differences in different sets of data and present that data visually in a way that makes comparisons easy. The analysis groups a set of objects that are more like each other than to objects in other groups. It is one of the main tasks of exploratory data mining.

5.3.1.5 Retention analysis

Retention analysis is an advanced analytics technique that studies the different cohorts of customers. These techniques can offer deeper insights into consumer behavioral patterns and what factors influence consumer growth and consumer retention.

5.3.2 ECOSYSTEM

FIGURE 30 ADVANCED ANALYTICS MARKET: ECOSYSTEM



Source: Articles, Press Releases, Company Website, MarketsandMarkets Analysis

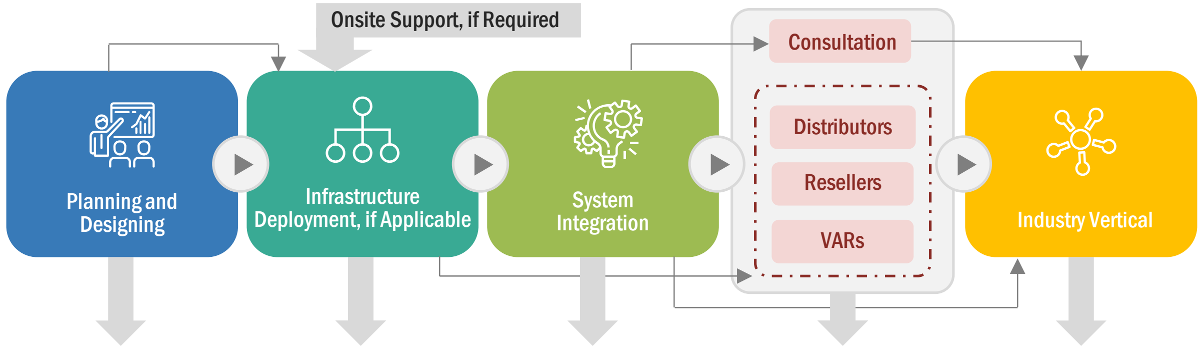
Vendors in the advanced analytics market are offering advanced analytics solutions and services. Services are categorized into professional and managed services. Professional services cover training and consulting, support and maintenance, and system integration and deployment services.

The key vendors providing these advanced analytics solution and services are given below:

- Solution: IBM, Oracle, SAP, SAS Institute, FICO, KNIME, Altair, Microsoft, and AWS
- Training and Consulting: IBM, Altair, Oracle, SAP, TIBCO Software, Databricks, and AWS
- System Integration and Testing: TIBCO Software
- Support and Maintenance: Kinetica, Microsoft, and AWS
- Managed Services: Teradata and Kinetica

5.3.3 SUPPLY CHAIN ANALYSIS

FIGURE 31 SUPPLY CHAIN ANALYSIS



MAJOR ROLE
- Solutions Vendors
- Resellers/ Channel Partners
- Service Providers

KEY PLAYERS
- IBM
- Oracle
- SAS Institute
- SAP
- FICO
- KNIME
- Microsoft
- Altair
- RapidMiner
- AWS
- Salesforce
- TIBCO Software
- Alteryx

- Teradata
- Adobe
- Absolutdata Analytics
- Moody's Analytics
- Qlik
- Databricks
- Dataiku
- H2O.ai
- Domino Data Lab
- DataRobot
- Data Chat
- Imply
- Kinetica
- Promethium
- Siren
- Tellius
- SOTA Solutions
- Vanti Analytics

MAJOR ROLE:
Deployment or Upgradation of Advanced Analytics solutions and services

MAJOR ROLE:
1. Consultation: Analysis and Advisory of Projects
2. VARs/Distributors/ Resellers: Distribution and Reselling of Advanced Analytics

- Retail
- Transportation and Logistics
- Government and Public Sector
- Media and Entertainment
- Manufacturing
- Travel and Hospitality
- Healthcare and Pharmaceuticals
- Other Verticals (Education, BFSI, constructions and buildings, automotive, and Utilities)

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The value chain of the advanced analytics market highlights the delivery methods of unique solutions/services to end users devised through collaborations between vendors and service providers. Companies independently offer both advanced analytics solutions and services to help organizations increase the overall productivity; improve performance; heighten security; extract business insights; and reduce IT operational costs. In the planning and designing phase, the requirements of customers are taken into consideration, and advanced analytics software is customized based on these requirements. In the infrastructure deployment phase, all elements required in the premises of customers are installed or are made available through the cloud. The system integration phase deals with the deployment of integrated solutions or the upgradation of the existing advanced analytics. The advanced analytics comprises different ML models; these models can be provided as a complete suite by companies or can be integrated with the deployed products, which are offered by third-party vendors.

IBM, Oracle, SAS Institute, SAP, FICO, KNIME, Microsoft, Altair, RapidMiner, AWS, Salesforce, TIBCO Software, Alteryx, Teradata, and Adobe are some well-established vendors in the advanced analytics market. Various verticals play a major role in this market, including BFSI, government and defense, retail and consumer goods, transportation and logistics, IT and telecom, travel and hospitality, healthcare and life sciences, and others (manufacturing, media and entertainment, and automotive).

TABLE 6 ADVANCED ANALYTICS MARKET: SUPPLY CHAIN

COMPANY	ROLE IN ECOSYSTEM
IBM	Technology and solution provider
Oracle	Solution and service provider
SAS Institute	Technology and solution provider
SAP	Technology provider
FICO	Solution provider
KNIME	Solution provider
Microsoft	Technology provider
Altair	Solution and service provider
RapidMiner	Solution and service provider
AWS	Solution provider
Salesforces	Solution provider
TIBCO Software	Solution provider
Teradata	Solution provider
Adobe	Solution provider
Absolutdata Analytics	Service provider
Moody's Analytics	Solution provider
Qlik	Solution and service provider
Databricks	Solution provider
Dataiku	Solution provider
H2O.ai	Solution provider
Domino Data Lab	Solution provider
DataRobot	Solution provider
DataChat	Solution provider
Imply	Solution and service provider
Kinetica	Solution provider
Promethium	Solution provider
Siren	Technology provider
Tellius	Solution provider
SOTA Solutions	Solution and service provider
Vanti Analytics	Solution provider

Source: Company Website, Annual Reports, and SEC Filings

5.3.4 ADVANCED ANALYTICS MARKET: COVID-19 IMPACT

The COVID-19 pandemic has impacted trading activities across regions. It has had a moderate impact on all elements of the technology sector. The hardware business is predicted to be the most impacted in the IT industry. Owing to the slowdown of hardware supply and reduced manufacturing capacity, the IT infrastructure growth has slowed down. Businesses providing solutions and services are also expected to slow down for a short period. However, the adoption of collaborative applications, analytics, security solutions, and AI is set to increase in the remaining part of the year. Verticals such as manufacturing, retail, and energy and utilities have witnessed a moderate slowdown, whereas BFSI, government, and healthcare and life sciences verticals have witnessed a minimal impact. With recovery, global ICT spending is estimated to increase by approximately 3.5%–4.5% from 2020 to 2021. The impact of COVID-19 is believed to be short-term; however, it may have a significant effect on businesses and forecasts to a significant extent for a minimum of 8–12 months.

The spread of COVID-19 pandemic has generated a huge disruption in daily activities. It has forced people to follow social distancing policy and several other restrictions. Under such circumstances the healthcare vertical has witnessed the increased use of analytics, which includes bridging gaps in data and changing capabilities needs. Having any analytics in place has allowed organizations to pivot and build what they need under pressure. These insights stand to shape the immediate future of healthcare data and analytics as pandemic-era patterns and practices carry over into a new industry landscape. Health care data and analytics has become inextricable from the COVID-19 response. From emergency department frontlines and hospital capacity planning to vaccine development and future emergency response, the industry has relied on COVID-19 data and analytics to understand the disease, keep populations safe, and eventually reopen communities. With the outbreak of the COVID-19 pandemic, everything is disrupted, from customer behavior to supply chains and eventually creating an economic slowdown, which is causing further changes. The pandemic has augmented the use of analytics and AI in various companies. Due to the COVID-19 pandemic, there is an increase in the deployment of advanced analytics in businesses to secure business continuity and process optimization. Advanced analytics is gaining popularity due to its deployment in the businesses along with the improvements in the latest technologies such as data mining, neural networks, ML, multivariate statistics, semantic analysis, and the growing data volumes generated by organizations.

The COVID-19 pandemic has accelerated various companies' uses of advanced analytics and AI. These strategies have helped engage customers through digital channels, manage fragile and complex supply chains, and support workers through disruption to their work and lives. Companies should focus on where they can obtain new insights rather than rely on lagging information. These can come from both from new sources of data and using the existing data in new ways. For instance, banks that traditionally use credit scores to analyze risk can instead turn to customer-account data, where they might identify gaps in deposits. The crisis has also exposed that one of the most advanced analytics techniques relies on principles of behaviors and patterns repeated periodically. But the patterns exposed are revealed through data and even suggested during identification, which changes the customer's mind. Many firms that had started benefitting its client with data analytics usage are growing in all departments. The departments most directly affected by the virus see the most incredible growth in marketing, finance, and customer service use cases for small businesses.

To mitigate the effect of COVID-19, the data scientists have considered both macro and micro views of data to understand the needs and design the right policies for an organization. Digital and social media analytics have offered a macro understanding of people's interests, behavior habits, and how they react to the evolving situation in different geographies and demographics. The statistics also provide epidemiological predictions that provide key insights into the organizational models and support. Analytics have also played a vital role in the determination of fraudulent activities with a system updated with fraudulent activities that help in determining security boundaries for an organization. For instance, IBM Cloud Pak for Security can help uncover hidden threats and make more informed risk-based decisions. An economic model has also been represented by data scientists to forecast the various insights about customers' probable future behavior, suppliers, and staff is a top priority. These predicted behaviors have helped the business in deriving the consequential financial effect through continuous, rapid decision-making based on accurate, data-driven analytics, and simulations.

COVID-19 is significantly affecting M&A activities across virtually all industries, particularly pertaining to new transactions that are expected to come. Industries such as media and entertainment, transportation and logistics, manufacturing, and retail have been affected the most. The COVID-19 may cause buyers and sellers to reconsider valuations, adjust pricing mechanisms, and implement new methods for interim operations and crisis response management. This situation is very delicate as investors are withdrawing their interest from ongoing deals. The advanced analytics market is expected to witness a growth dip in 2020, but it will pick up slowly from 2021 onward, owing to the growing awareness about how advanced analytics solutions can come to the rescue in the post-pandemic scenario.

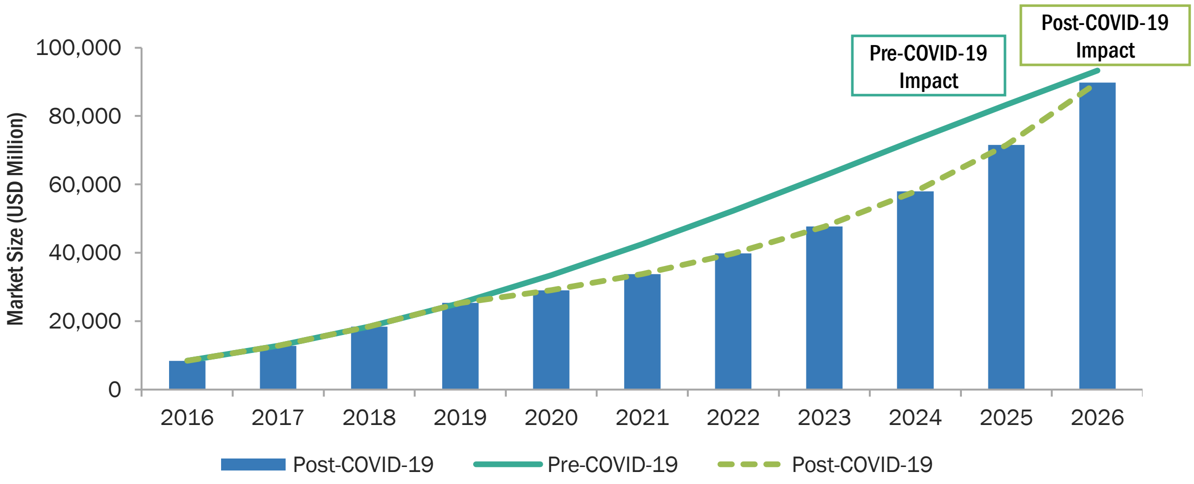
ASSUMPTIONS – RETURN TO "NORMALCY"

- There is no significant second wave of COVID-19, and the major impact is over by Q2 2020
- Economic recessionary conditions prevent a sharp increase in demand
- All major demand drivers fully kick-in by Q4 2020 in the US, the EU, and emerging markets
- Recovery trajectories: the US and the EU in Q4 2020, China in Q2, and India and others in Q3 2020

EARLY EVIDENCE OF REVENUE IMPACT

- According to CallMiner, call volumes in contact centers have dropped by about 25% during COVID-19, which reflects the overall slowdown in the economy because of the coronavirus pandemic. Text interactions are staying steady. In May 2020, the folks at CallMiner had theories about what their advanced analytics software would turn up when they launched an informal coronavirus customer research program in March.
- HGS develops advanced analytics tools to address the client's concerns about agent work-at-home security and customer satisfaction during the pandemic. Combining advanced analytics with webcam audits ensured agent adherence, data security, satisfaction, and home-office hygiene. The company created multiple queries to understand COVID-driven customer sentiment and call-mix changes and with the use of selected keyword search strings (COVID nineteen, pandemic, coronavirus, lockdown, and relief plan), the company's team members would be able to share real-time voice of the customer data with end users.

FIGURE 32 ADVANCED ANALYTICS MARKET TO WITNESS MINIMAL SLOWDOWN IN GROWTH IN 2021



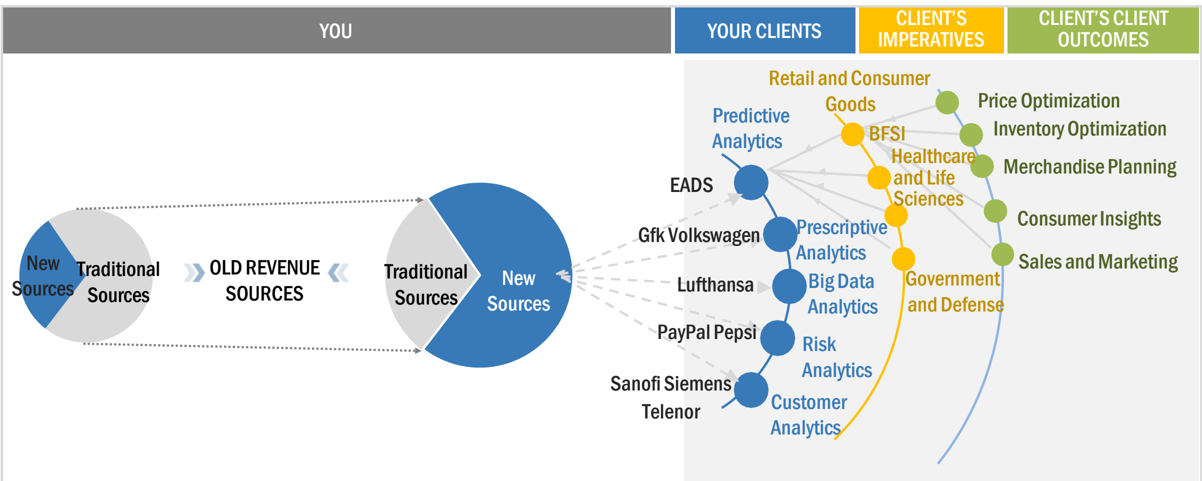
Source: Secondary Research and MarketsandMarkets Analysis

Prior to the COVID-19, the market for advanced analytics was anticipated to grow from USD 33,459 million in 2021 to USD 93,301 million by 2026, at a CAGR of 17.0% during 2021–2026, whereas, in the post-COVID-19 scenario, the market for the advanced analytics is projected to grow from USD 33,771 million in 2021 to USD 89,823 million by 2026, at a CAGR of 21.6% during 2021–2026. The postponement of scheduled project launches by companies across the world to mitigate the impact of the COVID-19 pandemic is the key factor affecting the growth of the advanced analytics market.

While the economic impact of the COVID-19 outbreak will be evident in the latter half of the year, it provides an opportunity for advanced analytics providers to become more robust and innovative. Advanced analytics providers are looking to provide various applications that can be useful during the pandemic. For instance, advanced analytics technology is being used by retail industry to manage or enhance multiple delivery options, such as in-store, online, and omnichannel distribution processes.

5.3.5 TRENDS/DISRUPTIONS IMPACTING BUYERS/CLIENTS OF THE ADVANCED ANALYTICS MARKET

FIGURE 33 ADVANCED ANALYTICS MARKET: TRENDS/DISRUPTIONS IMPACTING BUYERS/CLIENTS



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The advanced analytics market is estimated to grow at a CAGR of 21.6% during the forecast period. The rising adoption of big data and other related technologies, and the advent of ML and AI to offer personalized customer experiences, and the growing need to preclude fraudulent activities during the pandemic are expected to drive the growth of this market. The unprecedented and simultaneous advances in technologies, such as AI, ML, blockchain, and IoT, are blurring traditional boundaries and creating new business models. These technological developments have led businesses to monitor information in real time without worrying about dealing with huge amounts of insurmountable data.

5.3.6 REGULATORY IMPLICATIONS

Compliances and regulations ensure organizations adopt stringent information security solutions and services to protect confidential information. Today, cybersecurity is moving to the forefront in terms of compliance requirements, and regulatory bodies have introduced several standards and laws related to information rights management practices.

5.3.6.1 General Data Protection Regulation

The General Data Protection Regulation (GDPR) (Regulation [EU] 2016/679) is a regulation by which the European Parliament, the Council of the European Union (EU), and the European Commission ensure the safety of the EU citizens from privacy and data breaches. The GDPR extends the scope of the EU data protection law for all foreign organizations that process the data of EU citizens. It delivers data protection regulations throughout the EU, making it easier for non-European companies to comply with the regulations. However, if the regulations are not complied with, the company may pay four percent of its worldwide turnover as a penalty. Key privacy and data protection requirements of the GDPR include the following:

- Confidentiality of collected data to ensure data protection
- Safe transfer of data across borders
- Seeking consent of subjects for data processing
- Providing data breach notifications

GDPR would impose consistent data security laws for all EU members and the companies' marketing goods and services to EU residents, whatever the location may be. GDPR implementation maintains patient records to meet legal requirements.

5.3.6.2 Market Abuse Regulation

Effective since July 2016, the Market Abuse Regulation stipulates the keeping of records of all telephone and electronic communications. It is a crucial regulation, adherence to which might help establish the identity of persons responsible for the dissemination of false or misleading information, or individuals in contact at a certain time. In practice, this means all the communications of Direct Market Participants (Brokers and Dealers) must be recorded, to facilitate the identification of intent related to an event, and the actions behind the event.

5.3.6.3 Revision of Markets in Financial Instruments Directive

Markets in Financial Instruments (MiFID II) came into force in January 2018 and requires the recording and monitoring of all communications to detect market abuse. It is an important directive as the records of communications are crucial, and often the only, evidence of client relationships. Such recordings help prove and detect market abuse activities.

5.3.6.4 European Banking Regulations

Regulations are intended to control frameworks against potential market abuse conduct. This includes preventing the provision of false information related to the company's past financial records. To facilitate proper investigation, a company needs to provide regulatory bodies relevant communications, such as correspondence between submitting parties and others (internal and external traders, brokers, rate administrators, and calculation agents). The recordings need to be retained in secure and tamper-proof mediums that enable enterprises to access them for future reference, with documented audit trails. The recording submitters review sample records of businesses' voice communications that are recorded on electronic and phone communication systems, as opposed to personal devices and systems.

5.3.6.5 Health Insurance Portability and Accountability Act

The Health Insurance Portability and Accountability Act (HIPAA) compliance ensures the privacy of personal health information and gives patients the right to their own Protected Health Information (PHI). HIPAA defines the type of Electronic Health Records (EHR) used to identify patients uniquely. These records include names, birth dates, admission dates, telephone/fax numbers, social security numbers, medical record numbers, health plan beneficiary numbers, and biometric identifiers, such as fingerprints. As the healthcare vertical aggressively adopt electronic healthcare records accessible via web and mobile-based applications, it also exposes itself to hackers who can gain access to this sensitive data of healthcare customers/patients. According to the act, if litigation parties send, store, or share data related to doctors, patients, or other health-related information, they need to comply with HIPAA guidelines. Major violations of the HIPAA standards result from businesses not following policies and procedures to safeguard their healthcare records.

5.3.6.6 California Consumer Privacy Act

The California Consumer Privacy Act (CCPA) is a law designed to protect the data privacy rights of citizens living in California. In short, the law forces companies to provide more information to consumers about what is being done with their data and gives them more control over their data sharing. The law addresses the real issue that most consumers do not realize that their personal information is being shared or sold to others. Like the EU GDPR, the CCPA gives consumers important new rights called a right to know about "how the data is being used, a right to access, and a right to opt-out of having their data sold to third parties." In short, businesses must inform consumers about the categories of information collected and the purpose for which it is being collected. But if the consumer agrees to the data collection, they have additional rights. They can make an access request for their personal information to find out in more detail about the specific pieces of information held by the business and the third parties that received their information.

The CCPA applies to any business (regardless of location) that serves people in California and meets one of three criteria: annual revenue of USD 25 million or more (from any source), half of the annual revenue comes from selling personal data about California residents. It processes the personal data of about 50,000 or more California residents in a year. It gives consumers more control over the personal information that businesses collect about them, and the CCPA regulations guide how to implement the law.

5.3.6.7 Office of Communications

The Office of Communications (Ofcom) has been designed to regulate communication activities across commercial television and radio broadcasts in the UK. It collects and processes personal data to ensure people do not get scammed and are protected from bad practices. Ofcom obligates the channels to add subtitles to their broadcasted programming. According to Ofcom, subtitles and closed captioning are for everyone and not just for people with hearing loss. Similarly, with the proliferation of online educational courses, learners might need to read Same Language Subtitling (SLS) to enhance their understanding; hence, various universities and educational institutes are adding subtitles to their audio and video-based content.

5.3.6.8 Polish Civil Code

Under the Polish Civil Code, the conditions for legal protection of the voice as a personal right are to be viewed as a threat to personal rights. The person who provides his or her voice should get monetary compensation under this title. It appears that new technologies use subjects of personal rights (protected by given legal methods) in pioneering ways so that the effects of those activities require new concepts, i.e., applications editing an attorney's voice could be used for providing unfounded legal advice. The Polish Civil Code deals with a breach of the personal right related to the voice and related things, including image, scientific activity, freedom of conscience, or other implied legal consequences.

5.3.6.9 Federal Communications Commission

Federal Communications Commission (FCC) guidelines and rules state that the broadcasted content through various channels, such as TV, radio, and internet, should be captioned and subtitled in various languages to ensure viewers who are hard of hearing or deaf should get complete access to the necessary content. According to the guidelines, video programming should be understandable to the non-hearing person and accurate enough to communicate the exact meaning. The major required factors to comply with this regulation are accuracy, synchronicity, completeness, and proper placement.

5.3.6.10 Basel

It is a city in Switzerland and is the headquarters of the Bank of International Settlement (BIS), which nurtures cooperation among central banks with the shared objective of achieving financial stability and common standards of banking regulations. BIS hosts the meetings of the senior officials of the central banks of the member countries (currently 27 nations).

Basel Guidelines refer to the comprehensive administrative standards framed by a group of central banks called the Basel Committee on Banking Supervision (BCBS). The set of agreements mainly emphasizes the risks to banks and financial systems and is called the Basel Accords. The purpose of the Basel Accords is to ensure financial institutions have enough capital on account to meet obligations and absorb unexpected losses. Currently, the Basel 3 guidelines are accepted by the committee members. These guidelines were introduced in December 2010. The financial crisis of 2008 was the major reason for the introduction of these new norms.

5.3.7 CASE STUDY ANALYSIS

This section comprises use cases provided by vendors across verticals as a part of their advanced analytics offerings. Case studies include various solutions and their applications provided by vendors.

5.3.7.1 Education: Use Cases

5.3.7.1.1 Case Study 1: San Francisco University has selected Oracle Fusion Cloud EPM to streamline the budgeting and planning process

CATEGORY	DESCRIPTION
Problem	San Francisco State University needed a finance system that would provide real-time business insights and allow for strategic modeling by examining deficits, reserves, enrollment, tuition costs, buyout packages for employees, and many other factors.
Solution offered	San Francisco State University selected Oracle Fusion Cloud EPM for budgeting and planning due to its flexibility in modeling financial scenarios and handling new challenges that could potentially arise in the future. It has the ability to filter, analyze, and breakdown data has also allowed the university to meet its goal of removing reporting errors and giving the finance team the accurate data, it needs.
Benefits	▪ It has helped the finance team with tools to improve insights and enhance business and decision-making processes. ▪ It helps remove the tedious manual processes and replace them with simpler and more useful reporting and analytics methods. ▪ It saves time for the budget office, giving finance staffers the opportunity to analyze and think strategically about their data.
Solution Offered	Oracle Fusion Cloud EPM

Source: Company Website

5.3.7.1.2 Case Study 2: Shorelight used RapidMiner to create a workflow, automate the series of questions traditionally asked by the visa guides, and create a risk profile

CATEGORY	DESCRIPTION
Problem	Shorelight brings together universities and international students with the goal to help educate the world. Shorelight needs to build out a model to see what kind of students are at most risk for rejection of visa while studying abroad and also needs to help students strengthen their profiles.
Solution offered	Shorelight used RapidMiner to create a workflow, automate the series of questions traditionally asked by the visa guides, and create a risk profile. This all leads to a greater increase in operational predictability and growth in total profitable customers. It uses analytics to power growth and student success at a scale.
Benefits	▪ It helps the students by predicting the risk regarding visa rejection. ▪ It helps the student to strengthen their profiles. ▪ It efficiently resolves student questions.
Solution type	Future Bright Analytics

Source: Company Website

5.3.7.2 Retail and consumer goods: Use cases

5.3.7.2.1 Case Study 3: Fine Hygienic Holding choose Oracle Cloud for its data and application to transform and automate the core business processes

CATEGORY	DESCRIPTION
Problem	Fine Hygienic Holding is one of the world's leading wellness groups and manufacturers of sterilized tissue products. It. required a fundamental technology change to automate processes and use data for better decision-making, especially for supply chain and manufacturing for AI and robotics
Solution offered	Fine Hygienic Holding chooses Oracle Cloud for its data and application to transform and automate the core business processes. Oracle Cloud Applications and Infrastructure have helped it achieve its goals by driving integration across the organization.
Benefits	▪ It helps in reporting and analyzing the complex product categories. ▪ It helps in real-time decision-making. ▪ It helps predict consumer behavior.
Solution type	Oracle Cloud

Source: Company Website

5.3.7.2.2 Case Study 4: DEUTZ implemented SAP Analytics Cloud as a central analytics platform for BI

CATEGORY	DESCRIPTION
Problem	DEUTZ AG engine manufacturing company wants to replace three analytics solutions with one. As it was not running smoothly, data quality was insufficient, and preparing key figures took too much time.
Solution offered	DEUTZ implemented SAP Analytics Cloud as a central analytics platform for BI. It paves the way to greater transparency and better data quality. While implementing the solution, it has provided consistent data with accurate analysis and helps in maintaining the transparency of data.
Benefits	▪ Its analytics helps in improving decision-making in areas such as production control. ▪ It provides automated, centralized reports for managers.
Solution type	SAP Analytics Cloud

Source: Company Website

5.3.7.3 Media and entertainment: Use cases

5.3.7.3.1 Case Study 5: Scaling a small data team using ML

CATEGORY	DESCRIPTION
Problem	DAZN wanted a fast and low-maintenance way for enabling their small data team to handle predictive analytics and ML projects at scale to continue growing their business in the existing and new markets. It also wanted data analysts, not necessarily technical or experienced in ML, who can contribute to impactful data projects.
Solution offered	▪ Content Attribution – for determining the fixtures that drive sales and enable contextual information on key fixtures in each market. ▪ Propensity Modelling – for identifying customers that churn and enabling improved customer retention. ▪ Survival Analysis – for understanding customer stickiness and enabling calculation of the expected revenues for understanding customer RoI. ▪ Natural Language Processing – using NLP on social networks to perform market research. ▪ Advanced Customer Segmentation – to identify user behaviors, particularly regarding content and devices on which customers use the product.
Benefits	The solutions enabled AWS and Dataiku to shift the data culture at DAZN and highlight the innovations in advanced analytics and ML throughout the company.

Source: Company Website

5.3.7.4 Healthcare and life sciences: Use cases

5.3.7.4.1 Case Study 6: Innocens aims to reduce neonate mortality rates and helps speed sepsis diagnosis using hidden trends found in patient's data

CATEGORY	DESCRIPTION
Problem	Need to search for a technology solution that securely managed and analyzed personal patient data, automated and modernized the diagnosis process, predicted when an infant was trending toward sepsis and triggered an evaluation protocol.
Solution offered	Innocens has selected IBM to capture real-time data generated by the existing sensors. In addition, to the data modeling and analysis, it is facilitated through the open-source Red Hat OpenShift Container Platform and IBM Watson Studio on IBM Cloud Park for data. Using current and historical patient data integrated with AI, ML, and data modeling capabilities provided the ability to identify infants at risk of sepsis up to several hours, in some cases, before the standard protocol for monitoring and testing
Benefits	▪ Clinical trials are expected to confirm that early prediction leads to a rapid diagnosis. ▪ Timely treatment can improve clinical outcomes at a reduced cost.
Solution Type	IBM Watson Studio

Source: Company Website

5.3.7.4.2 Case Study 7: Fleury has selected FICO Forecaster to improve the demand forecasting and resource allocation process

CATEGORY	DESCRIPTION
Objective	Fleury is one of the largest diagnostic medical companies in Brazil. It needs to develop an analytic solution to improve the demand forecasting and resource allocation process.
Solution offered	Fleury has selected FICO Forecaster, built on the FICO Xpress Insight platform. This FICO solution provides an integrated demand view of the entire portfolio with a continuously updated forecast. It helps the company identify the most important pain points, and re-plan and prioritize internal actions to achieve its strategic goals.
Benefits	▪ It helps improve the demand forecasting and resource allocation process. ▪ It helps the company identify the most important pain points, re-plan and prioritize internal actions to achieve its strategic goals. ▪ It provides an integrated demand view of the entire portfolio with a continuously updated forecast.

Source: Company Website

5.3.7.5 Banking financial service and insurance: Use cases

5.3.7.5.1 Case Study 8: Danske Bank is modernizing with IBM Knowledge Watson Catalog to provide unified solutions across data quality, governance, and privacy

CATEGORY	DESCRIPTION
Objective	Danske banks need to store data and input it effectively. It also needs to make data easier to be used by every stakeholder, along with maintaining rigorous compliance standards.
Solution offered	Danske Bank is modernizing with IBM Knowledge Watson Catalog as the foundation of its metadata management practice. This unified solution across data quality, governance, and privacy helps the bank curate clean and secure data in an automated and agile way. It looks to establish proper data governance as well as consolidate data from disparate sources to achieve a single user experience that reduces data preparation and ensures data is trusted.
Benefits	▪ It helps secure the data in an automated and agile way. ▪ It helps in storing the data effectively. ▪ It helps reduce data preparations.
Solution	IBM Knowledge Watson Catalog

Source: Company Website

5.3.7.5.2 Case Study 9: KCB chose Oracle Exadata to consolidate all transaction processing and data warehouse workloads

CATEGORY	DESCRIPTION
Objective	Korea Credit Bureau's need to manage credit risk more effectively requires better evaluation of an individual consumer's credit worthiness. At the same time, credit bureaus strive for growth by advancing credit so that more consumers enjoy convenient financial services in their daily lives, such as loans, credit cards, or even telecom services.
Solution offered	KCB chose to consolidate all transaction processing and data warehouse workloads on Oracle Exadata to perform the crucial work of evaluating credit ratings. The bureau sought the increased performance, security, and availability that Oracle Exadata would provide.
Benefits	▪ It helps implement daily updates of financial data for all of KCB's clients, evaluating credit ratings based on the analysis. ▪ It provides a faster response time for credit analysis. ▪ It helps in performance improvement for data processing.
Solution	Oracle Exadata

Source: Company Website

5.3.7.6 IT and telecom: Use cases

5.3.7.6.1 Case Study 10: Worldpay uses FICO tools to gain better insights into onboarding decisions

CATEGORY	DESCRIPTION
Problem	Worldpay gains better insights into onboarding decisions through real-time analytics presented within dashboards and other intuitive user interfaces.
Solution offered	Worldpay uses FICO tools to gain better insights into onboarding decisions through real-time analytics presented within dashboards and other intuitive user interfaces. It has cut merchant onboarding time from days to minutes, which helps facilitate the company's rapid growth and opens new possibilities for partnerships with independent sales organizations and merchant banks.
Benefits	▪ It helps reduce merchant onboarding times. ▪ It results in improved customer experience and accelerated business growth.
Solution Offered	FICO Analytics Tools

Source: Company Website

5.3.7.6.2 Case Study 11: Improve customer segmentation due to multiple data sources and different business needs

CATEGORY	DESCRIPTION
Problem	The seamless integration of RapidMiner's lightning-fast data science platform and QlikView's strong visualization capabilities provides a Customer Segmentation solution to drive revenue optimization. The biggest challenge the IT companies are facing is customer segmentation due to multiple data sources and different business needs. As IT must prepare the data and modify the model to fit each business scenario manually, time and money are lost in the process.
Solution offered	RapidMiner and Qlik provided a solution to connect, combine, and analyze both structured and unstructured data coming from multiple data sources. It enabled the company to create an automated segmentation system based on users' specified parameters provided directly from the QlikView dashboard. The use of this QlikView connection enabled users to trigger a workflow directly from QlikView and evaluate different models based on those parameters.
Benefits	▪ Users are now able to visualize the customer base before and after segmentation in QlikView to quickly identify the most valued segments. ▪ A greater understanding of customer-based results reduced the cost of sales and improved revenue opportunities enabling them to immediately deliver accurate and timely decisions.

Source: Company Website

5.3.7.7 Transportation and logistics: Use cases

5.3.7.7.1 Case Study 12: Visualize and predict the status of the fleet

CATEGORY	DESCRIPTION
Objective	Schneider, one of the largest truckload carriers in North America, found that its legacy technology was placing limitations on its quest to reinvent itself as a fully digitized company. It needs to harness the data to visualize and predict the status of the fleet before a negative event can disrupt operations.
Solution offered	Schneider used Oracle's Digital Logistics capabilities to provide customers and employees with relevant, up-to-date information, such as delivery schedules, as well as predictive analytics to assist with planning for future loads and orders. Drivers received step-by-step assignments and managers were notified of deviations so they can correct them quickly. The solution also helped in detecting driver behavior patterns for immediate intervention or future coaching.
Benefits	It enabled Schneider to supply predictive analytics for delivering superior customer experiences and avoiding disruption that resulted in precise Estimated Time of Arrival (ETA) so that recipients downstream can be proactively notified with push-alerts.

Source: Company Website

5.3.7.8 Government and defense: Use cases

5.3.7.8.1 Case Study 13: HAN University implemented Qlik analytics solutions to promote BI skills within all the coursework

Description	HAN University of Applied Science wanted to improve the data literacy of its students by exploring the data analytics software and tools and to have access to a community of subject experts.
Challenge	▪ Encouraging the students to access various BI tools. ▪ Enabling the academics to integrate study materials into coursework. ▪ Provide additional learning materials at no cost to the university.
Solutions Offered	HAN joined the Qlik Academic Program and provided access to Qlik software, support, and training via a remote learning platform, qualifications, and community support.
Benefits	▪ Ensured access to various BI tools and expertise in the data analytics best practices. ▪ Allowed HAN to promote BI skills within all its coursework. ▪ Provided students with certifications and digital badges that can be shared on resumes and social networks.

Source: Company Website

5.3.7.9 Energy and utilities: Use cases

5.3.7.9.1 Case Study 14: Pöyry Choose FICO Xpress Optimization, part of Fico Decision Management Suite to provide customers with data-driven energy analyses and long-range predictions

CATEGORY	DESCRIPTION
Problem	Pöyry, a European energy consultancy, faces the challenge of the limitations of spreadsheet modeling to provide customers with data-driven energy analyses and long-range predictions.
Solution offered	Pöyry Choose FICO Xpress Optimization, part of Fico Decision Management Suite, to provide customers with data-driven energy analyses and long-range predictions. It has resulted in significantly faster runtimes and increased model accuracy. It has the ability to create complex, multi-country models to help energy companies stay ahead of changing industry dynamics.
Benefits	▪ It provides faster runtimes. ▪ It has resulted in increased model accuracy. ▪ It could create complex, multi-country models that give a competitive advantage.

Source: Company Website

5.3.7.9.2 Case Study 15: Rising need to optimize debt collection processes and improve recovery of receivables

CATEGORY	DESCRIPTION
Problem	The client intended to optimize its final debt collection processes to improve the recovery of receivables. The client also wanted to formulate focused debt management strategies for different customer segments to manage customer write-offs more effectively and decrease operational costs.
Solution offered	The client used the WNS solution to transform the collections process by embedding predictive analytics and making changes to the customer interaction strategy. The solution's key aspects included the Propensity-to-Pay predictive data model, customer classification based on scores, customer segment prioritization, cost-benefit analysis, and performance monitoring of pilot strategies.
Benefits	▪ Debt collection increased by 50% within three months. ▪ The modified process recorded an 8% rise in conversion rates compared to the standard process. ▪ Operational expenses decreased by 20%.

Source: Company Website

5.4 TECHNOLOGY ANALYSIS

Data is growing in variety and velocity in both structured and unstructured formats. Leveraging the data generated at ever-increasing speeds is often a difficult task. The rise in data complexity makes it challenging for advanced analytics to get the best from technologies such as AI and ML for digital transformation. With digital transformation reaching every aspect of work and personal lives, there is an automatic expectation of 24/7 anywhere availability when it comes to any organization with an online presence. This is a huge strain on call center and customer service professionals who must meet that expectation in a rapidly changing environment.

The advanced analytics market is also undergoing a large-scale digital transformation through various technologies. Advanced analytics solutions incorporating AI, ML, and Automated Speech Recognition (ASR) are becoming a priority to enable teams across organizations to enhance customer experiences and reduce failures. Although providers often implement change at slow pace, new information and analytics are enabling industry development and the redistribution of operational resources.

Various advanced analytics technologies are available to help solve problems across businesses. Most commercially proven applications focus on data cleansing to organize data or AI for better prediction. New technology players are leveraging AI and ML to augment the human problem-solving process through their ability to test large amounts of call data in minutes, as opposed to the months or years it could take the human brain. Advanced analytics is using big data technologies that enable a new generation of applications to analyze machine data and gain insight from it, which in turn improves business results.

5.4.1 ADVANCED ANALYTICS AND AI

AI and advanced analytics applications have an extremely real impact by making it possible to identify hidden patterns, provide personalized services, make predictions, and essentially bring the analysis the complex scenarios to the simple results, delivering unprecedented value. The most advanced technologies are used to analyze, enrich, manage, exploit, and display big data. The areas of applications are numerous, for example: automate human-fashioned activities to help support continuous presence (voice, sounds, figure, facial recognition), interact with users (bot, conversational interfaces), define the optimal use of assets predictive maintenance, traffic flow, logistic planning), understand documents (contract analysis, knowledge from unstructured records, character recognition), and many other. This technology involves the use of vast amounts of data to simulate nature-inspired behaviors, combined with traditional intelligence models and algorithms that are the shift in the evolution position AI and advanced analytics that have become the most enabling technology today. AI and advanced analytics can be applied to manage both the transition (Digital Transformation) and the “real target” of letting data drive the business. It is becoming a competitive advantage for businesses as they can add innovation, promote automation, along with cost reduction and intelligent decision-making. With the help of AI and advanced analytics, we can create a biometric (facial) recognition system for use in banking.

5.4.2 ADVANCED ANALYTICS AND ML

ML is a method of data analysis that automates analytical model building. It is a branch of AI based on the idea that systems can learn from data, identify patterns, and make decisions with minimal human intervention. It enables organizations to harvest a higher volume of insights from both structured and unstructured data. It is a form of predictive analytics that advances organizations up the BI maturity curve, moving from exclusive reliance on descriptive analytics focused on the past to include forward-looking, autonomous decision support. Analytic solutions based on ML often operate in real time, adding a new dimension to BI. Advanced analytics based on ML is used to deliver business benefits in every industry. For instance, when there are large amounts of data predictive models can be used with ML. ML when used with advanced analytics can provide companies with a competitive edge by solving problems and uncovering insights faster and more easily than with conventional analytics. It offers several advantages to IT, data scientists, various lines of business groups and their organizations.

5.4.3 ADVANCED ANALYTICS AND IOT

Analytics enables organizations to leverage the massive amounts of data generated by IoT devices, using analytics stacks. IoT analytics is often considered a subset of big data, involved with combining heterogeneous streams and transforming them into consistent and accurate insights. IoT analytics tools help organizations successfully leverage complex IoT datasets. When organizations are better equipped to understand data, they can improve products and increase earnings. Since customers are connected to the internet, organizations can optimize products according to customer needs. For example, IoT analytics can provide insights into the popularity of features, average usage patterns, optimal battery capacity. Additionally, this level of connectivity and analysis helps organizations fix issues, releasing software updates over the air. When fixes are released quickly and timely, customer engagement and user retention increase significantly. IoT when used in advanced analytics can save valuable time, reducing tasks related to the integration of data sources.

5.4.4 ADVANCED ANALYTICS AND BLOCKCHAIN

Blockchain and data science play an important role in industries that heavily rely on data, such as finance, retail and eCommerce, and telecommunications and IT. Blockchain technology facilitates a decentralized database where business is rendered transparent without the involvement of middlemen. It is expected to have a major impact on cybersecurity, IoT, supply chain management, market prediction, governance, information management, and financial transactions. Blockchain has redesigned the way in which people deal with their money due to its effectiveness, especially in terms of security. Hence, from the data analytics point of view, investigation of the application of blockchain technology in a wide range of domains is crucial. A blockchain is a growing list of records, called blocks, that are linked using cryptography. Blockchain technology has a huge potential in analytics. Modern businesses have been benefiting from data analytics for several years now.

5.5 PATENT ANALYSIS

5.5.1 METHODOLOGY

The patent analysis offers crucial insights into patent filing, business interests, and patenting activities yearly and country-wise. It is a technique used to understand the information contained in patents. This study defines patents claiming inventions linked to the advanced analytics market. For this study, a well-planned strategy is devised. Patents are categorized and analyzed using advanced analytics. Patents were categorized and analyzed using keywords, such as advanced analytics, predictive analytics, prescriptive analytics, and big data analytics. These keywords are used for 'Title or Abstract' to understand the patent filing profile. Vendors have been filing patents in the advanced analytics space. This section aims at providing an overview of patenting activities. Sources used for this analysis are The Lens—Patent, Scholarly Search, and Patentscope-WIPO.

5.5.2 DOCUMENT TYPE

TABLE 7 PATENTS FILED, 2018–2021

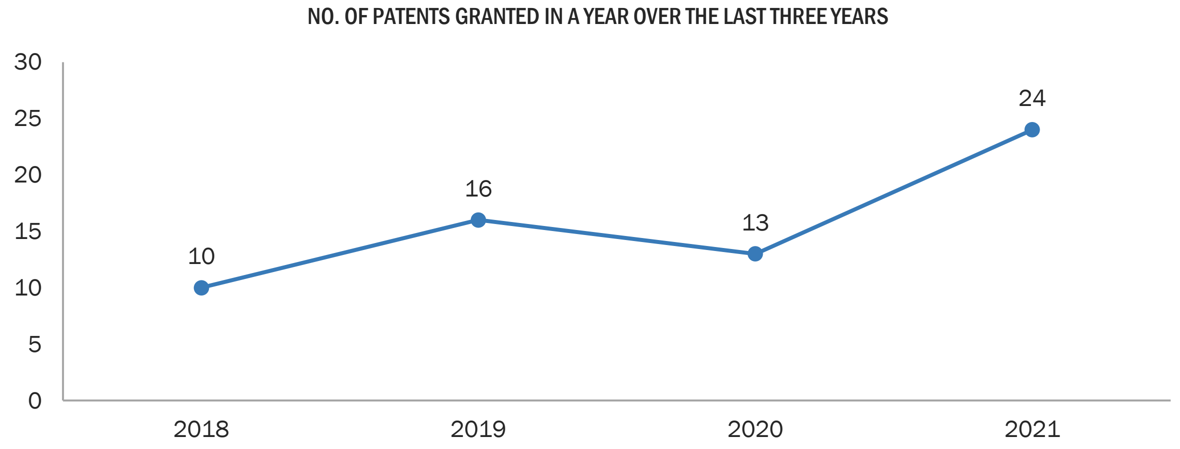
DOCUMENT TYPE	NO. OF PATENTS
Patent Applications	239
Granted Patents	104
Total	343

Source: The Lens—Patent, and Scholarly Search

5.5.3 INNOVATION AND PATENT APPLICATIONS

The analysis has been performed on patents filed and granted in the past three years specific to advanced analytics. An increasing trend has been observed in the number of patent grants in the past three years, with a total of 343 patents being granted in 2021. Grand Performance Online, IBM, and Advanced Elemental Tech are the top three companies that have applied for the maximum number of patents in the past three years.

FIGURE 34 TOTAL NUMBER OF PATENTS GRANTED IN A YEAR, 2018–2021

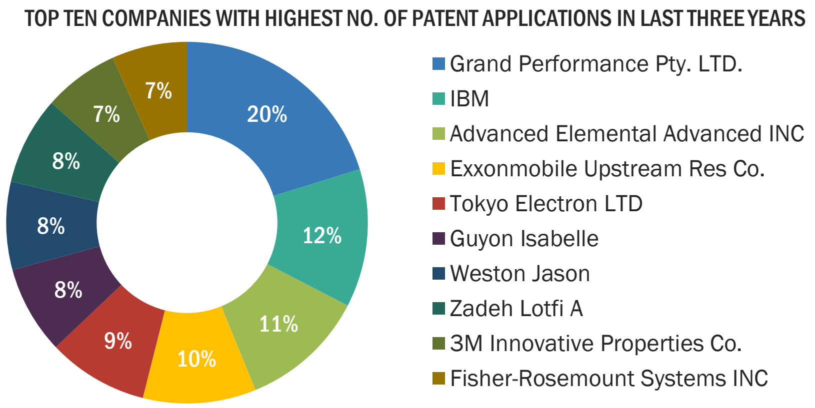


Source: Lens— Patent and Scholarly Search

5.5.4 TOP APPLICANTS

The top applicant's analysis helps know which organization possesses the most patents relative to the advanced analytics inventions. This, in turn, will help collaborators identify inventors and vice versa for the joint development of products.

FIGURE 35 TOP TEN COMPANIES WITH THE HIGHEST NUMBER OF PATENT APPLICATIONS, 2018–2021



Source: The Lens—Patent and Scholarly Search

5.6 PRICING MODEL ANALYSIS, 2021

Advanced analytics solutions and service providers suggest various pricing models. Some vendors have several fixed subscription plans, while others provide flexible pricing that depends on the business size and industry and annual sales volume. Providers may only charge per transaction. Companies share pricing information on requests. The pricing of solutions and services is subscription-based and tailor-based. Most advanced analytics offer free trial period days of subscription. The subscription is renewed every year. Service providers may offer tiered pricing based on the number of users and volume discounts. Advanced analytics solutions and services can be priced as per the method below:

- Pay-As-You-Go model – It is one of the pricing models for cloud services that encompass both subscription-based and consumption-based models. Pay-as-you-go models work differently as customers pay a one-time cost for products or services to get access to them. Then, if they need the products or services again, they will pay again when they choose. For instance, Microsoft offers Azure Data Lake Analytics, the first cloud serverless job-based analytics service where users can easily develop and run massively parallel data transformation and processing programs in U-SQL, R, Python, and. Net over petabytes of data.
- Subscription-based Pricing Model – It is one of the pricing methods where customers can purchase solutions or services from vendors for a specific period, monthly or annually, with a defined value. Subscription-based pricing offers a flat rate or a tier ranking system. Subscriptions automatically charge or renew a membership, even if a customer has forgotten that they have that membership, and businesses have that guaranteed revenue.
- Free Trial Period – Startup companies offer a free trial period-based subscription, after which users can subscribe to the paid services of their choice.

TABLE 8 ADVANCED ANALYTICS MARKET: PRICING MODEL ANALYSIS, 2021

COMPANY	OFFERING	PRICING	DETAILS
SAP	SAP Analytics Cloud	The company offers flexible licensing options to fulfill organization's needs.	▪ It offers three categories ▪ 30-day free trial ▪ BI starting from USD 35 /User/Month ▪ Planning: price given upon request
AWS	Amazon EMR	AWS offers users a pay-as-you-go approach for pricing for over 160 cloud services. With AWS users will pay only for the individual services as per need.	▪ Amazon EMR Pricing is based on requested vCPU and memory resources for the Task or Pod ▪ per vCPU per hour for USD 0.01012 ▪ per GB per hour for USD 0.00111125
TIBCO Software	TIBCO Cloud Spotfire	TIBCO Software offers immediate, anywhere access, and unlimited scalability without IT infrastructure and maintenance concerns services.	▪ The company offers five types of plans ▪ 30-day free trial ▪ Analyst: USD 125/month or USD 1250/year ▪ Business Author: USD 65/month or 650/year ▪ Consumer: USD 25/month or 250/year ▪ Library storage: USD 25/month or USD 250/year for 25 GB
Oracle	Oracle Analytics Cloud enterprise	The company offers a licensed based pricing model	▪ It offers standard edition at USD 80/User/Month and professional edition at USD 16/User/Month

Salesforce	Tableau's	Tableau's is a powerful analytics platform that empowers everyone across the organization with data.	▪ It offers three types of plans ▪ Tableau Creator ▪ USD 70 per user per month ▪ Tableau Explorer ▪ USD 42 per user per month ▪ Tableau Viewer ▪ USD 15 per user per month

Source: Company Websites

5.7 PORTER'S FIVE FORCES ANALYSIS

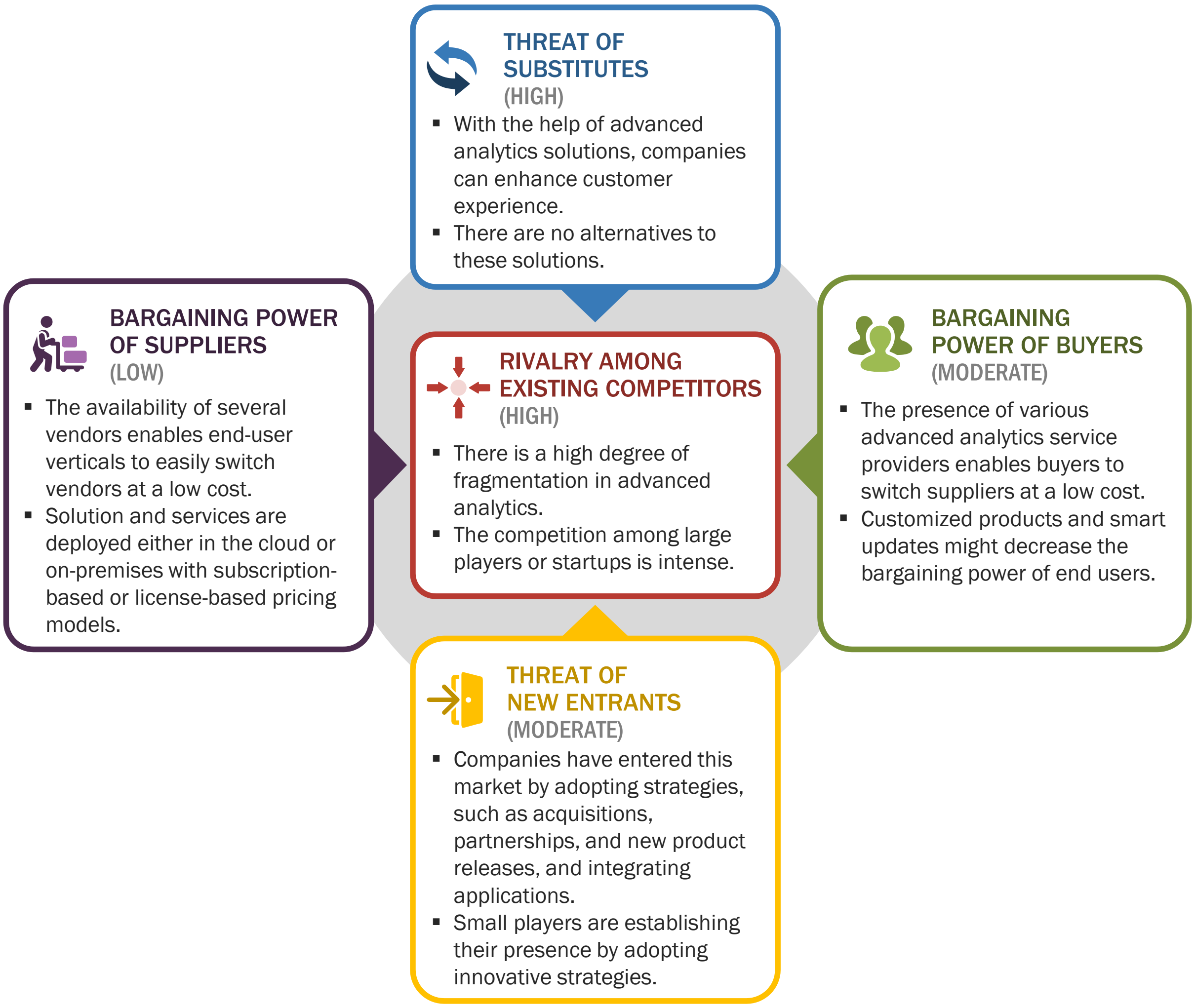
Porter's five forces analysis is a framework that determines the competitive intensity of a market. This section analyzes the advanced analytics market from five different perspectives: the intensity of competitive rivalry within industries, the threat of new entrants, the bargaining power of suppliers, the bargaining power of buyers, and the threat of substitutes. It draws insights from organizations' economics to find five forces that determine the competitive intensity and, therefore, the attractiveness of an advanced analytics market in terms of profitability.

TABLE 9 IMPACT OF EACH FORCE ON THE ADVANCED ANALYTICS MARKET

PORTER'S FIVE FORCES	IMPACT
Intensity of Competitive Rivalry	High
Bargaining Power of Suppliers	Low
Bargaining Power of Buyers	Moderate
Threat of Substitutes	High
Threat of New Entrants	Moderate

Source: Expert Interview and MarketsandMarkets Analysis

FIGURE 36 PORTER'S FIVE FORCES ANALYSIS



Source: MarketsandMarkets Analysis

5.7.1 THREAT OF NEW ENTRANTS

The threat of new entrants refers to the force of new potential competitors in any market who can capture the market share of the existing companies. Currently, the threat from new entrants in the advanced analytics market is moderate, as there is low R&D expenditure, and a moderate initial investment is required. There are several global and domestic players that have an established base in the market. These market players have a global presence and offer a wide range of advanced analytics solutions for various applications. Various companies enter this market by adopting strategies, such as acquisitions, partnerships, new product releases, and integrating applications. Proprietary technologies and stringent government regulations are the entry barriers in the market. The growing demand for advanced analytics solutions in emerging applications is enabling startups to set up their presence in the market. Hence, the threat of new entrants in the advanced analytics market is moderate.

5.7.2 THREAT OF SUBSTITUTES

The threat of substitutes refers to the possibility of alternative products replacing the existing ones in the market. Switching to substitutes will not be much expensive, especially for bigger corporations who have standardized their systems with specific advanced analytics solutions regardless of an update. Some advanced analytics business applications are available in the market work across applications, such as predictive analytics, prescriptive analytics, and text analytics. These advanced functionalities enable an efficient implementation of advanced analytics solutions and support businesses in making appropriate decision-making. An increase in innovation would translate to an increase in a better experience, which further would be worthy for the users. Loyal customers may approve or be able to afford the new prices, which would, in turn, provide better features. Hence, the threat of substitutes in the advanced analytics market is high.

5.7.3 BARGAINING POWER OF SUPPLIERS

The bargaining power of suppliers refers to the extent to which the solutions and services suppliers/providers pressurize end users of advanced analytics solutions and services. The suppliers of advanced analytics solutions and services offer various pricing models, such as subscription-based pricing or license-based pricing models, to the end users. There are several suppliers or low switching costs between rival suppliers, enabling users of advanced analytics solutions and services to switch suppliers/providers at a low cost. As a result, the bargaining power of suppliers is low.

5.7.4 BARGAINING POWER OF BUYERS

The bargaining power of buyers refers to the pressure that buyers exert on manufacturers of advanced analytics solutions. The customer base for the advanced analytics market is expanding day by day due to the presence of established players and upcoming startups in this market. Several unorganized vendors provide advanced analytics solutions and services at a low rate compared to other major players in the market. This increases the bargaining power of buyers in the advanced analytics market. However, buyers lose their power when there is a requirement of customized advanced analytics—a requirement-based platform to be integrated with the existing system or to cater to a specific business requirement. This makes advanced analytics solution providers price setters, decreasing the bargaining power of buyers. Hence, the bargaining power of buyers in this market is moderate.

5.7.5 INTENSITY OF COMPETITIVE RIVALRY

The competitive rivalry within the industry refers to the extent to which market players exert pressure on each other. The presence of various well-established players has intensified the level of competitive rivalry in the advanced analytics market. Some of these players IBM (US), Oracle (US), SAS Institute (US), SAP (Germany), FICO (US), KNIME (Switzerland), Microsoft (US), Altair (US), RapidMiner (US), AWS (US), Salesforce (US), TIBCO Software (US), Alteryx (US), Teradata (US), Adobe (US), Absolutdata (US), Moody's Analytics (US), Qlik (US), Databricks (US), Dataiku (US), Kinetica (US), H2O.ai (US), Domino Data Lab (US), DataRobot (US), DataChat (US), Imply (US), Promethium (US), Siren (Ireland), Tellius (US), SOTA Solutions (Germany), and Vanti Analytics (Israel). These solution providers have a strong foothold in the local and global markets due to their strong product portfolio, integration capabilities, and well-established strategic alliances. Besides these, there are several manufacturers in the unorganized sector, which collectively hold a prominent share in the advanced analytics market. Hence, the intensity of competitive rivalry in the market is high.

5.7.6 SCENARIO

TABLE 10 CRITICAL FACTORS TO IMPACT THE GROWTH OF THE ADVANCED ANALYTICS MARKET

ELEMENTS OF UNCERTAINTY
▪ Data-related policies and regulations fueling the growth, especially in North America and Europe, followed by emerging regions ▪ Increasing demand for online shopping and transactions due to the COVID-19 pandemic ▪ Growing demand for speech technology to gain meaningful insights across verticals ▪ Decrease in the number of customer visits in stores ▪ Consumer sentiment varies greatly across countries ▪ Organizations have gained an opportunity to accelerate the pivot to digital commerce ▪ Organizations globally are experiencing unprecedented workforce disruption ▪ Internal and external customer behavior transformation due to COVID-19

Source: MarketsandMarkets Analysis

6 ADVANCED ANALYTICS MARKET, BY COMPONENT

KEY FINDINGS

- The advanced analytics market, by component, is expected to grow from USD 33,771 million in 2021 to USD 89,823 million by 2026, at a CAGR of 23.4% during the forecast period.
- The services segment is expected to grow at a higher CAGR of 22.7% during the forecast period and projected to reach a market size of USD 34,287 million by 2026.
- The solution segment is estimated to have a larger market share of 64.5% by 2021.
- The services segment plays a vital role in the functionality of advanced analytics solutions. These services are an integral step in deploying technology solutions and are taken care of by the solution and service providers.
- The demand for advanced analytics solutions is increasing globally due to the increasing demand for offering enhanced customer support across major verticals such as BFSI, IT and telecom, and transportation and logistics.

6.1 INTRODUCTION

The global advanced analytics market by component has been segmented into solution and services. Advanced analytics employs predictive modeling, statistical methods, ML, and process automation techniques beyond the capacities of traditional BI tools to analyze data or business information. It provides the ability to forecast future trends or outcomes for a deeper understanding of the business. Advanced analytics solutions offer a wider set of capabilities and enable a stronger strategic decision-making process for the future. IBM, Google, Alteryx, RapidMiner, SAP, SAS Institute, and Oracle are some of the leading vendors that offer advanced analytics solutions in the market.

The use of advanced analytics solutions in demand forecasting allows organizations to make data-based informed decisions and augment profitability. The solutions in the advanced analytics market are customized to meet the unique needs of each business therefore several businesses are adopting progressive analytics solutions. Organizations can boost customer acquisition and retention with the help of advanced analytics solutions. The major adoption of the analytics solutions is in the developed countries of North America and Europe due to increased smart projects and IoT connectivity data. The service providers for advanced analytics help in implementing intelligent automation and technology for efficient data analysis. Services are used in various industries and verticals, such as manufacturing, banking, and professional services; government agencies are also investing aggressively in big data analytics, thereby driving the demand for advanced analytics. The solutions and services offered in the advanced analytics market have been described in this section.

6.1.1 COMPONENTS: COVID-19 IMPACT

- Managers have begun to provide the fullest security to their employees in the face of the COVID-19 pandemic. To avoid infection, it is critical to always keep track of contacts and adhere to social distancing, which is nearly hard to achieve through standard monitoring and inspection. In this case, RTLS technologies are more effective against COVID-19. These solutions can aid in the creation of a healthy work environment, the establishment of working spaces that consider the required distance, and the development of protocols that will aid in the rapid detection of infection cases.

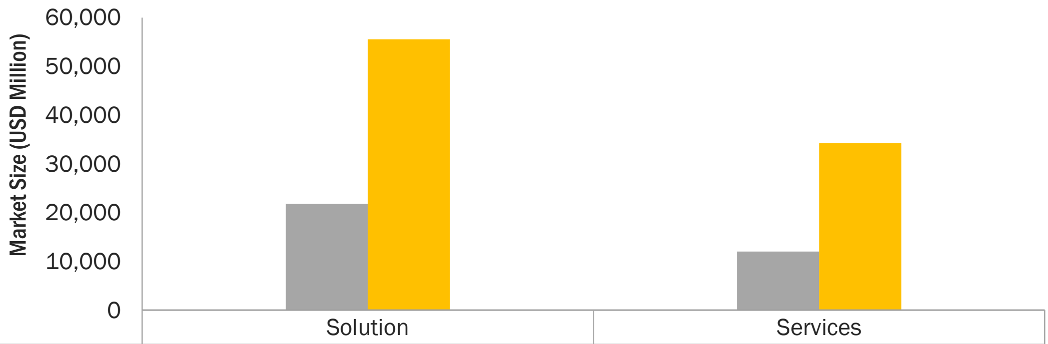
- The reality of physical proximity was rapidly brought to the forefront of everyone's attention by COVID-19. Personal proximity has now become a severe barrier in day-to-day life for most individuals all over the world. As a result, various businesses and governments are turning to technology to aid in the safe navigation and location of buildings and sites that people frequent.

- Fake news and disinformation through social media have been a pressing issue in several countries across the world, especially in the recent period of the COVID-19. Approximately 51% of the survey respondents by Statista agreed that they are concerned when facts are spun and twisted, then pushed for a particular agenda through fake news media. On the other hand, 32% of the respondents seemed to think satire was a concern in terms of fake news. AI-assisted virtual assistants have been proven to be the most effective tool to deal with this issue. Government organizations and enterprises have been leveraging digital assistants as a key strategy for communication.

- COVID-19 would have an impact on all the elements of the technology sector. Global ICT spending is estimated to decline by 4%–5% by the end of 2020. The hardware business is predicted to have the most impact on the IT industry. Owing to the slowdown of hardware supply and reduced manufacturing capacity, the IT infrastructure growth has slowed down. Businesses providing platforms and services are also expected to slow down for a short span of time. However, the adoption of collaborative applications, analytics, security solutions, and AI is set to increase in the remaining part of the year.

- Due to COVID-19 impact modern business applications leverage ML models to analyze real-world and large-scale data and predict or react intelligently to events. Unlike data analysis for research purposes, the models deployed in production are required to handle data at scale and often in real time and must provide accurate results and predictions for end users.
- Several big organizations, such as AWS and Google Cloud, have offered researchers free access to open datasets and analytics tools to help them develop COVID-19 solutions faster. A team developed an AI tool at UTHealth. This has shown the need for stricter, immediate interventions in the Greater Houston area. At Stanford University, researchers have launched a data-driven model that predicts the possible outcomes of various intervention strategies.
- The outbreak of the COVID-19 pandemic has prompted organizations to reconfigure and revamp their business operations. Progressive analytics is particularly allowing businesses to revamp their analytical models and data management processes in line with the changing business environments.
- Analytics professionals, BI professionals, and professionals providing expertise in more advanced analytics such as AI and ML have been called for their expertise to help executives make business decisions on how to respond to the new business challenges caused by the COVID-19 outbreak.

FIGURE 37 SERVICES SEGMENT TO REGISTER A HIGHER CAGR DURING THE FORECAST PERIOD



	Solution	Services
2021	21,766	12,005
2026	55,536	34,287
CAGR (2021-2026)	20.6%	23.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 11 ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2017–2020)
Solution	5,645	8,511	12,168	16,587	18,871	35.2%
Services	2,755	4,265	6,263	8,773	10,192	38.7%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 12 ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	21,766	25,467	30,249	36,492	44,668	55,536	20.6%
Services	12,005	14,348	17,411	21,462	26,848	34,287	23.4%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the advanced analytics market size by component. The market size of the solution segment is expected to hold a larger market share in 2021, while the services segment is projected to grow at a higher CAGR during the forecast period. This growth can be attributed to the need for determining the time and cost required to install the solution that requires fully managed advanced analytics services. The high growth is attributed to the higher adoption of advanced analytics solutions across key verticals such as BFSI, IT and telecom, and retail and consumer goods.

6.2 SOLUTION

6.2.1 INCREASING POPULARITY OF ONLINE SHOPPING AND RISING SOCIAL NETWORK PENETRATION TO BOOST THE DEMAND FOR THE ADVANCED ANALYTICS SOLUTION ACROSS THE GLOBE

Advanced analytics solution is becoming more widespread, as enterprises continue to create new data at a rapid rate. As a result, the amount of data generated worldwide is increasing immensely. Data generated, owing to the increasing social penetration, contains hidden patterns that traditional analytics tools are unable to find. If implemented in the correct form, advanced analytics solutions can help businesses unlock critical hidden information, which can then be used by organizations to customize their offerings to their consumers. With the increasing popularity of online shopping and the rising social network penetration, the demand for advanced analytics solutions is anticipated to rise substantially during the forecast period. Digitization acts as a major factor in the industry that was able to create an ecosystem that promotes the business to adopt these advanced solutions for improved business insights. For instance, American Express is using advanced analytics solutions to improve speed and performance in its service delivery. The company uses sophisticated predictive modeling to analyze historical data and over a hundred variables to predict potential customer churn, enabling it to speed up and optimize marketing efforts to retain these accounts. These solutions consist of software applications, which can be used to develop various business applications that include fraud detection, cyber risk management, consumer behavior analysis, and workforce management. Organizations using advanced analytics solutions gain insights from that data, they can form multilateral, incisive strategic plans for all areas and functions of the business, from customer service to digital marketing. According to industry experts, 62% of organizations reported themselves as planning to integrate advanced analytics into their decision-making models almost immediately. Major players offering advanced analytics solutions are IBM, Oracle, SAP, SAS Institute, Alteryx, Altair, and RapidMiner.

TABLE 13 SOLUTION: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	1,976	2,932	4,121	5,518	6,195	33.1%
Europe	1,515	2,267	3,217	4,352	4,918	34.2%
APAC	1,093	1,689	2,476	3,462	4,012	38.4%
MEA	602	919	1,331	1,838	2,118	37.0%
Latin America	460	704	1,023	1,418	1,627	37.2%
Total	**5,645**	**8,511**	**12,168**	**16,587**	**18,871**	**35.2%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 14 SOLUTION: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	7,048	8,129	9,513	11,300	13,611	16,598	18.7%
Europe	5,636	6,551	7,729	9,263	11,264	13,879	19.8%
APAC	4,714	5,619	6,799	8,356	10,420	13,243	22.9%
MEA	2,474	2,933	3,528	4,312	5,346	6,718	22.1%
Latin America	1,894	2,236	2,679	3,261	4,028	5,098	21.9%
Total	**21,766**	**25,467**	**30,249**	**36,492**	**44,668**	**55,536**	**20.6%**

e: estimated; p: projected

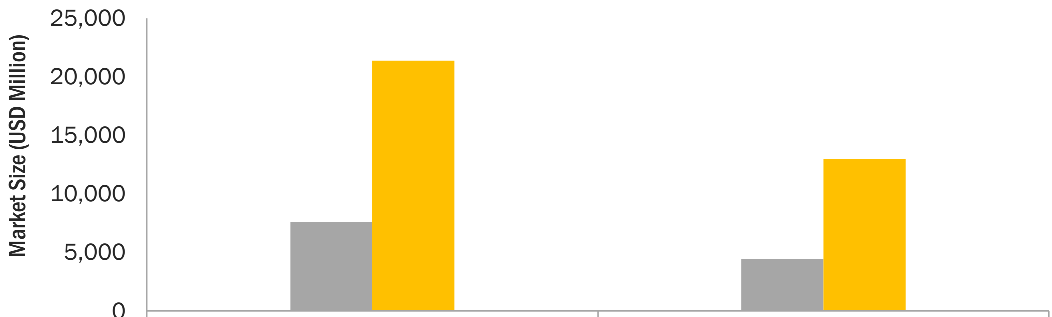
Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

6.3 SERVICES

The services segment has been further divided into professional services and managed services. These services play a vital role in the functioning of advanced analytics solutions. Advanced analytics vendors require technical support and consulting services to deploy their solutions quickly and smoothly in the market. These services help maximize the value of enterprise investments. Service providers ensure end-to-end deployment and maintenance of advanced analytics solutions and address pre-and post-deployment queries. These services play a vital role in the proper functioning of advanced analytics solutions. Advanced analytics vendors require technical support services and consulting services to deploy their solutions quickly and smoothly in the market. The amount of data generated is increasing day-by-day due to the factors such as the increasing adoption of IoT-enabled technologies across verticals, which have encouraged organizations to adopt advanced analytics services. The adoption of services helps organizations enhance their productivity by providing support for the delivery of solutions and associated training. Enterprises are continuously seeking services through support and maintenance service providers to deploy efficient advanced analytics solutions.

The overall cost of the advanced analytics technology installation depends on the complexity of the application and the type of technology used. The services play an important part in the overall advanced analytics implementation process, especially for asset tracking and navigation. Knowing the installation and maintenance details before deploying advanced analytics technology is crucial for determining the exact time and cost investments required to install that solution. For instance, some advanced analytics technology installations may require high energy consumption and can affect the production process and create interference and obstruction in work. On the other hand, some solutions such as RFID can be integrated with the existing wireless network infrastructure as it requires low energy than other technologies and can be easily integrated at a lesser cost. Support and maintenance services include the assistance provided during the installation of the solutions and maintenance activities.

FIGURE 38 MANAGED SERVICES TO GROW AT A HIGHER CAGR DURING FORECAST PERIOD



	Professional Services	Managed Services
2021	7,592	4,424
2026	21,383	12,976
CAGR (2021-2026)	23.0%	24.0%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 15 ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	1,767	2,728	3,993	5,576	6,460	38.3%
Managed Services	987	1,537	2,270	3,197	3,732	39.4%
Total	**2,755**	**4,265**	**6,263**	**8,773**	**10,192**	**38.7%**

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 16 ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	7,589	9,046	10,946	13,457	16,787	21,364	23.0%
Managed Services	4,416	5,303	6,464	8,005	10,060	12,923	24.0%
Total	**12,005**	**14,348**	**17,411**	**21,462**	**26,848**	**34,287**	**23.4%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the advanced analytics market size by service. The professional services segment is expected to hold a larger market size in 2021, while the managed services segment to have a higher CAGR during the forecast period due to the increasing outsourcing of advanced analytics solutions across firms. The growth of the professional services segment is said to be directly proportional to the complexities of the operations in the installed advanced analytics solutions. The growth of the services segment can be attributed to the provision of technical expertise, which helps companies enhance their focus on core business processes.

TABLE 17 SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	1,026	1,563	2,256	3,104	3,559	36.5%
Europe	719	1,105	1,612	2,242	2,589	37.8%
APAC	542	861	1,297	1,864	2,207	42.0%
MEA	260	408	608	865	1,019	40.7%
Latin America	208	328	490	698	819	40.8%
Total	**2,755**	**4,265**	**6,263**	**8,773**	**10,192**	**38.7%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 18 SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	4,134	4,871	5,824	7,069	8,702	10,901	21.4%
Europe	3,031	3,600	4,342	5,320	6,615	8,377	22.5%
APAC	2,649	3,226	3,989	5,010	6,386	8,340	25.8%
MEA	1,217	1,475	1,815	2,269	2,878	3,720	25.0%
Latin America	974	1,176	1,441	1,794	2,266	2,948	24.8%
Total	**12,005**	**14,348**	**17,411**	**21,462**	**26,848**	**34,287**	**23.4%**

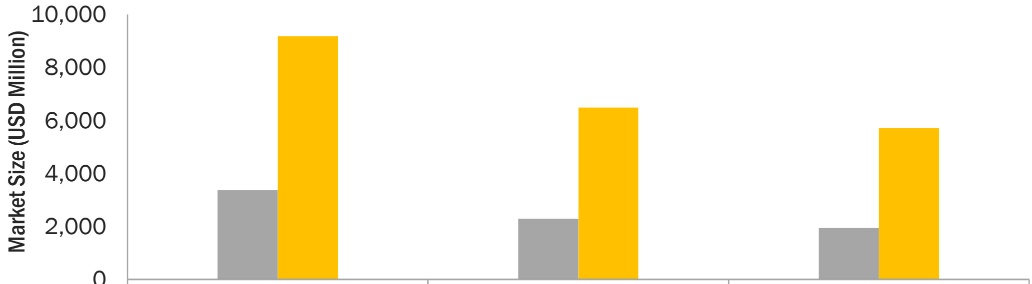
e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.1 PROFESSIONAL SERVICES

Professional services are mostly post-sales services delivered to customers after the purchase of the product. Customized, knowledge-based services for clients, such as solutions deployment, onboarding, training, and software handling, are considered within the ambit of professional services. Such services also include designing, planning, upgrades, and a host of consulting services offered to clients. Companies are increasingly engaging the services of consultants, solutions experts, and dedicated project management teams to design and deliver critical decision-support solutions and services. The professional service team ensures smooth project management and precise on-site installation of analytics solutions and services. The growth of the professional services segment is governed by the complexity of new business models and process efficiencies in the face of new imperatives, such as increasing demand, new technological interventions, and changing regulatory policies in today's marketplace, professional services are an essential part of the growing trend of services. The customized, knowledge-based services that are offered to clients fall under the professional services category. Professional services are categorized into training and consulting, system integration and implementation, and support and maintenance services. A few of the vendors that offer professional services are Oracle, SAP, Altair, TIBCO Software, and Databricks.

FIGURE 39 TRAINING AND CONSULTING SEGMENT TO GROW AT THE HIGHEST CAGR DURING THE FORECAST PERIOD



	System Integration and Implementation	Support and Maintenance	Training and Consulting
2021	3,363	2,293	1,937
2026	9,181	6,480	5,723
CAGR (2021-2026)	22.2%	23.1%	24.2%

Source: Secondary Research, Experts Interviews, and MarketsandMarkets Analysis

TABLE 19 PROFESSIONAL SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	428	668	988	1,394	1,632	39.7%
System Integration and Implementation	809	1,240	1,802	2,500	2,878	37.3%
Support and Maintenance	530	820	1,203	1,682	1,950	38.5%
Total	**1,767**	**2,728**	**3,993**	**5,576**	**6,460**	**38.3%**

Source: Secondary Research, Experts Interviews, and MarketsandMarkets Analysis

TABLE 20 PROFESSIONAL SERVICES: ADVANCED ANALYTICS SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	1,936	2,330	2,848	3,534	4,451	5,718	24.2%
System Integration and Implementation	3,361	3,983	4,791	5,855	7,262	9,172	22.2%
Support and Maintenance	2,292	2,733	3,308	4,067	5,073	6,474	23.1%
Total	**7,589**	**9,046**	**10,946**	**13,457**	**16,787**	**21,364**	**23.0%**

e: estimated, p: projected

Source: Secondary Research, Experts Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the professional services segment by type. The training and consulting segment is expected to grow at the highest CAGR during the forecast period. This growth is due to the adoption of professional services by organizations to integrate advanced analytics solutions into the existing business applications for enhanced productivity and customer experience.

TABLE 21 PROFESSIONAL SERVICES: ADVANCED ANALYTICS SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	663	1,006	1,448	1,987	2,272	36.1%
Europe	450	690	1,002	1,390	1,601	37.3%
APAC	345	546	820	1,174	1,386	41.6%
MEA	170	267	397	562	660	40.3%
Latin America	140	219	327	464	543	40.5%
Total	**1,767**	**2,728**	**3,994**	**5,577**	**6,462**	**38.3%**

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 22 PROFESSIONAL SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	2,632	3,093	3,688	4,465	5,482	6,844	21.1%
Europe	1,869	2,214	2,663	3,254	4,035	5,092	22.2%
APAC	1,660	2,016	2,486	3,115	3,960	5,152	25.4%
MEA	787	951	1,168	1,456	1,842	2,373	24.7%
Latin America	644	775	948	1,176	1,482	1,923	24.5%
Total	**7,592**	**9,050**	**10,953**	**13,466**	**16,801**	**21,383**	**23.0%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the market size of the professional services segment by region. North America is expected to hold the largest market size in 2021. APAC is projected to grow at the highest CAGR during the forecast period, due to the increasing organizations' demand for outsourcing services to improve operations and reduce operational costs.

6.3.1.1 Training and consulting services

6.3.1.1.1 Training and consulting services are needed in the initial phase of implementing advanced analytics solutions based on their business needs and going market trends

Training and consulting services are needed in the initial phase of implementing advanced analytics solutions or looking for customized solutions based on their business needs and market trends. These services help reduce complexities while operating and upgrading advanced analytics solutions. Consulting services also help raise the as they can be easily customized and can deliver maximum product assurance. Consulting services steer around the critical issues and opportunities related to strategy, marketing, operations, technology, mergers and acquisitions, and corporate finance, and handhold enterprises to help increase efficiency, boost performance, reduce expenses, and enhance resilience. The growing need for organizations to comply with various labor laws and regulations across the globe has boosted the demand Training and support services provide information about the functioning of solutions so that companies can use them in the right manner and gain maximum benefits from them. For instance, IBM data analytics and consulting services offer the strategic value of enterprise data and build an insight-driven organization. It creates a unified view of enterprise-wide data to build analytical and operational views for decision-making and automate end-to-end intelligent workflows. Various vendors offering training and consulting services are IBM and Altair in the advanced analytics market.

TABLE 23 TRAINING AND CONSULTING SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	159	245	357	495	573	37.7%
Europe	108	168	248	347	404	38.9%
APAC	88	140	211	304	362	42.5%
MEA	39	62	93	133	157	41.2%
Latin America	33	53	80	115	137	42.5%
Total	**428**	**668**	**988**	**1,395**	**1,632**	**39.7%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 24 TRAINING AND CONSULTING SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	672	799	964	1,180	1,467	1,849	22.4%
Europe	477	570	693	854	1,069	1,366	23.4%
APAC	436	533	661	834	1,067	1,397	26.2%
MEA	188	229	283	355	452	586	25.5%
Latin America	164	201	249	313	400	525	26.1%
Total	**1,937**	**2,332**	**2,849**	**3,537**	**4,455**	**5,723**	**24.2%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.2 System integration and implementation services

6.3.1.2.1 Advanced analytics solutions must be integrated with existing systems; this requires the right level of connectors and back-end integration capabilities

System integration and deployment services allow experts to integrate appliances or devices with the applied software and deploy it into the system. Integrating the advanced analytics platform with other data management tools plays a vital role in managing huge amounts of customer data generated by the conversations. Interoperability issues are a major concern in deploying the platform and solution with hardware components from different vendors. Hence, integrating the solution or platform with various hardware components and testing the functioning of the entire system is of the utmost importance. This is expected to encourage the adoption of system integration and testing services in the coming years. System integration and implementation services are gaining wide acceptance by end users globally, as these services ensure systems communicate with each other efficiently to enhance the working of the entire production process. System integration and implementation services begin with collecting customers' requirements, and then deploying, integrating, testing, and rolling out solutions. System integration and implementation services integrate location-based software and system regardless of technology. System integration provides facilities and frameworks for the integration of various platforms with third-party environments. This integration allows applications to run on other platforms and make advanced analytics-based solutions quick and efficient.

TABLE 25 SYSTEM INTEGRATION AND IMPLEMENTATION SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	306	461	657	894	1,014	34.9%
Europe	203	309	445	612	699	36.2%
APAC	156	247	369	526	618	41.0%
MEA	80	125	184	259	302	39.1%
Latin America	63	99	148	210	247	40.8%
Total	**809**	**1,240**	**1,803**	**2,500**	**2,879**	**37.4%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 26 SYSTEM INTEGRATION AND DEPLOYMENT SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,167	1,361	1,610	1,935	2,357	2,912	20.1%
Europe	810	952	1,136	1,377	1,695	2,116	21.2%
APAC	736	890	1,093	1,363	1,724	2,233	24.9%
MEA	357	428	521	645	811	1,033	23.7%
Latin America	293	354	434	540	682	887	24.8%
Total	**3,363**	**3,985**	**4,794**	**5,860**	**7,269**	**9,181**	**22.2%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.1.3 Support and maintenance services

6.3.1.3.1 Growing need to upgrade the existing system and assistance to the installed solutions to drive the growth of support and maintenance services across the globe

Support and maintenance services are adopted by organizations that use the on-premises deployment model for advanced analytics solutions. These services include the functional and technical support rendered by companies to their end customers with the help of advanced analytics vendors. Support and maintenance services involve the assistance provided to the installed software. It includes 24/7 troubleshooting assistance, upgradation of the existing software, problem-solving, repairing, replacing failure components, emergency response management, software maintenance, proactive services, technical support by technicians, and test scenario management. These services are offered by third-party service providers, as well as vendors, who facilitate clients to upgrade and maintain their advanced analytics ecosystem post-implementation. Support and maintenance services include functions of providing upgradations to the advanced analytics platform and assistance for solving issues in products. These services also include inspection, repair, planned maintenance, replacement, and maintenance of the acceptable standards of systems. As the number of advanced analytics deployment increases, the demand for support and maintenance services is also expected to increase. Some of the vendors, such as SAP and SAS Institute, offer support services to the advanced analytics vendors in the market.

TABLE 27 SUPPORT AND MAINTENANCE SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	197	301	434	597	684	36.5%
Europe	138	212	310	431	498	37.8%
APAC	101	160	240	344	407	41.8%
MEA	51	80	120	171	202	41.3%
Latin America	44	67	99	139	160	38.4%
Total	**530**	**820**	**1,203**	**1,682**	**1,950**	**38.5%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 28 SUPPORT AND MAINTENANCE SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	794	934	1,114	1,350	1,658	2,083	21.3%
Europe	582	692	834	1,022	1,271	1,609	22.5%
APAC	488	593	732	918	1,168	1,523	25.6%
MEA	242	294	363	455	579	753	25.5%
Latin America	186	221	265	324	400	512	22.4%
Total	**2,293**	**2,734**	**3,310**	**4,069**	**5,077**	**6,480**	**23.1%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

6.3.2 MANAGED SERVICES

6.3.2.1 Enterprises must ensure the provision of certain services for their clients to maintain their market position

It becomes troublesome for enterprises to concentrate on their core business procedures and simultaneously handle services. They often rely on third parties to offer managed services that specifically deal with client experiences and fit perfectly into the client's environment. Managed service providers offer technical expertise, service consistency, and flexibility, regardless of the client's geographical location. They deal with all the pre-and post-deployment questions and needs of clients. Hence, enterprises offer managed services to clients to maintain their market position. Managed services offer business benefits that give users enhanced security and compliance and improved system scalability, performance, and availability. It also provides change management services including product updates and upgrade processes that are engineered to give users access to the latest product innovations and capabilities related to advanced analytics. For instance, Teradata offers managed IT services that provide users with reliable, sustainable, highly available analytic ecosystems used in the advanced analytics market.

TABLE 29 MANAGED SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	363	556	808	1,117	1,287	37.2%
Europe	269	416	609	852	988	38.5%
APAC	197	315	477	690	820	42.8%
Latin America	89	142	213	306	363	42.0%
MEA	69	109	164	235	278	41.9%
Total	**987**	**1,538**	**2,271**	**3,200**	**3,737**	**39.5%**

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 30 MANAGED SERVICES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,502	1,778	2,135	2,604	3,220	4,057	22.0%
Europe	1,162	1,386	1,679	2,066	2,580	3,286	23.1%
APAC	989	1,210	1,502	1,896	2,427	3,188	26.4%
MEA	438	536	665	839	1,074	1,401	26.2%
Latin America	333	405	500	627	797	1,045	25.7%
Total	**4,424**	**5,315**	**6,482**	**8,031**	**10,098**	**12,976**	**24.0%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7 ADVANCED ANALYTICS MARKET, BY BUSINESS FUNCTION

KEY FINDINGS

- The marketing and sales segment is expected to have the largest market size and is estimated to grow from USD 9,192 million in 2021 to USD 23,825 million by 2026, at a CAGR of 21.0% during the forecast period.
- Increased use of structured and unstructured data across marketing and sales to understand the customer experience is expected to drive the adoption of advanced analytics solutions and services.
- The operations segment is expected to grow from USD 6,733 million in 2021 to 20,415 million in 2026 at the highest CAGR of 24.8%, due to the increased focus on improving the supply chain, IT, logistics, and business processes.
- The ever-increasing data of customers based on online purchases, web clicks, social media activities, and smart connected devices would drive the adoption of advanced analytics solutions to enhance marketing and sales business functions within the organization.
- The finance segment is growing at a CAGR of 19.6% during the forecast period. It is expected to grow from USD 7,604 million in 2021 to USD 18,590 million in 2026, due to the increasing adoption of data analytics solutions that can access data transactions and detect frauds to reduce data breaches.

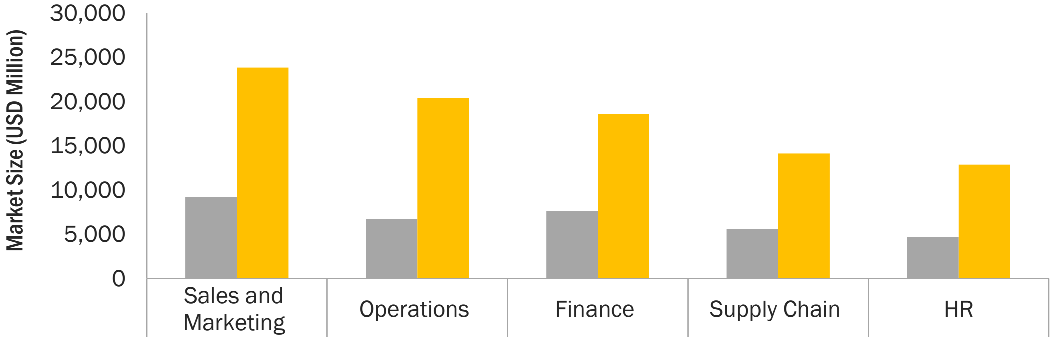
7.1 INTRODUCTION

The global advanced analytics market by business function has been segmented into finance, sales and marketing, operations, HR, and supply chain. Advanced analytics employs predictive modeling, statistical methods, and ML and process automation techniques beyond the capacities of traditional BI tools to analyze data or business information. Many industries, new entrants, and established competitors use data-driven strategies to compete, capture, and innovate in the market. The adoption of advanced analytics solutions can create new growth opportunities for enterprises. The advanced analytics market is gaining recognition worldwide due to the advantage of the gain meaningful insights from the large varieties of data and helping organizations make critical business decisions based on the analysis. IBM, Google, Alteryx, RapidMiner, SAP, SAS Institute, and Oracle are some of the leading vendors that offer advanced analytics solutions in the market. The operations business function is expected to witness rapid growth in the adoption of advanced analytics solutions, as businesses are leveraging these solutions to enhance internal processes and optimize operations. Advanced analytics solutions are finding applications in the marketing and sales business function for buyer segmentation and analysis of buyer needs.

7.1.1 BUSINESS FUNCTIONS: COVID-19 IMPACT

- The need to reduce wastage of time in asset and equipment tracking across the logistics and healthcare vertical leads to the adoption of predictive asset maintenance. After the rise of COVID-19, this application has taken a long jump.
- COVID-19 has changed the overall work environment and has enforced working from home strategies, overburdening HR and IT teams with requests from employees across locations and time zones. Employees turn to HR teams as trusted corporate sources for accurate and up-to-date information on COVID-19 and any changes that are impacting the organization.
- With businesses overwhelmed by customer requests during COVID-19, analytics has proven a natural choice, particularly in the finance department to provide the agility needed to deal with the business impact of the pandemic. In Europe, during the early COVID-19 lockdown, various banks were inundated by calls in what amounted to an increase of nearly 20–30 times the ordinary customer call volume.
- Due to COVID-19, SMEs are experiencing cash flow issues. Business is more reserved with spending and focuses on safety and avoiding risk. 59% of consumers say that coronavirus has impacted their shopping behavior week over a week.

FIGURE 40 OPERATIONS SEGMENT TO REGISTER THE HIGHEST CAGR DURING THE FORECAST PERIOD



	Sales and Marketing	Operations	Finance	Supply Chain	HR
2021	9,192	6,733	7,604	5,559	4,683
2026	23,825	20,415	18,590	14,136	12,857
CAGR (2021-2026)	21.0%	24.8%	19.6%	20.5%	22.4%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 31 ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2017–2020)
Finance	2,049	3,067	4,355	5,896	6,650	34.2%
Sales and Marketing	2,346	3,550	5,095	6,974	7,951	35.7%
Operations	1,453	2,274	3,375	4,778	5,632	40.3%
HR	1,128	1,726	2,506	3,471	4,004	37.3%
Supply Chain	1,424	2,159	3,100	4,241	4,826	35.7%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 32 ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Finance	7,604	8,823	10,393	12,438	15,104	18,590	19.6%
Sales and Marketing	9,192	10,781	12,839	15,531	19,067	23,825	21.0%
Operations	6,733	8,166	10,056	12,580	15,969	20,415	24.8%
HR	4,683	5,557	6,695	8,194	10,177	12,857	22.4%
Supply Chain	5,559	6,489	7,677	9,212	11,198	14,136	20.5%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the advanced analytics market size by business function. The market size of the sales and marketing segment is expected to hold the largest market share in 2021, while the HR segment is projected to grow at the highest CAGR during the forecast period. Businesses are adopting advanced analytics to take a deep dive into historical process data, identify patterns and relationships among discrete process steps and inputs, and then optimize the factors that prove to have the greatest effect on business yield. Many global manufacturers have abundant real-time shop-floor data and the capability to conduct such sophisticated statistical valuations. They are taking previously isolated data sets, aggregating them, and analyzing them to reveal important insights.

7.2 MARKETING AND SALES

7.2.1 WITH THE EVER-INCREASING DATA OF CUSTOMERS BASED ON ONLINE PURCHASES, WEB CLICKS, SOCIAL MEDIA ACTIVITIES, AND SMART CONNECTED DEVICES TO DRIVE THE ADOPTION OF ADVANCED ANALYTICS SOLUTION TO ENHANCE MARKETING AND SALES BUSINESS FUNCTION WITHIN THE ORGANIZATION

The adoption of advanced analytics solutions is changing the opportunity for marketing since the emergence of the internet and mobile technologies. Advanced analytics offers fundamental consequences for the marketing landscape due to the increasing trend of the digital world. Marketing and sales include other functions, such as customer support and after-sales service. Companies implement advanced analytics solutions to enhance customer experience and maintain future relationships with customers through customer support teams offering after-sales services to customers. Advanced analytics is a powerful tool that helps marketing campaigns and facilitates purchase decisions. It can help address issues and questions for customers at every stage of the customer lifecycle, from discovery to checkout. Organizations deploy advanced analytics solutions and services to optimize sales teams' productivity by analyzing bottlenecks in the sales cycle to improve sales resource deployment. Moreover, advanced analytics solutions support marketing teams to segment and profile their customers to target prospects with the most potential and improve campaign performance and promotional effectiveness.

TABLE 33 MARKETING AND SALES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	838	1,249	1,763	2,371	2,669	33.6%
Europe	624	937	1,335	1,813	2,054	34.7%
APAC	457	709	1,043	1,465	1,701	38.9%
MEA	241	369	536	743	858	37.4%
Latin America	187	287	418	582	669	37.6%
Total	**2,346**	**3,550**	**5,095**	**6,974**	**7,951**	**35.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 34 MARKETING AND SALES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	3,044	3,520	4,131	4,923	5,949	7,294	19.1%
Europe	2,359	2,749	3,252	3,908	4,767	5,903	20.1%
APAC	2,004	2,395	2,906	3,582	4,481	5,725	23.4%
MEA	1,005	1,193	1,439	1,764	2,193	2,768	22.5%
Latin America	781	924	1,110	1,355	1,678	2,134	22.3%
Total	**9,192**	**10,781**	**12,839**	**15,531**	**19,067**	**23,825**	**21.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

7.3 FINANCE

7.3.1 ADVANCED ANALYTICS SUPPORTS FINANCIAL FIRMS TO TRACK OUTSTANDING INVOICES AND AUTOMATE THE FOLLOW-UP PROCESS TO ENSURE INVOICES ARE PAID AND CLOSED ON TIME

Digitization in the finance industry has enabled technologies, such as advanced analytics, ML, AI, and cloud, to penetrate and transform financial institutions to remain competitive in the market. Finance business functions further include corporate finance, legal services, insurance services, accounting, and book-keeping functions for better data management. Industries, such as BFSI, are maintaining day-to-day operations of customer transactions. The adoption of predictive analytics solutions will help industries maintain insurance services, such as setting premium plans and helping credit officers to make lending decisions. By creating a structured table with all real-time or traditional information, more information can be generated from that data, which can be put to better use. For example, every financial organization has customer data, bank details, and customized services, such as mutual funds details, mobile banking, and home banking. By storing the information using a single data set, the information can be easily retrieved, and a more customized service platform can be created for consumers. With the adoption of advanced analytics solutions, organizations streamline and accelerate their traditional compliance work, expand strategic partnerships, and help drive their companies' profitability and growth.

TABLE 35 FINANCE: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	732	1,079	1,507	2,005	2,232	32.1%
Europe	545	810	1,141	1,533	1,718	33.2%
APAC	399	612	892	1,238	1,423	37.4%
MEA	210	319	458	628	718	35.9%
Latin America	163	248	358	492	560	36.1%
Total	**2,049**	**3,067**	**4,355**	**5,896**	**6,650**	**34.2%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 36 FINANCE: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	2,518	2,881	3,345	3,942	4,712	5,691	17.7%
Europe	1,951	2,249	2,633	3,130	3,776	4,606	18.7%
APAC	1,658	1,960	2,353	2,869	3,549	4,467	21.9%
MEA	831	977	1,165	1,412	1,737	2,160	21.0%
Latin America	646	756	899	1,085	1,329	1,665	20.9%
Total	**7,604**	**8,823**	**10,393**	**12,438**	**15,104**	**18,590**	**19.6%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7.4 HUMAN RESOURCES

7.4.1 INCREASING NEED TO PREDICT EMPLOYEE PERFORMANCE AND IDENTIFY HIDDEN TALENT FOR ORGANIZATIONS WOULD DRIVE THE ADOPTION OF ADVANCED ANALYTICS SOLUTION FOR THE HUMAN RESOURCES BUSINESS FUNCTION IN AN ORGANIZATION

A growing number of companies are trying to apply a data-driven approach to managing staff. A massive collection of employee data is used as the foundation for all Human Resource (HR) analytics projects. Today, problems faced by human resource managers include the need to focus on investments for various human resources functions within organizations. The adoption of advanced analytics solutions helps human resource managers to improve employee retention and identify people outside the organization that can contribute to the growth of the company. Advanced analytics solutions help recognize the strengths and vulnerabilities of the workforce, predict leadership needs, employee sentiment analysis, assess risk on an organizational level, and analyze critical skills. Major analytics companies, such as Microsoft, Oracle, IBM, and SAP, offer various tools, including predictive analysis, human resources analytics, and prescriptive analytics to help human resource teams improve their training and development programs, analyze candidates for the recruitment process, and analyze performances of each employee within an organization. Advanced analytics can be leveraged by HR teams to positively impact both talent and business decisions in the organization. HR can evolve from being just a people management function to playing a more transformational role in Human Capital Management (HCM) and being a strategic business partner in the company.

TABLE 37 HUMAN RESOURCES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	403	607	867	1,180	1,344	35.1%
Europe	300	456	657	902	1,034	36.3%
APAC	219	345	513	729	857	40.6%
MEA	116	179	264	370	432	39.0%
Latin America	90	139	206	290	337	39.2%
Total	**1,128**	**1,726**	**2,506**	**3,471**	**4,004**	**37.3%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 38 HUMAN RESOURCES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,550	1,814	2,154	2,597	3,175	3,936	20.5%
Europe	1,202	1,417	1,696	2,062	2,544	3,186	21.5%
APAC	1,021	1,234	1,515	1,890	2,392	3,089	24.8%
MEA	512	615	751	930	1,170	1,494	23.9%
Latin America	398	476	579	715	896	1,152	23.7%
Total	**4,683**	**5,557**	**6,695**	**8,194**	**10,177**	**12,857**	**22.4%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7.5 OPERATIONS

7.5.1 INCREASING NEED TO IMPROVE LOCATION AND DISTRIBUTION CENTER PERFORMANCE USING ALL THE RELEVANT DATA, RANGING FROM EXCEL SPREADSHEETS TO BOOST THE GROWTH OF ADVANCED ANALYTICS SOLUTIONS

Advanced analytics in operations enables companies to strike the right balance between operational cost, speed, flexibility, and quality. Many companies today are facing challenges in conducting operational management analysis. Today, almost every organization has started deploying analytics technologies, such as IoT, ERP systems, cloud computing, and social media for operation management. Every organization has various functional units carrying numerous activities, such as IT, manufacturing, supply chain, and logistics. The data generated by these units can be put to better use by sharing them indeterminately with other units. The growing responsibility of managing assets is driving IT teams to adopt advanced analytics solutions.

TABLE 39 OPERATIONS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	519	800	1,168	1,624	1,890	38.1%
Europe	387	600	884	1,242	1,455	39.3%
APAC	283	454	691	1,003	1,205	43.7%
MEA	149	236	355	509	608	42.1%
Latin America	116	184	277	399	474	42.3%
Total	**1,453**	**2,274**	**3,375**	**4,778**	**5,632**	**40.3%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 40 OPERATIONS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	2,229	2,666	3,236	3,987	4,982	6,250	22.9%
Europe	1,728	2,082	2,547	3,166	3,992	5,058	24.0%
APAC	1,468	1,814	2,276	2,901	3,753	4,906	27.3%
MEA	736	904	1,127	1,428	1,836	2,372	26.4%
Latin America	572	700	869	1,097	1,405	1,829	26.2%
Total	**6,733**	**8,166**	**10,056**	**12,580**	**15,969**	**20,415**	**24.8%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

7.6 SUPPLY CHAIN

7.6.1 INCREASING NEED TO ADDRESS THE CHALLENGES OF ANALYZING PRODUCTION PROCESSES IN ORGANIZATIONS THAT MAY INCREASE ROI TO DRIVE THE ADOPTION OF ADVANCED ANALYTICS SOLUTION

Optimizing the supply chain is also a challenge for manufacturers. As data is compiled from several sources ranging from ERP systems within the enterprise to the supplier's business, orders and shipment information, weblogs for customer shopping patterns logistics, Global Positioning System (GPS), RFID and recorders, mobile devices, and social channels, the adoption of advanced analytics solutions will provide a 360-degree view of the customer to create a unique brand experience. Major advanced analytics use cases for operations include data warehouse optimization, forecasting, and supply chain optimization. Supply chains are often one of the largest sources of risk for the enterprise. Proactively identifying risk factors and addressing them is key to supply chain resiliency. Advanced analytics provides complete point-and-click visual analytics solutions on real-time and historical supply chain data for complete contextual awareness and deeper, faster insights.

TABLE 41 SUPPLY CHAIN: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	509	759	1,073	1,442	1,620	33.6%
Europe	379	570	812	1,103	1,246	34.7%
APAC	277	431	635	891	1,033	38.9%
MEA	146	224	326	452	521	37.4%
Latin America	113	174	255	354	406	37.6%
Total	**1,424**	**2,159**	**3,100**	**4,241**	**4,826**	**35.7%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 42 SUPPLY CHAIN: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,841	2,119	2,470	2,920	3,494	4,328	18.6%
Europe	1,427	1,654	1,944	2,318	2,800	3,502	19.7%
APAC	1,212	1,441	1,738	2,125	2,632	3,397	22.9%
MEA	608	718	861	1,046	1,288	1,643	22.0%
Latin America	472	556	664	803	985	1,266	21.8%
Total	**5,559**	**6,489**	**7,677**	**9,212**	**11,198**	**14,136**	**20.5%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

8 ADVANCED ANALYTICS MARKET, BY TYPE

KEY FINDINGS

- The big data analytics segment is expected to hold the largest market size and projected to grow from USD 10,362 million in 2021 to USD 28,603 million by 2026, at a CAGR of 22.5% during the forecast period.
- Big data analytics helps organizations harness the data and use it to identify new opportunities. That, in turn, leads to smarter business moves, more efficient operations, higher profits, and happier customers.
- The risk analytics segment is expected to grow at the highest CAGR of 23.8% during the forecast period. The growth is attributed to the increasing need to improve operational and cyber resilience and meet evolving regulations across BFSI firms.
- Advanced analytics describes the analysis of data using complex techniques to forecast trends and predict events.

8.1 INTRODUCTION

Advanced analytics allows businesses to understand customers on a deeper level, predict future outcomes, reduce risks, and enhance overall productivity. It employs predictive modeling, statistical methods, and ML and process automation techniques beyond the capacities of traditional business BI tools to analyze data or business information. Advanced analytics leverages data science techniques that involve mature methods of analysis to project future trends and anticipate the likelihood of potential events. Advanced analytics offers a wider set of capabilities to deal with challenges that traditional rearview BI cannot, enabling stronger strategic decision-making for the future. Advanced analytics enables companies to optimize their operations and innovate to gain a competitive advantage. With better customer analysis, predictive analytics, and statistical modeling, advanced analytics supports companies to improve decision-making and keep pace with the extremely competitive, quick-changing markets.

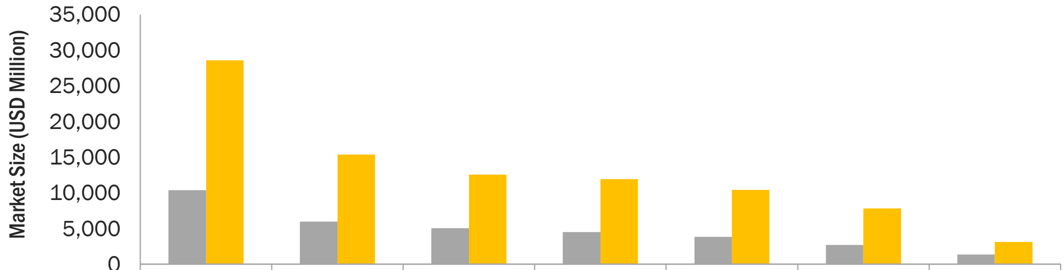
The advanced analytics market is segmented based on types such as big data analytics, customer analytics, predictive analytics, statistical analytics, prescriptive analytics, risk analytics, and other types (business analytics, augmented analytics, multimedia analytics, and text analytics). All these types can be used to explore and model a company's data to improve the business based upon actionable insights. These advanced analytics types are used for various applications across key verticals to predict the future, anticipate and solve problems, reduce risk, and enhance productivity.

8.1.1 TYPES: COVID-19 IMPACT

- Customer experience has taken on a new level of significance since the global economic downturn triggered by the COVID-19 pandemic and worldwide introduction of remote working. Uncertainty has touched every business on every continent, resulting in the increasing importance of focusing on customer experience.
- Customer retention is driven by customer experience and leads to revenue protection. To deliver an excellent customer experience, customer interactions need to be monitored, internal performance and processes analyzed, and service levels maintained; these must be continually evaluated to meet the evolving business needs through these challenging times and beyond.
- As businesses globally evoke business continuity plans in the wake of the COVID-19 pandemic, advanced analytics solutions are mobilizing teams that are now required to work from home, maintaining effective team collaboration, and supporting continued customer interaction.
- Considering the COVID-19 pandemic, companies must be empowered to shift their operations to alternate locations quickly and efficiently. While the use of advanced analytics solutions was multiplying even before the COVID-19 pandemic, it has assumed more significance in these times.
- As the primary driver of customer satisfaction and NPS scores, it is now more important than ever to ensure the service delivery team can efficiently assist customers in a time of need. Working from home will increasingly become the new norm for knowledge workers, contact center agents, and customer service functions.
- The newly developed COVID-19-focused applications provide businesses with real-time insights into topics such as job loss, social distancing, work from home, and home education.
- The COVID-19 pandemic has various many companies' uses of advanced analytics and AI. These strategies have helped engage customers through digital channels, manage fragile and complex supply chains, and support workers through disruption to their work and lives.
- Due to the impact of the COVID-19 pandemic, leaders have identified a major weakness in their analytics strategy: the reliance on historical data for algorithmic models. From customer behavior to supply and demand patterns, historical patterns and the assumption of continuity are what give predictive models their power.

- The use of AL, ML, and NLP by researchers and developers has increased at a rapid pace to track and contain COVID-19. Researchers at Rensselaer Polytechnic Institute (RPI) are using big data solutions and analytics to better comprehend COVID-19 from different angles.
- The COVID-19 pandemic has put a hold on some companies' investment plans in analytics technology, for other companies, analytics has become even more critically important, helping enterprises navigate fast-changing customer behaviors and supply chain disruptions.

FIGURE 41 RISK ANALYTICS SEGMENT TO REGISTER THE HIGHEST CAGR DURING THE FORECAST PERIOD



	Big Data Analytics	Customer Analytics	Predictive Analytics	Statistical Analytics	Prescriptive Analytics	Risk Analytics	Other Type
2021	10,362	5,978	5,051	4,513	3,821	2,689	1,357
2026	28,603	15,373	12,578	11,937	10,402	7,810	3,120
CAGR (2021-2026)	22.5%	20.8%	20.0%	21.5%	22.2%	23.8%	18.1%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 43 ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	2,482	3,803	5,528	7,664	8,850	37.4%
Predictive Analytics	1,338	2,010	2,863	3,890	4,402	34.7%
Customer Analytics	1,537	2,322	3,328	4,549	5,179	35.5%
Statistical Analytics	1,129	1,715	2,472	3,397	3,889	36.2%
Risk Analytics	609	944	1,388	1,946	2,271	39.0%
Prescriptive Analytics	928	1,419	2,056	2,843	3,273	37.0%
Other Types	377	563	796	1,072	1,199	33.6%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 44 ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	10,362	12,310	14,848	18,193	22,622	28,603	22.5%
Predictive Analytics	5,051	5,880	6,950	8,345	10,168	12,578	20.0%
Customer Analytics	5,978	7,002	8,326	10,058	12,329	15,373	20.8%
Statistical Analytics	4,513	5,315	6,355	7,719	9,515	11,937	21.5%
Risk Analytics	2,689	3,229	3,938	4,877	6,131	7,810	23.8%
Prescriptive Analytics	3,821	4,527	5,444	6,651	8,246	10,402	22.2%
Other Types	1,357	1,553	1,800	2,111	2,505	3,120	18.1%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the advanced analytics market size by type. The big data analytics segment is estimated to hold the largest market share in 2021. Advanced analytics uses state-of-the-art tools, such as ML and AI, and complex statistical analyses, and algorithms to examine big data and identify patterns to discover deeper insights. The risk analytics segment is expected to have the highest growth rate during the forecast period. The growth is attributed to the increasing demand for advanced credit risk analytics enabling institutions to improve underwriting decisions and increase revenues while reducing risk costs.

8.2 BIG DATA ANALYTICS

8.2.1 BIG DATA ANALYTICS USES ADVANCED ANALYTICS TECHNIQUES TO UNCOVER TRENDS, PATTERNS, AND CORRELATIONS IN LARGE AMOUNTS OF RAW DATA, MAKING DATA-INFORMED DECISIONS

Big data refers to massive complex structured and unstructured data sets that are rapidly generated and transmitted from a wide variety of sources. It refers to data sets that are so large, complex, and fast-moving that cannot be stored or processed by traditional data analytics and management tools. Big data analytics describes the process of using advanced analytics techniques such as clustering and data mining to uncover trends, patterns, and correlations in large amounts of raw data to help make data-informed decisions. The analytical techniques are similar, but due to the size of the data sets, highly specialized software backed by powerful cloud-based computing systems are required to process, manage, and analyze the data. Big data analytics are frequently leveraged by financial services organizations looking to mine massive amounts of stock market data to identify and capitalize off previously unknown trends. Public health organizations are also increasingly leveraging huge amounts of population health data to develop better policies, treatment, and healthcare practices. Big data has changed and revolutionized the way businesses and organizations work. A lot of enterprises from different industries are benefited from big data techniques and processing methods. Big data analytics has various applications in healthcare, banking, manufacturing, and more. For instance, Food and Drug Administration (FDA) which runs under the jurisdiction of the Federal Government of the US leverages the analysis of big data to discover patterns and associations to identify and examine the expected or unexpected occurrences of food-based infections.

TABLE 45 BIG DATA ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	895	1,351	1,931	2,630	2,998	35.3%
Europe	666	1,013	1,462	2,012	2,307	36.4%
APAC	488	766	1,143	1,625	1,911	40.7%
MEA	257	399	587	824	964	39.2%
Latin America	175	274	405	572	668	39.7%
Total	**2,482**	**3,803**	**5,528**	**7,664**	**8,850**	**37.4%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 46 BIG DATA ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	3,463	4,057	4,822	5,819	7,121	8,836	20.6%
Europe	2,684	3,168	3,795	4,620	5,706	7,151	21.7%
APAC	2,280	2,760	3,392	4,234	5,364	6,935	24.9%
MEA	1,143	1,375	1,680	2,085	2,625	3,354	24.0%
Latin America	791	950	1,159	1,436	1,805	2,327	24.1%
Total	**10,362**	**12,310**	**14,848**	**18,193**	**22,622**	**28,603**	**22.5%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.3 PREDICTIVE ANALYTICS

8.3.1 PREDICTIVE ANALYTICS HELPS ORGANIZATIONS EFFECTIVELY ANTICIPATE AND PROACTIVELY PLAN FOR FUTURE EVENTS FOR ENHANCED PRODUCTIVITY

Predictive analytics is a type of statistical modeling in which computers can forecast unknown future events based on historical data. It encompasses a variety of statistical techniques (ML, predictive modeling, and data mining) and leverages statistics (historical and current) to estimate or predict future outcomes. Companies use predictive analytics to effectively anticipate and proactively plan for future events. For instance, Nike uses predictive analytics to forecast customer demand on a hyper-local level. This helps Nike stores optimize their inventory accordingly and develop more targeted marketing campaigns. Predictive analytics mines data from systems such as CRM, ERP, marketing automation stacks, and other databases. The results are then visualized in a way so key business users can interpret them It is important for gaining the competitive advantage, understanding a variety of customer interests, finding new business opportunities, reducing costs and risks, and most importantly, addressing problems before they occur. For example, a European firm changed its retailer-specific assortments and planograms by using Predictive Analytics to analyze consumer data to find which Stock Keeping Units (SKUs) sold well in certain retail formats and which SKUs to swap in and out to best serve customer preferences. There are various industries such as healthcare firms that leverage predictive analytics solutions to accurately predict negative health events. Human Resources (HR) business function uses analytics to identify pain points and productivity spikes for predicting the future performance of employees.

TABLE 47 PREDICTIVE ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	478	707	991	1,323	1,477	32.6%
Europe	356	531	750	1,011	1,137	33.7%
APAC	260	401	586	817	942	37.9%
MEA	137	209	301	415	475	36.4%
Latin America	106	162	235	325	371	36.6%
Total	**1,338**	**2,010**	**2,863**	**3,890**	**4,402**	**34.7%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 48 PREDICTIVE ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,672	1,920	2,236	2,645	3,172	3,851	18.2%
Europe	1,296	1,499	1,760	2,100	2,542	3,116	19.2%
APAC	1,101	1,306	1,573	1,925	2,389	3,022	22.4%
MEA	552	651	779	948	1,169	1,462	21.5%
Latin America	429	504	601	728	895	1,127	21.3%
Total	**5,051**	**5,880**	**6,950**	**8,345**	**10,168**	**12,578**	**20.0%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.4 CUSTOMER ANALYTICS

8.4.1 CUSTOMER ANALYTICS USES ADVANCED ANALYTICS TECHNIQUES TO GAIN CUSTOMER INSIGHTS IN REAL TIME AND OPTIMIZE THE CUSTOMER INTERACTIONS

Customer analytics refers to processes and technologies that give organizations the customer insights necessary to deliver offers that are anticipated, relevant, and timely. Customer analytics is becoming important to understand how the customer will behave when interacting with the organization so that we can respond accordingly. The deeper the understanding of customers' buying habits and lifestyle preferences, the more accurate is the predictions of future buying behaviors of the customer will be more successful company will be at delivering relevant offers that will attract rather than alienate customers. With customer analytics there is an increase in response rates, customer loyalty, and ultimately, ROI by contacting the right customers with highly relevant offers and messages. It reduces campaign costs by targeting those customers most likely to respond, it decreases attrition by accurately predicting customers most likely to leave and developing the right proactive campaigns to retain them, it delivers the right message by segmenting customers more effectively and better understanding of target population. Organizations today need to operationalize advanced analytics and ML to deliver seamless customer experiences and optimal business outcomes. To succeed with digital transformation, this must be done in a transparent, trustworthy, and predictable way to win the customers' trust. For instance, major Canadian grocery retailers use FICO analytics to enable the retailer to understand individual customers more deeply and thus to make all offers and contacts extremely relevant. Systematic, ongoing analytic learning pinpoints what each customer cares about and where incentive investments will not only encourage continued loyalty but produce profitable changes in behavior.

TABLE 49 CUSTOMER ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	549	817	1,151	1,547	1,738	33.4%
Europe	409	613	872	1,183	1,338	34.5%
APAC	299	464	681	955	1,108	38.7%
MEA	158	241	350	485	559	37.2%
Latin America	122	188	273	380	436	37.4%
Total	**1,537**	**2,322**	**3,328**	**4,549**	**5,179**	**35.5%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 50 CUSTOMER ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,979	2,286	2,679	3,188	3,847	4,706	18.9%
Europe	1,534	1,785	2,109	2,531	3,082	3,809	19.9%
APAC	1,303	1,555	1,885	2,320	2,897	3,694	23.2%
MEA	653	775	933	1,142	1,418	1,786	22.3%
Latin America	508	600	720	877	1,085	1,377	22.1%
Total	**5,978**	**7,002**	**8,326**	**10,058**	**12,329**	**15,373**	**20.8%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.5 STATISTICAL ANALYTICS

8.5.1 GROWING NEED TO UNCOVER PATTERNS AND TRENDS USING DATA INTERPRETATION TECHNIQUES TO DRIVE THE DEMAND FOR STATISTICAL ANALYTICS IN THE MARKET

Statistical analysis is the collection and interpretation of data to uncover patterns and trends. It is a component of data analytics. In the context of BI, statistical analysis involves collecting and scrutinizing every data sample in a set of items from which samples can be drawn. It is the science of collecting, exploring, and presenting large amounts of data to discover underlying patterns and trends. Statistics are applied every day in research, industry, and government to become more scientific about decisions that need to be made. Researchers keep children healthy by using statistics to analyze data from the production of viral vaccines, which ensures consistency and safety. Traditional methods for statistical analysis from sampling data to interpreting results have been used by scientists for thousands of years. However, today’s data volumes make statistics ever more valuable and powerful. Affordable storage, powerful computers, and advanced algorithms have all led to increased use of computational statistics. For instance, SAS offers SAS Analytics Pro that helps organizations using manipulating, analyzing, and presenting information techniques with the help of comprehensive analytical toolset that combines statistical analysis, reporting, and high-impact visuals.

TABLE 51 STATISTICAL ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	403	603	855	1,155	1,305	34.1%
Europe	300	453	647	883	1,004	35.2%
APAC	220	342	506	713	832	39.5%
MEA	116	178	260	362	420	38.0%
Latin America	90	139	203	283	327	38.2%
Total	**1,129**	**1,715**	**2,472**	**3,397**	**3,889**	**36.2%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 52 STATISTICAL ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,494	1,736	2,045	2,447	2,969	3,655	19.6%
Europe	1,158	1,355	1,610	1,943	2,379	2,958	20.6%
APAC	984	1,181	1,439	1,780	2,236	2,868	23.9%
MEA	493	588	713	877	1,094	1,387	23.0%
Latin America	383	455	549	673	837	1,069	22.8%
Total	**4,513**	**5,315**	**6,355**	**7,719**	**9,515**	**11,937**	**21.5%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.6 RISK ANALYTICS

8.6.1 RISING NEED TO IMPROVE RISK ASSESSMENTS AND DETECT FRAUD TO DRIVE THE GROWTH OF RISK ANALYTICS IN THE MARKET

Risk analysis is the process of identifying and analyzing potential issues that could negatively impact key business initiatives or projects. An important part of risk analysis is identifying the potential for harm from these events, as well as the likelihood that they will occur. With the rise of computing power and new analytical techniques, banks can now extract deeper and more valuable insights from their ever-growing data. And they can do it quickly, as many key processes are now automated and many more. For risk departments, which have been using data analytics for decades, these trends present unique opportunities to better identify, measure, and mitigate risk. The recent increases in computing power have allowed banks to deploy advanced analytical techniques at an industrial scale. ML techniques, such as deep learning, are now common at top risk-analytics departments. The new tools radically improve banks' decision models. Techniques, such as NLP and geospatial analysis, expand the database from which banks can derive insights. These advances have allowed banks to automate more steps within currently manual processes such as data capture and cleaning. With automation, straight-through processing of most transactions becomes possible, as well as the creation of reports in near real time. This means that risk teams can increasingly measure and mitigate risk more accurately and faster. For instance, Italy's Ministry of Economy and Finance uses advanced analytics on SAS Viya to quickly calculate risk on financial guarantees. It has improved systemic liquidity and risk management as it enables the MEF to easily monitor fund performance to provide up-to-date reporting.

TABLE 53 RISK ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	208	317	459	633	729	36.8%
Europe	155	238	348	484	561	38.0%
APAC	113	180	272	391	465	42.3%
MEA	71	111	165	233	275	40.2%
Latin America	62	97	145	205	241	40.4%
Total	**609**	**944**	**1,388**	**1,946**	**2,271**	**39.0%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 54 RISK ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	852	1,010	1,214	1,482	1,835	2,294	21.9%
Europe	661	789	956	1,177	1,471	1,856	23.0%
APAC	561	687	854	1,079	1,382	1,800	26.3%
MEA	328	398	490	613	777	998	24.9%
Latin America	286	346	424	527	666	861	24.6%
Total	**2,689**	**3,229**	**3,938**	**4,877**	**6,131**	**7,810**	**23.8%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.7 PRESCRIPTIVE ANALYTICS

8.7.1 PRESCRIPTIVE ANALYTICS USES OPTIMIZATION AND SIMULATION ALGORITHMS TO ALLOW COMPANIES TO ASSESS A NUMBER OF POSSIBLE OUTCOMES BASED UPON THEIR ACTIONS

Prescriptive analyses are complex in the world of data science, often using both structured and unstructured data to form its insights. Applied statistics, deep learning, computer vision, and other advanced methods are used in prescriptive analytics. There are several challenges that need to be overcome while using prescriptive analytics as prescriptive analytics is not commonly used by various businesses today. Prescriptive analytics provides calculated next steps to take. It uses a combination of ML, business rules, AI, and algorithms to simulate various approaches to numerous potential outcomes. In another word, prescriptive analytics provides users with a recommended course of action based on informed simulations. One common application of this technology is Google maps. Using this approach, Google maps not only predicts how long a trip will take based on traffic and driving conditions, but it also suggests the optimal route. It is used in various verticals; for instance, CVS Health also leverages prescriptive analytics in their Transform Diabetes Care platform, a new tool designed to boost health outcomes for diabetes sufferers. This tool not only helps diabetes patients track their medication and habits, but also provides them with personalized recommendations based on their individual habits and behaviors.

TABLE 55 PRESCRIPTIVE ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	332	499	711	967	1,099	34.9%
Europe	247	375	539	739	845	36.0%
APAC	181	283	421	597	700	40.3%
MEA	95	147	216	303	353	38.8%
Latin America	74	115	169	237	276	39.0%
Total	**928**	**1,419**	**2,056**	**2,843**	**3,273**	**37.0%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 56 PRESCRIPTIVE ANALYTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,265	1,478	1,752	2,108	2,573	3,185	20.3%
Europe	981	1,154	1,379	1,674	2,061	2,577	21.3%
APAC	833	1,006	1,232	1,534	1,938	2,499	24.6%
MEA	418	501	610	755	948	1,209	23.7%
Latin America	325	388	471	580	726	932	23.5%
Total	**3,821**	**4,527**	**5,444**	**6,651**	**8,246**	**10,402**	**22.2%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

8.8 OTHER TYPES

Other types cover business analytics, augmented analytics, multimedia analytics, and text analytics. Advanced analytics is used to enhance decision-making and boost competitive advantage now as well as for years to come as advanced analytics use cases and applications are incredibly vast. Companies that seek to dominate the global digital economy in years to come, such as Amazon and Apple, are already investing massive amounts of resources into advanced analytics tools and infrastructure. Advanced analytics are being utilized differently from one organization to the next.

Business analytics is a set of disciplines and technologies for solving business problems using data analysis, statistical models, and other quantitative methods. It involves an iterative, methodical exploration of an organization's data, with an emphasis on statistical analysis, to drive decision-making The benefits of business analytics are many as it helps in interpreting data and predicting and forecasting. It also gives a deeper vision and meaning of the data and how best it can be used. It helps various processes within a business such as sales, finances, and internal processes. For instance, Starbucks can predict the purchases and offers a customer is likely to be interested in. This big coffee company used its loyalty card program to gather individualized purchase data of millions of its customers. With this information and business analytics tools, it was able to achieve its goal of identifying the pattern in a customer’s purchase and then suggest them offers through mobile devices which the company believed the customer may take up.

Augmented analytics is a class of analytics powered by AI and ML that expands a human's ability to interact with data at a contextual level. Augmented analytics consists of tools and software that bring analytical capabilities whether it be recommendations, insights, or guidance on a query to more people. Business users and executives get incredible value from augmented analytics because these technologies help them get value from their data quickly without the need for deep, technical skills or expertise in working with data. Travel and hospitality organizations may use augmented analytics to find the optimal, personalized offers to upsell or cross-sell customers.

Multimedia analytics is a new and exciting research area that combines techniques from multimedia analysis, visual analytics, and data management, with a focus on creating systems for analyzing large-scale multimedia collection. The size and complexity of media collections are ever-increasing to harvest useful information from these collections, with expected impacts ranging from the advancement of science to increased company profits. Analytics will combine cloud-enabled industrialized analytics environments with embedded AI to help organizations enhance overall productivity.

Text analytics combines a set of ML, statistical, and linguistic techniques to process large volumes of unstructured text or text that does not have a predefined format to derive insights and patterns. It enables businesses, governments, researchers, and media to exploit the enormous content at their disposal for making crucial decisions. Text analytics uses a variety of techniques such as sentiment analysis, topic modeling named entity recognition, term frequency, and event extraction. Text analytics is a sub-set of NLP that aims to automate the extraction and classification of actionable insights from unstructured text disguised as emails, tweets, chats, tickets, reviews, and survey responses scattered all over the internet. It is used in various industries; for instance, in sales and marketing. Text analytics techniques help reduce manual work with automation while providing valuable and specific insights to nurture the marketing funnel.

TABLE 57 OTHER TYPES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	136	200	278	369	407	31.6%
Europe	101	150	211	282	314	32.7%
APAC	74	113	165	228	260	36.9%
MEA	28	42	59	81	91	34.6%
Latin America	38	58	83	113	128	35.3%
Total	377	563	796	1,072	1,199	33.6%

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 58 OTHER TYPES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	455	514	588	680	796	973	16.4%
Europe	353	402	463	540	637	788	17.4%
APAC	300	350	414	495	599	764	20.6%
MEA	103	119	138	162	192	242	18.6%
Latin America	146	169	197	234	280	353	19.3%
Total	**1,357**	**1,553**	**1,800**	**2,111**	**2,505**	**3,120**	**18.1%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9 ADVANCED ANALYTICS MARKET, BY DEPLOYMENT MODE

KEY FINDINGS

- The on-premises segment is estimated to account for a larger market size of USD 24,879 million in 2021 and projected to reach a market size of USD 63,813 million by 2026, at a CAGR of 20.7% during the forecast period.
- Organizations with more reliance on data security and privacy move toward the on-premises deployment mode to gain more control over their data and its security.
- The cloud segment is expected to hold a market size of USD 26,010 million by 2026 and is projected to grow at a higher CAGR of 23.9% during the forecast period.
- Using cloud-based applications, SMEs can collect and analyze data easily to improve their customers' experience. Cloud-based advanced analytics solutions allow users to explore data in-depth and use advanced modeling and insight visualization techniques.
- The cloud segment offers benefits, such as scalability and cost-effectiveness, which are expected to be instrumental in propelling the growth of the overall advanced analytics.
- APAC is expected to witness the highest growth rate in the cloud segment due to the increasing adoption of advanced analytics solutions and services among SMEs in the region.

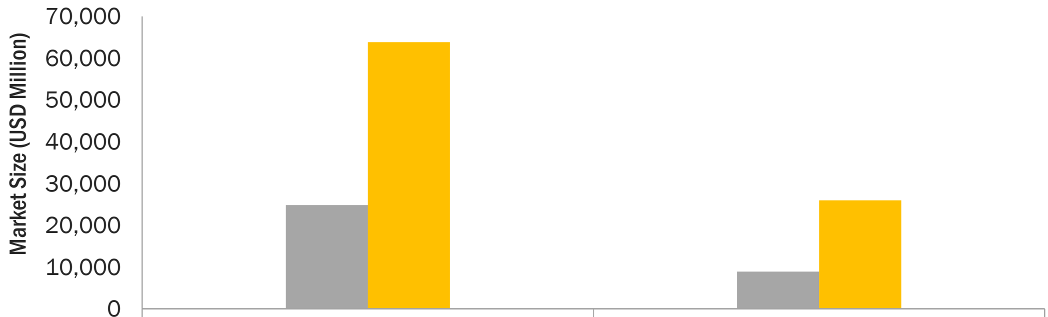
9.1 INTRODUCTION

The deployment mode in the advanced analytics market includes on-premises and cloud. Deployment refers to the setting up of an IT infrastructure with hardware, operating systems, and applications that are required to manage the IT ecosystem. An advanced analytics solution can be deployed on any one deployment mode based on security, availability, and scalability. Mostly advanced analytics solutions are getting deployed on the cloud as it offers advantages, such as pay-per-use and low installation and maintenance costs. This deployment mode is expected to show high growth soon. Advanced analytics workbenches are cloud-based solutions that allow users to explore data in-depth and use advanced modeling and insight visualization techniques.

9.1.1 DEPLOYMENT MODES: COVID-19 IMPACT

- The decline in IT spending by core verticals, such as retail and eCommerce, manufacturing and food and beverages, and their data-sensitive work environments has resulted in the increased adoption of on-premises advanced analytics solutions.
- The COVID-19 outbreak is encouraging numerous companies for prioritizing their IT strategies. Several organizations are emphasizing the importance of investing in the right cloud-based advanced analytics solutions, enabling employees to work remotely to limit the pandemic impact on productivity.
- The cloud technology market is nearing its maturity stage, and despite the end of support threats, advanced analytics decision-makers are opting for on-premises solutions to control social distancing. However, with the global lockdown and work-from-home practice gaining traction as the need of the hour, companies are opting for cloud-based infrastructure to manage and monitor employees, and facilities working remotely.
- The COVID-19 outbreak is forcing numerous companies to review their IT strategies as a top priority. Several organizations emphasize the importance of having the right cloud-based collaboration tools, enabling employees to work remotely to limit the impact on productivity.
- Most of the advanced analytics solution providers still run on-premises server hardware for backup as well as other purposes. Though the cloud technology is mature, and despite the end of support threat, analytics decision-makers still choose on-premises solutions to control security. However, with the global lockdown and work from home practice picking up as need of the hour, companies are moving to cloud-based infrastructure for managing and monitoring customer data and analyzing the same.
- The COVID-19 outbreak is forcing numerous companies to review their IT strategies as a top priority. Several organizations are emphasizing the importance of having the right cloud-based collaboration tools, enabling employees to work remotely to limit the pandemic impact on productivity.

FIGURE 42 CLOUD SEGMENT TO REGISTER A HIGHER CAGR DURING THE FORECAST PERIOD



	On-premises	Cloud
2021	24,879	8,892
2026	63,813	26,010
CAGR (2021-2026)	20.7%	23.9%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 59 ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	1,995	3,100	4,567	6,417	7,502	39.3%
On-premises	6,405	9,677	13,864	18,943	21,560	35.5%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 60 ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	8,892	10,692	13,051	16,181	20,356	26,010	23.9%
On-premises	24,879	29,124	34,609	41,773	51,159	63,813	20.7%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The market size and CAGR of the cloud segment are estimated to be higher than the on-premises segment during the forecast period. The cloud technology benefit of easy deployment and minimal capital requirement facilitates the adoption of the cloud deployment model. The adoption of cloud-based advanced analytics solutions is expected to be supported by the COVID-19 pandemic, as lockdowns and social distancing practices are encouraging companies to move to cloud solutions that can be managed remotely. The increasing demand for scalable, easy-to-use, and cost-effective advanced analytics solutions is expected to accelerate the growth of the cloud segment in the advanced analytics market.

9.2 ON-PREMISES

9.2.1 DATA-SENSITIVE ORGANIZATIONS TO ADOPT THE ON-PREMISES DEPLOYMENT MODE TO DEPLOY ADVANCED ANALYTICS SOLUTIONS

Data-sensitive enterprises prefer on-premises deployment of advanced analytics solutions. The on-premises deployment mode refers to the installation of advanced analytics solutions and associated services on the premises of a person or organization. This is the most secure type of deployment mode and is mostly adopted by large enterprises, as they have well-established physical servers, expert IT technicians, and a sufficient budget. Vendors in the market help securely deploy advanced analytics solutions into clients' premises with easy to manage, upgrade, and scale functionality. Vendors deliver on-premises solutions with a one-time license fee and an annual service agreement, which also includes free upgrades and new functionalities. Organizations with more reliance on data security and privacy move toward the on-premises deployment mode to gain more control over the data and its security. However, the emergence of cloud-based advanced analytics solutions has impacted the growth of on-premises solutions, which can be validated from the high growth rate of cloud-based solutions during the forecast period.

TABLE 61 ON-PREMISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	2,553	3,802	5,366	7,216	8,119	33.5%
Europe	1,593	2,393	3,409	4,632	5,246	34.7%
APAC	1,196	1,855	2,728	3,826	4,439	38.8%
MEA	567	866	1,253	1,729	1,987	36.8%
Latin America	495	761	1,108	1,540	1,769	37.5%
Total	**6,405**	**9,677**	**13,864**	**18,943**	**21,560**	**35.5%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 62 ON-PREMISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	9,257	10,704	12,559	14,960	18,073	22,143	19.1%
Europe	6,025	7,019	8,301	9,973	12,158	15,062	20.1%
APAC	5,222	6,231	7,549	9,288	11,596	14,809	23.2%
MEA	2,314	2,735	3,279	3,994	4,935	6,207	21.8%
Latin America	2,061	2,435	2,920	3,558	4,398	5,592	22.1%
Total	**24,879**	**29,124**	**34,609**	**41,773**	**51,159**	**63,813**	**20.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

9.3 CLOUD

9.3.1 CLOUD DEPLOYMENT MODE PROVIDES MULTIPLE BENEFITS, SUCH AS REDUCED OPERATIONAL COSTS, SIMPLE DEPLOYMENTS, AND HIGHER SCALABILITY

The cloud deployment mode provides multiple benefits, such as reduced operational costs, simple deployments, and higher scalability. The cloud deployment for the advanced analytics solution is expected to grow with an increasing awareness of the benefits of cloud-based solutions. The solution providers are focusing on the development of robust cloud-based solutions for their clients, as several organizations have migrated to either a private or a public cloud. Cloud-as-a-service is enabling organizations to manage not only costs but also achieve better agility. SMEs have highly preferred cloud-based solutions, as they are budget-friendly and easy to deploy. It helps in maintaining a competitive edge by eliminating the administrative roadblocks of the supporting infrastructure and allows organizations to focus on improving their competencies. Cloud-based advanced analytics solutions are significantly gaining traction due to their cost-effective benefits and global availability. The cloud technology provides additional flexibility for business operations and real-time deployment ease to companies that implement advanced analytics. Cloud solutions make it easy for users to apply analytics capabilities to organizations and help maintain a competitive edge by eliminating the administrative roadblocks of supporting infrastructure. Various businesses can design their personalized solutions to meet the needs that provide them the flexibility with the cloud-based solutions. These solutions also allow organizations to focus on improving their competencies. Major vendors in the advanced analytics market using cloud deployments are Alteryx, AWS, Dataiku, DataRobot, IBM, Oracle, Dataiku, Microsoft, Qlik, SAP, Teradata, and TIBCO Software.

TABLE 63 CLOUD: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	448	692	1,011	1,406	1,635	38.2%
Europe	641	979	1,419	1,962	2,261	37.0%
APAC	439	696	1,046	1,500	1,780	41.9%
MEA	294	462	686	973	1,150	40.6%
Latin America	173	272	405	576	677	40.7%
Total	**1,995**	**3,100**	**4,567**	**6,417**	**7,502**	**39.3%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 64 CLOUD: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,925	2,297	2,778	3,409	4,239	5,356	22.7%
Europe	2,642	3,132	3,770	4,610	5,721	7,194	22.2%
APAC	2,141	2,614	3,239	4,078	5,210	6,774	25.9%
MEA	1,377	1,673	2,064	2,587	3,290	4,231	25.2%
Latin America	807	977	1,200	1,497	1,896	2,455	24.9%
Total	**8,892**	**10,692**	**13,051**	**16,181**	**20,356**	**26,010**	**23.9%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

10 ADVANCED ANALYTICS MARKET, BY ORGANIZATION SIZE

KEY FINDINGS

- The large enterprises segment is projected to grow from USD 22,465 million in 2021 to USD 58,601 million by 2026, at a CAGR of 21.1% during the forecast period.
- The growth can be attributed to the increasing demand for advanced analytics technologies to provide customers with the support needed to resolve issues and enhance customer satisfaction.
- The SMEs segment is projected to record a higher CAGR of 22.5% during the forecast period.
- Verticals’ focus on adopting technologically advanced solutions and processes at a lower cost would drive the adoption of advanced analytics solutions and services among SMEs.
- Large enterprises have already started implementing advanced analytics solutions and services to provide personalized recommendations to the customers for better engagement and enhance productivity.

10.1 INTRODUCTION

Based on organization size, the advanced analytics market has been segmented into large enterprises and SMEs. Currently, the market share of large enterprises is higher; however, the market for SMEs is expected to increase at a higher CAGR during the forecast period.

The organization size is identified based on the number of employees working in the organization as per global standards given below:

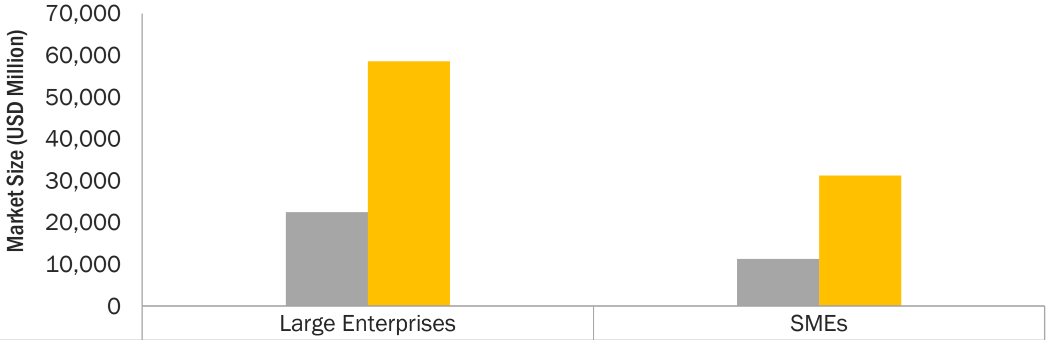
- SMEs: Less than 1,000 employees
- Large Enterprises: More than 1,000 employees

Both types of organizations are focusing on adopting advanced analytics solutions and services to better engage customers and deliver enhanced services. The advanced analytics solutions support businesses to enhance customer support services and optimize complex HR processes. Hence, the analysis of all kinds of data using sophisticated quantitative methods, for example, statistics, descriptive and predictive data mining, simulation, and optimization to produce insights that traditional approaches to BI. The deployment of advanced analytics solutions varies according to the needs of different end users. For SMEs, the advanced analytics market is expected to show tremendous growth during the forecast period, while the advanced analytics solutions and services for large enterprises are expected to show a stable growth rate during the forecast period. Among the Small and Medium-sized Enterprises (SMEs), the Software-as-a-Service (SaaS)-based advanced analytical solutions owing to lower cost have grown significantly. This is anticipated to boost the growth of the SMEs segment during the forecast period.

10.1.1 ORGANIZATION SIZE: COVID-19 IMPACT

- Countries are launching financial packages to restore their economies amid the COVID-19 pandemic gripping countries across the globe. For instance, the US is looking at a Senate vote for rolling out a rescue package worth USD 2 trillion.
- Italy, one of the most widely impacted countries by the crisis, has developed an emergency plan of USD 28 billion that could help the country inch slowly toward normalcy. As businesses across verticals are finding it a challenge to operate consistently, it has been a distressing time for SMEs.
- The GDP of countries is going to fall due to the COVID-19 pandemic. It would affect SMEs more as compared to large enterprises as they do not have sufficient capital reserves to bear losses, and there will be no or fewer new businesses for at least three to four months.
- The COVID-19 pandemic has significantly impacted global trade. Multinational companies have initially witnessed the supply shock, after which they started facing the demand shock as various countries introduced lockdown measures. Governments, businesses, and customers have struggled to procure basic products and materials.
- According to the data released by TradeShift, domestic and international trade transactions witnessed a decline of 56% since the mid of February 2020. The US, UK, and Europe have also witnessed a combined initial drop of 26% at the beginning of April 2020. As stated in the PowerhouseCooper (PwC) April CFO Pulse survey, 77% of CFOs focused on implementing cost-containment measures.

FIGURE 43 SMALL AND MEDIUM-SIZED ENTERPRISES SEGMENT TO REGISTER A HIGHER CAGR DURING THE FORECAST PERIOD



	Large Enterprises	SMEs
2021	22,465	11,306
2026	58,601	31,223
CAGR (2021-2026)	21.1%	22.5%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 65 ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	2,699	4,142	6,029	8,370	9,660	37.5%
Large Enterprises	5,701	8,634	12,402	16,990	19,402	35.8%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 66 ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	11,306	13,425	16,185	19,822	24,635	31,223	22.5%
Large Enterprises	22,465	26,391	31,475	38,132	46,880	58,601	21.1%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

The large enterprises segment is estimated to hold a larger market share in 2021. The growth of the segment is due to the increased competition in large enterprises from budding SMEs. Large enterprises are focusing on solutions to effectively manage complex business HR process to enhance customer engagement. Hence, these organizations are using advanced analytics solution to effectively manage complex operations. The SMEs segment is projected to register a higher CAGR during the forecast period due to the growing need to enhance business processes, reach new customers, and stay competitive and control their spending.

10.2 LARGE ENTERPRISES

10.2.1 EFFECTIVE USE OF ADVANCED ANALYTICS TECHNIQUES IS CAPABLE OF TRANSFORMING DATA INTO DELIVERABLE INFORMATION THAT WILL DRIVE THE ADOPTION OF ADVANCED ANALYTICS SOLUTION AMONG LARGE ENTERPRISES

Large enterprises are defined as business entities with 1,000 or more employees. These organizations have a large customer base that is to be managed via collaborative work teams to enhance their experience. With the help of advanced analytics solutions and services industries, such as BFSI, transportation and logistics, and retail industries, are deploying advanced analytics solutions and services to provide enhanced customer experience. Large enterprises collect customer and market data to help improve experiences and drive business growth. Today, most organizations adopt advanced analytics solutions to harness data to generate better business decisions. Business data analytics provides large enterprises with the opportunity to obtain administrative understanding from the hidden data to build an agile business model. Those firms with effective use of big data are more capable of transforming data into meaningful deliverable information in different parts of the organization began to recognize the ability to generate detailed insights into their business to create new business opportunities. Large enterprises are heavily investing in advanced technology to increase the company's overall productivity and efficiency. Large enterprises accumulate huge chunks of data that can be attributed to the widespread client base large enterprises are leveraging real-time data coming from various sources. For instance, social media feeds or sensors and cameras, each record needs to be processed in a way that preserves its relation to other data and sequence in time. Log files, eCommerce purchases, weather events, utility service usage, geo-location of people and things, server activity, and more allow them to take business decisions pro-actively as well as provide customers with better and more personalized services based on real-time insights. In large enterprises, the adoption of innovative technologies has produced more advantages and opportunities in the advanced analytics market. Large organizations are obligated to adhere to stringent compliance regulations such as General Data Protection Regulation (GDPR) and other banking-related regulations that include Basel III and Dodd-Frank Wall Street Reform and Consumer Protection Act; these regulations drive the adoption of advanced analytics among large enterprises.

TABLE 67 LARGE ENTERPRISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	2,008	2,993	4,228	5,691	6,414	33.7%
Europe	1,517	2,280	3,250	4,418	5,011	34.8%
APAC	1,118	1,737	2,559	3,595	4,182	39.1%
MEA	594	911	1,325	1,838	2,125	37.5%
Latin America	464	713	1,041	1,450	1,670	37.8%
Total	**5,701**	**8,634**	**12,402**	**16,990**	**19,402**	**35.8%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 68 LARGE ENTERPRISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	7,325	8,484	9,970	11,895	14,392	17,648	19.2%
Europe	5,764	6,726	7,968	9,589	11,711	14,505	20.3%
APAC	4,934	5,905	7,175	8,856	11,092	14,175	23.5%
MEA	2,492	2,964	3,580	4,393	5,469	6,907	22.6%
Latin America	1,951	2,312	2,781	3,400	4,217	5,365	22.4%
Total	**22,465**	**26,391**	**31,475**	**38,132**	**46,880**	**58,601**	**21.1%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

10.3 SMALL AND MEDIUM-SIZED ENTERPRISES

10.3.1 ADOPTION OF NEW TECHNOLOGIES TO FIX ISSUES FOR ENHANCING BUSINESS PROCESSES TO BOOST THE DEMAND FOR ADVANCED ANALYTICS AMONG SMALL AND MEDIUM-SIZED ENTERPRISES

SMEs are organizations with an employee strength of less than 1,000. SMEs are also implementing advanced analytics solutions and services to improve the efficiency of the call center and resolve customer queries in real time. Reduce operational costs, increase government support, and enhance IT infrastructure are the factors that would influence the adoption of advanced analytics solutions among SMEs. SMEs are at an early stage of advanced analytics adoption, with most small enterprises in the exploratory stage; however, they have started showing a greater interest in getting an analytics platform to reap the desired outcome. Most SMEs are now viewing advanced analytics more as a strategic initiative and an opportunity to increase business revenue. SMEs are looking for ways and methods to enrich their competitive positioning in the market and enhance their productivity through the adoption of technology such as analytics, big data, Artificial Intelligence and IoT. SMEs have also become active adopters of ICT as small and mid-size Indian companies have seen more big data deployments than the big competitors, which involve the use of advanced analytics markets. These are an early stage of big data adoption; businesses have started showing a greater interest in getting analytics platforms off the ground to reap the desired outcome. Vendors such as Oracle, IBM, and SAP are incorporating advanced analytics into their cloud offerings for automating business processes and operations across supply chains to enhance customer experiences. SMEs are making use of such offerings for improving their business outcomes. SMEs benefit from business values of business data analytics adoption. The adoption of business data analytics among SMEs is along with the integration of the Technology Organization Environment (TOE) model and Resource-based View (RBV) of the firm. Advanced analytics finds its applications across many other verticals as data has emerged as an important strategic asset for businesses development. With the arrival of in-memory analytics and advanced visualization techniques, organizations can uncover the true potential of their gathered data.

TABLE 69 SMALL AND MEDIUM-SIZED ENTERPRISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	993	1,501	2,149	2,932	3,340	35.4%
Europe	717	1,092	1,579	2,176	2,496	36.6%
APAC	516	813	1,215	1,731	2,036	40.9%
MEA	268	417	615	865	1,011	39.4%
Latin America	204	319	472	667	777	39.6%
Total	**2,699**	**4,142**	**6,029**	**8,370**	**9,660**	**37.5%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 70 SMALL AND MEDIUM-SIZED ENTERPRISES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	3,857	4,517	5,367	6,474	7,921	9,852	20.6%
Europe	2,903	3,425	4,103	4,994	6,168	7,751	21.7%
APAC	2,429	2,940	3,613	4,510	5,714	7,408	25.0%
MEA	1,199	1,443	1,763	2,188	2,755	3,530	24.1%
Latin America	918	1,100	1,339	1,655	2,077	2,682	23.9%
Total	**11,306**	**13,425**	**16,185**	**19,822**	**24,635**	**31,223**	**22.5%**

e: estimated; p: projected

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

11 ADVANCED ANALYTICS MARKET, BY VERTICAL

KEY FINDINGS

- The BFSI vertical is projected to hold the largest market share and is expected to grow from USD 6,219 million in 2021 to USD 15,701 million by 2026, at a CAGR of 20.3% during the forecast period.
- The growth of the BFSI vertical is attributed to the increased adoption of advanced analytics solutions by financial institutions, which helps them in flawlessly connecting with customers and shaping the business system with an improved customer experience and lower customer churn.
- The healthcare and life sciences vertical segment is expected to grow from USD 4,722 million in 2021 to USD 13,603 million by 2026, at the highest CAGR of 23.6% during the forecast period.
- This growth is attributed to the increasing need to improve patient management, monitoring, and experience and engagement. Healthcare firms leverage data analytics solutions to get actionable insights out of healthcare data to support fact-based decision-making.
- The IT and telecom vertical is projected to hold the second-largest market share and is expected to reach USD 14,367 million by 2026 during the forecast period. This growth can be attributed to the rising need to effectively help telecom sectors to track and manage data integrity, profitability, predictability, and mitigate risks in almost real time.

11.1 INTRODUCTION

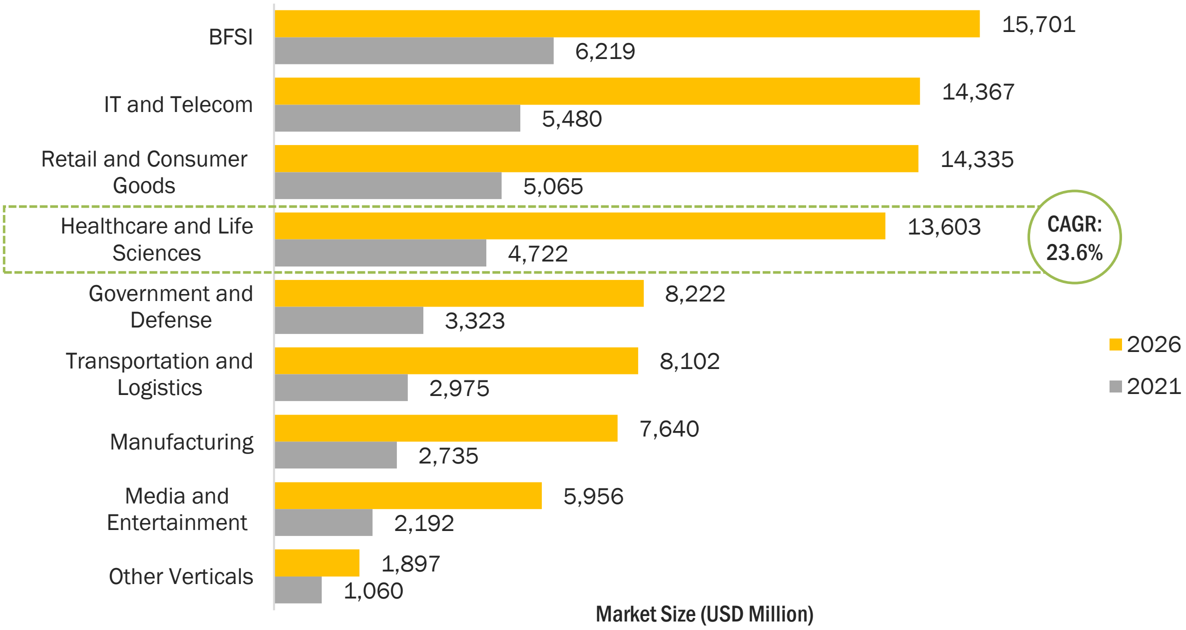
The section includes the segmentation of the advanced analytics market based on verticals into BFSI, healthcare and life sciences, IT and telecom, retail and consumer goods, manufacturing, government and defense, transportation and logistics, media and entertainment, and other verticals (education, energy and utilities, and travel and hospitality). The study helps in proposing the usage of advanced analytics solution in each sector and their respective benefits. Vendors in the advanced analytics market need to understand the end user's requirements within a particular industry and accordingly customize their solution offerings and services to make their virtual assistance more flexible, scalable, and manageable. Advanced analytics helps enterprise leaders to make data-based decisions, recognize opportunities, detect, and mitigate risk, spot, and stay ahead of trends, and understand customer demands. By using advanced analytics companies and business leaders can understand past events, predict possibilities for the future, and compare patterns. Moreover, using different types of advanced analytics solutions can improve their data-driven business decision-making and help bring value to the organization.

11.1.1 VERTICALS: COVID-19 IMPACT

- The goal of traffic control in hospitals used to be to optimize the movement of people and equipment. Contact tracing is used in the context of COVID-19 to analyze data on infected people's movements and the danger of them infecting others.
- Some issues arise because of medical equipment. The most crucial of them is ensuring immediate access to critical equipment in the event of an emergency. COVID-19 has demonstrated that patients require ventilators on occasion. It is critical to offer prompt delivery of equipment to save a patient's life, which is a major role of the advanced analytics solution.
- After COVID-19, the growing complexities in the healthcare vertical to generate real-time patients and equipment location is driving the adoption of advanced analytics solutions in the healthcare industry.
- Due to the outbreak of COVID-19, a complete 360° transformation in business operations and logistic processes can be observed. Processes have become more dynamic than before, while companies are focusing on the fulfillment of critical tasks.
- The COVID-19 outbreak is forcing numerous verticals to review their IT strategies as a top priority. COVID-19 is leading to drastic changes in customer purchase and viewing behaviors. The pandemic has resulted in the shutting down of retail stores and signaling commerce disruption. The retail vertical is facing multiple short-term challenges around health and safety, supply chain, labor force, cash flow, consumer demand, and marketing.
- The ongoing COVID-19 pandemic has adversely affected the manufacturing vertical. The manufacturing vertical is facing a slowdown all over the world, including the US, China, South Korea, and more than 15 other countries, due to country-wide lockdowns, strict international border controls, social distancing measures of workers, and supply chain issues.
- The BFSI vertical has been taking various measures, including providing reduced interest rates and offering moratorium benefits, to help credit payers and businesses fight and survive the financial crisis during the pandemic.
- Communication is playing a vital role across the globe during the COVID-19 crisis. Every individual and government, irrespective of federal, state, central, local, and provincial, is in constant touch with one another to provide and get real-time information on COVID-19.
- During the COVID-19 pandemic, the healthcare and life sciences vertical is facing immense pressure for enhancing and providing Personal Protective Equipment (PPE), ventilators, and prophylactic and anti-viral drugs across the world.

- In the COVID-19 response, organizations are identifying the new business challenges to overcome these challenges organizations stood up central nerve centers, mobilizing business and analytics resources to inform and address the challenges by building new data streams, reporting on business-critical issues to guide near-term decisions, and developing longer-term views of data to understand what the future may hold for their company, customers, and suppliers.

FIGURE 44 HEALTHCARE AND LIFE SCIENCES VERTICAL TO GROW AT THE HIGHEST CAGR DURING THE FORECAST PERIOD



Other verticals include education, energy and utilities, and travel and hospitality

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.1.2 MAJOR USE CASES: BY VERTICAL

VERTICAL	USE CASES
BFSI	▪ Virtual Assistance ▪ Customer Service ▪ Personal Financial Management ▪ Personalized Marketing ▪ Sales Notifications ▪ Fraud Detection ▪ Recommendation
IT and Telecom	▪ Customer Concierge ▪ Troubleshooting Advisors ▪ Recharge Assistance ▪ Field Service Assistant ▪ Business Performance Advisors ▪ Lead and Activity Manager

Healthcare and Pharmaceuticals	▪ Virtual Assistance ▪ Patient Engagement ▪ Customer Service ▪ Care Coordination ▪ Telematics ▪ Treatment Analytics
Retail and Consumer Goods	▪ Customer Service ▪ Inventory Tracking ▪ Personalized Push Ads ▪ Recommendation ▪ Auction
Manufacturing	▪ Product Recall ▪ Vendor Engagement ▪ Customer Service ▪ Supply Chain
Transportation and Logistics	▪ Accidental Assistance ▪ Insurance and Service Reminders ▪ Supply Chain Management ▪ Pre-Sales and Post Sales Services ▪ Cas Discovery
Education	▪ Personalized Language Content ▪ Student Feedback
Media and Entertainment	▪ Personalized Campaign
Government and Defense	▪ General Services Administration

Source: MarketsandMarkets Analysis

TABLE 71 ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	1,627	2,449	3,498	4,764	5,406	35.0%
IT and Telecom	1,385	2,099	3,019	4,140	4,730	35.9%
Retail and Consumer Goods	1,181	1,820	2,660	3,707	4,303	38.1%
Healthcare and Life Sciences	1,082	1,673	2,452	3,430	3,996	38.6%
Transportation and Logistics	723	1,104	1,601	2,213	2,548	37.0%
Government and Defense	886	1,329	1,891	2,566	2,900	34.5%
Manufacturing	647	994	1,448	2,013	2,330	37.8%
Media and Entertainment	535	817	1,182	1,633	1,879	36.9%
Other Verticals	335	491	680	894	970	30.5%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Other verticals include education, energy and utilities, and travel and hospitality

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 72 ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	6,219	7,260	8,603	10,355	12,649	15,701	20.3%
IT and Telecom	5,480	6,442	7,689	9,324	11,475	14,367	21.3%
Retail and Consumer Goods	5,065	6,049	7,335	9,035	11,294	14,335	23.1%
Healthcare and Life Sciences	4,722	5,660	6,890	8,520	10,694	13,603	23.6%
Transportation and Logistics	2,975	3,524	4,238	5,178	6,420	8,102	22.2%
Government and Defense	3,323	3,864	4,562	5,471	6,658	8,222	19.9%
Manufacturing	2,735	3,257	3,938	4,837	6,030	7,640	22.8%
Media and Entertainment	2,192	2,595	3,120	3,810	4,723	5,956	22.1%
Other Verticals	1,060	1,165	1,286	1,423	1,573	1,897	12.3%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Other verticals include education, energy and utilities, and travel and hospitality

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

The table above highlights the advanced analytics market size by vertical. The market size of the BFSI segment is expected to hold the largest market size in 2021. The growth of the BFSI vertical is attributed to the increased adoption of advanced analytics solutions by financial institutions, which helps them in flawlessly connecting with customers and shaping the business system with an improved customer experience and lower customer churn. The healthcare and life sciences vertical is projected to grow at the highest CAGR during the forecast period. Advanced analytics solutions provide care plans and educate patients to improve patient care and outcomes while decreasing healthcare costs to healthcare firms.

11.2 BANKING, FINANCIAL SERVICES, AND INSURANCE

11.2.1 ADVANCED ANALYTICS IDENTIFIES NEW BUSINESS OPPORTUNITIES, DESIGNING CUSTOMER VALUE PROPOSITIONS AND CONSTRUCTING NEW BUSINESS MODELS ACROSS THE BFSI VERTICAL

The banking, financial services, and insurance vertical is experiencing significant turmoil due to the heightened regulatory environment and changing customer behavior. All these are shrinking demands, globalization and competition are driving the industry to aggressively look at data, drive analytics, and ignite profitable growth. Analytics is applied in all areas of insurance organizations. They are used in various areas such as claims, marketing, customer experience, finance, and human resources. Advanced analytics solutions enable BFSI clients to outperform challenges such as rising customer acquisition costs and declining customer satisfaction. For instance, the risk analytics solution of WPN offers the entire spectrum of the BFS industry retail banks, commercial finance, capital markets, asset management and investment banks. Firms harness the power of AI to reinvent critical business processes everything from evaluating and pricing risk to personalizing customer experiences and much more.

In the BFSI vertical, the analytics solutions help in expanding customer relationships through targeted up-sell and cross-sell, reduce churn by predicting at-risk customers in time to proactively prevent attrition. For instance, PayPal uses RapidMiner's sentimental analysis solutions and text analytics to gain customer feedback and can identify, classify, and count customers as top promoters and top detractors, from various customers. Nowadays, customers' expectations from banks and financial institutions have gone beyond being mere enablers to also being counselors and advisors for their financial decisions. Modern data analytics have helped financial institutions to move the needle toward a more personalized approach for customers by supporting their financial journey through measures such as guidance on savings, along with individualized recommendations for products or services. This in return has helped financial institutions to market and sell products that are relevant to the needs of customers instead of having a one-size-fits-all approach.

In the BFSI vertical, the data, once analyzed, has the potential to generate business insights for customer acquisition and customer experience transformation and thereby generate monetization avenues in these industries. The use of advanced analytics has helped leverage analytics, AI, cloud, open banking APIs, FinTech, and RegTech technologies. It has also helped manage increasing regulatory demands, help to stop fraud in its tracks, and maximize return on capital. These analytics solutions have achieved operational excellence and have created personalized customer experiences that have increased customer engagement. Advanced analytics has led to growth in businesses in the BFSI vertical by identifying new business opportunities, designing customer value propositions, and constructing new business models across the financial services value chain. In the BFSI sector, companies are integrating analytics and ML into processes that trigger automated actions and more contextual, real-time decision-making. They automate certain processes such as underwriting, customer onboarding, branch operations, customer interactions, and customer intelligence. This includes constantly evaluating data such as transaction status, stakeholder details, transaction type, payments processing, compliance, documentation, and many other processes.

TABLE 73 BFSI: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	613	910	1,281	1,719	1,929	33.2%
Europe	456	683	970	1,315	1,485	34.3%
APAC	288	445	651	909	1,050	38.2%
MEA	152	231	334	461	529	36.7%
Latin America	118	180	261	361	413	36.9%
Total	**1,627**	**2,449**	**3,498**	**4,764**	**5,406**	**35.0%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 74 BFSI: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	2,194	2,530	2,961	3,519	4,240	5,179	18.7%
Europe	1,700	1,976	2,331	2,794	3,397	4,191	19.8%
APAC	1,230	1,461	1,763	2,162	2,689	3,410	22.6%
MEA	616	728	873	1,064	1,316	1,649	21.8%
Latin America	479	564	673	817	1,007	1,271	21.6%
Total	**6,219**	**7,260**	**8,603**	**10,355**	**12,649**	**15,701**	**20.3%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.3 IT AND TELECOM

11.3.1 GROWING PENETRATION OF SMARTPHONES, COUPLED WITH THE INCREASE IN THE NUMBER OF SERVICE PROVIDERS, HAVE CREATED THE DEMAND TO ADOPT ADVANCED ANALYTICS SOLUTION AMONG THE TELECOM INDUSTRY

The telecom and IT industry is highly competitive and continuously focuses on delivering enhanced services to acquire new customers and retain the existing ones. The factors such as the growing complexity of the telecom industry and the growth in the number of service providers have created the demand to adopt advanced analytics solutions. The fast use of smartphones and other connected mobile devices have resulted in an increase in the volume of data flowing across telecom networks. Operators must process, store, and extract information from the given data. In the IT and telecom vertical, advanced analytics plays a major role as it helps optimize the network operations along with maximizing network profitability. Moreover, advanced analytics helps support diverse data sources, formats, skill sets, methods, and infrastructure. It also scales easily using automated and semi-automated methods to dynamically tweak capacity and address problems of any size or complexity. It ensures superior performance and resilience with secure authentication and authorization governance and regulatory traceability all with centralized administration.

The IT and telecom vertical is constantly developing in technological advancements and competition. It involves a large amount of customer care and intense competition across the target segment. The unpredictable nature of the telecom marketplace and the high subscriber churn rate have compelled several telecom providers to become more responsive and integrate robust customer experience management tools in their business systems. The deployment of advanced analytics solutions helps forecast and manage network traffic with greater precision. By using advanced analytics in the IT and telecom vertical we can predict and prevent network issues before they affect the customer behavior, optimize network investment, help to know the customer experience throughout the network, and help in identifying customer consumption patterns across network elements. For instance, OTE Cosmote uses SAS for network analytics to facilitate analyzing and visualizing huge amounts of network and customer data quickly and efficiently. It has helped OTE Cosmote deliver the best customer experience and remain a leader in the Greek telecommunications market. This has helped it to analyzes vast amounts of data and enhance customer experience, service, and loyalty.

In the IT and telecom vertical, the goal of analytics is to analyze the growing volumes of network and customer data with greater speed, which helps in accelerating the decision-making process. In this vertical, there is a need for reducing churn as the competition is increasing with the incidence of new entrants, as they provide lucrative deals that are generally inexpensive than that of the incumbents. The advanced analytics market combines many sophisticated BI technologies, which satisfy the complex demand for the telecom industry. These include developing sales, reducing churn and deception, enhancing risk management, and decreasing operational costs. Technologies such as big data help telecom companies increase profitability by helping optimize network usage and services, enhancing customer experience, and improving security. Big data plays a major role to drive progress in the telecommunication industry. With the right data analytics approach, telecommunication companies can improve their services and make their subscribers happier. Companies and enterprises that implement big data analytics can reap several benefits such as informed decision-making, improved customer service, and efficient operations.

TABLE 75 IT AND TELECOM: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	503	749	1,058	1,422	1,600	33.5%
Europe	375	562	801	1,088	1,232	34.7%
APAC	274	425	626	879	1,020	38.9%
MEA	131	204	300	421	493	39.2%
Latin America	102	158	234	330	385	39.4%
Total	**1,385**	**2,099**	**3,019**	**4,140**	**4,730**	**35.9%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 76 IT AND TELECOM: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,825	2,110	2,476	2,949	3,563	4,366	19.1%
Europe	1,414	1,648	1,949	2,342	2,855	3,533	20.1%
APAC	1,202	1,436	1,742	2,146	2,684	3,427	23.3%
MEA	585	704	860	1,067	1,344	1,717	24.0%
Latin America	454	545	663	820	1,028	1,324	23.9%
Total	**5,480**	**6,442**	**7,689**	**9,324**	**11,475**	**14,367**	**21.3%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.4 RETAIL AND CONSUMER GOODS

11.4.1 RISING NEED TO ENHANCE CUSTOMER EXPERIENCE, EMPLOYEE EFFICIENCY, AND THE OVERALL FUNCTIONING OF THE BUSINESS TO BOOST THE DEMAND FOR ADVANCED ANALYTICS ACROSS RETAIL AND CONSUMER GOODS SECTORS

Advances in AI for retail and consumer goods have made it possible to automate complex tasks through constant learning, enabling retailers to improve operations and customer experiences by breaking down departmental silos and applying omnichannel analytics to every step of the customer journey. Understanding where customers are in their journeys and optimizing each interaction drives longevity, loyalty, and growth by turning data into action. The use of advanced analytics solutions enables retail and consumer goods sectors to deliver more personalized shopping experiences and increase revenue. This analytics help in every step of the customer journey for better connections and deeper insights. The use of advanced analytics in this vertical help by providing analytical optimization to gain better customer insights for improving the customer experience, boosting response rates, and lowering campaign costs. Today's customers are mobile and have more control over the purchasing process, which means retailers must use customer data and insights from analytics to improve the customer engagement program.

In these verticals, advanced analytics help in engaging and responding to customer needs instantly using contextual listening and advanced data integration to enhance both in-store and online shopping experiences. It also helps increase selling by understanding what actions drive transactions and customer value. It personalizes interactions to drive the best performance and integrate the experience across all touchpoints throughout the customer journey. It provides an omnichannel approach to enable the organization to spend less on customer acquisition. With the help of predictive modeling that forms the basis for comprehensive integrated marketing as it improves the outcomes of the marketing interactions. For instance, German electronics retailer, Conrad Electronic, uses SAS Predictive Analytics for analyzing the online and offline customer data in real time to align the offers to the interests of individual customers. AI-driven personalization gives a boast to electronics as it offers higher customer satisfaction. The ability to score online shoppers based on their current and historical behavior enables the retailer to learn their preferences and display relevant offers along the customer journey. With advanced analytics, the retailer can now identify 60% of visitors, enabling it to increase the relevancy of displayed banners and ultimately drive higher customer satisfaction and increase revenue.

In the retail and consumer goods vertical, the robust expansion of eCommerce sector, coupled with the growing need among retailers to offer distinguished in-store experiences to their customers, has strongly driven the adoption of analytics in the retail industry. The deployment of retail analytics by online and offline retailers has increased to a great extent due to the numerous advantages involved such as monitoring of supply chain movements, inventory levels, buying decisions, and consumer demand and insights. With the growing popularity of big data, AI and ML technologies act as a major driving force for the use of retail marketing analytics. These solutions offer better consumer insights which help retailers target potential buyers and deliver products preferred by them. The deployment of such advanced technologies will transform the retail sector in the future. Retail businesses specifically generate vast amounts of data. This data can be turned into a goldmine by using an advanced analytics technique, such as predictive analytics, to translate it into data-driven insights based on customer preferences and prevailing trends. Retailers can utilize predictive techniques to use structured and unstructured data to predict sales growth due to changes in consumer behaviors and market trends. This analysis has helped retailers stay ahead, gain a competitive advantage, increase revenue, and improve RoI overall. For instance, Amazon is an example of how predictive analytics can enable the personalized shopping experience online. Thus, technology innovation, digitalization, and ML models have fundamentally changed the landscape of retail and consumer products businesses. Making model-driven decisions that apply ML with best-of-breed tools is the key to success for these industries.

TABLE 77 RETAIL AND CONSUMER GOODS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	423	641	921	1,261	1,445	36.0%
Europe	315	481	697	964	1,112	37.1%
APAC	230	364	545	779	921	41.4%
MEA	120	188	279	394	464	40.1%
Latin America	93	146	217	308	362	40.4%
Total	**1,181**	**1,820**	**2,660**	**3,707**	**4,303**	**38.1%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 78 RETAIL AND CONSUMER GOODS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,677	1,974	2,358	2,860	3,518	4,381	21.2%
Europe	1,300	1,542	1,856	2,271	2,819	3,545	22.2%
APAC	1,104	1,343	1,659	2,081	2,650	3,438	25.5%
MEA	554	671	825	1,031	1,307	1,678	24.8%
Latin America	430	519	636	792	1,000	1,293	24.6%
Total	**5,065**	**6,049**	**7,335**	**9,035**	**11,294**	**14,335**	**23.1%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.5 HEALTHCARE AND LIFE SCIENCES

11.5.1 NEED TO IMPROVE PATIENT MANAGEMENT, MONITORING, AND ENHANCE PATIENT EXPERIENCE ARE DRIVING THE ADOPTION OF ADVANCED ANALYTICS ACROSS THE HEALTHCARE AND LIFE SCIENCES VERTICAL

Technological advancements have played a vital role in the healthcare and life sciences vertical. The vertical is looking for the right tool and technology to engage with patients in a better way. Advanced analytics solutions help healthcare and life sciences organizations to efficiently manage how data of the patients is collected, treated, processed, and presented for the process of testing and simulation of new treatments, scenarios, and devices. The healthcare stakeholders must adhere to the compliance policies and government regulations while deploying the virtual health assistant on various platforms. Data is transforming healthcare and life sciences as it helps create better patient outcomes while reducing costs. Healthcare providers that are taking advantage of the vast quantity of data are differentiating themselves now and leading the industry to the future. In these verticals, analytics insights provide value-based health care. Advanced analytics in this vertical helps combine health and non-health data to understand and predict future population health needs, it improves patient safety through the better determination of treatment criteria and risk factors, it helps discover new treatments and identify the right patient candidate profiles for each treatment, and it provides patients with precision care along with personalized treatment and healthier outcomes. For instance, Evidation is on a mission to empower everyone to participate in better health outcomes. To achieve this, the company looks at patients' lived experiences beyond healthcare facilities. It has to leverage with Domino Enterprise MLOps capabilities and a Snowflake cloud-based data warehouse that provide delivery of new production-grade models in as little as eight weeks. It has also reduced onboarding times from weeks to less than a day. It has led to increasing team productivity and scale to serve more clients.

Pressures on healthcare and life sciences companies worldwide to cut costs, enhance coordination and outcomes, give more with less, and be more patient-centric are growing. This is driving the growth of the advanced analytics market. Data and AI are revolutionizing how healthcare organizations treat patients and deliver value for the broader population. Advanced analytics empowers healthcare organizations by unifying data analytics and ML to unlock precision care, improve patient engagement, and streamline administration processes. Healthcare organizations use advanced analytics to provide innovation in patients as it predicts health risks with analytics and AI that scale for millions of patient records in the cloud. Life sciences companies transform data into life-changing insights. For instance, clinical research analytics helps to get better, safer products to market in a lot less time with faster, more efficient clinical trials. It has advanced medical research and discovery through greater transparency. In this vertical, AI and advanced analytics are used for unlocking new possibilities as embedded AI capabilities are tailored to the needs of life sciences organizations. IoT leverages data from medical devices and sensors to strengthen analytics insights and response times at every stage of the product journey.

TABLE 79 HEALTHCARE AND LIFE SCIENCES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	400	608	875	1,201	1,378	36.2%
Europe	298	456	663	918	1,061	37.4%
APAC	198	315	475	684	815	42.4%
MEA	103	162	241	343	407	41.0%
Latin America	83	132	198	283	335	41.7%
Total	**1,082**	**1,673**	**2,452**	**3,430**	**3,996**	**38.6%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 80 HEALTHCARE AND LIFE SCIENCES: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,604	1,892	2,264	2,752	3,392	4,229	21.4%
Europe	1,243	1,477	1,782	2,185	2,718	3,422	22.5%
APAC	984	1,205	1,499	1,894	2,428	3,164	26.3%
MEA	489	596	738	928	1,184	1,526	25.6%
Latin America	402	490	606	762	972	1,263	25.7%
Total	**4,722**	**5,660**	**6,890**	**8,520**	**10,694**	**13,603**	**23.6%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.6 TRANSPORTATION AND LOGISTICS

11.6.1 GROWING TO MANAGE TRAFFIC AND HANDLE COMPLEX SUPPLY CHAIN OPERATIONS WILL GIVE A BOOST TO THE GROWTH OF ADVANCED ANALYTICS ACROSS THE TRANSPORTATION AND LOGISTICS VERTICAL

The transportation and logistics business is undergoing massive transformations. Digital transformations, changing consumer expectations, evolving company models, and changing customer expectations are all examples of this. Advanced analytics in transport and logistics is used to improve operational efficiencies, reduce costs of critical transportation and infrastructure assets using predictive maintenance analytics. It improves insights into public safety across the traffic and transportation infrastructure by applying AI and ML techniques. It brings together disparate data including a flood of rich new real-time data from sensors, meters, and video and Unmanned Aerial Vehicles (UAVs) from other agencies to unlock new insights and make better decisions in real time. For instance, The Road Safety Commission of Western Australia uses AI and ML capabilities of SAS in the cloud to improve road safety as it uses AI and advanced analytics to bolster safety efforts. It has helped in curbing traffic accidents and saving lives with ML.

The transportation industry has never lacked data, as it is collected across every interaction point. Whether for a guest, passenger, or cargo, data helps balance the demand with supply to optimize the revenue and profitability of its inventory and capacity. From travel and hospitality across airlines, hotels, car rental, and cruise companies, each are capturing customer behaviors and experiences that contain valuable insights for the business. Transportation and logistics companies across trucking, freight rail, air delivery, maritime, and logistics service providers are trying to link disparate data sources to manage network and capacity planning as well as optimize routes for profitability. In these verticals, advanced analytics helps link similar data to gain the insights needed to improve revenue management. It also helps yield data-driven pricing strategies and provides visibility into real-time inventory. For instance, Transport for London uses Rapidminer for the operation of the road network, managing the traffic signals and ensuring safe, high-quality roadworks across the city. The transportation and logistics operators in the world use advanced analytics to capture the data, organize it, and find the exploitable connections between the long-term corporate strategy, the data strategy, and daily business operations. Advanced analytics helps this industry to meet consumer demand and overcome brutal competition. It helps make transportation and logistics safer, cleaner, and more reliable all while reducing costs as it applies predictive maintenance to fleets and infrastructure to reduce repair costs. It helps predict storm patterns to minimize passenger and cargo disruption.

TABLE 81 TRANSPORTATION AND LOGISTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	245	368	524	711	807	34.7%
Europe	183	276	397	544	621	35.8%
APAC	156	244	362	513	601	40.2%
MEA	78	121	178	250	292	38.9%
Latin America	61	94	139	196	227	39.1%
Total	**723**	**1,104**	**1,601**	**2,213**	**2,548**	**37.0%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 82 TRANSPORTATION AND LOGISTICS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	928	1,082	1,280	1,538	1,874	2,317	20.1%
Europe	719	845	1,008	1,221	1,502	1,875	21.1%
APAC	715	862	1,056	1,313	1,657	2,136	24.5%
MEA	345	414	505	625	785	1,002	23.8%
Latin America	268	321	389	480	601	772	23.6%
Total	**2,975**	**3,524**	**4,238**	**5,178**	**6,420**	**8,102**	**22.2%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.7 GOVERNMENT AND DEFENSE

11.7.1 RISING NEED TO MANAGE CRITICAL DATA AND DELIVER ENHANCED SERVICES TO THE CITIZENS TO DRIVE THE GROWTH OF ADVANCED ANALYTICS AMONG GOVERNMENT AND DEFENSE INDUSTRY

The government vertical includes all government organizations and agencies that deal with tax collection, safety, public interest, and education. The demand for transparency and accountability in the public sector is at an all-time high, and it starts with data. Governments are redefining how they deliver services and interact with the public. Improving the online experience for citizens is a top priority for public sector leaders establishing digital citizen services. Data and analytics play a major role in the understanding of constituent needs, improving customer service, and for many other purposes. With analytics solutions across the full range of government duties, including public safety, defense, national intelligence, social programs, cybersecurity, and finance and operations, Advanced analytics enables governments throughout the world to care for and ensure citizens' safety. For instance, The State of Tennessee’s Department of Economic and Community Development (TNECD) has a mission to develop strategies that help make Tennessee the top location in the Southeast for high-quality jobs. The department seeks to attract new corporate investment in Tennessee from the US and global companies and works with local companies to facilitate expansion and economic growth. It uses Tableau to share investment and programmatic information with a variety of audiences, including department executives, commissioners, and partners in the General Assembly. It helps in automated analysis of strategic plan performance.

In the government and defense vertical, advanced analytics helps in improving personnel readiness through analytics that delivers improved training, recruitment, and retention programs. It also enables better strategic, operational, and tactical situational awareness through enhanced analytics applied to intelligence. It helps in pinpointing the security, immigration, and health risks for cross-border movement of goods because of the use of enhanced analytics. For instance, Lockheed Martin uses AI and IoT analytics from SAS to revolutionize maintenance and keep aircraft operational for crucial missions. The company uses AI, IoT, and advanced analytics to predict when parts will fail, keeping more aircraft airborne for vital missions worldwide. Advanced analytics has been used to influence decision-making in other areas of government, including the intelligence community. In the private sector, corporations such as Google, Amazon, and Facebook have aided in the adoption of advanced predictive analytics by the general public. Using predictive and other advanced analytics to successfully drive budget decision-making can help defense organizations save money. Advanced analytics plays a major role in the government and defense vertical. Recently, in India, The Ministry of Corporate Affairs (MCA) launched a new version of its portal, version 3.0, MCA 21, which will leverage data analytics, AI, and ML, to simplify regulatory filings for companies. The idea behind the revamp is to promote ease of doing business and compliance monitoring.

Technological advancements have played a vital role in the government and defense vertical. The mission of the government agencies is to serve citizens, ensuring the safety and well-being of all people. Innovation is necessary to accomplish this. With advanced analytics, including AI and ML, governments can put data to work improving outcomes for citizens. It has helped governments at all levels to make better, faster, and more cost-effective decisions that make a difference in the lives of the people they serve. Advanced analytics has helped in discovering hidden insights across the spectrum of public sector data. It effectively manages all critical data regardless of size, type, or location that embraces efficiency and improves integration. Advanced analytics uses automated business rules, predictive modeling, text mining, exception reporting, and network link analysis that enables governments to detect, prevent, and more effectively manage both opportunistic and professional fraud.

TABLE 83 GOVERNMENT AND DEFENSE: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	321	474	664	886	990	32.5%
Europe	239	356	503	678	762	33.6%
APAC	175	269	393	548	631	37.8%
MEA	84	127	183	251	287	36.1%
Latin America	67	102	148	203	231	36.0%
Total	**886**	**1,329**	**1,891**	**2,566**	**2,900**	**34.5%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 84 GOVERNMENT AND DEFENSE: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	1,120	1,285	1,496	1,768	2,120	2,571	18.1%
Europe	868	1,003	1,177	1,404	1,698	2,081	19.1%
APAC	737	874	1,052	1,287	1,597	2,018	22.3%
MEA	333	392	468	568	699	871	21.2%
Latin America	266	311	369	445	544	681	20.7%
Total	**3,323**	**3,864**	**4,562**	**5,471**	**6,658**	**8,222**	**19.9%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.8 MANUFACTURING

11.8.1 ADVANCED ANALYTICS USED IN THE MANUFACTURING VERTICAL HELPS IMPROVE DECISION-MAKING AND ENHANCE PERFORMANCE

Advanced analytics plays a major role in the manufacturing vertical. As with advanced analytics, the process plants can use advanced analytics for large and complex datasets. The advanced analytics tools are powerful enough to cope with the complex plant data, but intuitive enough for non-data science experts to use independently. The process industry can benefit from accurate, real-time insights into opportunities, risks, trends, and the changing demand in their vertical. Advanced analytics effectively provides visibility into every aspect of the plant, its equipment, and processes; the target market and customer demands; and broader economic conditions. With the help of advanced analytics, process plants can drive innovation, improve customer experience, helps in monitoring market trends and fluctuations, and helps predict customer demand and concern. For instance, PG Dry Cooling uses advanced analytics from SAS to analyze the performance of its air-cooled condensers, enabling power plants to run more efficiently, improve maintenance planning, and better forecast energy production. It also helps in predicting energy production.

Implementing advanced analytics to process manufacturing plants gives visibility about the plant, market, supply chain, and customer demands, enabling us to cut costs, increase profitability, refine production quality, and improve customer experience. The use advanced analytics platforms in manufacturing vertical helps enhance business strategy and drive innovation to future-proof the organization and boost up the bottom line. Advanced analytics and other AI-powered platforms in the manufacturing vertical provide digital transformation that demands an organization-wide culture shift. It helps educate employees about the benefits and challenges of digital transformation to ensure everyone is on board with the digitalization journey. Data analytics used in the manufacturing vertical helps improve decision-making and enhance performance. Advanced analytics is becoming a permanent and critical element of today's manufacturing processes and product design and for maintaining and servicing equipment and products in the field. As companies re-examine methods and approaches for improving product design and production systems, analytics will become a requirement and be applied across the entire design, simulate, build, maintain the phase of its lifecycle. Next-generation analytics applications based on big data and ML will allow manufacturers to detect and determine best practices for their product design and manufacturing processes. Thus, advanced analytics serve as one of the necessary technologies that will enable advanced manufacturing and the factories of the future.

TABLE 85 MANUFACTURING: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	225	340	488	666	761	35.6%
Europe	168	255	369	509	586	36.7%
APAC	123	193	289	411	485	41.1%
MEA	74	115	170	239	280	39.4%
Latin America	57	90	133	187	218	39.6%
Total	**647**	**994**	**1,448**	**2,013**	**2,330**	**37.8%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 86 MANUFACTURING: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	881	1,035	1,233	1,491	1,830	2,274	20.9%
Europe	683	808	970	1,184	1,466	1,841	21.9%
APAC	580	704	867	1,085	1,378	1,785	25.2%
MEA	332	401	490	609	768	983	24.2%
Latin America	258	310	378	468	588	758	24.0%
Total	**2,735**	**3,257**	**3,938**	**4,837**	**6,030**	**7,640**	**22.8%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.9 MEDIA AND ENTERTAINMENT

11.9.1 MEDIA AND ENTERTAINMENT VERTICAL LEVERAGES ADVANCED ANALYTICS SOLUTION TO ENHANCE CONTENT AND CUSTOMER VIEWING EXPERIENCE

In the media and entertainment vertical, advanced analytics helps media companies to leverage audience, content, and viewing behavior data to build better, more personalized experiences for the audience as traditional business models continue to stagnate and decline. Media companies need to move faster to keep up with fickle audiences enjoying near-limitless entertainment options. For instance, Comcast media company struggled with massive data, fragile data pipelines, and poor data science collaboration. With Databricks including Delta Lake and ML flow, they can build performant data pipelines for petabytes of data and easily manage the lifecycle of 100s of models to create a highly innovative, unique, and award-winning viewer experience using voice recognition and ML. Thus, advanced analytics are increasingly addressing the needs of the media and entertainment vertical to support data processes and automate data analysis.

TABLE 87 MEDIA AND ENTERTAINMENT: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	196	294	417	565	640	34.4%
Europe	146	221	316	432	493	35.5%
APAC	103	162	241	342	402	40.5%
MEA	55	85	126	177	208	39.6%
Latin America	35	55	82	116	136	40.3%
Total	**535**	**817**	**1,182**	**1,633**	**1,879**	**36.9%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 88 MEDIA AND ENTERTAINMENT: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	734	855	1,009	1,210	1,472	1,815	19.8%
Europe	569	667	794	961	1,179	1,469	20.9%
APAC	479	579	710	886	1,121	1,448	24.8%
MEA	247	298	366	455	575	736	24.4%
Latin America	162	196	240	298	377	488	24.6%
Total	**2,192**	**2,595**	**3,120**	**3,810**	**4,723**	**5,956**	**22.1%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

11.10 OTHER VERTICALS

The other verticals include travel and hospitality, energy and utilities, and education. In the education vertical, data and analytics play an important role in anticipating the future and responding proactively to changing trends in the dynamic world of education. Advanced analytics helps education institutions to move beyond merely collecting and reporting on data to applying sophisticated analytics that can bring about systemic improvements. It has helped educational institutions to better serve the needs of their students with the power of data. For instance, Flinders University uses Databricks Lakehouse Platform, to maximize the power of data to gain actionable insights into their student population, respond to trends and transform the way they make decisions.

In the energy and utilities vertical, advanced analytics provides customer-centric digital transformation, which enables greater reliability, resource utilization, risk mitigation, and customer satisfaction with an open analytics platform that unifies increasingly diverse data and analytics ecosystem. For instance, the Enexis energy network operator needs to make data more accessible to employees to improve decision-making in grid operation. Enexis deployed SAS Visual Analytics to have better control over data and reports. It also provides faster insights through visually compelling reports.

TABLE 89 OTHER VERTICALS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2017–2020)
North America	75	108	147	190	204	28.4%
Europe	56	81	112	145	157	29.5%
APAC	88	134	192	261	293	35.0%
MEA	65	94	128	166	177	28.7%
Latin America	51	74	102	132	139	28.4%
Total	**335**	**491**	**680**	**894**	**970**	**30.5%**

Note: Numbers are rounded off to the nearest integer, resulting in rounding-off errors

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 90 OTHER VERTICALS: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	220	238	258	281	305	368	10.9%
Europe	170	186	203	223	244	298	11.8%
APAC	333	381	440	513	602	758	17.9%
MEA	190	204	219	234	246	276	7.8%
Latin America	148	157	165	173	176	197	5.9%
Total	**1,060**	**1,165**	**1,286**	**1,423**	**1,573**	**1,897**	**12.3%**

e: estimated; p: projected

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

12 ADVANCED ANALYTICS MARKET, BY REGION

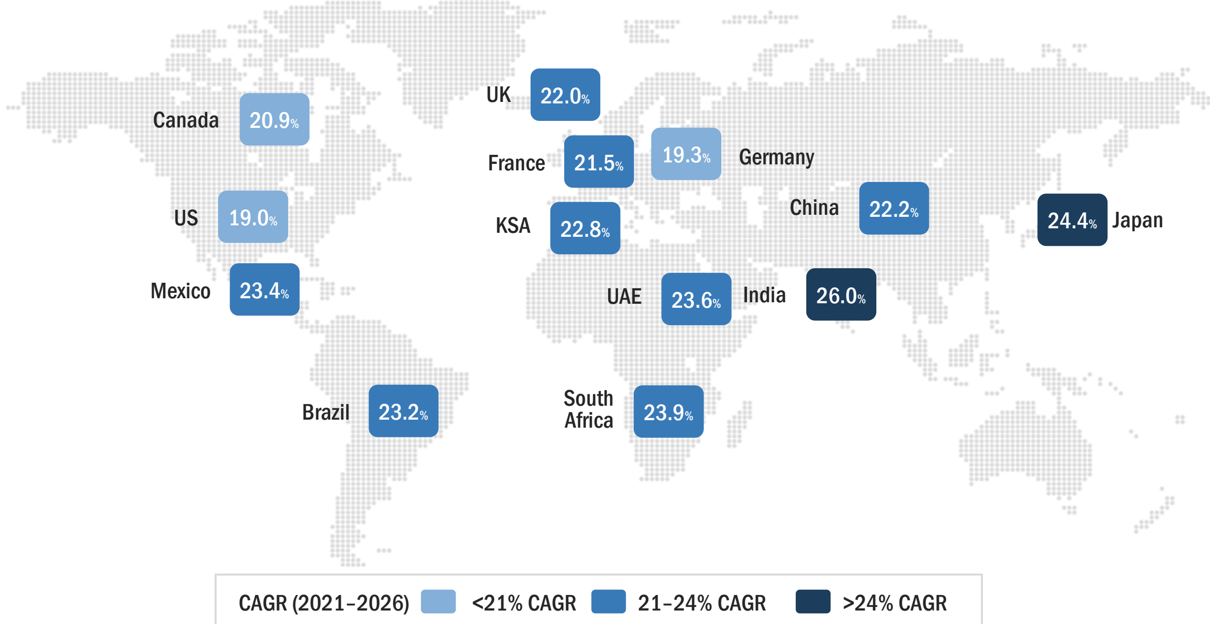
KEY FINDINGS

- North America is estimated to hold the largest market share in the global advanced analytics market in 2021 and is expected to grow at a CAGR of 19.7% during the forecast period. This growth can be attributed to the increasing use of the latest technologies and the rising standards of security and safety in work premises.
- The advanced analytics market in Europe is expected to grow from USD 8,666 million in 2021 to USD 22,256 million by 2026, at a CAGR of 20.8% during the forecast period.
- APAC is projected to grow from USD 7,363 million in 2021 to USD 28,584 million by 2026, at the highest CAGR of 24.0% during the forecast period. This growth can be attributed to the increasing penetration of smartphones and the need to enhance the customer experience in the region.
- In terms of growth, Latin America and MEA are expected to grow at CAGRs of 23.1% and 22.9%, respectively, during the forecast period.

12.1 INTRODUCTION

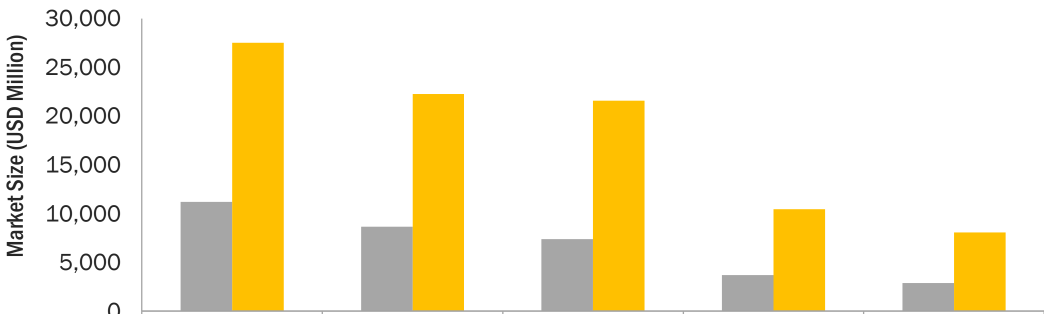
The advanced analytics market has been segmented into five regions: North America, Europe, APAC, MEA, and Latin America. North America is projected to hold the largest market size during the forecast period. The key factor favoring the growth of the advanced analytics market in North America is the increasing demand for enhanced customer support services to strengthen customer retention initiatives. Europe is expected to be in the second position in terms of market size and market share during the forecast period. The growing demand to reduce enterprise workloads related to internal and external communication monitoring is boosting the adoption of advanced analytics solutions in Europe. APAC is expected to record the fastest growth rate during the forecast period. The advanced analytics market is expected to witness considerable developments and adoption of solutions across APAC during the forecast period. The increasing number of advanced analytics vendors across regions is further expected to drive the advanced analytics market.

FIGURE 45 INDIA TO ACCOUNT FOR THE HIGHEST CAGR DURING THE FORECAST PERIOD



Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 46 ASIA PACIFIC TO WITNESS THE HIGHEST CAGR DURING THE FORECAST PERIOD



	North America	Europe	APAC	MEA	Latin America
2021	11,182	8,666	7,363	3,691	2,868
2026	27,499	22,256	21,584	10,438	8,047
CAGR (2021-2026)	19.7%	20.8%	24.0%	23.1%	22.9%

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 91 ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
North America	3,002	4,494	6,376	8,622	9,754	34.3%
Europe	2,234	3,372	4,828	6,594	7,507	35.4%
APAC	1,635	2,550	3,774	5,326	6,219	39.7%
MEA	862	1,327	1,939	2,702	3,136	38.1%
Latin America	668	1,032	1,513	2,116	2,447	38.3%
Total	**8,400**	**12,776**	**18,431**	**25,360**	**29,063**	**36.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 92 ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
North America	11,182	13,001	15,337	18,369	22,313	27,499	19.7%
Europe	8,666	10,151	12,072	14,584	17,879	22,256	20.8%
APAC	7,363	8,845	10,788	13,366	16,806	21,584	24.0%
MEA	3,691	4,408	5,343	6,581	8,224	10,438	23.1%
Latin America	2,868	3,412	4,120	5,055	6,293	8,047	22.9%
Total	**33,771**	**39,816**	**47,660**	**57,954**	**71,515**	**89,823**	**21.6%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

In North America, advanced analytics solutions and services are highly effective in most organizations and verticals due to the increasing need to provide businesses with a way to operationalize and get more value from data assets. Europe is gradually advancing toward incorporating advanced analytics within its market. APAC is showing a substantial rise in the adoption of advanced analytics solutions and services during the forecast period, while Latin America and MEA are slowly picking up advanced analytics due to its benefits for various industries to get user insights in real time.

12.2 NORTH AMERICA

North America is expected to be the largest contributor to the advanced analytics market. The top countries contributing to the growth of the advanced analytics market in North America are the US and Canada. Organizations in various countries of this region, especially in the US, have leveraged AI, ML, and deep learning technologies as a part of their ongoing business process to stay ahead in the market. Countries in this region have a well-established economy, which allows advanced analytics vendors to invest in new technologies. The dominance of the region is due to its increasing adoption of advanced technologies, presence of supporting infrastructure, and drive for early technological adoption, due to the competition from other businesses operating in the other regions. This region has been extremely open toward the adoption of any new and innovative technologies and is expected to provide market growth opportunities to advanced analytics vendors, as it is expected to witness an exponential growth of IT device-based generated data and stringent laws and policies for safeguarding the data. The acceleration of omnichannel service delivery and digital transformation, coupled with the inclusion of new technologies, such as AI, is driving future technology acquisitions and investments in the region. Conventional analytics techniques and tools fail to identify hidden patterns in data, whereas the advanced analytics market tools support the extraction of the hidden information, that is later used by organizations to have a prominent understanding of their customers' behavior. Advanced analytics also support companies to customize their offerings to customers by utilizing the extracted hidden information.

It has been the most promising region across verticals such as BFSI, IT and telecom, retail and consumer goods, government and defense, and healthcare and life sciences. For instance, American Express company uses advanced analytics techniques to improve speed and performance and enhance the service delivery process. The company uses sophisticated predictive modeling to analyze historical data and over a hundred variables to predict potential customer churn, enabling it to speed up and optimize marketing efforts to retain this account. The government and public sector have also joined the race to become technologically advanced to cater to a large customer base. North America is contributing significantly to the advanced analytics market and is expected to grow further. The use of advanced technologies, such as cloud computing, AI, big data and analytics, cybersecurity, and IoT, has led to innovation and transformation, thereby stimulating growth in the business ecosystem in North America. These technologies have transformed the legacy approach to business into a modern approach. The region is becoming a new hotspot in the digital transformation market due to the rising investments in the process of digitalization across potential economies. Such trends are expected to boost the adoption of the advanced analytics market across the industries in the region. For instance, Volvo Trucks North America strengthened its portfolio of uptime-boosting services by improving remote diagnostics with an advanced analytics platform from SAS Institute. Advanced analytics solution providers have a strong foothold in the region, which is anticipated to drive the market in the region. Some of the vendors of the region are SAS Institute, Oracle, FICO, Microsoft, and IBM.

12.2.1 NORTH AMERICA: COVID-19 IMPACT

- The COVID-19 pandemic has introduced challenges for various vendors in the region. These include developing agile workforce strategies to keep the global economy viable and helping people and their families survive financially now and in the future.
- The COVID-19 pandemic has increased the strain on various telecommunication service providers in the region; the work from home concept has increased the demand for data connectivity, such as Wi-Fi for voice and video conferences, and resulted in a surge in mobile data traffic.
- In the current pandemic scenario, technologies such as AI and robotics have created opportunities for innovation for companies, thus boosting their growth; however, reduced manpower efforts may affect the work culture in the region.
- North America, one of the world’s most developed regions, has heavily invested in technologies, such as analytics, AI, and ML. Most industries are using AI to create different predictive models to help them stay abreast with market dynamics during this pandemic.
- At the time of the COVID-19 pandemic, popular voice assistant applications offered straightforward insights on the rising case trends, safety instructions of the World Health Organization (WHO), Centers for Disease Control and Prevention (CDC), and National Institutes of Health (NIH).
- The global advanced analytics market has seen strong growth during the COVID-19 pandemic as professionals with expertise in analytics, BI, and more sophisticated analytics such as AI and ML have been enlisted to assist CEOs in making business decisions about how to respond to the new business difficulties posed by the COVID-19 pandemic.
- Owing to the COVID-19 impact, manufacturing companies in the US would see a tremendous opportunity through the adoption of emerging technologies (IoT, AI, and ML) to reduce risk failure in equipment’s working ability.

- HACERA, an enterprise blockchain startup, collaborated with IBM and Oracle to launch the MiPasa app to assist data integration and analytics for the COVID-19 situation.

12.2.2 NORTH AMERICA: REGULATIONS

12.2.2.1 Personal Information Protection and Electronic Documents Act (PIPEDA)

The PIPEDA is Canada's key private-sector privacy law. Compliance with PIPEDA is essential for private sector organizations operating in the country. The following organizations and people were deemed to fall under PIPEDA:

- A daycare center, even though it received government funding
- A non-profit that was engaged in commercial activities
- A public-sector doctor undertaking an examination for an insurance company

12.2.2.2 Gramm–Leach–Bliley Act

The GLB Act was developed in 1999 by the US federal government. It is also known as the Financial Services Modernization Act. The major objective of this Act was to maintain the security of private information pertaining to every individual that is handled by financial institutes. It includes provisions to protect the consumer's personal financial information held by banks, securities companies, and insurance companies. Under the jurisdiction of the Federal Trade Commission (FTC), these financial institutions are required to design, implement, and maintain an information security program to protect the privacy and integrity of customer data.

12.2.2.3 Health Insurance Portability and Accountability Act of 1996

The HIPAA was signed and brought into law by President Bill Clinton in 1996. The Act was passed onto US legislation to safeguard and protect medical information by maintaining data privacy. The five sections included in the act are as follows:

- HIPAA Title I protects the health insurance cover of individuals and prohibits group health plans from denying coverage to any individual.
- HIPAA Title II establishes national standards for processing electronic healthcare transactions.
- HIPAA Title III includes provisions related to tax and guidelines for medical care.
- HIPAA Title IV includes provisions for individuals with pre-existing conditions and those seeking continued coverage.
- HIPAA Title V includes provisions on company-owned life insurance and treatment of those who lose their US citizenship for income tax purposes.

12.2.2.4 Health Level Seven (HL7)

HL7 pertains to the healthcare industry. It helps organizations ensure electronic patient data always remains secure by standardizing the data and the way it is used. Advanced analytics come into play here because they may connect to other systems in a healthcare facility.

12.2.2.5 Occupational Safety and Health Administration (OSHA)

Congress created the Occupational Safety and Health Administration (OSHA) to ensure safe and healthful working conditions for workers by setting and enforcing standards and by providing training, outreach, education, and assistance. OSHA is part of the US Department of Labor. The administrator for OSHA is the Assistant Secretary of Labor for Occupational Safety and Health. OSHA's administrator answers to the Secretary of Labor, who is a member of the cabinet of the President of the US.

12.2.2.6 Federal Information Security Management Act

The Federal Information Security Management Act (FISMA) of 2002 enforces stringent standards to ensure the security, confidentiality, and integrity of US federal property and information. It sets the requirements for each federal agency to create, document, and implement programs that ensure security for the agencies' data and the systems that support the agencies' operations and assets. This extends to state governments that manage federal programs, contractors, and other non-governmental organizations working with and funded by the federal government. The examples of federal programs that states manage are Medicare, Medicaid, and unemployment insurance.

12.2.2.7 Federal Information Processing Standards

Created by the National Institute of Standards and Technology's (NIST's) Computer Security Division, the Federal Information Processing Standards (FIPS) established a data security and computer system standard that organizations must adhere to as per the Federal Information Security Management Act of 2002 (FISMA).

To become FIPS compliant, a US government agency or contractor's computer systems must meet the requirements outlined in the FIPS publications numbered 140, 180, 186, 197, 198, 199, 200, 201, and 202.

- FIPS 140 covers cryptographic module and testing requirements in both hardware and software.
- FIPS 180 specifies how organizations can be FIPS compliant when using secure hash algorithms for computing a condensed message.
- FIPS 186 is a group of algorithms for generating digital signatures.
- FIPS 197 is a standard that created the Advanced Encryption Standard, a publicly accessible cipher approved by the National Security Agency (NSA) for top secret information.
- FIPS 198 is about a mechanism for message authentication that utilizes cryptographic hash functions.
- FIPS 199 standardizes how federal agencies categorize and secure information.
- FIPS 200 is a standard that helps federal agencies with risk management through levels of information security based on risk levels.
- FIPS 201 specifies the standard for common identification for federal employees and contractors.
- FIPS 202 gives the specifications for the Secure Hash Algorithm-3 (SHA-3). It consists of four cryptographic hash functions and two extendable-output functions.

12.2.2.8 California Consumer Privacy Act

The CSPA of 2018 gives consumers more control over the personal information that businesses collect related to them. This landmark law secures new privacy rights for consumers in California, which are as follows:

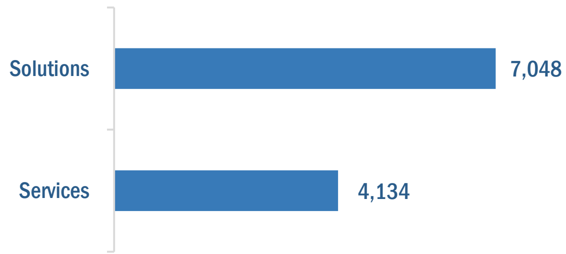
- The right to know about the personal information a business collects about them and how it is used and shared
- The right to delete personal information collected from consumers
- The right to opt-out of the sale of their personal information
- The right to non-discrimination for exercising their CCPA rights

FIGURE 47 NORTH AMERICA: MARKET SNAPSHOT

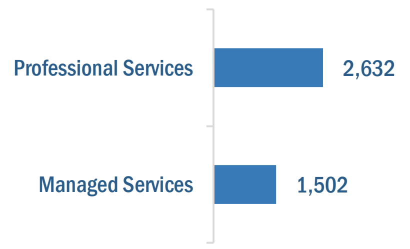
NORTH AMERICA

19.7% CAGR (2021–2026)	USD 11,182 Million Market Size (2021)	Canada Faster-growing country in the region	33.1% Share of the region in the global market

BY COMPONENT, 2021 (USD MILLION)



BY SERVICE, 2021 (USD MILLION)



BY COUNTRY (USD MILLION)

COUNTRY	MARKET SIZE, 2021 (USD MILLION)	CAGR (2021–2026)
US	7, 083	19.0%
Canada	4,099	20.9%

FACTORS SUPPORTING THE GROWTH OF THE ADVANCED ANALYTICS MARKET IN NORTH AMERICA

- North America is home to many leading advanced analytics solution vendors and service providers. The region is at the forefront in terms of smartphone adoption and latest technological advancements.
- In North America, verticals are adopting advanced analytics solution for effective management of unstructured data to gain meaningful insights.

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 93 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	1,976	2,932	4,121	5,518	6,195	33.1%
Services	1,026	1,563	2,256	3,104	3,559	36.5%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 94 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	7,048	8,129	9,513	11,300	13,611	16,598	18.7%
Services	4,134	4,871	5,824	7,069	8,702	10,901	21.4%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 95 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	663	1,006	1,448	1,987	2,272	36.1%
Managed Services	363	556	808	1,117	1,287	37.2%
Total	**1,026**	**1,563**	**2,256**	**3,104**	**3,559**	**36.5%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 96 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	2,632	3,093	3,688	4,465	5,482	6,844	21.1%
Managed Services	1,502	1,778	2,135	2,604	3,220	4,057	22.0%
Total	**4,134**	**4,871**	**5,824**	**7,069**	**8,702**	**10,901**	**21.4%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 97 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2016–2020 (USD MILLION)

Professional Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	159	245	357	495	573	37.7%
System Integration and Implementation	306	461	657	894	1,014	34.9%
Support and Maintenance	197	301	434	597	684	36.5%
Total	**663**	**1,006**	**1,448**	**1,987**	**2,272**	**36.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 98 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	672	799	964	1,180	1,467	1,849	22.4%
System Integration and Implementation	1,167	1,361	1,610	1,935	2,357	2,912	20.1%
Support and Maintenance	794	934	1,114	1,350	1,658	2,083	21.3%
Total	**2,632**	**3,093**	**3,688**	**4,465**	**5,482**	**6,844**	**21.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 99 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	448	692	1,011	1,406	1,635	38.2%
On-premises	2,553	3,802	5,366	7,216	8,119	33.5%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 100 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	1,925	2,297	2,778	3,409	4,239	5,356	22.7%
On-premises	9,257	10,704	12,559	14,960	18,073	22,143	19.1%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 101 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	993	1,501	2,149	2,932	3,340	35.4%
Large Enterprises	2,008	2,993	4,228	5,691	6,414	33.7%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 102 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	3,857	4,517	5,367	6,474	7,921	9,852	20.6%
Large Enterprises	7,325	8,484	9,970	11,895	14,392	17,648	19.2%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 103 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2016–2020)
Finance	732	1,079	1,507	2,005	2,232	32.1%
Sales and Marketing	838	1,249	1,763	2,371	2,669	33.6%
Operations	519	800	1,168	1,624	1,890	38.1%
HR	403	607	867	1,180	1,344	35.1%
Supply Chain	509	759	1,073	1,442	1,620	33.6%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 104 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Finance	2,518	2,881	3,345	3,942	4,712	5,691	17.7%
Sales and Marketing	3,044	3,520	4,131	4,923	5,949	7,294	19.1%
Operations	2,229	2,666	3,236	3,987	4,982	6,250	22.9%
HR	1,550	1,814	2,154	2,597	3,175	3,936	20.5%
Supply Chain	1,841	2,119	2,470	2,920	3,494	4,328	18.6%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 105 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	895	1,351	1,931	2,630	2,998	35.3%
Predictive Analytics	478	707	991	1,323	1,477	32.6%
Customer Analytics	549	817	1,151	1,547	1,738	33.4%
Statistical Analytics	403	603	855	1,155	1,305	34.1%
Risk Analytics	208	317	459	633	729	36.8%
Prescriptive Analytics	332	499	711	967	1,099	34.9%
Other Types	136	200	278	369	407	31.6%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 106 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	3,463	4,057	4,822	5,819	7,121	8,836	20.6%
Predictive Analytics	1,672	1,920	2,236	2,645	3,172	3,851	18.2%
Customer Analytics	1,979	2,286	2,679	3,188	3,847	4,706	18.9%
Statistical Analytics	1,494	1,736	2,045	2,447	2,969	3,655	19.6%
Risk Analytics	852	1,010	1,214	1,482	1,835	2,294	21.9%
Prescriptive Analytics	1,265	1,478	1,752	2,108	2,573	3,185	20.3%
Other Types	455	514	588	680	796	973	16.4%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 107 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	613	910	1,281	1,719	1,929	33.2%
IT and Telecom	503	749	1,058	1,422	1,600	33.5%
Retail and Consumer Goods	423	641	921	1,261	1,445	36.0%
Healthcare and Life Sciences	400	608	875	1,201	1,378	36.2%
Transportation and Logistics	245	368	524	711	807	34.7%
Government and Defense	321	474	664	886	990	32.5%
Manufacturing	225	340	488	666	761	35.6%
Media and Entertainment	196	294	417	565	640	34.3%
Other Verticals	75	108	147	190	204	28.4%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**53.6%**

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 108 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	2,194	2,530	2,961	3,519	4,240	5,179	18.7%
IT and Telecom	1,825	2,110	2,476	2,949	3,563	4,366	19.1%
Retail and Consumer Goods	1,677	1,974	2,358	2,860	3,518	4,381	21.2%
Healthcare and Life Sciences	1,604	1,892	2,264	2,752	3,392	4,229	21.4%
Transportation and Logistics	928	1,082	1,280	1,538	1,874	2,317	20.1%
Government and Defense	1,120	1,285	1,496	1,768	2,120	2,571	18.1%
Manufacturing	881	1,035	1,233	1,491	1,830	2,274	20.9%
Media and Entertainment	734	855	1,009	1,210	1,472	1,815	19.8%
Other Verticals	220	238	258	281	305	368	10.9%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 109 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2016–2020 (USD MILLION)

Country	2016	2017	2018	2019	2020	CAGR (2016–2020)
US	1,958	2,913	4,107	5,518	6,211	33.5%
Canada	1,044	1,581	2,269	3,104	3,543	35.7%
Total	**3,002**	**4,494**	**6,376**	**8,622**	**9,754**	**34.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 110 NORTH AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
US	7,083	8,192	9,613	11,451	13,834	16,931	19.0%
Canada	4,099	4,808	5,724	6,918	8,479	10,568	20.9%
Total	**11,182**	**13,001**	**15,337**	**18,369**	**22,313**	**27,499**	**19.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.2.3 UNITED STATES

12.2.3.1 Rapid adoption of technology and use of digitally innovative solutions drive the growth of the advanced analytics market in the US

The US market is inclined toward innovation and state-of-the-art infrastructure. Business establishments widely adopt technological advancements to cater to customers and improve business efficiencies more effectively. The country has the advanced infrastructure and initiatives necessary to evolve technology into robust solutions with innovative benefits. The US is the major contributor to North America in terms of data generation and technological adoption. In the US, organizations seeking more data-driven decision-making capabilities, the promise of advanced analytics is immense: collect more information, which yields more insights and can even predict the future as moving from traditional to advanced forms of analytics can help the organizations to get smarter about customers, processes, and products. Accurate BI and predictive modeling make it easier to explore new approaches and strategies. In this region, as organizations grow and develop their analytics strategy, analytics changes from examining historical data to understanding the present and future. In the US, every business is looking to reap the benefits of predictive analytics, AI, ML, and cognitive computing to build competitive advantage, faster decision-making, improved efficiency, and increased customer retention. For instance, Amazon is perhaps the best example of an organization that has mastered data analytics. Amazon provides one of the most seamless and personalized customer experiences as they use BI to make suggestions based on previous purchases, but also on what other customers have bought with certain items, browsing behavior and other factors. The regional and global players are involved in partnerships and expansions in this region, which further lead to increase in the competition in the market. For instance, recently, in December 2021 CVS Health and Microsoft announced a new strategic alliance to reimagine personalized care and accelerate digital transformation. It focuses on developing innovative solutions to help consumers improve their health.

In this region, advanced analytics plays a large role in determining which organizations thrive and which sink into oblivion. Companies that seek to dominate the global digital economy in years to come use advanced analytics, such as Amazon and Apple. They are already investing massive amounts of resources into advanced analytics tools and infrastructure. Advanced analytics techniques enable new solutions across a range of industries, from manufacturing to healthcare and many other industries. Advanced analytics proliferates with the increasing need to monitor communications and mitigate risk, and the US market is undergoing a major transformation in the implementation of AI, ML, and analytics-based solutions. Advanced analytical techniques include the design of self-service analytics tools for end users, which could provide end-to-end analytics with drill-down features from multiple data sources. The country's market growth is also driven by the presence of global mega-vendors and the high adoption of the cloud. These factors, in turn, are fueling the demand for advanced analytics cloud solutions across the country. The US accounts for the world's highest spending on advanced analytics solutions. It also recorded many mergers and acquisitions, and venture capitalist/private equity deals, making it the largest market for investors and vendors worldwide.

TABLE 111 UNITED STATES: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	1,289	1,900	2,654	3,532	3,945	32.3%
Services	669	1,013	1,453	1,987	2,266	35.7%
Total	**1,958**	**2,913**	**4,107**	**5,518**	**6,211**	**33.5%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 112 UNITED STATES: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	4,464	5,123	5,962	7,044	8,439	10,219	18.0%
Services	2,619	3,070	3,650	4,407	5,395	6,712	20.7%
Total	**7,083**	**8,192**	**9,613**	**11,451**	**13,834**	**16,931**	**19.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.2.4 CANADA

12.2.4.1 Digital transformation of businesses into data-driven organizations to drive the growth of the advanced analytics market in Canada

Canada, one of the major countries in North America, is witnessing the high adoption of advanced analytics solutions. Its broad and deep talent pool gives it an advantage over other countries. Though Canada is slightly behind in the adoption of advanced analytics solutions, it has shown tremendous adoption of other new technologies. Canada is often known as one of the leading countries in AI and analytics research, with companies such as Google and Cisco taking advantage of the highly specialized expertise coming out of the country's universities, especially in Toronto and Montreal. Despite the country's technological business attractiveness, data security concerns and regulations from the government are creating challenges as several companies are hesitant to move their existing communication platforms to intelligent advanced analytics solutions. Advanced analytics enables us to find deeper insights and drive real-time actions. It easily engages both business and technical users to uncover opportunities and address big issues. The country has also set up an institution, Canada.ai, which would provide information related to advancements in AI and ML research to various startups and enterprises. With updated regulations, businesses are now on the hook for big penalties because of non-compliance with laws. In Canada, the growing focus is on higher customer satisfaction is one of the essential factors for the growth of the advanced analytics market. Cloud-based advanced analytics solutions are anticipated to develop this market. That will also lead to growth opportunities in this market. The growth of advanced analytics in this region is driven by high penetration in the media and entertainment, healthcare, government, defense, and aerospace verticals.

TABLE 113 CANADA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	687	1,032	1,467	1,987	2,251	34.5%
Services	357	550	803	1,117	1,293	38.0%
Total	**1,044**	**1,581**	**2,269**	**3,104**	**3,543**	**35.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 114 CANADA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	2,583	3,007	3,550	4,255	5,172	6,379	19.8%
Services	1,515	1,802	2,174	2,662	3,307	4,189	22.6%
Total	**4,099**	**4,808**	**5,724**	**6,918**	**8,479**	**10,568**	**20.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.3 EUROPE

Europe holds the second-largest share of the global advanced analytics market. European customers demand better customer service and engagement from organizations. The major countries considered in the region for analysis are the UK, Germany, France, and the rest of Europe. Along with these countries, several other European countries are also incorporating advanced analytics solutions and services into their existing IT portfolios to deliver quick and actionable executive insights and keep pace with competitors from other regions. In Europe, there are a number of companies that have prototyped analytics solutions and sought to operationalize them. Besides data scientists, user groups such as business analysts are now exploring the potential of advanced analytics. The low cost of analytics components, cloud adoption, and the increasing usage of the web and higher speed broadband connectivity across the region are some of the driving factors that are expected to drive the adoption of advanced analytics in this region. Advanced analytics tools are increasingly addressing the needs of these different user groups and processes and even automating data analysis. It uses mathematical and statistical formulas and algorithms to generate new information, recognize patterns, and predict outcomes and their respective probabilities. It enables both “optimization” and “innovation.” It can support the improvement of existing processes, for example, in the form of more precise sales planning and for many other processes, the new insights driven from advanced analytics can highlight the potential of a business or even make new products and services possible. In this region, the adoption of advanced analytics by various organizations and the rise in the need to create meaningful insights from the unused data drive the growth of the advanced analytics market. The increase in the demand for cloud-based big data analytics software among SMEs fuels the growth of this market.

In Europe, the main challenges in the advanced analytics market are a lack of resources such as time and personnel as well as costs and a lack of analytical literacy. Overcoming these challenges, companies in Europe gain a competitive advantage by using advanced analytics in their operational processes. Advanced analytics techniques, such as the analysis of novel large and unstructured data sources, or the application of techniques from ML and AI, offer new insights into problems in economics and finance. Recently, in Europe, November 2021 Bank of England (BoE), the European Central Bank (ECB), and the Data Analytics for Finance and Macro Research Centre (DAFM) at King’s College London. Organized a conference to connect leading researchers in academia and policy institutions to present and discuss the latest advances in the interdisciplinary field of advanced analytics.

12.3.1 EUROPE: COVID-19 IMPACT

- Owing to COVID-19, the specifics of the extensive long-term impact of the coronavirus are currently unknown, but the pandemic has the potential to materially impact supply chain operations across a wide variety of companies in Europe.
- Prominent European countries, such as the UK, Germany, France, Italy, and Spain, have been adversely impacted by this global pandemic. The virus has infected millions, and the total number of fatalities is surging with each passing day. This severe impact on human life and health has resulted in a major economic slowdown; it is expected that the entire European region will witness a significant recession post the COVID-19 pandemic.
- In Europe, most companies have closed their offices and manufacturing hubs. This sudden lockdown of major economies in the world will impact the EU by downplaying Euro values; it is expected that businesses across European countries will witness a major loss.
- At the time of the COVID-19 outbreak, the potential end-use industries are adopting cutting-edge technology-enabled advanced analytics solutions, further boosting the market's growth. While call centers have long been a frontier of workplace automation, the pandemic has accelerated the process. This is further expected to modify the product offerings and combine them with virtual agents to speed up and enhance the advanced analytics results during the forecast period.

- Europe has been among the worst impacted regions due to COVID-19. The major industries in Europe have been affected. However, the software firms including the big data and business analytics market have adapted certain changes to continue amid the crisis. The demand for big data analytics has been increased from end-use sectors such as hospitals, education, and retail and eCommerce.
- Travel and tourism services have been among the first and most affected sectors, from both demand and supply sides. The standstill in tourism will affect small and less diversified countries in the Western Balkans, Turkey, and the Caucasus.

12.3.2 EUROPE: REGULATIONS

12.3.2.1 General Data Protection Regulation

GDPR (Regulation [EU] 2016/679) is a regulation by which the European Parliament, the Council of the European Union, and the European Commission ensure the safety of EU citizens from privacy and data breaches. GDPR extends the scope of the EU data protection law for all foreign organizations that process the data of EU citizens. It delivers the coordination of data protection regulations throughout the EU, thereby making it easier for non-European companies to comply with the regulations. However, if the regulations are not complied with, companies may have to pay four percent of their worldwide turnover as a penalty. Key privacy and data protection requirements of GDPR include the following:

- Confidentiality of collected data to ensure data protection
- Safe transfer of data across borders
- Seeking consent of subjects for data processing
- Providing data breach notifications

GDPR would impose a consistent data security law for all EU members and the companies marketing goods and services to EU residents, whatever may be the location.

12.3.2.2 Payment Card Industry Data Security Standard

PCI standards are used to help protect the information on payment cards. The PCI Standard is mandated by the card brands but administered by the Payment Card Industry Security Standards Council. Organizations that work with payment card data are required under the standard to restrict physical access to data or systems that house cardholder data to prevent unauthorized access to or removal of devices, data, systems, or hard copies. Organizations need to maintain a physical audit trail of their visitor information and activity and retain the logs for at least three months.

12.3.2.3 European Committee for Standardization

The European CEN is a government body that works along with ISO and was established to develop IoT standards in the region. It imposes rules and regulations for integrating the data collected from sensors with the existing RFID-driven barcode systems.

TABLE 115 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	1,515	2,267	3,217	4,352	4,918	34.2%
Services	719	1,105	1,612	2,242	2,589	37.8%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 116 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	5,636	6,551	7,729	9,263	11,264	13,879	19.8%
Services	3,031	3,600	4,342	5,320	6,615	8,377	22.5%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 117 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	450	690	1,002	1,390	1,601	37.3%
Managed Services	269	416	609	852	988	38.5%
Total	**719**	**1,105**	**1,612**	**2,242**	**2,589**	**37.8%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 118 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	1,869	2,214	2,663	3,254	4,035	5,092	22.2%
Managed Services	1,162	1,386	1,679	2,066	2,580	3,286	23.1%
Total	**3,031**	**3,600**	**4,342**	**5,320**	**6,615**	**8,377**	**22.5%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 119 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2016–2020 (USD MILLION)

Professional Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	108	168	248	347	404	38.9%
System Integration and Implementation	203	309	445	612	699	36.2%
Support and Maintenance	138	212	310	431	498	37.8%
Total	**450**	**690**	**1,002**	**1,390**	**1,601**	**37.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 120 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Services	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	477	570	693	854	1,069	1,366	23.4%
System Integration and Implementation	810	952	1,136	1,377	1,695	2,116	21.2%
Support and Maintenance	582	692	834	1,022	1,271	1,609	22.5%
Total	**1,869**	**2,214**	**2,663**	**3,254**	**4,035**	**5,092**	**22.2%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 121 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	641	979	1,419	1,962	2,261	37.0%
On-premises	1,593	2,393	3,409	4,632	5,246	34.7%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 122 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	2,642	3,132	3,770	4,610	5,721	7,194	22.2%
On-premises	6,025	7,019	8,301	9,973	12,158	15,062	20.1%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 123 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	717	1,092	1,579	2,176	2,496	36.6%
Large Enterprises	1,517	2,280	3,250	4,418	5,011	34.8%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 124 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	2,903	3,425	4,103	4,994	6,168	7,751	21.7%
Large Enterprises	5,764	6,726	7,968	9,589	11,711	14,505	20.3%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 125 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2016–2020)
Sales and Marketing	545	810	1,141	1,533	1,718	33.2%
Finance	624	937	1,335	1,813	2,054	34.7%
Operations	387	600	884	1,242	1,455	39.3%
HR	300	456	657	902	1,034	36.3%
Supply Chain	379	570	812	1,103	1,246	34.7%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 126 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Sales and Marketing	1,951	2,249	2,633	3,130	3,776	4,606	18.7%
Finance	2,359	2,749	3,252	3,908	4,767	5,903	20.1%
Operations	1,728	2,082	2,547	3,166	3,992	5,058	24.0%
HR	1,202	1,417	1,696	2,062	2,544	3,186	21.5%
Supply Chain	1,427	1,654	1,944	2,318	2,800	3,502	19.7%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 127 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	666	1,013	1,462	2,012	2,307	36.4%
Predictive Analytics	356	531	750	1,011	1,137	33.7%
Customer Analytics	409	613	872	1,183	1,338	34.5%
Statistical Analytics	300	453	647	883	1,004	35.2%
Risk Analytics	155	238	348	484	561	38.0%
Prescriptive Analytics	247	375	539	739	845	36.0%
Other Types	101	150	211	282	314	32.7%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 128 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	2,684	3,168	3,795	4,620	5,706	7,151	21.7%
Predictive Analytics	1,296	1,499	1,760	2,100	2,542	3,116	19.2%
Customer Analytics	1,534	1,785	2,109	2,531	3,082	3,809	19.9%
Statistical Analytics	1,158	1,355	1,610	1,943	2,379	2,958	20.6%
Risk Analytics	661	789	956	1,177	1,471	1,856	23.0%
Prescriptive Analytics	981	1,154	1,379	1,674	2,061	2,577	21.3%
Other Types	353	402	463	540	637	788	17.4%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 129 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	456	683	970	1,315	1,485	34.3%
IT and Telecom	375	562	801	1,088	1,232	34.7%
Retail and Consumer Goods	315	481	697	964	1,112	37.1%
Healthcare and Life Sciences	298	456	663	918	1,061	37.4%
Transportation and Logistics	183	276	397	544	621	35.8%
Government and Defense	239	356	503	678	762	33.6%
Manufacturing	168	255	369	509	586	36.7%
Media and Entertainment	146	221	316	432	493	35.5%
Other Verticals	56	81	112	145	157	29.5%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 130 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	1,700	1,976	2,331	2,794	3,397	4,191	19.8%
IT and Telecom	1,414	1,648	1,949	2,342	2,855	3,533	20.1%
Retail and Consumer Goods	1,300	1,542	1,856	2,271	2,819	3,545	22.2%
Healthcare and Life Sciences	1,243	1,477	1,782	2,185	2,718	3,422	22.5%
Transportation and Logistics	719	845	1,008	1,221	1,502	1,875	21.1%
Government and Defense	868	1,003	1,177	1,404	1,698	2,081	19.1%
Manufacturing	683	808	970	1,184	1,466	1,841	21.9%
Media and Entertainment	569	667	794	961	1,179	1,469	20.9%
Other Verticals	170	186	203	223	244	298	11.8%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 131 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2016–2020 (USD MILLION)

Country	2016	2017	2018	2019	2020	CAGR (2016–2020)
UK	905	1,349	1,908	2,572	2,897	33.8%
Germany	516	789	1,144	1,582	1,820	37.1%
France	437	665	959	1,319	1,510	36.3%
Rest of Europe	376	569	818	1,121	1,280	35.9%
Total	**2,234**	**3,372**	**4,828**	**6,594**	**7,507**	**35.4%**

Rest of Europe includes Italy, Spain, Sweden, and Denmark

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 132 EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
UK	3,309	3,834	4,508	5,385	6,526	8,003	19.3%
Germany	2,122	2,511	3,017	3,681	4,559	5,747	22.0%
France	1,753	2,065	2,470	3,002	3,701	4,641	21.5%
Rest of Europe	1,482	1,741	2,076	2,516	3,093	3,865	21.1%
Total	**8,666**	**10,151**	**12,072**	**14,584**	**17,879**	**22,256**	**20.8%**

e: estimated; p: projected

Rest of Europe includes Italy, Spain, Sweden, and Denmark

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.3.3 UNITED KINGDOM

12.3.3.1 Advanced IT infrastructure, coupled with the advent of disruptive technologies, will drive the growth of advanced analytics solutions in the UK

The UK is one of the strongest economies in Europe and is a major adopter of advanced analytics solutions due to various factors, such as legacy infrastructure, presence of global mega-vendors, high adoption of cloud technology, and the need to comply with stringent regulations. The adoption rate of advanced analytics solutions is higher in large enterprises than in SMEs due to the factors such as large employee strength, high budgets, and the need to maintain a competitive market position. However, SMEs are adopting advanced analytics solutions to improve employee productivity and detect errors and risks earlier. Many of the advanced analytics vendors in Europe are headquartered in the UK, making it one of the fastest-growing markets for advanced analytics in the region. The advanced analytics market growth is expected to be driven by the continued transition toward online services due to the relative ease and speed with which they can be accessed. The increasing user access and simplicity stem from technological advances. The UK and European policymakers are encouraging the governments of the countries to increase their data capture. Recently, a resolution was passed by the European Parliament to ensure a 'data-driven economy,' which would aggressively assert the implementation of advanced analytics for business houses and government verticals.

Advanced analytics is important for businesses because it helps facilitate innovation being able to predict, forecast, and plan for the changing market dynamics, improves business operations, and could hold the key to unlocking ongoing growth and success. Organizations have never had access to as much volume and complexity of data as they do these days, and most are seeking to interpret these increasingly complex data sets to make informed business decisions. Advanced analytics have helped organizations discover the actionable insights from the data, for instance, CGI's offer Text Analytics Services. It is a holistic, one-stop-shop advisory solution for organizations who want better insights into market and customer sentiment about their brands. It will help clients leverage the potential of AI to inform their customer experience, innovation, communications, and service delivery strategies.

TABLE 133 UNITED KINGDOM: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	614	907	1,271	1,697	1,898	32.6%
Services	291	442	637	874	999	36.1%
Total	**905**	**1,349**	**1,908**	**2,572**	**2,897**	**33.8%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 134 UNITED KINGDOM: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	2,152	2,474	2,887	3,420	4,111	4,991	18.3%
Services	1,157	1,360	1,622	1,965	2,415	3,012	21.1%
Total	**3,309**	**3,834**	**4,508**	**5,385**	**6,526**	**8,003**	**19.3%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.3.4 GERMANY

12.3.4.1 High skilled labor force and rising government initiatives are the key factors driving the adoption of advanced analytical solutions in Germany

Germany is expected to be the largest contributor to the advanced analytics market in Europe. A highly skilled labor force and strong infrastructure are the key factors driving the adoption of analytical technologies in the country. In December 2019, the German federal government announced plans to invest USD 3.41 billion in boosting AI capabilities, including setting up a new AI research center in the country. Germany witnessed outstanding growth in the IT and telecom, and retail and eCommerce verticals. The BFSI, and retail and eCommerce verticals are adopting advanced analytics solutions for various applications, including customer experience management, call monitoring, agent performance monitoring, sales intelligence, and risk and compliance management. Advanced analytics is used in numerous applications in the manufacturing industry, which includes forecasting, text mining for product development, predictive maintenance and logistics, risk management and prevention, and stream mining and real-time prediction. Various government-funded organizations, such as The German Research Center for Artificial Intelligence (DFKI) and non-profit Open Knowledge Foundation Germany, have partnered with corporations, such as Siemens and AGT International, in an initiative called the Big Project. Recently, in September 2021, Germany and World Health Organization Open Hub for Pandemic and epidemic Intelligence in berlin its mission is to provide the world with better data, analytics, and decisions to detect and respond to health emergencies. Moreover, the regional and global players are involved in partnerships, enhancement of analytics platform along with the expansion in this region, which further leads to an increase in competition in the market; for instance, recently, in September 2021, SAP released a new version of SAP Analytics Cloud 2021.20. It updates new dashboard and analytics services to enhance user experience.

TABLE 135 GERMANY: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	347	526	756	1,037	1,183	35.9%
Services	168	262	388	546	637	39.4%
Total	**516**	**789**	**1,144**	**1,582**	**1,820**	**37.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 136 GERMANY: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,370	1,608	1,916	2,320	2,849	3,555	21.0%
Services	753	903	1,100	1,361	1,710	2,192	23.8%
Total	2,122	2,511	3,017	3,681	4,559	5,747	**22.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.3.5 FRANCE

12.3.5.1 Increasing technological development in AI acts as a driving factor toward the adoption of advanced analytics technology in France

France is also inching toward the massive adoption of advanced analytics. The country is expected to grow at the second-highest CAGR in the advanced analytics market in the coming five years. France's technology startups are getting a boost because of the heavy inflow of capital from various investors, including Behemoth and angel investor firms. The high-potential client base could play a strong role in the adoption of analytics technology in the country. France is determined to become the superpower in technology, especially AI. France is already one of the leading countries for smart grids and smart cities, which has contributed to the adoption of the advanced analytics market. In May 2018, Alteryx expanded its presence by opening offices in France, which is expected to help vendors in the country to experience faster business decisions with the use of the modern Alteryx analytics platform. Alteryx offers pre-built tools and macros to quickly explore data and impact the future of the organizations with simulation capabilities such as predictive, prescriptive, and in-database analytics solutions.

TABLE 137 FRANCE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	299	450	644	877	997	35.1%
Services	138	215	315	442	513	38.7%
Total	**437**	**665**	**959**	**1,319**	**1,510**	**36.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 138 FRANCE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,149	1,343	1,594	1,922	2,350	2,917	20.5%
Services	604	722	876	1,080	1,351	1,724	23.3%
Total	**1,753**	**2,065**	**2,470**	**3,002**	**3,701**	**4,641**	**21.5%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.3.6 REST OF EUROPE

Italy, Spain, Sweden, Denmark, and other European countries are the key countries across the region that witness various advantages of deploying advanced analytics solutions and services. European governments have started taking initiatives for providing training and education programs to AI professionals across the Rest of Europe. Enterprises in this region are strongly reinforced by governments with the needed support for bringing in innovations and technological advancements. European countries have recognized the role of technologies in economic and social development and how they create ample opportunities for international businesses. With the increasing need to gather important information from unstructured data, companies are relying more on big data analytics. For instance, in the Netherlands, public sector implements data analytics solutions for the process of retrieving data, analysis of data, publication of the results as well as re-using the data by government organizations to improve their operations and enhance public policy. In this region, AI and ML have matured, and cloud technologies are providing solutions necessary to support advanced analytics. The advent of the cloud in this region is adding to the adoption of cloud-based solutions, which is said to be fueling the demand for advanced analytics solutions in this region.

TABLE 139 REST OF EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	257	386	549	745	845	34.7%
Services	119	184	269	376	435	38.3%
Total	**376**	**569**	**818**	**1,121**	**1,280**	**35.9%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 140 REST OF EUROPE: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	971	1,132	1,340	1,610	1,964	2,429	20.1%
Services	511	609	736	905	1,129	1,435	23.0%
Total	**1,482**	**1,741**	**2,076**	**2,516**	**3,093**	**3,865**	**21.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.4 ASIA PACIFIC

APAC has a significant technology adoption rate and is expected to record the highest growth rate in the advanced analytics market over the next few years. The region holds more than 50% of the world's population, and any major technological shift is expected to shape its future. Several Asian countries, such as Japan, China, and India, are leveraging information-intensive technologies, and advanced analytics is one of the leading technology trends.

With new growth opportunities declining in conventional, strong markets such as North America and Europe, several vendors are showing an interest in APAC. China, Japan, and India are technology-driven countries and present major opportunities in terms of investments and revenues. Companies operating in APAC will benefit from flexible economic conditions, the industrialization-and globalization-motivated policies of governments, as well as the expanding digitalization and technological adoption, all of which are expected to have a huge impact on the business community in the region. Hong Kong, Malaysia, Australia, Singapore, and South Korea are also looking to integrate new technologies with their businesses.

APAC majorly comprises China, India, and Japan. In APAC, the technology advancements among key industries are rapidly developing and disrupting. Key vendors, such as IBM, FICO, SAP, and Microsoft, are strategizing to optimize operations and enhance customer experiences across the region. Data analytics has seen rapid advances in recent years. The process of analyzing raw data to automate data into mechanical processes and algorithms, which helps how data can be used to optimize business performance. The use of big data analytics has grown drastically in bank and telecom companies in this region. In September 2022, APAC is organizing the virtual conference on big data and data analytics to see the rapid advancement of this technology in the recent year.

APAC has started to grow in the advanced analytics market and is expected to witness the highest growth rate in the next few years as organizations in this region are putting huge amounts of money into big data and business analytics solutions to improve business performance, identify fraud, and keep a competitive edge in the worldwide economy. In the APAC region, China is taking a clear stand in advocating faster construction of a 5G network, AI, big data centers, and other infrastructure. With the successful construction of China's 5G network, more advanced analytics will be possible, involving IoT, intelligent medical care, long-distance education, automated driving, AI, the smart city, intelligent agriculture, and more. For instance, in China, Alibaba Cloud's Data Analytics and AI solutions help build a unified platform with full data analytic capabilities to streamline the data pipeline and create a consistent user experience throughout the complete data lifecycle. Alibaba Cloud provides industry solutions and applications to embed these data analytics capabilities into the business processes and professionals in terms of national governance, the role of science and technology will become more and more prominent. In the foreseeable future, there will be explosive growth of the advanced analytics market in this region. In APAC, analytics is gaining traction in countries such as South Korea, Australia, Japan, and New Zealand. Players in the IT sector, such as IBM, have currently formed collaborations with the Government of India to help develop smart cities.

12.4.1 ASIA PACIFIC: COVID-19 IMPACT

- While China was the epicenter of the outbreak, the economic impact was felt across APAC. The COVID-19 pandemic has slowed down a majority of the economies, such as China, India, Singapore, Japan, and Australia; however, these countries are expected to see increased adoption of AI and analytics.
- As companies struggle to overcome travel restrictions, they are relying on various technologies to help keep staff safe and healthy, and ensure they remain efficient and productive. For instance, in Japan, a tech startup, Bespoke, has launched an AI-powered chatbot called Bebot to provide the latest and reliable updates to travelers on the COVID-19 outbreak.
- Several countries, such as China, India, and Japan, are leveraging the benefits of information-intensive AI and ML technologies for different verticals. For instance, Shenzen-based WeBank is applying analytics and AI techniques to data from satellite imagery for developing indexes and other investment tools related to environmental risks for Chinese companies.
- In India, the Telangana state government has implemented the country's first automated COVID-19 live monitoring app to identify and track infected persons and provide real-time analytics on the COVID-19 outbreak.
- The Government of Telangana from India has implemented India's first automated COVID-19 live monitoring app to help users in identifying, undertaking live surveillance, tracking, monitoring, and providing real-time analytics.
- In APAC, governments have also introduced countercyclical fiscal and monetary policies, along with the measures taken to protect jobs and businesses to combat the effect of the COVID-19 pandemic.
- In APAC, central government budgetary commitments to health system responses to COVID-19 ranged from around 1.1% of GDP in Hong Kong, China to around 0.01% in Myanmar and Papua New Guinea.
- Due to the COVID-19 pandemic, in this region, most of the people are adopting voice technology for searching and assisting in the entertainment, communication, and medical field as a way of contactless human interaction.
- In China, high-tech and big data analysis have been widely used to support government policy decisions, and that has continued in the fight against COVID-19. A mobile app was created that has allowed us to gain insights into the dynamic COVID-19 situation and monitor the number of people infected and their global location.
- The rising demand for analytics solutions, which is cloud-driven and cloud-supported, has resulted in the increase in the demand for analytics solutions in APAC, thereby resulting in increasing investments and technological advancements across industries.

12.4.2 ASIA PACIFIC: REGULATIONS

12.4.2.1 Personal Data Protection Act

PDPA establishes a data protection law that comprises various rules governing the collection, use, disclosure, and care of personal data in Singapore. It recognizes both the rights of individuals to protect their personal data, including rights of access and correction, and the need of organizations to collect, use, or disclose personal data for legitimate and reasonable purposes. The PDPA provides a national Do Not Call (DNC) registry for establishments. The DNC Registry enables individuals to register their Singaporean telephone numbers to opt out of receiving marketing phone calls and mobile text messages, such as Short Message Service (SMS) or Multimedia Messaging Service (MMS), and faxes from organizations.

The PDPA covers personal data stored in electronic and non-electronic forms. The data protection provisions in PDPA (parts III to VI) generally do not apply to the following:

- Any individual acting on a personal or domestic basis.
- Any employee acting during his or her employment with an organization.
- Any public agency or an organization while acting on behalf of a public agency in relation to the collection, use, or disclosure of the personal data; companies can refer to the Personal Data Protection (Statutory Bodies) Notification 2013 for the list of specified public agencies.
- Business contact information: This refers to an individual's name, position name or title, business telephone number, business address, business electronic mail address or business fax number, and any other similar information related to the individual, not provided by the individual solely for his/her personal purposes.

These rules act as a baseline law operating as part of the law of Singapore. They do not supersede the existing statutes, such as the Banking Act and Insurance Act, but work in conjunction with them and the common law.

12.4.2.2 Act on the Protection of Personal Information (APPI)

The purpose of APPI is to protect the rights and interests of individuals while ensuring due consideration for the usefulness of personal information by basic principles for the proper handling of personal information. The new version of APPI came into law in 2017. With it, the Personal Information Protection Commission (PPC) was also brought into effect. If a company handles the personal data of individuals in Japan, then it must comply with these regulation sets.

12.4.2.3 International Organization for Standardization 27001

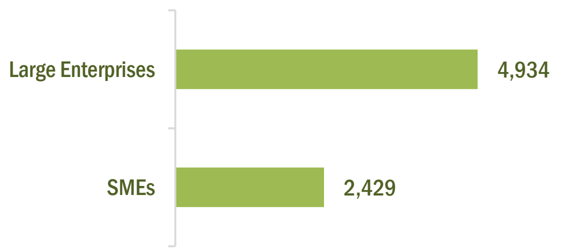
The ISO 27001 compliance provides a risk-based framework of policies and procedures for establishing, implementing, operating, monitoring, reviewing, maintaining, and improving the Information Security Management System (ISMS). The procedures and policies include defining a security policy and the scope of ISMS, conducting risk assessments, managing identified risks, selecting control objectives and controls to be implemented, and preparing a statement of applicability. ISO 27001 guidelines cover various areas, such as access control, audit, incident response, and system and information integrity. The key 114 controls required as a part of ISO 27001 compliance are security policy; organization of information security; asset management; HR security; physical and environmental security; communications and operations management; access control; information systems acquisition, development, and maintenance; information security incident management; business continuity management; and compliance.

FIGURE 48 ASIA PACIFIC: MARKET SNAPSHOT

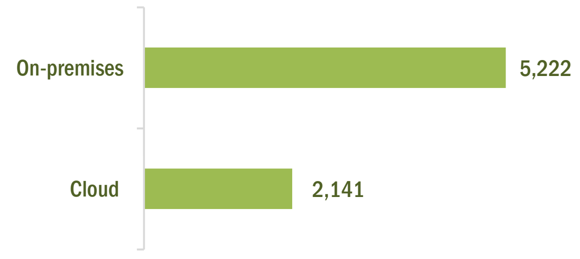
ASIA PACIFIC

24.0% CAGR (2021–2026)	USD 7,363 Million Market Size (2021)	India The fastest-growing country in the region	21.8% Share of the region in the global market

BY ORGANIZATION SIZE, 2021 (USD MILLION)



BY DEPLOYMENT MODE, 2021 (USD MILLION)



BY COUNTRY (USD MILLION)

COUNTRY	MARKET SIZE 2021 (USD MILLION)	CAGR (2021–2026)
China	2,045	22.2%
Japan	1,924	24.4%
India	1,527	26.0%
Rest of APAC	1,866	23.8%

FACTORS SUPPORTING THE GROWTH OF THE ADVANCED ANALYTICS MARKET IN ASIA PACIFIC

- The region is witnessing proliferation of smartphones and need to offer enhanced customer experience.
- Cloud-based solutions are creating opportunities for the growth of the advanced analytics market across the region due to scalability and security reasons.
- The region is witnessing rapid growth in technological advancements and innovations across major verticals such as BFSI, retail and consumer goods, and IT and telecom sectors.

Rest of APAC includes ANZ, Singapore, Malaysia, and Hong Kong

Source: Secondary Literature, Expert Interviews, and MarketsandMarkets Analysis

TABLE 141 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	1,093	1,689	2,476	3,462	4,012	38.4%
Services	542	861	1,297	1,864	2,207	42.0%
Total	1,635	2,550	3,774	5,326	6,219	39.7%

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 142 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	4,714	5,619	6,799	8,356	10,420	13,243	22.9%
Services	2,649	3,226	3,989	5,010	6,386	8,340	25.8%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 143 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	345	546	820	1,174	1,386	41.6%
Managed Services	197	315	477	690	820	42.8%
Total	**542**	**861**	**1,297**	**1,864**	**2,207**	**42.0%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 144 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	1,660	2,016	2,486	3,115	3,960	5,152	25.4%
Managed Services	989	1,210	1,502	1,896	2,427	3,188	26.4%
Total	**2,649**	**3,226**	**3,989**	**5,010**	**6,386**	**8,340**	**25.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 145 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2016–2020 (USD MILLION)

Professional Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	88	140	211	304	362	42.5%
System Integration and Implementation	156	247	369	526	618	41.0%
Support and Maintenance	101	160	240	344	407	41.8%
Total	**345**	**546**	**820**	**1,174**	**1,386**	**41.6%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 146 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	436	533	661	834	1,067	1,397	26.2%
System Integration and Implementation	736	890	1,093	1,363	1,724	2,233	24.9%
Support and Maintenance	488	593	732	918	1,168	1,523	25.6%
Total	**1,660**	**2,016**	**2,486**	**3,115**	**3,960**	**5,152**	**25.4%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 147 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	439	696	1,046	1,500	1,780	41.9%
On-premises	1,196	1,855	2,728	3,826	4,439	38.8%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 148 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	2,141	2,614	3,239	4,078	5,210	6,774	25.9%
On-premises	5,222	6,231	7,549	9,288	11,596	14,809	23.2%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 149 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	516	813	1,215	1,731	2,036	40.9%
Large Enterprises	1,118	1,737	2,559	3,595	4,182	39.1%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 150 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	2,429	2,940	3,613	4,510	5,714	7,408	25.0%
Large Enterprises	4,934	5,905	7,175	8,856	11,092	14,175	23.5%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 151 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2016–2020)
Finance	399	612	892	1,238	1,423	37.4%
Sales and Marketing	457	709	1,043	1,465	1,701	38.9%
Operations	283	454	691	1,003	1,205	43.7%
HR	219	345	513	729	857	40.6%
Supply Chain	277	431	635	891	1,033	38.9%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 152 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Finance	1,658	1,960	2,353	2,869	3,549	4,467	21.9%
Sales and Marketing	2,004	2,395	2,906	3,582	4,481	5,725	23.4%
Operations	1,468	1,814	2,276	2,901	3,753	4,906	27.3%
HR	1,021	1,234	1,515	1,890	2,392	3,089	24.8%
Supply Chain	1,212	1,441	1,738	2,125	2,632	3,397	22.9%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 153 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	488	766	1,143	1,625	1,911	40.7%
Predictive Analytics	260	401	586	817	942	37.9%
Customer Analytics	299	464	681	955	1,108	38.7%
Statistical Analytics	220	342	506	713	832	39.5%
Risk Analytics	113	180	272	391	465	42.3%
Prescriptive Analytics	181	283	421	597	700	40.3%
Other Types	74	113	165	228	260	36.9%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 154 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	2,280	2,760	3,392	4,234	5,364	6,935	24.9%
Predictive Analytics	1,101	1,306	1,573	1,925	2,389	3,022	22.4%
Customer Analytics	1,303	1,555	1,885	2,320	2,897	3,694	23.2%
Statistical Analytics	984	1,181	1,439	1,780	2,236	2,868	23.9%
Risk Analytics	561	687	854	1,079	1,382	1,800	26.3%
Prescriptive Analytics	833	1,006	1,232	1,534	1,938	2,499	24.6%
Other Types	300	350	414	495	599	764	20.6%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 155 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	288	445	651	909	1,050	38.2%
IT and Telecom	274	425	626	879	1,020	38.9%
Retail and Consumer Goods	230	364	545	779	921	41.4%
Healthcare and Life Sciences	198	315	475	684	815	42.4%
Transportation and Logistics	156	244	362	513	601	40.2%
Government and Defense	175	269	393	548	631	37.8%
Manufacturing	123	193	289	411	485	41.1%
Media and Entertainment	103	162	241	342	402	40.5%
Other Verticals	88	134	192	261	293	35.0%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 156 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	1,230	1,461	1,763	2,162	2,689	3,410	22.6%
IT and Telecom	1,202	1,436	1,742	2,146	2,684	3,427	23.3%
Retail and Consumer Goods	1,104	1,343	1,659	2,081	2,650	3,438	25.5%
Healthcare and Life Sciences	984	1,205	1,499	1,894	2,428	3,164	26.3%
Transportation and Logistics	715	862	1,056	1,313	1,657	2,136	24.5%
Government and Defense	737	874	1,052	1,287	1,597	2,018	22.3%
Manufacturing	580	704	867	1,085	1,378	1,785	25.2%
Media and Entertainment	479	579	710	886	1,121	1,448	24.8%
Other Verticals	333	381	440	513	602	758	17.9%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 157 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2016–2020 (USD MILLION)

Country	2016	2017	2018	2019	2020	CAGR (2016–2020)
China	488	750	1,092	1,518	1,750	37.6%
Japan	420	657	976	1,382	1,619	40.1%
India	313	496	745	1,069	1,269	42.0%
Rest of APAC	415	648	960	1,357	1,580	39.7%
Total	**1,635**	**2,550**	**3,774**	**5,326**	**6,219**	**39.7%**

Rest of APAC includes ANZ, Singapore, Malaysia, and Hong Kong

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 158 ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
China	2,045	2,425	2,917	3,565	4,420	5,574	22.2%
Japan	1,924	2,319	2,839	3,530	4,454	5,739	24.4%
India	1,527	1,865	2,313	2,913	3,723	4,841	26.0%
Rest of APAC	1,866	2,236	2,719	3,359	4,210	5,430	23.8%
Total	**7,363**	**8,845**	**10,788**	**13,366**	**16,806**	**21,584**	**24.0%**

e: estimated; p: projected

**Rest of APAC includes ANZ, Singapore, Malaysia, and Hong Kong*

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.4.3 INDIA

12.4.3.1 Increase in technology innovations and digitalization drives the growth of the advanced analytics market in India

India's large base of internet users is generating data that can be used for fine-tuning advanced analytics solutions. As more communications come under regulatory scrutiny, companies in India realize the need for appropriate technology solutions. According to Capgemini, approximately 58% of the tech companies in India have adopted AI-based technology on a large scale, increasing the number of internet users in the country. China and India are among the top 10 countries in terms of the number of AI companies. India is among the top five countries globally when it comes to the number of AI-driven startups. Most of the businesses, the government, and individuals have witnessed multiple enterprises advanced analytics solutions that are built to solve internal and external enterprise use cases. India already plays a major role in outsourcing analytics services, but now local companies are actively investing in advanced analytics capabilities AI and advanced analytics are transforming the way organizations interact with their clients. These organizations understand that advanced analytics has the potential to transform the business strategy and improve the performance of business. Recently, in August 2021, enterprises in India are investing in big data analytics to base their decisions on real-time data and spending on such solutions is poised to grow by 11.5% in 2021. India to capitalize on advanced analytics, country must not only add to the ranks of those with advanced analytics skills but also develop true depth of expertise. In 2015, the government has established the Digital India program to help the country become a digitally empowered knowledge economy. The 2018–2019 budget doubled Digital India's allocation to more than USD 4 billion, and a new program focused on AI was introduced. In a recent white paper, the National Institution for Transforming India cited AI's potential to help address domestic challenges, including access to healthcare, agricultural output, and efficiency of cities. India National missions such as Digital India and Make in India have sharpened the focus of businesses on developing technologies that address not only India's challenges but also serve the global market venture capital investment that will strengthen India's analytics ecosystem. A diverse group of Venture Capital and private equity investors had begun to fund this sector, ranging from investments in analytics service providers such as Mu Sigma and Fractal to data science and AI companies such as Locus and Vue.ai. Thus, India is a vital hub of analytics expertise and has fueled the early growth of its analytics industry.

India has the potential to become the dominant global provider of advanced analytics expertise by focusing on data security and privacy. The first step is for India's analytics industry, in conjunction with the government, to adopt data security protocols, standards, and certifications to minimize the risk of data misuse and privacy or security breaches. Already India has seen an influx of demand from large global corporations for human review of the results of ML and AI algorithms. The need will only grow over time, and with the appropriate policies in place, India could leverage this work into broader expertise in data use and curation. India is one of the fastest-growing countries adopting AI, big data analytics, and IoT. The demand for data analytics in India is on the rise. It has led to an increase in the demand for data scientists in the country. Several industries in India, such as eCommerce, manufacturing, and retail, have taken up big data to ensure customer satisfaction and business growth. India is proven to be a rich ground for analytics services dissemination, with several local entrepreneurs working with huge global corporations and functioning as outsource supports for more established markets. Analytics has long found various applications in businesses, such as simulating manufacturing processes to predicting sales across the region. For instance, in January 2019, Rolls Royce announced its partnerships with Indian startups through its R2 Data Labs. The company would mentor and provide technical support to startups that specialize in IoT, AI, advanced analytics, quantum computing, and blockchain.

TABLE 159 INDIA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	209	328	489	695	817	40.7%
Services	104	168	257	375	452	44.5%
Total	**313**	**496**	**745**	**1,069**	**1,269**	**42.0%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 160 INDIA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	974	1,179	1,448	1,808	2,289	2,947	24.8%
Services	553	686	864	1,105	1,433	1,895	27.9%
Total	**1,527**	**1,865**	**2,313**	**2,913**	**3,723**	**4,841**	**26.0%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.4.4 JAPAN

12.4.4.1 Advancements in innovative technologies, coupled with strong IT infrastructure, to drive the growth of the advanced analytics market in Japan

Japan has seen a meteoric rise in the adoption of AI, IoT, big data, smart cities, and connected devices. The world's third-largest economy is taking a systematic approach toward investments in AI and analytics-powered solutions and R&D. It has impressive telecom growth and is adopting technological advancements by leaps and bounds. Japan owes its IT success to the Ministry of Internal Affairs and Communications (MIC), which developed the u-Japan Policy in early 2004 to address the growing demand for telecom network access and gave guidelines and instructions for propelling smart devices. According to Japan Digital Landscape, 2018, Japan is among the top six countries globally in terms of internet users. The rapid development of AI with cutting-edge robotics technology and a strong economy with the aim of providing unparalleled solutions and services to the world population will drive market growth and development in Japan. For instance, Japan's largest eCommerce site, Rakuten, continues to invest in AI to better predict customer behavior. It is estimated that by 2050 labor population will fall by 25% in Japan. This sustained drop in population poses a critical challenge for Japanese businesses. This provides an enormous opportunity for advanced analytics. Moreover, harnessing data enables organizations to refine the way they work, how processes perform, and the value that can be derived from the available assets. Thus, organizations in Japan witness a huge opportunity to become innovators by putting data and analytics solutions at the center of future transformation strategies. Japan represents a perfect storm for advanced analytics solutions to deliver growth Japan has previously proven itself to be quick to adapt and adopt new technology when required, and the near future is promising as out of top 20 AI developers applied for international patents 12 are Japanese. In February 2020, the Japanese government agreed to a deal with AWS to help in moving HR systems and document management tools in the cloud environment by 2025. Being a developed economy, with impressive growth in the telecom and other sectors, Japan is expected to rapidly adopt advanced analytics solutions and services.

TABLE 161 JAPAN: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	273	416	599	839	972	37.4%
Services	147	241	377	543	648	44.9%
Total	**420**	**657**	**976**	**1,382**	**1,619**	**40.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 162 JAPAN: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,141	1,360	1,645	2,022	2,522	3,133	22.4%
Services	783	960	1,194	1,508	1,932	2,606	27.2%
Total	**1,924**	**2,319**	**2,839**	**3,530**	**4,454**	**5,739**	**24.4%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.4.5 CHINA

12.4.5.1 Technological development and rising government support to drive the growth of the advanced analytics market in China

In China, advanced analytics solutions are highly popular in the financial, healthcare, and educational sectors. China is one of the leading global hubs of technological development. Its vast population and diverse industry mix have the potential to generate huge volumes of data and provide an enormous market. Researchers are infusing advanced analytics elements into daily applications, such as Skype, and breaking down linguistic barriers by translating languages in real time. Chinese players, such as Baidu, Tencent, and Alibaba, have achieved breakthroughs in developing AI algorithms used in voice recognition and targeted advertising. The State Council of the People's Republic of China has recently issued guidelines on AI development. It is aiming at becoming a global innovation center in the field by 2030. In China, high-tech and big data analysis have been widely used to support government policy decisions, and that has continued in the fight against COVID-19. A mobile app was created that has allowed to gain insights into the dynamic COVID-19 situation and monitor the number of people infected and their global location. In China, advanced analytics solutions are used across verticals such as in banking and telecom. For instance, during the COVID-19 pandemic, the Chinese Ministry of Industry and Information Technology set up a working group for preventing and regulating the outbreak to provide data services, which was responsible for coordinating the information sharing between departments and provinces. Based on the Teradata data platform, advanced analytics created a path map for people flow, especially for those who came from and passed through Hubei Province or other outbreak areas. These maps tracked the flow of people who had tested positive for the coronavirus in the past 14 days, to find out who these people had come into close contact with. These results helped local government and communities to implement defensive measures. Owing to the increased number of SMEs in the country, the demand for cloud-based predictive analytics software and services is expected to increase in China.

Chinese government has passed a number of policies such as Made in China 2025, and Next Generation AI development plan, to give a clear signal to different AI stakeholders, including entrepreneurs, investors, and even researchers, that AI is a field that is being backed by the government in China. Recently, in October, China medical workers have entered into Collaborations with AI researchers to combat diseases such as diabetes and COVID-19. China is taking a clear stand in advocating faster construction of a 5G network, AI, big data centers, and other infrastructure. Thus, in the foreseeable future, there will be explosive growth of advanced analytics in China.

TABLE 163 CHINA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	317	475	671	921	1,050	34.9%
Services	171	275	422	597	700	42.3%
Total	**488**	**750**	**1,092**	**1,518**	**1,750**	**37.6%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 164 CHINA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,213	1,421	1,691	2,042	2,503	3,043	20.2%
Services	832	1,003	1,227	1,523	1,917	2,531	24.9%
Total	**2,045**	**2,425**	**2,917**	**3,565**	**4,420**	**5,574**	**22.2%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.4.6 REST OF ASIA PACIFIC

The Rest of APAC includes ANZ, Malaysia, Singapore, South Korea, and other countries in APAC. The growing threat landscape in these countries is leading to increased spending on AI and analytics-based solutions. Various government initiatives are accelerating technology adoption in this region. The growing economy, rising number of businesses, and the presence of top vendors are driving the demand for advanced analytics solutions in these countries. The strong focus on reducing business costs and growing mobile workforce requirements among SMEs have further increased the demand for advanced analytics solutions in the region. The rest of APAC is witnessing collaborations and partnerships and attracting investments from global organizations to promote R&D in big data, AI, and IoT. The utilization of big data analytics across the Rest of APAC region to take care of an assortment of business issues and different difficulties has been developing and is relied upon to quicken amid the pandemic. Telcos are the next greatest spenders on big data analytics, typically to obtain experiences related to marketing, sales, and customers. Telcos and banks contributed almost 33% of spending on big data analytics. Big data analytics is a flourishing business sector where an ever-increasing number of organizations are advancing toward higher phases of technological advancements, and it gives an incredible platform to the supplier to provide their solution. The increasing government spending would boost IoT and ML infrastructure, which acts as a major factor driving the growth of the advanced analytics market in South Korea. The rest of APAC is witnessing collaborations and partnerships for promoting R&D in data analytics, IoT, and AI.

TABLE 165 REST OF ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	277	430	631	885	1,022	38.5%
Services	137	218	329	472	558	42.0%
Total	**415**	**648**	**960**	**1,357**	**1,580**	**39.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 166 REST OF ASIA PACIFIC: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,196	1,421	1,714	2,100	2,610	3,335	22.8%
Services	670	814	1,005	1,259	1,600	2,095	25.6%
Total	**1,866**	**2,236**	**2,719**	**3,359**	**4,210**	**5,430**	**23.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.5 MIDDLE EAST AND AFRICA

MEA has a diverse economic landscape and is one of the top growing economies in the world in terms of the number of industries and the presence of global mega-vendors, which provide a larger demand pool for advanced analytics monitoring solutions. The adoption of disruptive technologies has fueled the growth of the advanced analytics market in the region. Businesses in these parts of the region have been investing heavily in new technologies and are supported by governments as early consumers of technology. However, technology adoption outside the Gulf economies has been slower. The differences in adoption levels are driven by differences in infrastructure and access to skilled labor. Data shortage and sharing restrictions imposed by large enterprises also contribute to the growth of advanced analytics solutions. The rapid digitalization in countries such as the UAE and South Africa has resulted in several linked devices in the area in recent years. Data security has become a prime concern for businesses in the United Arab Emirates (UAE) due to the availability of various financial services and their transactions in MEA. In this region, various governments have conducted long-term plans for using AI and ML as an economic stimulus or enhancing workforce efficiency and government structure agility. Now, AI programs are carried out with governments' direct support, but companies in the region would have their own AI departments in the short term. The UAE government is the first country in the Middle East to launch its AI strategy and the first in the world to create a ministry of AI. The government is planning to invest in AI technologies in about nine sectors. With the help of rapid digitization in countries such as Saudi Arabia and the UAE, it has increased the number of Internet-enabled, connected devices in recent years. The regional and global players are involved in partnerships and expansions in this region, which further lead to increase in competition in the market. For instance, in January 2022, Qlik announced that La Mutuelle Générale, expert in health insurance and personal protection, has chosen Qlik to support the evolution of its management processes. The democratization of analytics within the company responds to a desire to measure and actualize operations, at all levels of responsibility to support managers in their decision-making. For instance, in October 2020, Cisco Systems accelerated secure cloud adoption with its new WAN Edge Platform to deliver secure and automated access to applications. Cisco Catalyst 8000 Edge Platform accelerates cloud adoption by providing businesses with flexible options for secure connectivity and visibility to applications across cloud, data center, and edge. The Middle East's traditional business mindset has taken a significant detour from oil, retail, and property to IT and digital advancements. With the latest investments, strategies, and products in fields such as IoT, cloud, and eCommerce, the Middle East countries have shown their intent to stay ahead in the game of technological advancements. They are ready to embrace developing technologies, such as ML by creating efficient plans and capital investments. Organizations in both regions focus heavily on customer relationships, evidenced through innovations including ML-based credit scoring, often supported by homegrown startups tailoring products to local conditions. For instance, MyBucks, the first African FinTech to be listed on the Frankfurt Stock Exchange, uses ML algorithms to estimate loan repayment risk and proactively reduce the likelihood of missed payments by raising flags based on behavioral and transaction trend analysis.

12.5.1 MIDDLE EAST AND AFRICA: COVID-19 IMPACT

- Kaggle, a data science competition platform provider, has issued a data competition, 'COVID-19 Open Research Dataset Challenge,' based on COVID-19 data. Similarly, Zindi, Africa's largest data competition platform provider, has announced a competition to 'accurately predict the spread of COVID-19 around the world over the next few months.
- The UAE's Ministry of Economy has set up an interactive online tool to help companies address the impact of the COVID-19 pandemic, support business continuity, and promote sound business practices as part of efforts to shore up the private sector.
- Healthcare providers in the region have adopted advanced technologies to diagnose diseases more accurately and administer medical treatment with more efficiency. Big data and AI are being utilized in the healthcare sector for contact tracing and vaccine development to curb the effects of COVID-19.
- Gulf Cooperation Council (GCC) members, such as Saudi Arabia, Bahrain, Oman, Qatar, Kuwait, and the UAE, are increasing the use of AI tools to stop the spread of the virus. They are increasingly deploying sophisticated speed cameras, drones, and robots to ensure movement is limited, and social distancing is in place.
- MENA governments should use the momentum created by the COVID-19 pandemic to jumpstart digital transformation and tear down the digital divide. Without it, its recovery will remain slow, and the region will linger in a low-growth, no-innovation cycle for years to come.
- The COVID-19 pandemic presents a puzzling challenge tangled with an opportunity. As companies have now capitalized on the boost in online consumption and internet services. Governments have complemented the process by formalizing affordability packages that bring people online, amplifying the trend of spreading connectivity, and encouraging investments that lower the digital divide and benefit the region's already thriving startups.
- South Africa has established the COVID-19 ZA South Africa Dashboard that is maintained by the Data Science for Social Impact Research Group at the University of Pretoria. Big data and AI have been majorly utilized in the healthcare sector for contact tracking and vaccine development to curb the effects of the COVID-19.

12.5.2 MIDDLE EAST AND AFRICA: REGULATIONS

The section includes the laws and regulations implemented by government bodies to safeguard personal data and privacy in MEA. Some of these regulations are ISO 27001 and PDPL. These regulations help maintain data quality and prevent fraudulent activities by ensuring data transparency among defined entities. These regulations and acts help define standards for advanced analytics solutions and services in organizations. Companies further acknowledge the importance of complying with data safety regulations and try to adhere to them.

12.5.2.1 Israeli Privacy Protection Regulations (Data Security), 5777-2017

- The Israeli Privacy Protection Regulations (Data Security), 5777-2017 entered into force concurrently with the EU General Data Protection Regulation in May 2018. The regulations significantly expanded the privacy protection obligations that apply to most companies operating in Israel. These regulations impose detailed information security requirements upon any organization that is in possession of a database containing personal data, including data relating to employees, candidates for new positions, consumers, and other personal information.

12.5.2.2 GDPR Applicability in KSA

- From May 2018, the KSA companies that process personal data of EU-based data subjects in the context of activities of a European establishment, offer goods or services to, or monitor the behavior of EU-based data subjects must comply with GDPR.
- Businesses in the KSA that fall within the scope of GDPR must appoint an EU representative located in one of the European countries of the individuals who are offered products or subject to behavioral monitoring. The representative acts on behalf of the companies in the KSA and may be addressed by any EU data protection supervisory authority and the data subject.
- The enforcement of the GDPR legislation leads to increased reliance on real-time analytics. Big data needs to be stored securely. GDPR enables customers to remove their data from one vendor's database and switch it to another vendor, enabling greater customer control over data.

12.5.2.3 Protection of Personal Information Act

- POPIA 4 of 2013 is the comprehensive data protection legislation enacted in South Africa. It aims to give effect to the constitutional right to privacy while balancing this against competing rights and interests, particularly the right of access to information. POPIA was signed into law on November 19, 2013.
- The purpose of POPIA is to protect people from harm by protecting their personal information, to stop their money and their identity from being stolen, and generally to protect their privacy, which is a fundamental human right.
- The legislation affects any natural or juristic person who processes personal information, including large corporations and governments. The data protection laws of several other countries exempt SMEs, but not currently in South Africa.

TABLE 167 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	602	919	1,331	1,838	2,118	37.0%
Services	260	408	608	865	1,019	40.7%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 168 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	2,474	2,933	3,528	4,312	5,346	6,718	22.1%
Services	1,217	1,475	1,815	2,269	2,878	3,720	25.0%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 169 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	170	267	397	562	660	40.3%
Managed Services	89	142	213	306	363	42.0%
Total	**260**	**409**	**610**	**868**	**1,024**	**40.9%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 170 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	787	951	1,168	1,456	1,842	2,373	24.7%
Managed Services	438	536	665	839	1,074	1,401	26.2%
Total	**1,225**	**1,487**	**1,833**	**2,295**	**2,916**	**3,773**	**25.2%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 171 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2016–2020 (USD MILLION)

Professional Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	39	62	93	133	158	41.5%
System Integration and Implementation	80	125	184	259	302	39.1%
Support and Maintenance	51	80	119	170	201	41.1%
Total	**170**	**267**	**397**	**562**	**660**	**40.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 172 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	190	232	287	361	460	598	25.8%
System Integration and Implementation	357	428	521	645	811	1,033	23.7%
Support and Maintenance	240	292	360	450	571	741	25.3%
Total	**787**	**951**	**1,168**	**1,456**	**1,842**	**2,373**	**24.7%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 173 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	294	462	686	973	1,150	40.6%
On-premises	567	866	1,253	1,729	1,987	36.8%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 174 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	1,377	1,673	2,064	2,587	3,290	4,231	25.2%
On-premises	2,314	2,735	3,279	3,994	4,935	6,207	21.8%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 175 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	268	417	615	865	1,011	39.4%
Large Enterprises	594	911	1,325	1,838	2,125	37.5%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 176 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	1,199	1,443	1,763	2,188	2,755	3,530	24.1%
Large Enterprises	2,492	2,964	3,580	4,393	5,469	6,907	22.6%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 177 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2016–2020)
Finance	210	319	458	628	718	35.9%
Sales and Marketing	241	369	536	743	858	37.4%
Operations	149	236	355	509	608	42.1%
HR	116	179	264	370	432	39.0%
Supply Chain	146	224	326	452	521	37.4%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 178 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Finance	831	977	1,165	1,412	1,737	2,160	21.0%
Sales and Marketing	1,005	1,193	1,439	1,764	2,193	2,768	22.5%
Operations	736	904	1,127	1,428	1,836	2,372	26.4%
HR	512	615	751	930	1,170	1,494	23.9%
Supply Chain	608	718	861	1,046	1,288	1,643	22.0%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 179 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	257	399	587	824	964	39.2%
Predictive Analytics	137	209	301	415	475	36.4%
Customer Analytics	158	241	350	485	559	37.2%
Statistical Analytics	116	178	260	362	420	38.0%
Risk Analytics	71	111	165	233	275	40.2%
Prescriptive Analytics	95	147	216	303	353	38.8%
Other Types	28	42	59	81	91	34.6%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 180 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	1,143	1,375	1,680	2,085	2,625	3,354	24.0%
Predictive Analytics	552	651	779	948	1,169	1,462	21.5%
Customer Analytics	653	775	933	1,142	1,418	1,786	22.3%
Statistical Analytics	493	588	713	877	1,094	1,387	23.0%
Risk Analytics	328	398	490	613	777	998	24.9%
Prescriptive Analytics	418	501	610	755	948	1,209	23.7%
Other Types	103	119	138	162	192	242	18.6%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 181 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	152	231	334	461	529	36.7%
IT and Telecom	131	204	300	421	493	39.2%
Retail and Consumer Goods	120	188	279	394	464	40.1%
Healthcare and Life Sciences	103	162	241	343	407	41.0%
Transportation and Logistics	78	121	178	250	292	38.9%
Government and Defense	84	127	183	251	287	36.1%
Manufacturing	74	115	170	239	280	39.4%
Media and Entertainment	55	85	126	177	208	39.6%
Other Verticals	65	94	128	166	177	28.7%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 182 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	616	728	873	1,064	1,316	1,649	21.8%
IT and Telecom	585	704	860	1,067	1,344	1,717	24.0%
Retail and Consumer Goods	554	671	825	1,031	1,307	1,678	24.8%
Healthcare and Life Sciences	489	596	738	928	1,184	1,526	25.6%
Transportation and Logistics	345	414	505	625	785	1,002	23.8%
Government and Defense	333	392	468	568	699	871	21.2%
Manufacturing	332	401	490	609	768	983	24.2%
Media and Entertainment	247	298	366	455	575	736	24.4%
Other Verticals	190	204	219	234	246	276	7.8%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 183 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2016–2020 (USD MILLION)

Region	2016	2017	2018	2019	2020	CAGR (2016–2020)
KSA	174	268	390	542	628	37.8%
UAE	135	208	305	427	497	38.6%
South Africa	157	243	358	501	585	39.0%
Rest of MEA	396	608	887	1,232	1,426	37.8%
Total	**862**	**1,327**	**1,939**	**2,702**	**3,136**	**38.1%**

Rest of the MEA includes Israel, Kuwait, Qatar, and Egypt

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 184 MIDDLE EAST AND AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY REGION, 2021–2026 (USD MILLION)

Region	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
KSA	737	879	1,063	1,306	1,628	2,061	22.8%
UAE	587	704	856	1,058	1,327	1,691	23.6%
South Africa	693	833	1,015	1,258	1,582	2,020	23.9%
Rest of MEA	1,674	1,993	2,409	2,959	3,687	4,666	22.8%
Total	**3,691**	**4,408**	**5,343**	**6,581**	**8,224**	**10,438**	**23.1%**

e: estimated; p: projected

Rest of the MEA includes Israel, Kuwait, Qatar, and Egypt

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.5.3 KINGDOM OF SAUDI ARABIA

12.5.3.1 Adoption of technology and transformation to data-driven economy drives the growth of advanced analytics solution in the country

The KSA is one of the technological powerhouses in the region, with products that are impacting the world. The adoption of advanced technologies, a large mobile workforce, and a growing economy are driving the adoption of advanced analytics solutions in the country. The leading advanced analytics solution vendors in KSA are targeting higher revenues and expanding business due to the high competition and demand prevailing across the region. The government provides complete support to SMEs for technological adoption. These factors are driving the adoption of advanced analytics solutions in the KSA at a rapid pace. Saudi Arabia is diversifying its economy by becoming a global technological hub as driven by its 'Vision of 2030' initiative, it has embarked on the most ambitious and far-reaching transformation plan this transformation focusing on the investment and development of AI in this region. For instance, Saudi Arabia has become the first country in the world to grant citizenship to an AI-driven humanoid robot called Sophia. As AI technology opens a pathway for authoritarian persistence and regime survival. The KSA is one of the prominent locations across the Middle East for data center investments. The improvement in network connectivity, government support, and rapid growth in the adoption of cloud, big data, and IoT services have been strong enablers for the growth of the Saudi Arabian data center industry. The growing adoption of smart devices and the increasing demand for analytics, cloud adoption, and the growth of wireless networking technologies have led several organizations in Saudi Arabia to invest in big data and IoT technology.

TABLE 185 KINGDOM OF SAUDI ARABIA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	118	180	260	358	411	36.6%
Services	56	88	130	184	217	40.2%
Total	**174**	**268**	**390**	**542**	**628**	**37.8%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 186 KINGDOM OF SAUDI ARABIA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	480	567	680	829	1,026	1,286	21.8%
Services	258	312	382	476	602	776	24.6%
Total	**737**	**879**	**1,063**	**1,306**	**1,628**	**2,061**	**22.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.5.4 UNITED ARAB EMIRATES

12.5.4.1 Country witnesses the increasing trend toward the adoption of AI and analytics technologies that boost the growth of the market

The UAE is one of the major countries in this region, along with Bahrain, KSA, and Israel. Advanced analytics solutions have been used in the UAE for a while now. In this region, using advanced analytics techniques such as text analytics, ML, predictive analytics, data mining, statistics, and NLP, businesses can analyze previously untapped data sources independently or together with their existing enterprise data to gain new insights, resulting in significantly better and faster decisions. The UAE is investing in and becoming a global pioneer in the use of AI technology in various sectors of its economy. With the rapidly changing technology, businesses must not only adapt but also master new advancements. The new technological advancements are bound to change the business's position in the marketplace unless strategically managed. Data analytics proves to be such a strategically managing tool for businesses. The Government of UAE is exploring the potential of AI, analytics, and IoT technologies to improve industry efficiency in terms of equipment expense and production costs. The UAE government is spending on digital transformation for utilities and is looking for reliable and secure solutions to manage assets efficiently and optimize their operations.

TABLE 187 UNITED ARAB EMIRATES: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	91	140	203	282	326	37.4%
Services	43	68	102	145	171	41.1%
Total	**135**	**208**	**305**	**427**	**497**	**38.6%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 188 UNITED ARAB EMIRATES: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	382	454	548	672	836	1,054	22.5%
Services	205	250	308	386	491	636	25.4%
Total	**587**	**704**	**856**	**1,058**	**1,327**	**1,691**	**23.6%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.5.5 SOUTH AFRICA

12.5.5.1 Growing digitalization and rising government support to drive the growth of advanced analytics solution in the African market

The South African region is showing promising growth in terms of technological solutions and service adoption. Advanced analytics is gaining ground in Africa, and South Africa is perfectly poised to capture a significant portion of this market potential. In South Africa, advanced analytics solutions will add significant value related to ease of use for the healthcare and life sciences sectors. In South Africa, there is a considerable need for training in ML, data science, and analytics. It is a big market with numerous investment opportunities for multinationals. The growth of the country is attributed to various industry verticals, from manufacturing to technology, and energy and mining, which are supporting economic growth for the entire country. In this region, advanced data analytics can help drive innovative business decision-making. Advanced analytics and reporting use sophisticated tools for data mining, big data, and predictive analytics to mine data for important trends, patterns, and performance. As the amount of valuable data any company gathers increases, so will the need to use that data for insights that provide a competitive advantage. It helps identify large data sets to find patterns and insights that help guide clients in measuring and managing where they are and predicting where they will be in the future. In this region, technology has been disrupting in the past having lagged in innovation this region now has been able to leverage a better place in technology, allowing it to leapfrog for several development stages. The African government is driving the country toward the implementation of a National Development Plan (NDP) and vision for 2023 across public and private sectors to improve the digital transformation of the organizations in the region. NDP is considered a blueprint document for African vendors. The aim of NDP is to eliminate poverty and reduce inequality by 2023 via the adoption of advanced technologies, such as big data analytics, IoT, AI, ML, social media, and cloud that will enhance citizen engagement and drive operational efficiencies and improve flexibility inevitably, creating massive amounts of data. Extracting actionable intelligence from this data requires advanced analytics to support various line-of-business functions, which, in turn, requires appropriate skills and funding.

TABLE 189 SOUTH AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	106	164	238	331	384	37.8%
Services	50	80	119	170	202	41.4%
Total	**157**	**243**	**358**	**501**	**585**	**39.0%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 190 SOUTH AFRICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	451	537	650	799	997	1,259	22.8%
Services	242	295	365	459	585	760	25.7%
Total	**693**	**833**	**1,015**	**1,258**	**1,582**	**2,020**	**23.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.5.6 REST OF MIDDLE EAST AND AFRICA

The Rest of MEA includes Israel, Qatar, Egypt, and Kuwait. Israel has always been at the forefront of adopting innovative technologies in every field. For instance, in February 2019, Voicesense, an innovative provider of voice-based predictive analytics solutions, announced the addition of predictive analytics to its call center offering. The region comprises numerous technology startups and other SMEs, which rely on advanced technologies to improve work efficiency and business productivity. The rise in the use of advanced digital technologies for the thorough and secured delivery of confidential information is a major driver for the growth of advanced analytics solutions in the region. Moreover, the regional and global players are involved in partnerships and expansion in this region, which further leads to an increase in competition in the market.

Vodafone and Qatar National Tourism Council (QNTC) have signed an agreement for the implementation of Vodafone’s Big Data & Advanced Analytics solutions across the country. As per the agreement, the partnership will accelerate the country’s goal of being a world-class tourist destination. Moreover, Vodafone’s tailored Big Data & Advanced Analytics solution is a clear choice for QNTC, due to its ability to deliver real-time, anonymized, aggregated data, in accordance with Qatar regulations and global best practices.

In October 2020, Ericsson’s AI & Analytics Hub in Egypt achieves a software milestone by shipping its first Cognitive Software to be used by Ericsson customers worldwide to design and optimize networks. The hub focuses on Research and Development (R&D) in AI and Automation, leveraging cutting-edge technologies to create data-driven, intelligent, and robust systems for automation, evolution, and growth.

TABLE 191 REST OF MEA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	269	409	591	813	934	36.6%
Services	127	199	296	419	492	40.2%
Total	**396**	**608**	**887**	**1,232**	**1,426**	**37.8%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 192 REST OF MEA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,088	1,286	1,543	1,879	2,323	2,909	21.7%
Services	585	707	867	1,079	1,364	1,756	24.6%
Total	**1,674**	**1,993**	**2,409**	**2,959**	**3,687**	**4,666**	**22.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.6 LATIN AMERICA

The advanced analytics market size in Latin America is small; the challenge is not a lack of jobs, but a shortage of qualified, skilled experts. The most significant challenges to the adoption of advanced analytics solutions are the quality of education systems and weak research and innovation ecosystems at regional levels. Innovation gaps due to the absence of investments and the lack of coordination among key players are the factors restricting market growth. The lack of advanced analytics vendors in the region is also limiting the growth of the market. However, multinational companies are trying to bring advanced analytics solutions into the market through mergers and acquisitions. Mexico and Brazil are the fastest-growing countries in terms of advanced analytics solution penetration in this region.

In the coming years, it would be increasingly important for healthcare providers to be able to attract and retain top-tier talent. Hence, it is time for Latin American policymakers to pay attention to the potential impacts of AI and analytics technologies, especially on the education and healthcare verticals, which are the largest employers in the region. Latin America is moving at a slower pace in adopting advanced analytics solutions as compared to other regions in the market; however, the region holds immense growth opportunities due to the rising IT awareness and adoption of solutions. Companies in this region are investing substantially in advanced analytics solutions and services for increasing business productivity, detecting risks and errors earlier, speeding up investigations, and improving overall performance. The region holds high-level opportunities for advanced analytics solutions, which could be the solution to the region's long-term low economic productivity. Governments and legal entities are enforcing new laws and upgrading existing compliances due to the increasing expansion of verticals in the region.

The Latin American region includes major countries, such as Brazil, Mexico, Argentina, Colombia, Peru, and other countries. The region is slowly gaining traction in the adoption of major technologies, such as IoT, cloud, and AI. Due to the staggering growth of digitalized infrastructure across the region, the generation of large data volumes has become limited. Big Data and analytics help organizations make proactive real-time decisions using data. This enables companies to improve customer experience, especially as customer loyalty is now low and access to information is high to better understand customers and be able to quantify it this provides competitive advantage. Technological adoption is low in Latin America, but it is expected to grow in the coming years, primarily because of many organizations investing in implementing various big data analytics solutions. For instance, IBM expanded its global cloud footprint by launching the IBM Cloud Multizone Region (MZR) in Latin America by late 2020. Brazil is among the countries that have an enormous dependence on the manufacturing sector and is witnessing increasing internet adoption as well. Advanced analytics as perspective and cognitive are the reality worldwide, but in Latin America they tend toward more simple solutions such as descriptive and predictive analytics impacting the size of the market. In 2019, Latin America's data and analytics market generated USD 2.9 billion is expected to grow rapidly over the next 3 to 5 years.

12.6.1 LATIN AMERICA: COVID-19 IMPACT

- The COVID-19 pandemic has resulted in the reduction of international trade, a fall in commodity prices, and a decrease in the demand for tourism services. These factors will negatively affect the economy of Latin America.
- As per a recent survey conducted by DNA Human Capital in Chile, Columbia, and Peru, more than two-thirds (about 68.7%) of business executives believe COVID-19 will greatly affect the bottom-line of their companies.
- Latin America has also developed the potential to leverage big data and AI for better and healthier lifestyles. According to Telco, mobile access is a powerful tool that can support the entire region to control the spread of the virus.
- On April 02, 2020, Telefonica Brazil promised to offer cellphone observations to analyze the quarantined population for slowing down the virus outbreak. The company is offering big data capabilities and management to aggregated network data, mobility data, and cloud data processing centers, as well as providing digital assistance capabilities for public administration and health institutions.
- Latin America Invented new ways to detect COVID-19 with voice recognition. Using an advanced AI-based voice-recognition methodology, research teams analyzed several hours of audio recordings of both healthy and sick individuals to identify patterns that are consistent with COVID-19 infections. With their deep-learning algorithm to recognize COVID-related sound patterns, Cyberlaws was able to identify three out of four sick people correctly (75.9% of the time) when using audio recordings of their coughing, and 70.3% of the time when relying on recordings of speech.
- During the pandemic, the major Latin American countries such as Brazil, Mexico, and Argentina are undergoing digital transformations, as evidenced by high mobile penetration; a growing IoT ecosystem; and soaring adoption of cloud computing, social media, and eGovernance platforms in these countries. As a result, the amount of digital data generated in the region is massive.

12.6.2 LATIN AMERICA: REGULATIONS

The section includes the laws and regulations implemented by government bodies to safeguard personal data and privacy in Latin America. Some of these regulations are ISO 27001 and LFPDPPP. These regulations help maintain data quality and prevent fraudulent activities by ensuring data transparency among defined entities. These regulations and acts help define standards for advanced analytics solutions and services in organizations. Companies further acknowledge the importance of complying with data safety regulations and try to adhere to them.

12.6.2.1 Brazil Data Protection Law

- Brazil's data protection law, Lei Geral de Proteção de Dados (LGPD), came into force in August 2020. Before the LGPD, the data protection legislation in Brazil was sector-based and primarily regulated by the country's civil rights framework for the Internet (Internet Act) and Consumer Protection Code.
- The LGPD attempts to unify over 40 different statutes that currently govern personal data, both online and offline, by replacing certain regulations and supplementing others.
- The LGPD imposes detailed rules for the collection, use, processing, and storage of personal data. This new regulation replaces more than 40 different regulations that deal with data protection. The final version of the LGPD also introduced important revisions to the original version of the law, such as the creation of the National Data Protection Authority that will be responsible for overseeing and enforcing the LGPD.

12.6.2.2 Argentina Personal Data Protection Law No. 25.326

- Argentina was the first Latin American country to implement data protection laws and the first non-European country to be recognized by the European Commission as having adequate levels of data protection. The need to revisit the current legislation is a result of technological advances and the changing international landscape with the introduction of the GDPR since the Argentinian Personal Data Protection Act 2000 came into force.
- Argentina’s Personal Data Protection Law No. 25.326 has guided how data is processed and protected. For the past year, lawmakers have debated a replacement bill that would more closely align with GDPR. Among other mandates, it requires that employers obtain consent from individuals before data is collected. The new bill expressly permits and regulates background checks and provides individuals with the right to access and request the deletion of their personal data. The replacement bill will go before the Senate, where it has a high likelihood of passing. If the bill passes into law, organizations will have a two-year period to prepare before enforcement.
- Argentina’s new draft data protection bill proposes further changes to bring the country’s data protection law in line with the GDPR. The bill acknowledges the right to be forgotten and the right to data portability. The other changes include stricter provisions in the area of cross-border transfers to countries with inadequate levels of data protection, new legal bases for data processing other than data subject consent, including legitimate interests, and new definitions of biometric and genetic data.

12.6.2.3 Federal Law on Protection of Personal Data Held by Individuals

The main data protection legislation in Mexico is the Ley Federal de Protección de Datos Personales en Posesión de los Particulares or Federal Law on Protection of Personal Data Held by Individuals (LFPDPPP). The law came into force in July 2010 and was followed in December 2011 by secondary regulations that clarified the obligations of personal data controllers under the LFPDPPP. The regulations apply to all personal data processing when:

- Processed in a facility of the data controller located in Mexican territory.
- Processed by a data processor, regardless of its location, if the processing is performed on behalf of a Mexican data controller.
- Where the Mexican legislation is applicable because of Mexico’s adherence to an international convention or the execution of a contract (even where the data controller is not located in Mexico).
- Where the data controller is not located in Mexican territory but uses means located in Mexico to process personal data unless such means are used only for transit purposes.

The Law only applies to private individuals or legal entities that process personal data and does not apply to the government. Credit reporting companies are governed by the Law Regulating Credit Reporting Companies for persons carrying out the collection and storage of personal data exclusively for personal use where it is not disclosed for commercial use.

TABLE 193 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	460	704	1,023	1,418	1,627	37.2%
Services	208	328	490	698	819	40.8%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 194 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	1,894	2,236	2,679	3,261	4,028	5,098	21.9%
Services	974	1,176	1,441	1,794	2,266	2,948	24.8%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 195 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2016–2020 (USD MILLION)

Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Professional Services	140	219	327	464	543	40.5%
Managed Services	69	109	164	235	278	41.9%
Total	**208**	**328**	**491**	**700**	**821**	**40.9%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 196 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY SERVICE, 2021–2026 (USD MILLION)

Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Professional Services	644	775	948	1,176	1,482	1,923	24.5%
Managed Services	333	405	500	627	797	1,045	25.7%
Total	**977**	**1,180**	**1,447**	**1,803**	**2,279**	**2,968**	**24.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 197 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2016–2020 (USD MILLION)

Professional Service	2016	2017	2018	2019	2020	CAGR (2016–2020)
Training and Consulting	33	53	80	115	137	42.5%
System Integration and Implementation	63	99	148	210	247	40.8%
Support and Maintenance	44	67	99	139	160	38.4%
Total	**140**	**219**	**327**	**464**	**543**	**40.5%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 198 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY PROFESSIONAL SERVICE, 2021–2026 (USD MILLION)

Professional Service	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Training and Consulting	164	201	249	313	400	525	26.1%
System Integration and Implementation	293	354	434	540	682	887	24.8%
Support and Maintenance	186	221	265	324	400	512	22.4%
Total	**644**	**775**	**948**	**1,176**	**1,482**	**1,923**	**24.5%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 199 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2016–2020 (USD MILLION)

Deployment Mode	2016	2017	2018	2019	2020	CAGR (2016–2020)
Cloud	173	272	405	576	677	40.7%
On-premises	495	761	1,108	1,540	1,769	37.5%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 200 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY DEPLOYMENT MODE, 2021–2026 (USD MILLION)

Deployment Mode	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Cloud	807	977	1,200	1,497	1,896	2,455	24.9%
On-premises	2,061	2,435	2,920	3,558	4,398	5,592	22.1%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 201 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2016–2020 (USD MILLION)

Organization Size	2016	2017	2018	2019	2020	CAGR (2016–2020)
SMEs	204	319	472	667	777	39.6%
Large Enterprises	464	713	1,041	1,450	1,670	37.8%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 202 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY ORGANIZATION SIZE, 2021–2026 (USD MILLION)

Organization Size	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
SMEs	918	1,100	1,339	1,655	2,077	2,682	23.9%
Large Enterprises	1,951	2,312	2,781	3,400	4,217	5,365	22.4%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 203 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2016–2020 (USD MILLION)

Business Function	2016	2017	2018	2019	2020	CAGR (2016–2020)
Finance	163	248	358	492	560	36.1%
Sales and Marketing	187	287	418	582	669	37.6%
Operations	116	184	277	399	474	42.3%
HR	90	139	206	290	337	39.2%
Supply Chain	113	174	255	354	406	37.6%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 204 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY BUSINESS FUNCTION, 2021–2026 (USD MILLION)

Business Function	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Finance	646	756	899	1,085	1,329	1,665	20.9%
Sales and Marketing	781	924	1,110	1,355	1,678	2,134	22.3%
Operations	572	700	869	1,097	1,405	1,829	26.2%
HR	398	476	579	715	896	1,152	23.7%
Supply Chain	472	556	664	803	985	1,266	21.8%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 205 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2016–2020 (USD MILLION)

Type	2016	2017	2018	2019	2020	CAGR (2016–2020)
Big Data Analytics	175	274	405	572	668	39.7%
Predictive Analytics	106	162	235	325	371	36.6%
Customer Analytics	122	188	273	380	436	37.4%
Statistical Analytics	90	139	203	283	327	38.2%
Risk Analytics	62	97	145	205	241	40.4%
Prescriptive Analytics	74	115	169	237	276	39.0%
Other Types	38	58	83	113	128	35.3%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 206 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY TYPE, 2021–2026 (USD MILLION)

Type	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Big Data Analytics	791	950	1,159	1,436	1,805	2,327	24.1%
Predictive Analytics	429	504	601	728	895	1,127	21.3%
Customer Analytics	508	600	720	877	1,085	1,377	22.1%
Statistical Analytics	383	455	549	673	837	1,069	22.8%
Risk Analytics	286	346	424	527	666	861	24.6%
Prescriptive Analytics	325	388	471	580	726	932	23.5%
Other Types	146	169	197	234	280	353	19.3%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Other types include augmented analytics, multimedia analytics, and social and visual analytics

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 207 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2016–2020 (USD MILLION)

Vertical	2016	2017	2018	2019	2020	CAGR (2016–2020)
BFSI	118	180	261	361	413	36.9%
IT and Telecom	102	158	234	330	385	39.4%
Retail and Consumer Goods	93	146	217	308	362	40.4%
Healthcare and Life Sciences	83	132	198	283	335	41.7%
Transportation and Logistics	61	94	139	196	227	39.1%
Government and Defense	67	102	148	203	231	36.0%
Manufacturing	57	90	133	187	218	39.6%
Media and Entertainment	35	55	82	116	136	40.3%
Other Verticals	51	74	102	132	139	28.4%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 208 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY VERTICAL, 2021–2026 (USD MILLION)

Vertical	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
BFSI	479	564	673	817	1,007	1,271	21.6%
IT and Telecom	454	545	663	820	1,028	1,324	23.9%
Retail and Consumer Goods	430	519	636	792	1,000	1,293	24.6%
Healthcare and Life Sciences	402	490	606	762	972	1,263	25.7%
Transportation and Logistics	268	321	389	480	601	772	23.6%
Government and Defense	266	311	369	445	544	681	20.7%
Manufacturing	258	310	378	468	588	758	24.0%
Media and Entertainment	162	196	240	298	377	488	24.6%
Other Verticals	148	157	165	173	176	197	5.9%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Other verticals include education, travel and hospitality, and energy and utilities

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 209 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2016–2020 (USD MILLION)

Country	2016	2017	2018	2019	2020	CAGR (2016–2020)
Brazil	242	374	550	771	894	38.7%
Mexico	203	315	465	654	760	39.2%
Rest of Latin America	224	343	498	691	792	37.2%
Total	**668**	**1,032**	**1,513**	**2,116**	**2,447**	**38.3%**

Rest of Latin America includes Argentina, Chile, Colombia, and Peru

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 210 LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COUNTRY, 2021–2026 (USD MILLION)

Country	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Brazil	1,051	1,253	1,517	1,866	2,329	2,984	23.2%
Mexico	897	1,073	1,303	1,608	2,014	2,589	23.6%
Rest of Latin America	921	1,086	1,300	1,581	1,951	2,474	21.8%
Total	**2,868**	**3,412**	**4,120**	**5,055**	**6,293**	**8,047**	**22.9%**

e: estimated; p: projected

Rest of Latin America includes Argentina, Chile, Colombia, and Peru

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.6.3 BRAZIL

12.6.3.1 Rising need to enhance the business efficiency, coupled with the growing utility of IoT analytics, to drive the growth of the advanced analytics market in Brazil

Brazil is one of the major countries responsible for shaping the technological landscape of Latin America. It has established trade ties with the US and is experiencing growth in the wake of the financial downturn. A recent survey from Accenture found that 83% of Brazilian financial service consumers would trust banking advice that was entirely generated by computers. IBM is set to launch the first Latin American research institution of its AI Horizons Network in Brazil. Sectors that would potentially benefit from the research center include agribusiness, financial services, and healthcare. Brazilian startups are making significant investments in emerging technologies such as advanced analytics and process automation. Security and privacy are the two top concerns of organizations that are considering the use of intelligent communication solutions. The growth in the advanced analytics market in the region can be accredited to various factors such as the need to enhance the business efficiency, coupled with the growing utility of IoT analytics, big data analytics, and AI, in numerous end-user industries such as BFSI, retail, manufacturing, and healthcare. Brazil is slowly moving toward Industry 4.0, and industries including oil and gas, mining, and agriculture are expected to be the significant adopters of big data solutions and services. Over the next five years, the country will take initiatives to achieve significant efficiency advantages in smart connected networks, digitization, and automated operations. According to Software.org, 21% to 24% of SMEs in Brazil expect digitization to aid in the development of more personalized products and services. In this region, advanced analytics is used across verticals, for instance, in the banking sector, it is benefitted from faster processing and data management at a minimal cost. Brazil is shaping the technological landscape of Latin America. Brazilian organizations are aware of the advantages of advanced analytics. For instance, during the COVID-19 pandemic, the need to extract, visualize, and execute this intelligence in near-real-time, including measures to restrict its spread and assist firms to stay afloat, is fast becoming a mission-critical aim, boosting the predictive analytics market. Moreover, advanced analytics solution providers have a strong foothold in the region, which is anticipated to drive the market in the region. For instance, in May 2020, Qlik announced the general availability of Qlik Alerting, an intelligent alerting platform for Qlik Sense. Qlik Alerting delivers actionable, self-service, and centralized alerting that enhances customers' ability to proactively monitor their business data and take timely action based on insights.

TABLE 211 BRAZIL: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	164	252	367	509	586	37.5%
Services	78	123	184	262	308	41.1%
Total	**242**	**374**	**550**	**771**	**894**	**38.7%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 212 BRAZIL: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	683	809	971	1,185	1,467	1,861	22.2%
Services	367	444	546	681	862	1,123	25.0%
Total	**1,051**	**1,253**	**1,517**	**1,866**	**2,329**	**2,984**	**23.2%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.6.4 MEXICO

12.6.4.1 Government initiatives toward technological development to drive the growth of the advanced analytics market in Mexico

Mexico is increasingly adopting advanced analytics solutions and services. The country has witnessed a major digital transformation of the telecom sector, which is expected to create huge opportunities for advanced analytics solution providers in the coming years. The country needs a robust communication monitoring solution to enhance employee experience and increase organizational productivity. Mexico has ample opportunities to become a digitally advanced country in terms of technical expertise, due to its proximity to the US. It demonstrates the use of emerging technologies to achieve social goals, such as increasing financial inclusion, combating corruption, improving public health, and reducing crimes. The cloud-based adoption of advanced analytics solutions in Mexico is expected to increase, due to benefits such as lower costs, and improved innovation and productivity. The Mexican government is actively supporting and encouraging enterprises to adopt emerging technologies. The growing advanced analytics trend in Mexico is due to factors such as cost-effectiveness, growth in big data, the integration of mobile technology, and the development of the social media landscape. Mexico's economy has moved into an open and diverse economy over the past 10 years. The stable economic growth indicates that the Mexican economy still has a high potential and would continue to grow dynamically in the future. In Mexico, companies in the advanced analytics market focus on data analytics and insights, including data and tech consulting, data governance and ML and forecasting, predictive analysis, and predictive analytics and strategy. The increasing trade and rising customer base are expected to fuel the overall market growth in Mexico. Foreign trade comprises a larger percentage of Mexico’s economy due to the benefits of free trade agreements. Government initiatives related to the development of Mexico are expected to fuel the advanced analytics market.

TABLE 213 MEXICO: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solution	138	212	310	431	498	38.0%
Services	65	103	155	222	262	41.6%
Total	**203**	**315**	**465**	**654**	**760**	**39.2%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 214 MEXICO: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	583	692	834	1,021	1,269	1,614	22.6%
Services	314	380	469	587	745	974	25.5%
Total	**897**	**1,073**	**1,303**	**1,608**	**2,014**	**2,589**	**23.6%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

12.6.5 REST OF LATIN AMERICA

Rest of Latin America comprises Chile, Columbia, Argentina, and Peru. These countries have witnessed a notable shift in technological investments in the last few years. In fact, Argentina has become a breeding ground for young businesses. New opportunities are rising in several other countries and funding initiatives are making the region attractive for startups. Technological advancements are driving the adoption of advanced analytics solutions in the region. Businesses throughout the Rest of Latin America are eager to embrace technology, and they need proven solutions that can help them bring their intelligent communication solutions and services to market faster and with more control. With the power of AI, machines will also be able to predict users' needs based on their habits and preferences. In this region, most of the verticals, such as BFSI, healthcare and life sciences, manufacturing, energy and utilities, retail and consumer goods, and transportation and logistics, have realized that the implementation of advanced analytics solution shows the strength toward more advanced AI-and IoT-based solutions. Businesses throughout the rest of Latin America are eager to embrace AI-infused IoT, and they need proven solutions that can help businesses to improve business operations and customer experience.

TABLE 215 REST OF LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2016–2020 (USD MILLION)

Component	2016	2017	2018	2019	2020	CAGR (2016–2020)
Solutions	152	230	332	456	519	36.0%
Services	72	112	166	235	273	39.6%
Total	**224**	**343**	**498**	**691**	**792**	**37.2%**

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

TABLE 216 REST OF LATIN AMERICA: ADVANCED ANALYTICS MARKET SIZE, BY COMPONENT, 2021–2026 (USD MILLION)

Component	2021-e	2022	2023	2024	2025	2026-p	CAGR (2021–2026)
Solution	599	701	832	1,004	1,229	1,542	20.8%
Services	322	385	468	577	722	931	23.7%
Total	**921**	**1,086**	**1,300**	**1,581**	**1,951**	**2,474**	**21.8%**

e: estimated; p: projected

Source: Secondary Research, Expert Interviews, and MarketsandMarkets Analysis

13 COMPETITIVE LANDSCAPE

13.1 OVERVIEW

The competitive landscape analyzes the growth strategies implemented by various major players in the advanced analytics market globally. This section includes the market evaluation framework, revenue analysis, and market ranking of the vendors in the advanced analytics market. It also highlights the strategies undertaken by the companies to expand their market shares and increase their offerings in the market.

13.2 KEY PLAYER STRATEGIES

KEY PLAYER	PRODUCT NAME	DEVELOPMENT	REGION
IBM	IBM Cognos Analytics	IBM partnered with Boston Dynamics to deliver data analysis at the edge to help industrial companies improve worker safety, optimize field operations, and boost maintenance productivity. This partnership aims to help organizations with data and analytics that identify problems in real-time, improve decision-making, and perform tasks more efficiently.	Americas contributes more than 45% of its company's revenue.
Oracle	Oracle Analytics	Oracle announced the availability of the Oracle Exadata X9M platforms, the latest version of the industry's fastest and most affordable systems for running Oracle Database. It will lead to increase in analytic SQL throughput and ML workloads.	Americas contributes more than 52% to the company's revenue.
SAS Institute	SAS Advanced Analytics Software	Happy Mental Health has collaborated with SAS Institute to harness the power of SAS Health Analytics to identify individuals more intelligently in need of mental health support and target the most vulnerable populations for access to effective, low-cost care.	Americas to contributes most of 50% its revenue.
SAP	SAP Analytics Cloud	SAP partnered with HPE to deliver SAP HANA Enterprise Cloud with HPE GreenLake Cloud Services. This partnership would enable customers to keep their SAP software landscape and data on-premises.	Americas contributes more than 40% of its revenue
FICO	FICO Analytics Workbench	FIS collaborated with FICO to build an advanced anti-money laundering solution for the US financial institutions. The collaboration would support FIS AML Compliance Manager was integrated with the FICO Falcon X technology to help financial institutions keep ahead of financial crimes.	North America to contribute more than 70% of its revenue.

Source: Company Annual Reports, Press Releases, and MarketsandMarkets Analysis

The strategies of various advanced analytics players are assessed, along with their choices listed below:

- Product Portfolio: Key players in the advanced analytics market offer a comprehensive product portfolio. These players offer various advanced analytics and services to end users. Advanced analytics solutions in form of platform or API or software tools to make faster business decisions.
- Regional Focus: Players in the advanced analytics market have a well-diversified presence across the globe.
- Inorganic play/partnerships: Key players have been primarily deploying inorganic development strategies. They are leveraging partnerships focused on technology enhancements to grow their revenues.

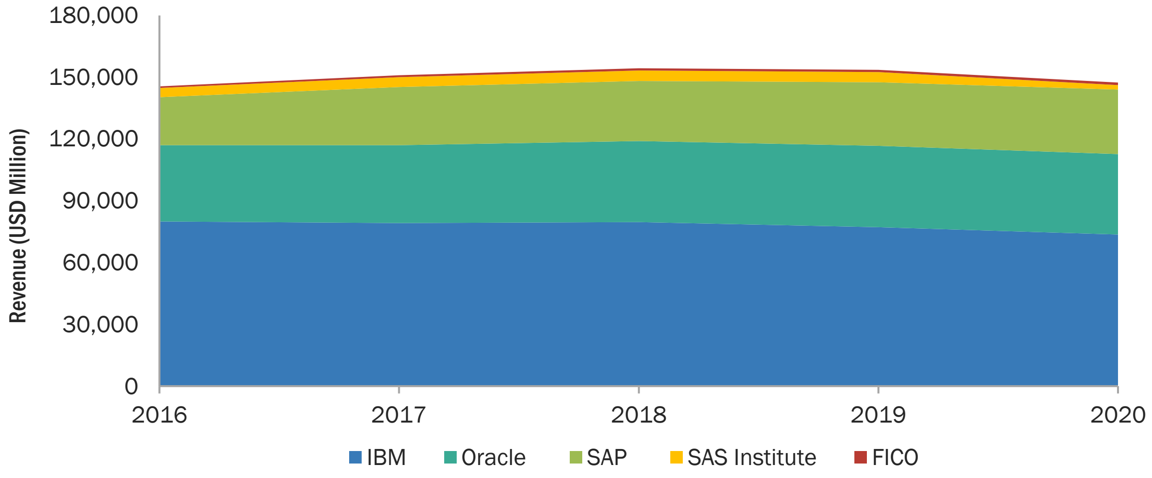
Keeping in mind the above considerations, MarketsandMarkets has concluded that different companies are deploying different strategies to win in the advanced analytics market. While IBM, Oracle, and SAS Institute have a diverse geographic revenue mix. In general, key companies are all strong in their home regions and explore geographic diversification alternatives to grow their businesses.

IBM, Oracle, FICO, SAP, and SAS Institute have been focusing on increasing their market shares through partnerships and acquisitions in the last few years. Companies are adopting growth strategies, but their focus is on inorganically expanding their presence.

13.3 REVENUE ANALYSIS

Players in the advanced analytics market were identified through secondary research. The share of each market player, in terms of value, was inferred from annual reports, presentations, and secondary research sources. Primary research interviews were conducted with key opinion leaders, such as CEOs, directors, industry experts, and other executives to validate revenue and market share.

FIGURE 49 REVENUE ANALYSIS FOR KEY COMPANIES IN THE PAST FIVE YEARS



Source: Company Website, Annual Reports, and SEC Filings

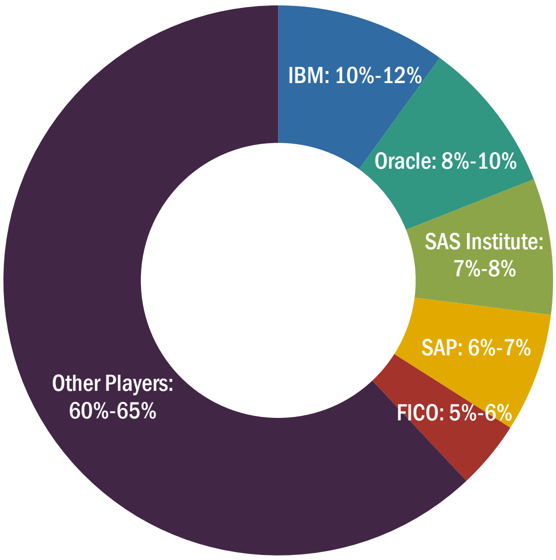
The global advanced analytics market is a consolidated market with five players: FICO, Oracle, IBM, SAP, and SAS Institute, holding over 40% market share. From the figure provided above, it can be understood that the business revenue for most of the major players has slightly declined in the last two years, and it is expected to fall further attributed to the sudden outbreak of the COVID-19 pandemic. Some of the reasons for the decline in revenue growth could be the serious recession faced by the global economy, along with the slow demand and supply in 2020.

13.4 MARKET SHARE ANALYSIS

In 2021, the advanced analytics market was consolidated with the presence of several global and regional key market players, such as FICO, Oracle, IBM, SAP, and SAS Institute. These companies develop advanced analytics solutions and services based on extensive R&D, ensuring large market shares. The other players in this market are Microsoft, KNIME, Salesforce, Altair, RapidMiner, AWS, TIBCO Software, Alteryx, Teradata, Adobe, Absolutdata Analytics, Moody's Analytics, Qlik, Databrick, Dataiku, H2O.ai, Domino Data Lab, DataRobot, DataChat, Imply, Kinetica, Promethium, Siren, Tellius, SOTA Solutions, and Vanti Analytics.

The degree of competition in the advanced analytics market is consolidated in 2021, with the top five companies expecting to register a combined market share of more than 40%. This segment studies growth strategies such as product launches, partnerships, acquisitions, expansions, contracts, and collaborations adopted by market players from January 2018 to June 2021 to expand their global presence and increase their shares in the advanced analytics market. The chart below gives a detailed ranking of key market players. Factors such as the market presence of companies, their client/customer base, total revenue generated by them from the advanced analytics segment, and key developments undertaken by them have been considered for ranking the top five players in the advanced analytics market. The key manufacturers and technology providers that are a part of the advanced analytics market ecosystem have been considered under the company profiles section.

FIGURE 50 MARKET SHARE ANALYSIS FOR KEY COMPANIES



Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

TABLE 217 ADVANCED ANALYTICS MARKET: DEGREE OF COMPETITION

DEGREE OF COMPETITION	CONSOLIDATED
Total market share of top five players	35%–40%
IBM	10%–12%
Oracle	8%–10%
SAS Institute	7%–8%
SAP	6%–7%
FICO	4%–6%
Other Key Players	60%-65%

THE DEGREE OF COMPETITION IS DEFINED AS BELOW:

- Fragmented: When top five players have a total market share <25%
- Competitive: When top five players have a total of 25%–50% market share
- Consolidated: When top five players have a total market share >50%

Note: The market share of the above companies is calculated based on the relevant segment revenues generated by companies

Source: Annual Reports, Press Releases, Expert Interviews, and MarketsandMarkets

13.5 COMPANY EVALUATION QUADRANT

13.5.1 STARS

Vendors who fall into this category receive high scores for most of the evaluation criteria. They have a strong and established product portfolio, strong market presence, and effective business strategies. They provide mature and reputable advanced analytics solutions and services and serve most of the regions worldwide. Star players primarily focus on acquiring the leading market position through their strong financial capabilities and well-established brand equity. IBM, SAS Institute, Oracle, SAP, FICO, Microsoft, TIBCO Software, KNIME, and RapidMiner are the vendors who fall in this category.

13.5.2 EMERGING LEADERS

Emerging leaders are vendors who have demonstrated substantial product innovations as compared to their competitors. They have a broad product portfolio. However, they do not have a very strong growth strategy for their overall business. Altair, Alteryx, Absolutdata Analytics, Moody's Analytics, and Qlik are the vendors who fall in this category.

13.5.3 PERVASIVE

Pervasive players are established vendors with very strong business strategies. They offer innovative solutions and services portfolios and have an extensive network of channel partners and resellers to increase the deployment of their solutions across a multitude of vertical markets. They focus on specific types of technologies related to products. The vendors who fall under this category are AWS, Salesforce, Adobe, and Teradata.

13.5.4 PARTICIPANT

These are vendors with niche product offerings and are starting to gain their position in the market. They do not have strong business strategies as compared to other established vendors. They might be new entrants in the market and require some time before gaining significant traction. Vendors who fall under this category are Dataiku, Databricks, and Kinetica.

FIGURE 51 KEY ADVANCED ANALYTICS MARKET PLAYERS, COMPANY EVALUATION QUADRANT, 2021

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

13.6 COMPETITIVE BENCHMARKING

13.6.1 COMPANY PRODUCT FOOTPRINT

FIGURE 52 PRODUCT PORTFOLIO ANALYSIS OF TOP PLAYERS IN THE ADVANCED ANALYTICS MARKET

COMPANY	PRODUCT OFFERING	INDUSTRY REACH	FOCUS ON PRODUCT INNOVATION
FICO	●●●○	●●●○	●●●○
IBM	●●●○	●●●○	●●●○
SAS Institute	●●○○	●●○○	●●○○
Oracle	●●●○	●●○○	●●○○
Salesforce	●●●○	●●●○	●●●○
SAP	●●○○	●●○○	●●○○
Microsoft	●●○○	●●○○	●●○○
RapidMiner	●●○○	●●○○	●●○○
Alteryx	●●○○	●○○○	●○○○
Qlik	●●○○	●●○○	●●○○
AWS	●●●○	●●○○	●●○○
Adobe	●●○○	●●○○	●●○○
Absolutdata Analytics	●●○○	●●●○	●●○○
Moody's Analytics	●●○○	●●○○	●●○○
TIBCO Software	●●○○	●●○○	●●○○
Dataiku	●●○○	●●○○	●●○○
KNIME	●●○○	●●○○	●●○○
Databricks	●●○○	●●○○	●●○○
Teradata	●●○○	●●○○	●●○○
Altair	●●○○	●●○○	●●○○
Kinetica	●●○○	●●○○	●●○○

RATING : ●●●● EXCELLENT ●●●○ GOOD ●●○○ AVERAGE ●○○○ BELOW AVERAGE ○○○○ NOT APPLICABLE

Note: The rating of the "breadth and depth of product offerings" parameter is given based on the company's offerings in the market; the "product features and functionalities" parameter rating highlights the company's products characteristics; the "focus on product innovations" parameter is rated based on the company's product launches/developments; the" branding" parameter is rated based on the company's brand value in the market

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 53 BUSINESS STRATEGY EXCELLENCE OF TOP PLAYERS IN THE ADVANCED ANALYTICS MARKET

COMPANY	REVENUE	GEOGRAPHIC FOOTPRINT	BREADTH OF APPLICATIONS	INORGANIC (M&A, JV, PARTNERSHIPS)	ORGANIC (NPD, EXPANSIONS)
FICO	●●●○	●●●●	●●●○	●●○○	●●●○
IBM	●●●○	●●●●	●●●○	●●●○	●●○○
SAS Institute	●●●○	●●●●	●●●○	●○○○	●●○○
Oracle	●●●○	●●●●	●●●○	●●○○	●●○○
Salesforce	●●●○	●●●●	●●●○	●●○○	●●●○
SAP	●●●○	●●●●	●●○○	●●○○	●○○○
Microsoft	●●●○	●●●○	●●○○	●○○○	●●○○
RapidMiner	●○○○	●●●○	●●○○	●○○○	●○○○
Alteryx	●○○○	●●●○	●●○○	●○○○	●●○○
Qlik	●○○○	●○○○	●●○○	●○○○	●●○○
AWS	●○○○	●○○○	●●●○	●●○○	●●○○
Adobe	●●○○	●●○○	●●○○	●○○○	●●○○
Absolutdata Analytics	●●○○	●●○○	●●○○	●○○○	●○○○
Moody's Analytics	●●○○	●●●○	●●○○	●○○○	●○○○
TIBCO Software	●●○○	●●●○	●●○○	●○○○	●○○○
Dataiku	●●○○	●●●○	●●○○	●○○○	●○○○
KNIME	●●○○	●●●○	●●○○	●○○○	●○○○
Databricks	●●○○	●●●○	●●○○	●○○○	●○○○
Teradata	●●○○	●●●○	●●○○	●○○○	●○○○
Altair	●●○○	●●●○	●●○○	●○○○	●○○○
Kinetica	●●○○	●●●○	●●○○	●○○○	●○○○

RATING : ●●●● EXCELLENT ●●●○ GOOD ●●○○ AVERAGE ●○○○ BELOW AVERAGE ○○○○ NOT APPLICABLE

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

TABLE 218 COMPANY OFFERING FOOTPRINT

COMPANY NAME	SOLUTION	SERVICE	TOTAL SCORE
IBM	Y	Y	4
Oracle	Y	Y	4
SAS Institute	Y	Y	4
SAP	Y	Y	4
FICO	Y	N	4
KNIME	Y	N	3
Microsoft	Y	Y	2
Altair	Y	Y	2
Rapid Miner	Y	N	3
AWS	Y	Y	2
Salesforce	Y	Y	2
TIBCO Software	Y	Y	3
Alteryx	Y	N	3
Teradata	Y	Y	4
Adobe	Y	N	3
Absolutdata Analytics	Y	Y	2
Moody’s Analytics	Y	N	2
Qlik	Y	N	2
Databricks	Y	Y	2
Dataiku	Y	Y	2

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

TABLE 219 COMPANY REGION FOOTPRINT

COMPANY NAME	NORTH AMERICA	EUROPE	APAC	MEA	LATIN AMERICA	TOTAL SCORE
IBM	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Oracle	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
SAS Institute	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
SAP	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
FICO	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3
KNIME	Yes (✓)	Yes (✓)	No (✓)	No	No	2
Microsoft	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Altair	Yes (✓)	Yes (✓)	No	No	Yes (✓)	3
Rapid Miner	Yes (✓)	Yes (✓)	No	No	No	2
AWS	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Salesforce	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	No	4
TIBCO software	Yes (✓)	No	No	No	No	1
Alteryx	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	5
Teradata	Yes (✓)	Yes (✓)	No	No	No	2
Adobe	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3
Absolutdata Analytics	Yes (✓)	Yes (✓)	Yes (✓)	Yes (✓)	No	4
Moody's Analytics	Yes (✓)	Yes (✓)	No	Yes (✓)	No	3
Qlik	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3
Databricks	No	Yes (✓)	No	No	No	1
Dataiku	Yes (✓)	Yes (✓)	Yes (✓)	No	No	3

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

FIGURE 54 ADVANCED ANALYTICS: COMPETITIVE BENCHMARKING OF TOP 10 KEY PLAYERS

	VERTICAL					GEOGRAPHY				
COMPANY NAME	GOVERNMENT AND DEFENSE	TRANSPORTATION AND LOGISTICS	BFSI	RETAIL AND CONSUMER GOODS	OTHERS	NORTH AMERICA	EUROPE	APAC	MEA	LATIN AMERICA
IBM		✔		✔	✔	✔	✔	✔	✔	✔
Oracle	✔		✔	✔	✔	✔	✔	✔	✔	✔
SAS Institute	✔	✔	✔	✔	✔	✔	✔	✔	✔	✔
SAP			✔	✔	✔	✔	✔	✔	✔	✔
FICO	✔		✔	✔	✔	✔	✔			
KNIME					✔	✔	✔			
Microsoft		✔	✔		✔	✔	✔	✔	✔	✔
Altair	✔		✔	✔	✔	✔	✔			✔
Rapid Miner	✔	✔		✔	✔	✔	✔			
AWS		✔	✔	✔	✔	✔	✔	✔	✔	✔

Source: Press Releases, and MarketsandMarkets Analysis

13.7 STARTUP/SME EVALUATION QUADRANT

The startup/SME evaluation matrix provides information related to the major players offering advanced analytics solutions and outlines findings and analysis on how well each market vendor performs within the pre-defined competitive leadership mapping criteria. The startups/SMEs selected for the matrix were founded after 2014. Vendor evaluations are based on two broad categories: strength of product portfolio and business strategy excellence. Each category carries various criteria based on which the vendors have been evaluated. The evaluation criteria considered under the strength of the product portfolios include breadth of offering, delivery (based on industries that the vendors cater to, deployment modes, and subscriptions), and support (based on pre-and post-sales support services). The evaluation criteria considered under the business strategy excellence include reach (based on geographic presence), channel (based on channel partners that the vendors cater to), and inorganic growth (comprising partnerships, collaborations, and acquisitions).

13.7.1 PROGRESSIVE COMPANIES

H2O.ai, SOTA Solutions, and DataRobot are progressive companies in the advanced analytics market. They offer strong solution portfolios in the market. These vendors have been marking their presence by offering customized solutions as per requirements and adopting growth strategies to achieve advanced infrastructure in the advanced analytics market.

13.7.2 RESPONSIVE COMPANIES

Tellius, Imply, Siren, and Domino Data Lab is recognized as a responsive company in the advanced analytics market. It possesses innovative solutions to cater to future advanced analytics demands. They are concerned related to their product portfolio and have a robust potential to build strong business strategies to expand their business and stay at par with progressive companies. These vendors have been offering advanced analytics solutions to their customers for fulfilling their demands. Responsive companies have been at the forefront of the deployment of their solutions.

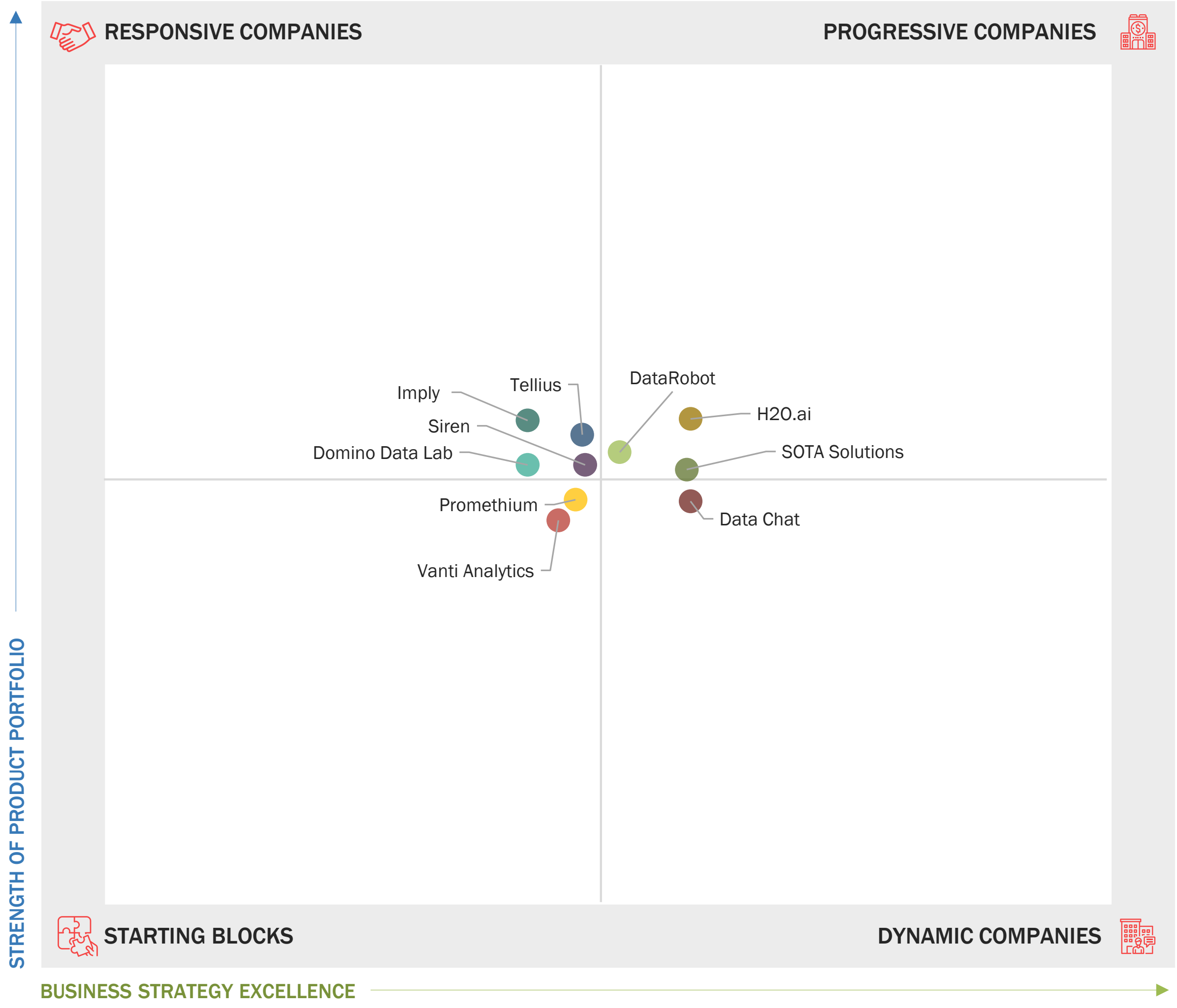
13.7.3 DYNAMIC COMPANIES

DataChat is recognized as a dynamic company in the advanced analytics market. They offer innovative solution portfolios and have an extensive network of channel partners and resellers to increase the deployment of their solutions across a multitude of vertical markets. The dynamic companies have been consistently generating positive revenue growth in the advanced analytics market, and their market position is boosted by organic and inorganic ventures undertaken by them.

13.7.4 STARTING BLOCKS

Promethium and Vanti Analytics are recognized as the starting blocks in the advanced analytics market. Most starting blocks have been undertaking multiple acquisitions for boosting their sales capabilities across regions to offer integrated solutions and services to a wide range of clients.

FIGURE 55 STARTUP/SME ADVANCED ANALYTICS MARKET EVALUATION QUADRANT, 2021



Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

13.8 STARTUP/SME COMPETITIVE BENCHMARKING

13.8.1 COMPANY PRODUCT FOOTPRINT

FIGURE 56 PRODUCT PORTFOLIO ANALYSIS OF STARTUP/SME IN THE ADVANCED ANALYTICS MARKET

COMPANY	PRODUCT OFFERING	INDUSTRY REACH	FOCUS ON PRODUCT INNOVATION
Domino Data Lab	●●●○	○○○○	●●●○
DataRobot	●●●○	●○○○	●●●○
Promethium	●●●○	○○○○	●●○○
Vanti Analytics	●●●○	○○○○	●●●○
Siren	●●●○	○○○○	●●●○
Tellius	●●●○	●●○○	●●●○
Imply	●●●○	●●○○	●●●○
SOTA Solutions	●●○○	○○○○	●●○○
H2O.ai	●●●○	●●●○	●●●○
Data Chat	●●●○	●●●○	●●●○

RATING : ●●●● EXCELLENT ●●●○ GOOD ●●○○ AVERAGE ●○○○ BELOW AVERAGE ○○○○ NOT APPLICABLE

Note: The rating of the "breadth and depth of product offerings" parameter is given based on the company's offerings in the market; the "product features and functionalities" parameter rating highlights the company's products characteristics; the "focus on product innovations" parameter is rated based on the company's product launches/developments; the "branding" parameter is rated based on the company's brand value in the market

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

FIGURE 57 BUSINESS STRATEGY EXCELLENCE OF STARTUP/SME IN THE ADVANCED ANALYTICS MARKET

COMPANY	REVENUE	GEOGRAPHIC FOOTPRINT	BREADTH OF APPLICATIONS	INORGANIC (M&A, JV, PARTNERSHIPS)	ORGANIC (NPD, EXPANSIONS)
Domino Data Lab	●●○○	●○○○	●●○○	●●○○	●●○○
DataRobot	●●○○	●○○○	●●●○	●●●○	●●●○
Promethium	●○○○	●○○○	●●○○	●●○○	●●○○
Vanti Analytics	●○○○	●○○○	●●○○	●○○○	●○○○
Siren	●●○○	●○○○	●●○○	●●○○	●●○○
Tellius	●●○○	●○○○	●●●○	●●●○	●●●○
Imply	●●○○	●○○○	●●●○	●●○○	●●○○
SOTA Solutions	●●○○	●○○○	●●○○	●●○○	●○○○
H2O.ai	●●●○	●●○○	●●●○	●●●○	●●●○
Data Chat	●●●○	●●○○	●●●○	●●●○	●●●○

RATING : ●●●● EXCELLENT ●●●○ GOOD ●●○○ AVERAGE ●○○○ BELOW AVERAGE ○○○○ NOT APPLICABLE

Source: Press Releases, Expert Interviews, and MarketsandMarkets Analysis

TABLE 220 STARTUP/SME COMPANY OFFERING FOOTPRINT

COMPANY NAME	SOLUTION	SERVICE	TOTAL SCORE
H2O.AI	Y	N	3
Domino Data Lab	Y	N	3
DataChat	Y	N	3
Imply	Y	Y	3
DataRobot	Y	Y	2
Promethium	Y	N	3
Siren	Y	N	2
Tellius	Y	N	2
SOTA Solutions	Y	Y	2
Vanti Analytics	Y	N	4

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

TABLE 221 STARTUP/SME COMPANY REGION FOOTPRINT

COMPANY NAME	NORTH AMERICA	EUROPE	APAC	MEA	LATIN AMERICA	TOTAL SCORE
H2O.AI	Yes (✓)	Yes (✓)	No	No	No	2
Domino Data Lab	Yes (✓)	No	No	No	No	1
DataRobot	Yes (✓)	No	Yes (✓)	No	No	2
DataChat	Yes (✓)	No	No	No	No	1
Imply	Yes (✓)	No	No	No	No	1
Promethium	Yes (✓)	No	No	No	No	1
Siren	Yes (✓)	Yes (✓)	No	No	No	2
Tellius	Yes (✓)	No	Yes (✓)	No	No	2
SOTA Solutions	No	Yes (✓)	No	No	No	2
Vanti Analytics	No	No	No	Yes	No	1

Source: Annual Reports, Press Releases, Investor Presentations, Primary Interviews, and MarketsandMarkets Analysis

TABLE 222 ADVANCED ANALYTICS MARKET: DETAILED LIST OF KEYS STARTUP/SMES

COMPANY NAME	CATEGORY	OWNERSHIP STATUS	HQ LOCATION	YEAR FOUNDED	EMPLOYEES	FINANCING STATUS	LATEST FUNDING ROUND	TOTAL FUNDING (USD MN)
H2O.AI	Computer Software	Private	Mountain View, United States	2012	201-500	Venture Capital	USD 100M, Series E	USD251 M
Domino Data Lab	Computer Software	Private	San Francisco, United States	2013	201-500	Venture Capital	USD100M, Series F	- USD228 M
DataRobot	IT and services	Private	Boston, United States	2012	1001-5000	Venture Capital	- USD300M, Series G,	USD1.1 B
DataChat	IT and services	Private	Madison, United States	2017	11-50	Venture Capital	USD25M, Series A,	USD29 M
Imply	Computer Software	Private	San Francisco, United States	2015	51-200	Capital Markets	-USD70M, Series C	USD116 M
Promethium	Computer Software	Private	Menlo Park, California, United States	2018	11-50	Venture Capital	USD 6M, Series A,	USD8.5 M
Siren	Computer Software	Private	San Francisco, United States	2015	11-50	-	USD9M, Series B	USD30.9M
Tellius	Computer Software	Private	Herndon, United States	2016	11-50	Venture Capital	USD8M, Series A	17M
SOTA Solutions	IT and services	Private	Berlin, Germany	2012	51-200	Venture Capital	Undisclosed, Seed,	
Vanti Analytics	Computer Software	Private	San Francisco, United States	2019	11-50		USD4.5M, Seed	USD6M

Source: Company Website and MarketsandMarkets Analysis

FIGURE 58 ADVANCED ANALYTICS: COMPETITIVE BENCHMARKING OF KEY STARTUP/SMES

	VERTICAL					GEOGRAPHY				
COMPANY NAME	GOVERNMENT AND DEFENSE	TRANSPORTATION AND LOGISTICS	BFSI	RETAIL AND CONSUMER GOODS	OTHERS	NORTH AMERICA	EUROPE	APAC	MEA	LATIN AMERICA
H2O.ai		✔		✔	✔	✔	✔			
Domino Data Lab		✔	✔			✔				
DataRobot	✔			✔	✔	✔				
Data Chat						✔		✔		
Imply	✔			✔		✔				
Promethium		✔	✔			✔				
Siren						✔	✔			
Tellius						✔		✔		
SOTA Solutions			✔		✔		✔			
Vanti Analytics		✔			✔				✔	

Source: Press Releases, and MarketsandMarkets Analysis

13.9 COMPETITIVE SCENARIO AND TRENDS

Key market developments include the recent developments of the key players in the global advanced analytics market. The market size analysis of the key industry players has been done based on various parameters, including the number of advanced analytics stakeholders, the depth of product offerings, and the revenue generated.

13.9.1 PRODUCT LAUNCHES

Companies have adopted the strategy of new product launches and product enhancements to enhance their product portfolios and meet the growing demand from the advanced analytics market globally. Leading companies, such as IBM, FICO, Microsoft, Oracle, Altair, SAP, and SAS Institute have adopted this strategy.

TABLE 223 ADVANCED ANALYTICS MARKET: PRODUCT LAUNCHES, APRIL 2020–AUGUST 2021

YEAR- MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Dec	IBM	Solution	IBM Streaming Analytics	IBM integrated streaming analytics on its Cloud Park for Data as a Service (DaaS). This service adds a tool or other type of interface that runs in IBM Cloud and provides APIs that can use outside of Cloud Pak for Data as a Service.
2021-Nov	FICO	Solution	Fico Origination Solution	Fico announced the launch of Fico Origination Solution Powered by the Fico platform. The solution is powered by the industry-leading FICO Platform and delivered globally on Amazon Web Services (AWS) Cloud. The solution empowers businesses to deliver exceptional, personalized customer experiences by automating a frictionless customer journey.

2021-Nov	Microsoft	Solution	Azure Stream Analytics	Azure Stream Analytics operation can directly feed data into an Azure Data Explorer table using this new integration. The direct integration of Azure Data Explorer as an output for Azure Stream Analytics has increased the power of Azure-powered real-time analytics architecture.
2021-Sept	Oracle	Solution	Oracle Exadata X9M platforms	Oracle announced the availability of the Oracle Exadata X9M platforms, the latest version of the industry's fastest and most affordable systems for running Oracle Database. It will lead to increase in analytic SQL throughput and ML workloads.
2021-Oct	Altair	Solution	Toggled iQ Smart	Altair announced significant enhancements to its integrated portfolio of simulation and design tools. This update to Altair's simulation software suite is focused on accelerating simulation-driven design.
2021-Sept	SAP	Solution	SAP Analytics Cloud	SAP released a new version of SAP Analytics Cloud 2021.20. It updates new dashboard and analytics platform services to enhance user experience.
2021-Aug	SAP	Solution	SAP Asset Intelligence Network	SAP added contextual geographical information of the work order and operations in a map view. It would allow maintenance of geospatial coordinates for an owned work order header and operation.
2021-May	SAS Institute	Solution	SAS Viya	SAS will reinforce the foundation for data and analytic success by incorporating new data management solutions into its powerful, cloud native SASViya platform.

Source: Company Websites and Secondary Sources

13.9.2 DEALS

The inorganic growth strategies of acquisitions, partnerships, collaborations, and investments have been used by leading market players to broaden their customer base and enhance their technological capabilities. Key market players, such as IBM, SAS Institute, Oracle, FICO, Microsoft, Altair, Oracle, and AWS, have adopted this strategy.

TABLE 224 ADVANCED ANALYTICS MARKET: DEALS, MAY 2019–OCTOBER 2021

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Dec	Agreement	IBM	AIB	AIB has signed a three-year agreement with IBM that will help it to accelerate its ongoing digital transformation, meet its customer-first commitment, and provide a clear pathway to a hybrid cloud technology environment. The technology portfolio which includes data analytics and process automation tools provides market-leading capabilities, selected specifically to complement AIB's existing technology roadmap and business requirements. It will also enable enhanced system availability capabilities which will support the bank's approach to operational resilience.

2021-Dec	Partnership	SAS Institute	PERSOWN	SAS is partnering with PERSOWN, to bring affordable, lifesaving technologies to the billions of traditionally underserved people of the world. From visualizing disease mutation and progression in the current COVID-19 pandemic to monitoring treatment efficacy and compliance. The partnership will empower global health leaders, clinicians, and researchers to make more informed decisions.
2021-Nov	Partnership	Oracle	Samsung Securities	Samsung Securities has selected Oracle Cloud Infrastructure (OCI) to underpin its digital innovation strategy. The company has secured the flexibility, reliability, and high-performance capabilities it required to effectively run advanced analytics in the rapidly changing financial market environment. It is now able to analyze the derivatives in near real-time to maximize customer profits. Through its transition to the cloud, Samsung Securities can gain access to flexible, secure, and high-performance computing resources.
2021-Nov	Partnership	FICO	Vodafone	FICO partnered with Vodafone to leverage FICO's expertise in tackling fraud risk across digital channels. The FICO advisors and consultancy team will assess Vodafone's operations across 19 countries and advise on tackling new fraud risks at a country level.
2021-Sept	Partnership	Microsoft	Truveta	Microsoft and Truveta announced a strategic partnership to apply the power of Microsoft Azure and AI to help Truveta achieve its vision of saving lives with data. It will create the world's largest identified health data platform.
2021-Aug	Acquisition	Altair	S-FRAME	Altair announced the acquisition of S-FRAME Software, a structural analysis software platform used by engineers worldwide to evaluate a structure's ability to endure external loads (such as wind, water, and snow) and meet design code requirements.
2021-Jun	Collaboration	Oracle	Wipro	Wipro collaborated with Oracle to launch Wipro Zero Cost Transformation, which would help organizations migrate their workloads to Oracle Cloud Infrastructure (OCI) at a lower cost. Wipro is a member of the Oracle Partner Network (OPN).
2020-Dec	Collaboration	AWS	Novartis	AWS collaborated with Novartis to accelerate the digital transformation of its business operations. This collaboration enables Novartis to reimagine its core pharmaceutical manufacturing, supply chain, and delivery operations using AWS' portfolio of cloud services.

Source: Company Websites and Secondary Source

14 COMPANY PROFILES

14.1 INTRODUCTION

The section provides a brief overview of the major players operating in the advanced analytics market. It comprises information related to solutions and services; key strengths, weaknesses, and competitive threats; and recent developments during the last three years of the top players in the advanced analytics market. Advanced analytics is the process of data analysis methodology that uses predictive modeling, ML algorithms, deep learning, business process automation, and other statistical methods to analyze business information from a variety of data sources. Advanced analytics includes techniques such as data/text mining, ML, pattern matching, forecasting, visualization, semantic analysis, sentiment analysis, network and cluster analysis, multivariate statistics, graph analysis, simulation, complex event processing, neural networks. The major players of advanced analytics generate a significant portion of their revenues from North America, followed by Europe, APAC, MEA, and Latin America. The top 31 vendors have been evaluated in this evaluation matrix, including IBM (US), Oracle (US), SAS Institute (US), SAP (Germany), FICO (US), KNIME (Switzerland), Microsoft (US), Altair (US), RapidMiner (US), AWS (US), Salesforce (US), TIBCO Software (US), Alteryx (US), Teradata (US), Adobe (US), Absolutdata (US), Moody's Analytics (US), Qlik (US), Databricks (US), Dataiku (US), Kinetica (US), H2O.ai (US), Domino Data Lab (US), DataRobot (US), DataChat (US), Imply (US), Promethium (US), Siren (Ireland), Tellius (US), SOTA Solutions (Germany), and Vanti Analytics (Israel).

14.2 IBM

14.2.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 225 IBM: BUSINESS AND FINANCIAL OVERVIEW

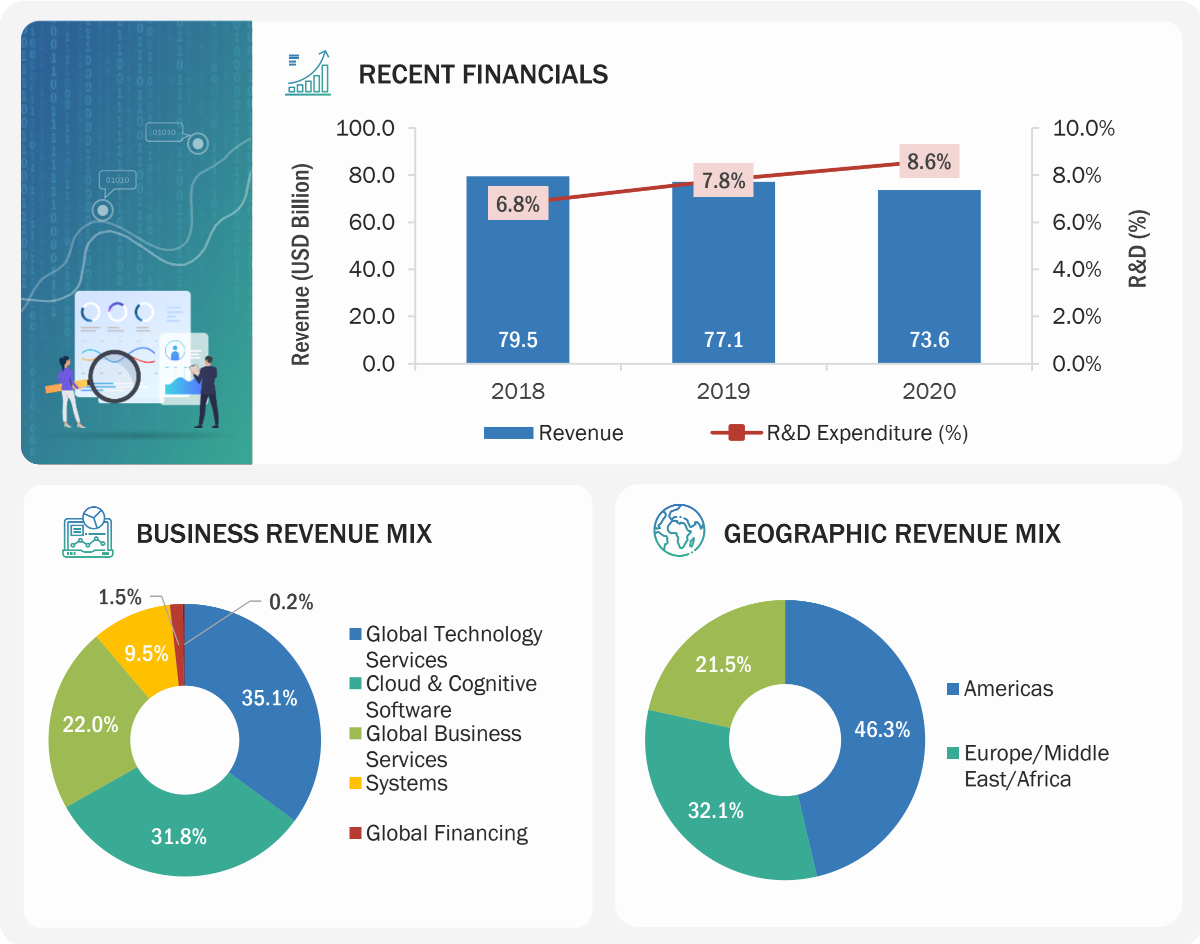
COMPANY AT A GLANCE-2021	
Founded	1911
Country	US
City	New York
No. of Employees	430,000+
Overall Annual Revenue	USD 73.6 billion
R&D Expenditure	USD 6.3 billion
Ownership	Public (NASDAQ: IBM)

Source: Company Website, Annual Reports, and SEC Filings

IBM is a multinational technology and consulting corporation that offers infrastructure, hosting, and consulting services. The company operates through five major business segments: cloud and cognitive software, global business services, global technology services, systems, and global financing. It caters to various verticals that include aerospace and defense, education, healthcare, oil and gas, automotive, electronics, insurance, retail and consumer products, banking and finance, energy and utilities, life sciences, telecommunications, media and entertainment, chemicals, government, manufacturing, travel and transportation, construction, and metals and mining. The company has a robust presence in the Americas, Europe, MEA, and APAC and clients in more than 175 countries.

IBM offers IBM Advanced Analytics in the advanced analytics market. This solution helps discover actionable data-driven insights to drive customer engagement, reveal revenue-increasing opportunities, and outsmart frauds for protecting revenue and reputation.

FIGURE 59 IBM: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on December 31

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.2.2 SOLUTIONS OFFERED

TABLE 226 IBM: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION NAME	DESCRIPTION
Solution	IBM Predictive Analytics	The solution unlocks data-driven insights that help businesses make better decisions. IBM Predictive Analytics is a combination of advanced analytics capabilities spanning ad-hoc statistical analysis, predictive modeling, data mining, text analytics, entity analytics, optimization, real-time scoring, and ML. The predictive analytics products offered under this solution are IBM Statistical Package for Social Sciences (SPSS) Modeler and IBM SPSS Statistics.
Solution	IBM Decision Optimization	IBM Decision Optimization represents a family of optimization software that delivers prescriptive analytics capabilities to help make better decisions and deliver improved business outcomes.
Solution	IBM Data Science	Uncover patterns and build predictions using data, algorithms, ML, and AI techniques.
Solution	IBM Planning Analytics	Planning analytics provides insights into historical data to inform current plans and optimize for the future.
Solution	IBM Streaming Analytics	IBM Streaming Analytics for IBM Cloud assesses a wide range of streaming data, including unstructured text, video, audio, geospatial, and sensor data, to help businesses identify opportunities and risks and make real-time decisions. Streaming Analytics on IBM Cloud offers most of the features of IBM Streams on an agile, cloud-based platform.
Solution	IBM Cloud Pak for Data	IBM Cloud Pak for Data simplifies administration and includes advanced governance and data privacy features to safeguard data while enhancing usage for analytics and AI.

Source: Company Website

14.2.3 SERVICES OFFERED

TABLE 227 IBM: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
consulting services	IBM's data and analytics	The data and analytics consulting services help organizations integrate enterprise data for operational, analytical, data science, and ai models to build an insights-driven organization.
consulting services	Artificial intelligence services	AI services can help drive the smart evolution of workflows, technologies, and entire organizations.

14.2.4 RECENT DEVELOPMENTS

14.2.4.1 IBM: Product launches and enhancements

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Dec	IBM	Solution	IBM streaming analytics	IBM integrated streaming analytics on its Cloud Park for Data as a Service (DaaS). This service adds a tool or other types of interfaces that runs in IBM Cloud and provides APIs that can use outside of Cloud Pak for DaaS.
2021-Nov	IBM	Solution	IBM Watson Discovery	IBM announced new NLP enhancements planned for IBM Watson Discovery. These planned updates are designed to help business users in industries such as financial and legal to enhance customer care and accelerate business processes by uncovering insights and synthesizing information from complex documents.

Source: Press Releases

TABLE 228 IBM: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Dec	Agreement	IBM	AIB	AIB signed a three-year agreement with IBM that will help it accelerate its ongoing digital transformation, meet its customer-first commitment, and provide a clear pathway to a hybrid cloud technology environment. The technology portfolio which includes data analytics and process automation tools provides market-leading capabilities, selected specifically to complement AIB's existing technology roadmap and business requirements. It will also enable enhanced system availability capabilities, which will support the bank's approach to operational resilience.
2021-Oct	Partnership	IBM	Deloitte	DAPPER, an AI-enabled managed analytics solution, is a new offering from IBM and Deloitte. The product enables enterprises in accelerating hybrid cloud and AI adoption across the enterprise.
2021-Oct	Acquisition	IBM	Adobe Workfront Consultancy	To advance its hybrid cloud and AI strategy, IBM announced the acquisition of Rego Consulting Corporation's Adobe Workfront consulting team and assets. The company focuses on work management software consulting for commercial clients, leveraging IBM's strategic partnership with Adobe.
2021-Oct	Partnership	IBM	Boston Dynamics	The partnership aims to help organizations with data and analytics that identify problems in real-time, improve decision-making, and perform tasks more efficiently.

2021-Oct	Agreements	IBM	Mc Donald's	IBM and McDonald's to further accelerate the development and deployment have entered into an agreement of its Automated Order Taking (AOT) technology. IBM will acquire Mc Donald's Tech Lab, which was created to advance employee and customer-facing innovations. This agreement will accelerate McDonald's efforts to provide an even more convenient and unique customer and crew experience.

Source: Press Releases

14.2.5 IBM: COVID-19 DEVELOPMENT

In July 2021, Amadeus Integrated IBM Digital Health Pass into Its Digital Health Verification technology. The integration was intended to help simplify the validation of COVID-19 health paperwork for air travelers, decreasing check-in friction and assisting airlines as they attempt to expand international travel volumes. IBM launched Watson Works solutions that use data, AI technology, and blockchain to help enterprises make decisions to call back employees on-site, manage resources for them, and ensure workplace safety. The company constantly monitors developments related to the COVID-19 outbreak. In March 2020, the company developed a new, interactive global dashboard to show the spread of COVID-19 across the world. The COVID-19 data reflected on the dashboard is obtained from state and local governments and WHO. The dashboard is designed to offer the public, researchers, and government officials detailed, localized, and current information on COVID-19. IBM has also launched Watson Works to address the challenges of returning to workplaces. Watson Works is a curated set of products embedded with the Watson AI model and applications to help companies navigate many aspects of the return-to-workplace challenge.

14.2.6 MNM VIEW

14.2.6.1 Key strengths/right to win

IBM creates value for clients by enabling new capabilities, leveraging products and solutions, and integrating technology and expertise in business processes with a broad ecosystem of partners and alliances. The data offerings of IBM help clients organize, collect, analyze, and embed their data into their workflow and span areas, ranging from data management and discovery to reporting, governance, compliance, and risk management. IBM has clients in more than 175 countries, with a presence in North America, Europe, APAC, MEA, and Latin America.

14.2.6.2 Strategic choices made

IBM has chosen to enhance its portfolio by launching new solutions, particularly in the cloud. This has resulted in IBM having a strong portfolio of solutions and services in the cloud. The investments made by the company in data and AI have strengthened its position of enterprise market leadership. It has invested USD 45 billion in R&D, forging the futures of cloud, AI, blockchain, and quantum computing. The company continues to invest, for a long-term opportunity, in key growth countries, such as India and China, and the regions of Eastern Europe, MEA, and Latin America to expand its product and service portfolios. In India, Vodafone Idea, a joint venture of Vodafone Group, has signed an IT outsourcing partnership with IBM to use IBM's hybrid cloud, analytics, and security capabilities for accelerating Vodafone Idea's progression to an open, agile, and secure IT environment.

14.2.6.3 Weaknesses and competitive threats

IBM's revenue is highly dependent on the American market as the Americas contribute to 47% of its total revenue. As IBM is continuously investing in new technologies and key strategic areas for driving its revenue growth, the risk of failure of innovative initiatives is also increasing, which can impact the company's competitiveness. As the company has a global presence, its business and operations could be impacted by local legal, economic, political, and health conditions.

14.3 ORACLE

14.3.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 229 ORACLE: BUSINESS OVERVIEW

COMPANY AT A GLANCE-2021	
Founded	1977
Country	Austin
City	Texas, US
No. of Employees	132,200
Overall Annual Revenue	USD 40.5 Billion
R&D Expenditure	USD 6.5 Billion
Ownership	Public (NYSE: ORCL)

Source: Company Website, Annual Reports, and SEC Filings

Oracle is one of the leading providers of cloud and database solutions. The company's business is divided into three major segments: cloud and license, hardware, and services. It is present in more than 145 countries, with a workforce of 206,815 employees serving more than 420,000 customers. It has a presence in North America, Europe, APAC, MEA, and Latin America. Oracle is an established and leading vendor of enterprise-grade products and solutions. The company specializes in developing, manufacturing, and marketing hardware systems, database and middleware software, and application software. It offers analytics software designed to leverage big data and enterprise data for enabling organizations to analyze data and discover new ways for strategizing and planning business operations. Oracle works on various emerging technologies, including IoT, AI, blockchain, ML, chatbots and digital assistance, and data science, which help in developing new business models and delivering SaaS applications embedded with these emerging technologies. Its mission is to enable financial institutions to excel through the effective use of IT.

In the advanced analytics market, it offers Oracle Analytics, which is a complete platform with ready-to-use services for a wide variety of workloads and data. It provides actionable insights from all types of data in the cloud, on-premises, or in a hybrid deployment. It empowers business users, data engineers, and data scientists to access and process relevant data, evaluate predictions, and make quick, accurate decisions.

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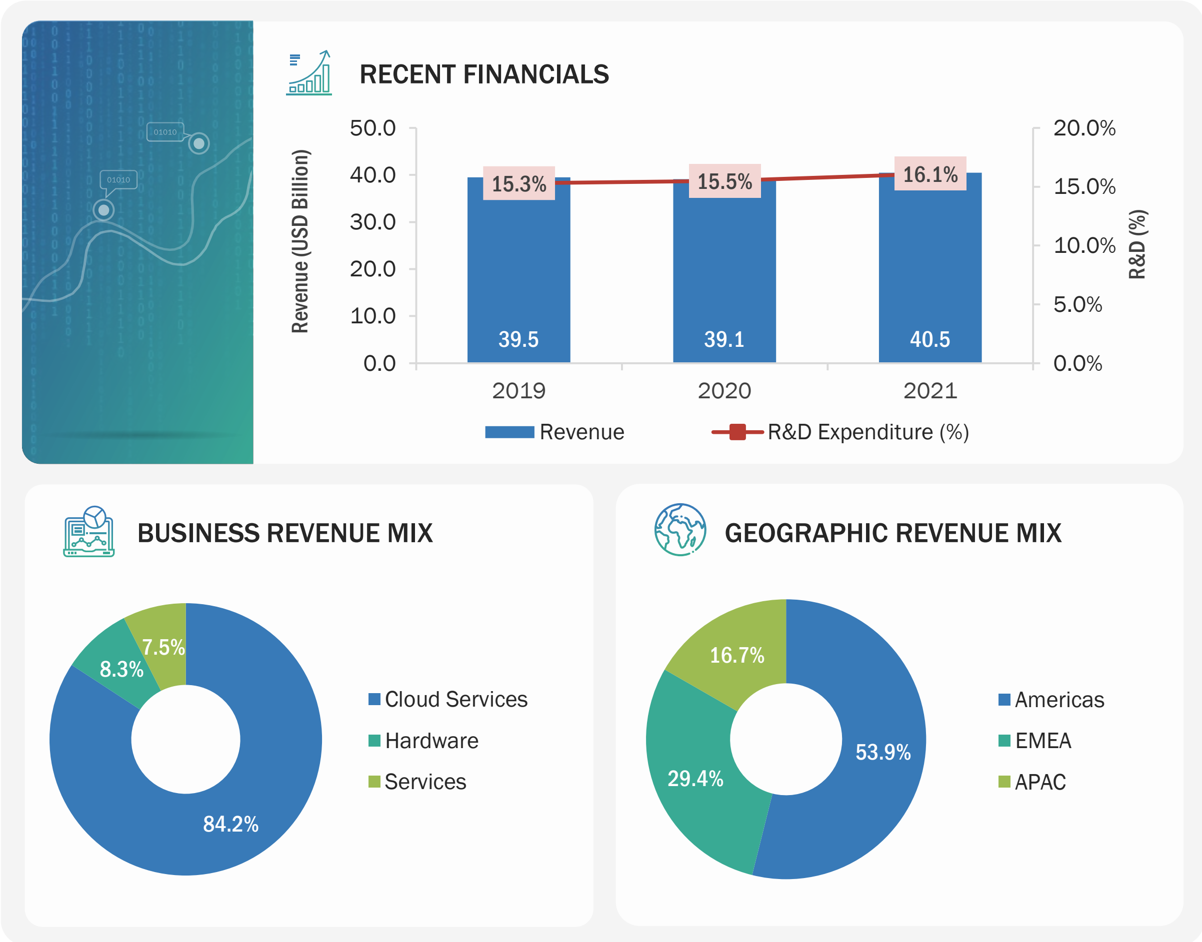
FIGURE 60 ORACLE: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on January 31

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.3.2 SOLUTIONS OFFERED

TABLE 230 ORACLE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION NAME	DESCRIPTION
Solution	Oracle Analytics Platform	It empowers leaders, analysts, and IT to access data from wherever they are including mobile devices. It is embedded with ML, which helps organizations discover unique insights faster with automation and intelligence.
Solution	Oracle Fusion ERP Analytics	Oracle Fusion ERP Analytics is a prebuilt solution for Oracle Cloud ERP that helps finance and procurement professionals uncover underlying drivers of profitability, improve the use of working capital, and better control business expenditures. Delve deep into sophisticated analytics covering topics such as general ledger, accounts receivables, accounts payables, spend, employee expenses, and procurement.

Solutions	Oracle Fusion HCM Analytics	Oracle Fusion HCM Analytics is a prebuilt solution for Oracle Cloud HCM, which provides human resource professionals with ready-to-use workforce insights to improve their decisions related to diversity, employee attrition and retention, talent acquisition, compensation, performance, skills, internal mobility, and more.
Solutions	Oracle Fusion SCM Analytics	Oracle Fusion SCM Analytics is a prebuilt solution for Oracle Cloud SCM, which helps supply chain professionals uncover underlying drivers to improve efficiency, reduce costs, and ensure customer satisfaction. SCM teams can enrich their analytics without coding using embedded ML and additional data from other sources beyond Oracle Cloud SCM.
Solutions	Oracle Analytics mobile app	The Oracle Analytics mobile app personalizes content for each employee by learning from their data interests and patterns. This enables intelligent recommendations and real-time alerts based on various triggers. It uses NLP technology to help employees query data verbally or listen to the results of their queries.
Solutions	Oracle Cloud Infrastructure Data Science	With Oracle Cloud Infrastructure Data Science, enterprises can quickly leverage predictive analytics to drive positive business outcomes. The platform is a part of Oracle's Data and AI Platform that enables data scientists to easily build, train, and manage ML models on Oracle Cloud using Python, JupyterLab, and open-source ML libraries.
Solutions	Oracle Data Mining (ODM)	ODM helps build and apply predictive models in Oracle Database for predicting customer behavior, targeting best customers, developing customer profiles, identifying cross-selling opportunities, and detecting anomalies and potential frauds. Oracle Data Miner creates predictive models that can be integrated with applications to automate the discovery and distribution of new BI predictions, patterns, and discoveries throughout the enterprise.

Source: Company Website

14.3.3 SERVICES OFFERED

TABLE 231 ORACLE: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
▪ Financial ▪ Services	Oracle financial services	It delivers world-class customer experiences that build loyalty and drive growth.
▪ Professional ▪ Services	Oracle Professional Services	It helps increase client satisfaction and improve operations that help drive growth.

Source: Company Website

14.3.4 RECENT DEVELOPMENTS

14.3.4.1 ORACLE: Product launches and enhancements

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Sept	Oracle	Solution	Oracle Exadata X9M platforms	Oracle announced the availability of the Oracle Exadata X9M platforms, the latest version of the industry's fastest and most affordable systems for running Oracle Database. It will lead to increase in analytic SQL throughput and ML workloads.
2020-March	Oracle	Solution	Therapeutic Learning System	Oracle built and donated COVID-19 Therapeutic Learning System to the US government. Therapeutic Learning System helps physicians record patients' daily progress.

Source: Company Website and Press Releases

TABLE 232 ORACLE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Nov	Partnership	Oracle	Samsung Securities	Samsung Securities selected Oracle Cloud Infrastructure (OCI) to underpin its digital innovation strategy. The company has secured the flexibility, reliability, and high-performance capabilities it required to effectively run advanced analytics in the rapidly changing financial market environment. It is now able to analyze the derivatives in near real-time, to maximize customer profits. Through its transition to the cloud, Samsung Securities is able to gain access to flexible, secure, and high-performance computing resources.
2021-Jun	Collaboration	Oracle	Wipro	Wipro collaborated with Oracle to launch Wipro Zero Cost Transformation, which would help organizations to migrate their workloads to Oracle Cloud Infrastructure (OCI) at a lower cost. Wipro is a member of the Oracle Partner Network (OPN).
2021-May	Agreement	Oracle	The Premier League	The Premier League has chosen Oracle as its official cloud provider. It provides a deeper understanding of the live action on the pitch. Oracle's data and analytics and ML technologies will deliver these groundbreaking statistics in real-time to a global audience of billions each season.

2021-Apr	Partnership	Oracle	SailGP	Oracle and SailGP, sailing's premier racing league, have expanded their successful data analytics platform leading into the second season. SailGP has used Oracle cloud infrastructure to deliver real-time data to each of the league's eight national teams as well as to broadcast partners and fans worldwide. New features added this season will improve team performance and give fans a thrilling crew-eye view of the action.

Source: Company Website and Press Releases

14.3.5 ORACLE: COVID-19 DEVELOPMENT

In response to the COVID-19 pandemic, in November 2020, cloud technology has been taken to Africa by the Tony Blair Institute (TBI) and Oracle to handle public health programs. As soon as the COVID-19 vaccine is supplied to Africa, Ghana, Rwanda, and Sierra Leone will use the new Oracle Health Management System to produce electronic health records for their vaccination programs for yellow fever, HPV, polio, measles, and COVID-19. More than thirty other countries in Africa, Asia, Europe, and North America are in discussion with TBI and Oracle about using the same cloud solution to run their COVID-19 immunization campaigns. In March 2020, Oracle built and donated the COVID-19 Therapeutic Learning System to the US government. The new system was developed with a unique collaboration of the National Institute of Health, Food and Drug Administration. Oracle is helping education institutions to provide better learning for students. Oracle offers free access to online learning content and certifications to a broad array of users on Oracle cloud infrastructure and Oracle autonomous database. The system can be accessed by various users, including developers, technical professionals, architects, students, and professors, who can have quick and easy access to more than 50 hours of online training and six certification exams.

14.3.6 MNM VIEW

14.3.6.1 Key strengths/right to win

Oracle Cloud software-as-a-service and infrastructure-as-a-service offerings provide opportunities for Oracle to expand its cloud and license business and market them to small and medium-sized businesses and the non-IT lines of business purchasers. It has strategically balanced product offerings which help it to reduce its business risks and enable it to tap the opportunities across digital platforms. It offers a broad portfolio of offerings to address the big data requirements of organizations. It also includes cloud-based services for data integration, data management, data science, and analytics and integrated ML.

14.3.6.2 Strategic choices made

Oracle has partnered and collaborated with high-tech companies. It is focused on increasing its business by partnering with industry leaders. In the last few years, it has partnered with various leading players. It aims at constantly expanding its portfolios of products and services by adopting different organic as well as inorganic growth strategies. As the world continues to navigate the risks and uncertainties associated with the COVID-19 pandemic, Oracle is committed to providing critical technologies, programs, and support to individuals and organizations to navigate, adjust, and continue their operations even in the pandemic situation.

14.3.6.3 Weaknesses and competitive threats

The company faces intense competition from players, such as IBM and Microsoft, in the advanced analytics market. The company faces challenge to cope with the rapidly changing product demand in a dynamic technological landscape. It has not been able to tackle the challenges presented by the new entrants in the segment and has lost major market share.

14.4 SAS INSTITUTE

14.4.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 233 SAS INSTITUTE: BUSINESS OVERVIEW

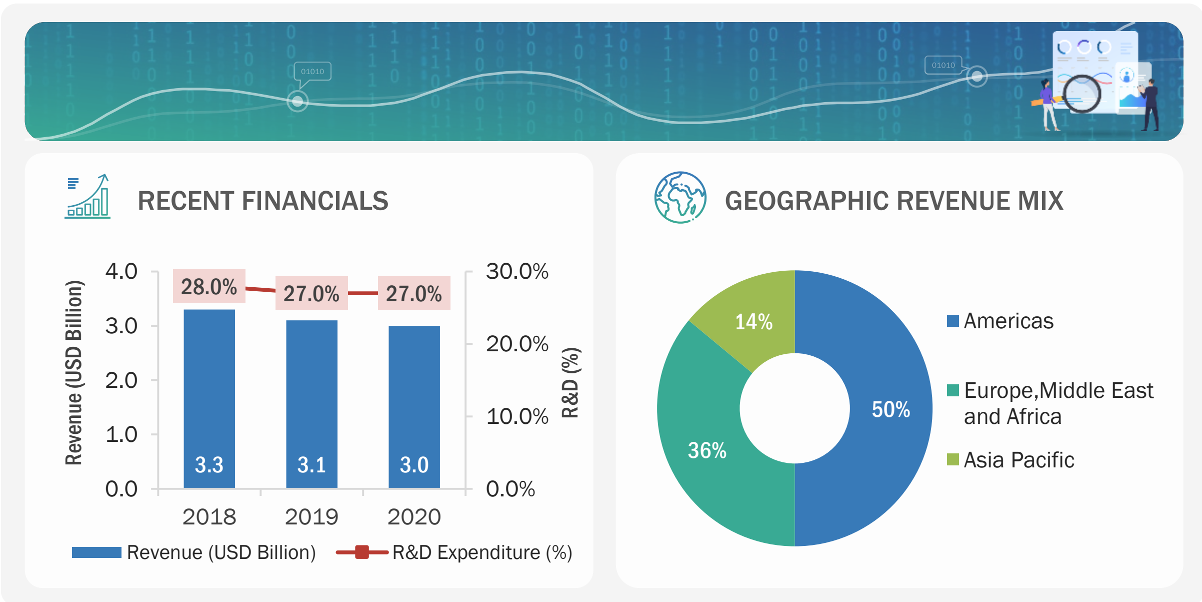
COMPANY AT A GLANCE-2021	
Founded	1976
Country	US
City	North Carolina
No. of Employees	13,939
Overall Annual Revenue	USD 3 Billion
R&D Expenditure	USD 0.8 billion
Ownership	Private

Source: Company Website, Annual Reports, and SEC Filings

SAS Institute is the largest independent vendor in the advanced analytics market. The main objective of SAS Institute is to meet the customer requirements of each industry and provide new innovations in analytics. The company caters to customers across industries, such as retail, BFSI, government, healthcare, technology, and media and telecom. It helps customers improve performance and deliver value by making better decisions faster.

In the advanced analytics market, the company offers Data Mining, OLAP, Business Analytics, Predictive Analytics Statistics Customer Intelligences, and many other solutions. The solution helps turn data into timely insights for taking better and faster decision-making.

FIGURE 61 SAS INSTITUTE: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on Dec 31

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.4.2 SOLUTIONS OFFERED

TABLE 234 SAS INSTITUTE: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION
Solutions	SAS Enterprise Miner	Streamline the data mining process to develop models quickly. Understand key relationships. Find the patterns that matter most.
Solutions	SAS Visual Data Mining and Machine Learning	It solves the most complex analytical problems with a single, integrated, collaborative solution along with its automated modeling API.
Solutions	SAS Visual Text Analytics	It helps find the information using NLP. It scales the human act of reading, organizing, and extracting useful information from huge volumes of textual data.
Solutions	SAS Visual Analytics	The solution helps visually explore all data, discover new patterns, and publish reports to the web and mobile devices. It enables building predictive models using the raw data and generate model outputs.
Solutions	SAS Predictive Analytics	This solution helps derive accurate insights at the right time to increase the reach and value of data. It helps make evidence-based decisions every day. The different functionalities of SAS Predictive Analytics include data preparation, visualization, descriptive and predictive analysis, model deployment, and governance and operationalization.

Source: Company Website

14.4.3 RECENT DEVELOPMENTS

14.4.3.1 SAS INSTITUTE: Product launches and enhancements

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-May	SAS INSTITUTE	Solution	SAS Viya	SAS will reinforce the foundation for data and analytics success by incorporating new data management solutions into its powerful, cloud native SAS Viya platform.

Source: Company Website

TABLE 235 SAS INSTITUTE: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Dec	Partnership	SAS INSTITUTE	PERSOWN	SAS is partnering with PERSOWN, to bring affordable, lifesaving technologies to the billions of traditionally underserved people of the world. From visualizing disease mutation and progression in the current COVID-19 pandemic to monitoring treatment efficacy and compliance. The partnership will empower global health leaders, clinicians, and researchers to make more informed decisions.
2021-Oct	Collaboration	SAS INSTITUTE	The Happy Mental Health	The Happy mental health platform has collaborated with SAS. This new collaboration allows The Happy Mental Health to harness the power of SAS Health Analytics to identify individuals more intelligently in the need of mental health support and target the most vulnerable populations for access to effective, low-cost care.
2021-Sept	Collaboration	SAS INSTITUTE	University of North Carolina	SAS and the University of North Carolina at Chapel Hill are collaborating to revolutionize the medication development process to avoid infectious disease threats such as COVID-19 from becoming pandemics. The collaboration is centered on the work of the University's Rapidly Emerging Antiviral Drug Development Initiative (READDI), which is developing broad-spectrum antiviral medications to keep on hand in case of future pandemics.
2021-Jan	Acquired	SAS	Boemska	SAS acquired Boemska for accelerating its AI integration into cloud marketplace and third-party applications.

14.4.4 SAS INSTITUTE: COVID-19 DEVELOPMENT

To counter COVID-19 disruptions and improve demand planning for organizations, SAS worked with organizations to offer a “quick start” 90-day fee-waived portfolio of supply chain analytics software and services. It delivers business results rapidly, for instance, the leading Russian online grocery chain Utkonos turned to SAS’ intelligent product substitution solution delivered through the quick start program in the Microsoft Azure cloud environment. Harnessing the flexibility, ease of use, and sophisticated depth of the demand planning solution, Utkonos, was able to power reliable and scalable demand forecasts and anticipates revenue and profit margin increases of up to 0.8% and order fulfillment increases of up to 75%. Recently, SAS has also leveraged into partnership with Happy Mental Health. This new collaboration allows Happy to harness the power of SAS Health Analytics to more intelligently identify individuals in the need of mental health support and target the most vulnerable populations for access to effective, low-cost care.

14.4.5 MNM VIEW

14.4.5.1 Key strengths/right to win

SAS Institute has adapted to the demand for analytics in the cloud by introducing SAS Viya 4 to become a cloud native company. The company helps customers operationalize their analytics and modernize cloud to achieve the full potential of their analytics and AI investments, and ultimately be successful in their digital transformation. SAS Platform is an engine that powers its analytics technology and handles the entire analytics life cycle, helping businesses operationalize analytics and get models into production faster.

14.4.5.2 Strategic choices made

SAS Institute is focusing on advanced AI technology, education, and services. A significant part of its annual revenue is reinvested in R&D to optimize the analytics life cycle for its customers. In 2019, SAS Institute announced AgTech, a new business unit, to help farmers secure a safe and sustainable food supply for the growing world population. The company has invested USD 1 billion in AI over three years through software innovation, education, and expert services.

14.4.5.3 Weaknesses and competitive threats

The company's revenue is highly dependent on the American market as the Americas contribute to 49% of its total revenue. As SAS Institute highly depends on partnerships and business expansions, its business and operations could be impacted by local legal, economic, political, and health conditions.

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14.5 SAP

14.5.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 236 SAP: BUSINESS OVERVIEW

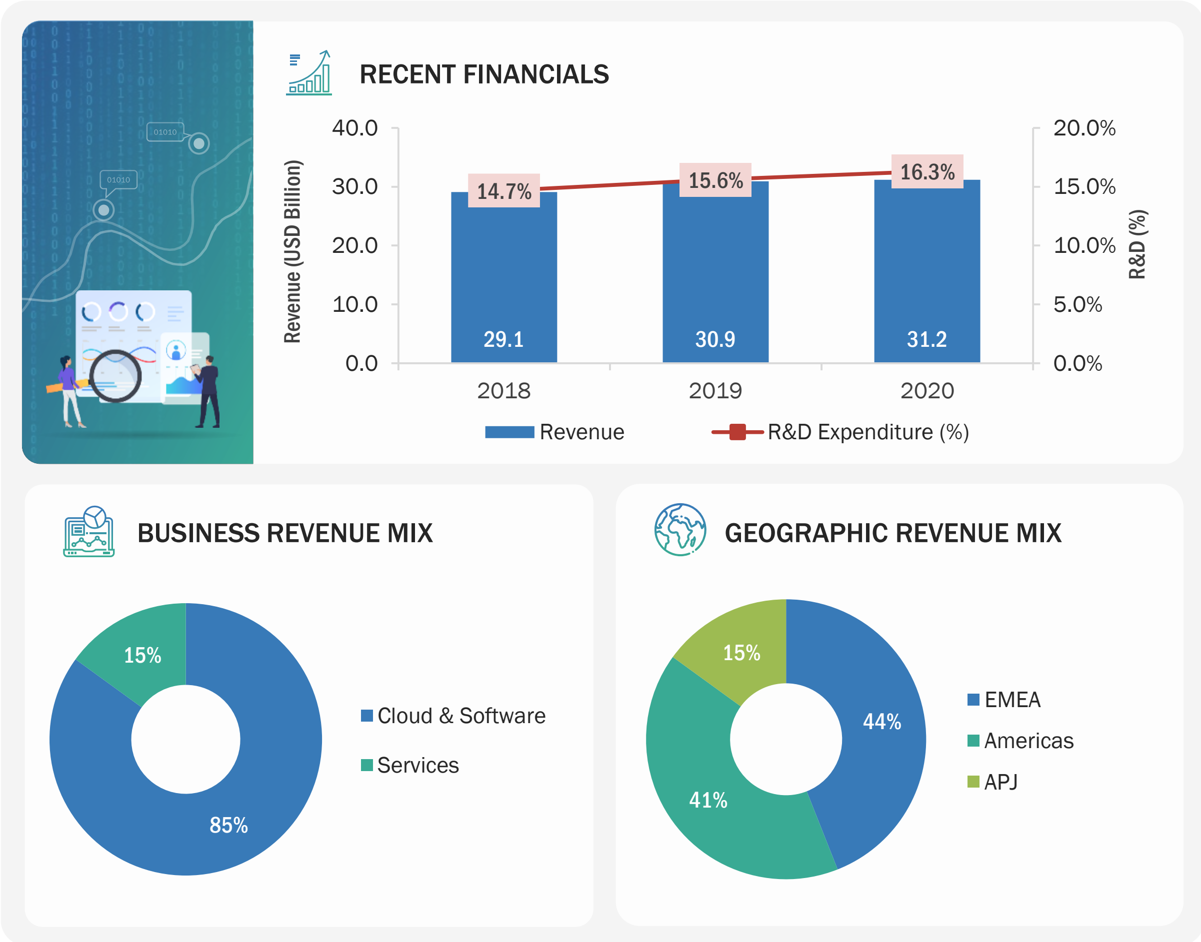
COMPANY AT A GLANCE-2021	
Founded	1972
Country	Germany
City	Walldorf
No. of Employees	102,430
Overall Annual Revenue	USD 31.2 Billion
R&D Expenditure	USD 5.1 Billion
Ownership	Public (NYSE: SAP)

Source: Company Website, Annual Reports, and SEC Filings

SAP is a leading provider of enterprise application software. The company's business areas are cloud and software and services. It specializes in analytics and BI services and has expertise in various technologies, including analytics, application platform and infrastructure, data management, IT management, and security software. SAP operates through segments, such as applications, technology, and support, Concur, Qualtrics, and services. It caters to various verticals that include energy and natural resources, financial services, consumer industries, discrete industries, service industries, and public services. The company is present across the Americas, EMEA, and APAC.

In the advanced analytics market, SAP offers the SAP Analytics Cloud, SAP Predictive Analytics, and many others. This solution inherits its data acquisition and data manipulation functionalities from SAP Lumira.

FIGURE 62 SAP: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on Dec 31

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.5.2 SOLUTIONS OFFERED

TABLE 237 SAP: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION
Solutions	SAP BusinessObjects	This solution helps make informed decisions with access to real-time insights on a single platform. It allows to expand BI, predictive analytics, and planning capabilities with SAP Analytics Cloud.
Solutions	SAP Predictive Analytics	It is used for statistical analytics, data mining, and predictive analytics. It enables building predictive models for discovering hidden insights and relationships in data to make accurate predictions about future occurrences. It helps gain unprecedented insights for customer acquisition through cross-sell, up-sell, and churn prevention. It takes the best action for every interaction across customer channels.
Solutions	SAP Analytics Cloud	The SAP Analytics Cloud solution combines BI, augmented, and predictive analytics, and planning capabilities into one cloud environment. As the analytics layer of SAP's Business Technology Platform, it supports advanced analytics enterprise-wide.
Solutions	SAP HANA Streaming Analytics	SAP HANA streaming analytics is used for instances where data is generated as events occur and there is a benefit in gathering, analyzing, and acting on this data as soon as possible. Some of the data sources that generate real-time streams of events include sensors, smart devices, web pages, IT systems, and social media platforms.

Source: Company Website

14.5.3 SERVICES OFFERED

TABLE 238 SAP: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
▪ Professional ▪ Service	▪ Essential Business Services ▪ Innovation Services and Solutions ▪ Advisory Services ▪ Implementation Services ▪ Cloud Services ▪ Support Services ▪ Premium Engagements ▪ SAP HANA Services	It defines and supports the best communications with help from skilled professionals. Assess and plan solutions, map the customer journey, analyze performance, plan optimization, and even augment the staff.

Source: Company Website

14.5.4 RECENT DEVELOPMENTS

TABLE 239 SAP: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Sept	SAP	Solution	SAP Analytics Cloud	SAP released a new version of SAP Analytics Cloud 2021.20. It updates new dashboard and analytics platform services to enhance user experience.
2021-Aug	SAP	Solution	SAP Asset Intelligence Network	SAP added contextual geographical information of the work order and operations in a map view. It would allow maintenance of geospatial coordinates for an owned work order header and operation.
2021-Jun	SAP	Software	Corona Warn App	SAP launched the German government's Corona Warn App. The app was developed in collaboration between SAP and Deutsche Telekom as well as other partners. It was developed in an open-source mode, and the program code was continuously visible to the public on the development platform, GitHub.

TABLE 240 SAP: DEALS AND OTHERS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Aug	Partnership	SAP	Microsoft	SAP India and Microsoft launched a joint skilling program, TechSaksham, for empowering young women students from underserved communities to build careers in technology. Through this joint initiative, SAP India and Microsoft will skill 62,000 women students in AI, cloud computing, web design, and digital marketing.
2021-Apr	Partnership	SAP	ESRI	ESRI extended its partnership with SAP by announcing the certification and support for SAP's DaaS offering and SAP HANA Cloud. With this capability, companies can accelerate customer adoption of cloud-based location intelligence platforms with a high-performance environment.
2020-Aug	Strategic Partnership	SAP	HPE	SAP partnered with HPE to deliver SAP HANA Enterprise Cloud with HPE GreenLake Cloud Services. This partnership would enable customers to keep their SAP software landscape and data on-premises.

Source: Press Releases

14.5.5 SAP: COVID-19 DEVELOPMENT

SAP established a USD 3.6 million COVID-19 Emergency Fund to support the needs of the World Health Organization (WHO), the CDC Foundation, and smaller nonprofits and social enterprises that work on the front lines to support local communities. SAP CSR partnered with more than 1,200 nonprofits and social enterprises that are mostly small to mid-sized organizations in the areas, such as quality education and youth entrepreneurship, to help them shift to remote operations effectively. The company established a global pandemic task force to work with local crisis teams and health authorities, coordinate all efforts around the COVID-19 situation, and provide guidance to the company's staff, partners, and suppliers. The company's business, support, and cloud delivery teams developed business continuity plans to provide seamless services to its customers.

14.5.6 MNM VIEW

14.5.6.1 Key strengths/right to win

SAP helps companies to enhance feedback with analytics to offer end-to-end customer experiences. Its strategy is to be an experienced company powered by Intelligent Enterprise through Intelligent Suite of Line of Business (LoB) applications, Business Technology Platform, and the Experience Management (XM) platform. Its business analytics tools help customers get insights, enabling them to adapt businesses in real time. SAP's cloud and software revenue includes cloud revenue and software licenses and support revenue. Cloud revenue and software revenue are the key revenue drivers of SAP as they affect the other revenue streams of SAP.

14.5.6.2 Strategic choices made

To expand its business, the company is focusing on acquiring businesses, products, and technologies. The company is focusing on organic investments in technology and innovations for ensuring the sustainable growth of its product portfolio. SAP supports the entrepreneurs through venture capital funds managed by Sapphire Ventures, which manages over USD 3.5 billion and has invested in more than 160 countries.

14.5.6.3 Weaknesses and competitive threats

As SAP operates globally in more than 180 countries and territories, there are numerous risks subject to international business operations. Uncertainty in the global economy, financial markets, social and political instability caused by state-based conflicts, terrorist attacks, and war could lead to SAP's business disruption and impose a negative impact on the business, financial position, profit, and cash flows of the company. As SAP expands into new countries and markets, including emerging and high-risk markets, risks may intensify due to the application of respective local laws and regulations.

14.6 FICO

14.6.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 241 FICO: BUSINESS OVERVIEW

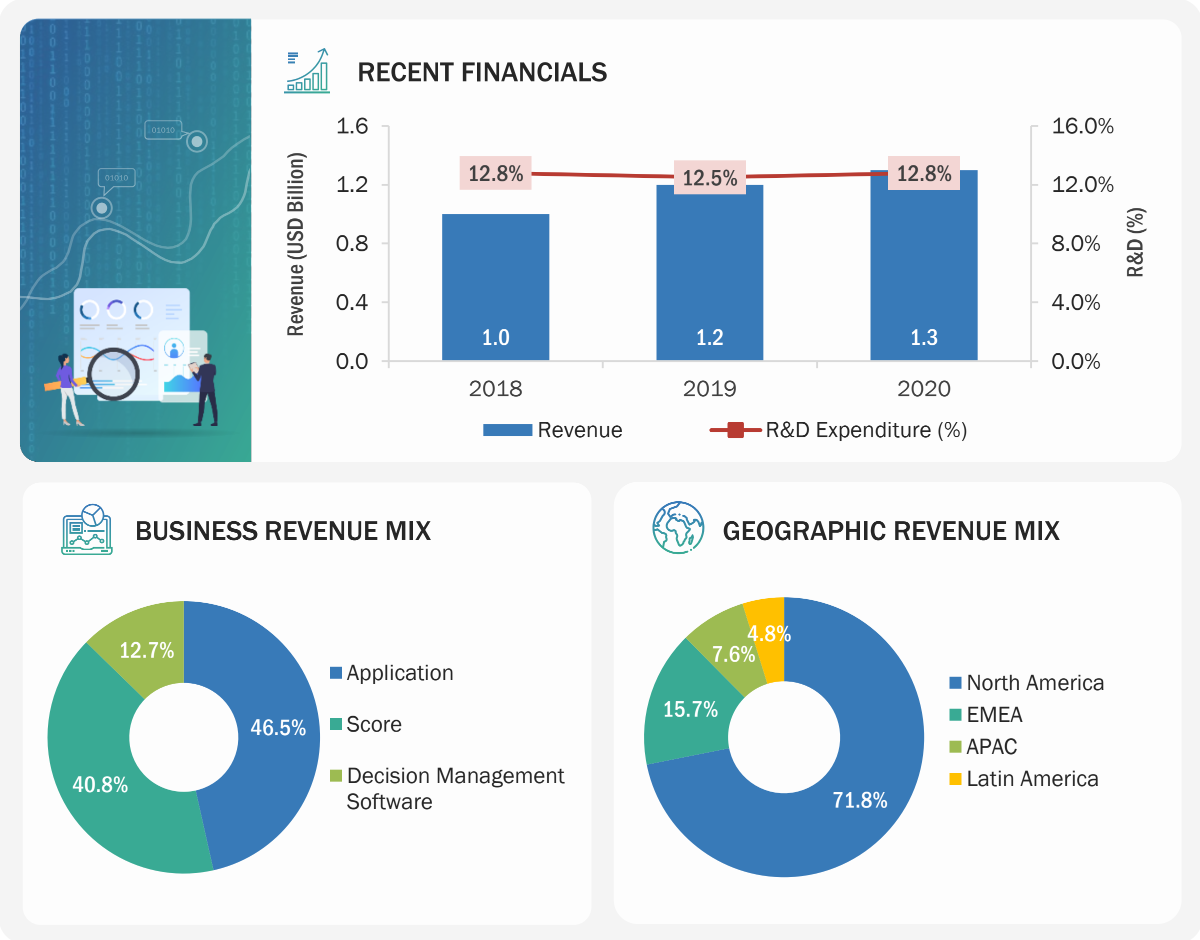
COMPANY AT A GLANCE-2021	
Founded	1956
Country	US
City	California
No. of Employees	4,000+
Overall Annual Revenue	USD 1.3 billion
R&D Expenditure	USD 0.2 billion
Ownership	Public (NYSE: FICO)

Source: Company Website, Annual Reports, and SEC Filings

Fair Issac Corporation (FICO) is a leading analytics software company. The company's regional offices are in San Rafael, San Diego, and New York City (US). It is spread across London (UK), Birmingham (UK), Moscow (Russia), Bangalore (India), Beijing (China), and Singapore. The company operates in more than 90 countries, helping client organizations make better decisions and attaining a higher level of growth, profits, and customer satisfaction. It offers various products, including Fico Debt Manager Solution, Fico Falcon Fraud Manager, Fico Engagement Analyzer, Fico Blaze Advisor Decision Rules Management Systems, Fico Score, Fixo Decision Management Suite, Fico Score Open Access, and Fico Xpress Optimization Suite. FICO caters to various industries, including financial services, health care, insurance, automotive, public sector, retail, pharmaceuticals, telecommunications, travel and hospitality, media and entertainment, and high tech and utilities.

In the advanced analytics market, FICO offers FICO Predictive Analytics and FICO Analytics Workbench. These solutions are used across industries to manage risks, fight frauds, build more profitable customer relationships, and for many other uses. Its analytics solutions such as credit scoring have made credit more widely available.

FIGURE 63 FICO: COMPANY SNAPSHOT



Note 1: The company's fiscal year ends on December 31

Note 2: Recent financials, business revenue mix, and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.6.2 SOLUTIONS OFFERED

TABLE 242 FICO: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION
Solutions	Customer Analytics	It helps gain deep customer insights and mathematically optimize customer interactions. Organizations need to operationalize advanced analytics and ML to deliver seamless customer experiences and optimal business outcomes.
Solutions	FICO Predictive Analytics	FICO Predictive Analytics helps develop the best predictive models for business needs. Some of the predictive modeling techniques and approaches used by FICO are FICO Scorecard Technology, Neural Networks, Time-to-Event Models, Action Effect/Uplift Models, and Economic Impact Models.

Solutions	FICO Analytics Workbench	It is an integrated suite of state-of-the-art advanced analytics capabilities that enable companies to improve business decisions across the customer life cycle. It offers decision-making capabilities using tools such as AI and ML. The platform is built on the leading FICO Decision Management Platform to allow new predictive models and strategies to be deployed into production with ease.

Source: Company Website

14.6.3 RECENT DEVELOPMENTS

TABLE 243 FICO: PRODUCT LAUNCHES AND ENHANCEMENTS

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Dec	Fico	Solution	BSafe Platform SaaS	FICO and Sistemas Críticos announced the launch of BSafe Platform SaaS, which combines the fraud prevention technologies of FICO Falcon Fraud Manager and TenS a proprietary Sistemas Críticos platform, on the Cloud for any type of company/institution, notwithstanding of size and industry. It combines with predictive analytics and AI to offer comprehensive fraud prevention, security, and compliance management which focus on the customer lifecycle.
2021-Nov	Fico	Solution	Fico Origination Solution	Fico announced the launch of the Fico Origination Solution Powered by the Fico platform. The solution is powered by the industry-leading FICO platform and delivered globally on Amazon Web Services (AWS) Cloud. The solution empowers businesses to deliver exceptional, personalized customer experiences by automating a frictionless customer journey throughout the loan origination process.

Source: Company Website

TABLE 244 **FICO: DEALS**

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Nov	Leverage	FICO	Vodafone	Vodafone to leverage FICO's Expertise in tackling fraud risk in digital channels. FICO Advisors and the consultancy team will assess Vodafone's operations across 19 countries and advise on tackling new fraud risks at a country level.
2021-Sept	Partnered	FICO	FinTecSystems	FICO partnered with FinTecSystems (FTS) to optimize decisioning across the customer life cycle. The initial focus of the partnership is on combining the FTS technology with the market-leading decision platform of FICO. The FICO Platform provides the ideal decisioning foundation the companies need to successfully achieve digital transformation. It provides unprecedented insights into customers' immediate and future needs by eliminating data silos and enabling interoperability between enterprise applications.
2020-Jun	Collaboration	FICO	FIS	FIS Teams collaborated with FICO. The two organizations collaborated to build an advanced anti-money laundering solution for the US financial institutions. Following this collaboration, FIS AML Compliance Manager was integrated with the FICO Falcon X technology to help financial institutions keep ahead of financial crimes. The collaboration would help North American financial institutions keep ahead of the increasingly sophisticated money launderers and other financial criminals.

Source: Press Releases

14.6.4 FICO: COVID-19 DEVELOPMENT

As a result of the COVID-19 pandemic, it had temporarily closed the majority of our offices and implemented travel restrictions, both of which have disrupted how it operates its business. Its operational flexibility and financial stability allowed to successfully manage the initial impact of COVID-19 while protecting its cash flow and liquidity. However, certain areas of the business have been adversely impacted as a result of the pandemic's global economic impact. For example, COVID-19 has been adversely affecting certain purchasing decisions made by our customers in the applications and decision management software segments.

14.6.5 MNM VIEW

14.6.5.1 Key strengths/right to win

FICO is the world's leading provider of analytics software, solutions, and services that transform the way organizations understand their customers and optimize business processes around them. It empowers businesses across multiple industries to use insights gained through data to make complex, critical decisions at scale with precisions and confidence.

14.6.5.2 Strategic choices made

Fico applied innovative data science with AI to solve the complex business problem of upgrading decision-making and driving the outcome. Its analytics use available data to produce the best decisions possible. It has proven science leverages that use the latest technology to drive consumer behavior insights and optimize business processes. It focuses on becoming the pre-eminent platform player in decisioning analytics solutions.

14.6.5.3 Weaknesses and competitive threats

Even though Fico is one of the leading organizations. It has faced challenges in moving to other product segments with its present culture. Fico is heavily spending on R&D but has not been able to compete with the leading players in the industry in terms of innovation. It has been lacking in the proper financial planning system also few products have large market share while others have low that led to its dependence more on a large share market product.

14.7 KNIME

14.7.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 245 KNIME: BUSINESS OVERVIEW

COMPANY AT A GLANCE - 2021	
Founded	2008
Country	Europe
City	Zurich
No. of Employees	200
Ownership	Private

Source: Company Website, Annual Reports, and SEC Filings

KNIME Software bridges the worlds of dashboards and advanced analytics through an intuitive interface, appropriate for anybody working with data. It empowers more business experts to be self-sufficient and more data experts to push the business to the bleeding edge of modern data science, integrating the latest AI and ML techniques. It is distinct in its open approach, which ensures easy adoption and future-proof access to new technologies.

In the advanced analytics market, it offers specialties in data mining, BI, data analysis, data integration, predictive models, statistics, workflow tool, reporting, data science, ML, and data analytics.

14.7.2 SOLUTIONS OFFERED

TABLE 246 KNIME: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION
Solutions	KNIME Analytics Platform	KNIME Analytics Platform is the open-source software for creating data science. It is continuously integrating new developments, KNIME makes understanding of data and designing data science workflows and reusable components accessible to everyone.

Source: Company Website

14.7.3 RECENT DEVELOPMENTS

14.7.3.1 KNIME: Product launches and enhancements

YEAR-MONTH	COMPANY NAME	PRODUCT TYPE	PRODUCT NAME	PRODUCT DESCRIPTION
2021-Jan	KNIME	Solution	Knime edge	Knime announced the launch of Knime edge it will help the data and IT teams as it provides inference workflows as a service. Applying models at the edge is a technical solution for several requirements, typically saving IT teams on costs, minimizing latency, increasing reliability, and enabling the deployment of inference services in multiple geographic regions.

14.8 MICROSOFT

14.8.1 BUSINESS AND FINANCIAL OVERVIEW

TABLE 247 MICROSOFT: BUSINESS OVERVIEW

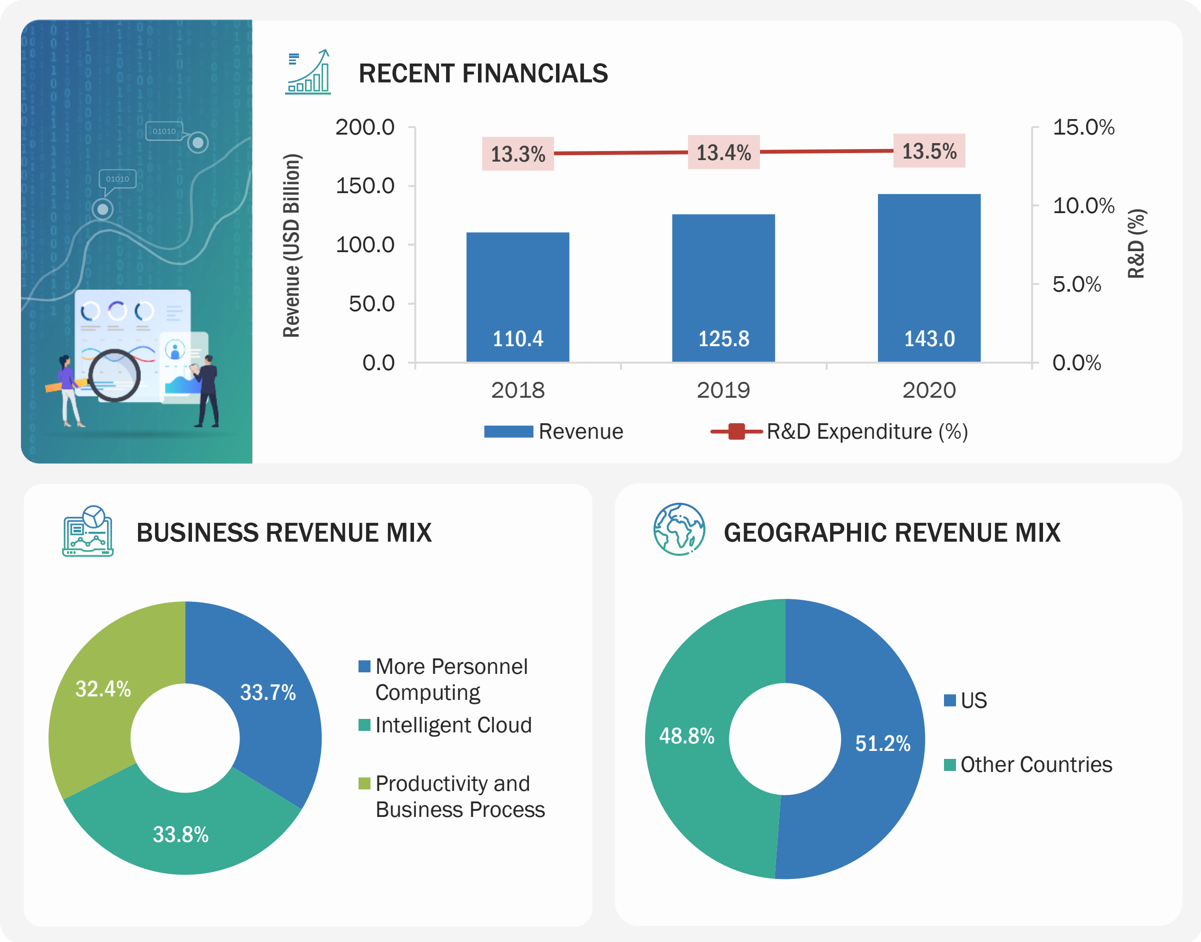
COMPANY AT A GLANCE-2021	
Founded	1975
Country	US
City	Washington
No. of Employees	18,100+
Overall Annual Revenue	USD 143.0 billion
R&D Expenditure	USD 19.3 billion
Ownership	Public (NASDAQ: MSFT)

Source: Company Website, Annual Reports, and SEC Filings

Microsoft develops software, services, devices, and solutions to compete in the era of an intelligent cloud and an intelligent edge. With the increase in investments in the mix-reality cloud, Microsoft enables its customers to digitalize their business processes. The company's offerings include cloud-based solutions that provide customers with software, platforms, and content. Its product offerings include Operating Systems (OS), cross-device productivity applications, server applications, business solution applications, desktop and server management tools, software development tools, and video games. Microsoft's platforms and tools help drive the productivity of small businesses, the competitiveness of large businesses, and the efficiency of the public sector. The company works on three pillars, which include privacy, cybersecurity, and responsible AI, to provide tools and frameworks for its customers to encourage policy change. It focuses on investing in data centers and other hybrid and edge infrastructure to enhance its service offerings. Microsoft operates in three segments: productivity and business processes, intelligent cloud, and more personal computing. The productivity and business processes offer dynamic business solutions, including Dynamics 365, a set of cloud-based applications across ERP and CRM; Dynamics ERP on-premises; and Dynamics CRM on-premises. It operates in more than 190 countries across North America, Europe, APAC, MEA, and Latin America.

In the Advanced analytics market, the company offers Microsoft offers Azure ML, Azure Stream Analytics, Azure Synapse Analytics, Azure Analysis Services, and many others for various use as for instance Azure Synapse Analytics is used to bring together data integration, enterprise data warehousing and big data analytics.

FIGURE 64 MICROSOFT: COMPANY SNAPSHOT



Note: The recent financials and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.8.2 SOLUTIONS OFFERED

TABLE 248 MICROSOFT: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION	DESCRIPTION
Solution	Azure ML	It offers end-to-end life cycle support for ML. It contains tools that help rapidly build and deploy ML models. It is a cloud-based service that enables detecting patterns while processing a huge amount of data to predict occurrences while processing new data. It gives built-in support for open-source tools and frameworks. These tools and frameworks allow ML model training and inferencing.

Source: Company Website

14.8.3 SERVICES OFFERED

TABLE 249 MICROSOFT: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
SERVICES	Azure Stream Analytics	Azure stream analytics, the easy-to-use, real-time analytics service designed for mission-critical workloads, is built upon an end-to-end serverless streaming pipeline.
Services	Azure synapse analytics	Azure synapse analytics is a limitless analytics service that brings together data integration, enterprise data warehousing, and big data analytics. It helps transform, manage, and serve data for immediate BI and ML needs.
Services	Azure Analysis Services	Azure analysis services help combine data from multiple sources into a single, trusted BI semantic model, which is easy to understand and use. It also reduces time-to-insights on large and complex datasets.

14.8.4 RECENT DEVELOPMENTS

TABLE 250 MICROSOFT: SOLUTION LAUNCHES AND ENHANCEMENTS

DATE	COMPANY NAME	SOLUTION NAME	DESCRIPTION
2021-Nov	Microsoft	Azure Stream Analytics	Azure Stream Analytics operation can directly feed data into an Azure Data Explorer table using this new integration. The direct integration of Azure Data Explorer as an output for Azure Stream Analytics has increased the power of Azure-powered real-time analytics architecture.
2019-Aug	Microsoft	Azure Synapse Analytics	Azure stream analytics adds MATCH_RECOGNIZE, which would significantly reduce the complexity and cost associated with building, modifying, and maintaining queries that match a sequence of events for alerts or further data computation.

Source: Press Releases

TABLE 251 MICROSOFT: DEALS AND OTHERS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Dec	Collaboration	Microsoft	CVS Health	CVS Health and Microsoft announced a new strategic alliance to reimagine personalized care and accelerate digital transformation. It focuses on developing innovative solutions to help consumers improve their health.
2021-Sept	Partnership	Microsoft	Truveta	Microsoft and Truveta announced a strategic partnership to apply the power of Microsoft Azure and AI to help Truveta achieve its vision of saving lives with data. It will create the world's largest identified health data platform.

2021-Jul	Partnership	Microsoft	Qlik	With a scalable, high-performance, real-time data integration and automation solution, Qlik data integration would help Microsoft clients speed analytics and ML initiatives. Thousands of Microsoft clients have used Qlik to transfer their data to Azure fast and intelligently. With automated data pipelines, Qlik extends beyond migration tools to help Microsoft customers achieve agility throughout the data and analytics process. Qlik and Microsoft Azure work together to improve the accessibility of data and related insights that are now locked away in legacy systems, warehouses, or corporate apps.
2021-May	Partnership	Microsoft	KX	The partnership aims to launch KX Insights, a cloud-first platform for streaming analytics that leverages the benefits of cloud architecture natively to deliver fast, scalable real-time data insights via the Microsoft Azure Marketplace.

Source: Press Releases

14.9 ALTAIR

14.9.1 BUSINESS OVERVIEW

TABLE 252 ALTAIR: BUSINESS OVERVIEW

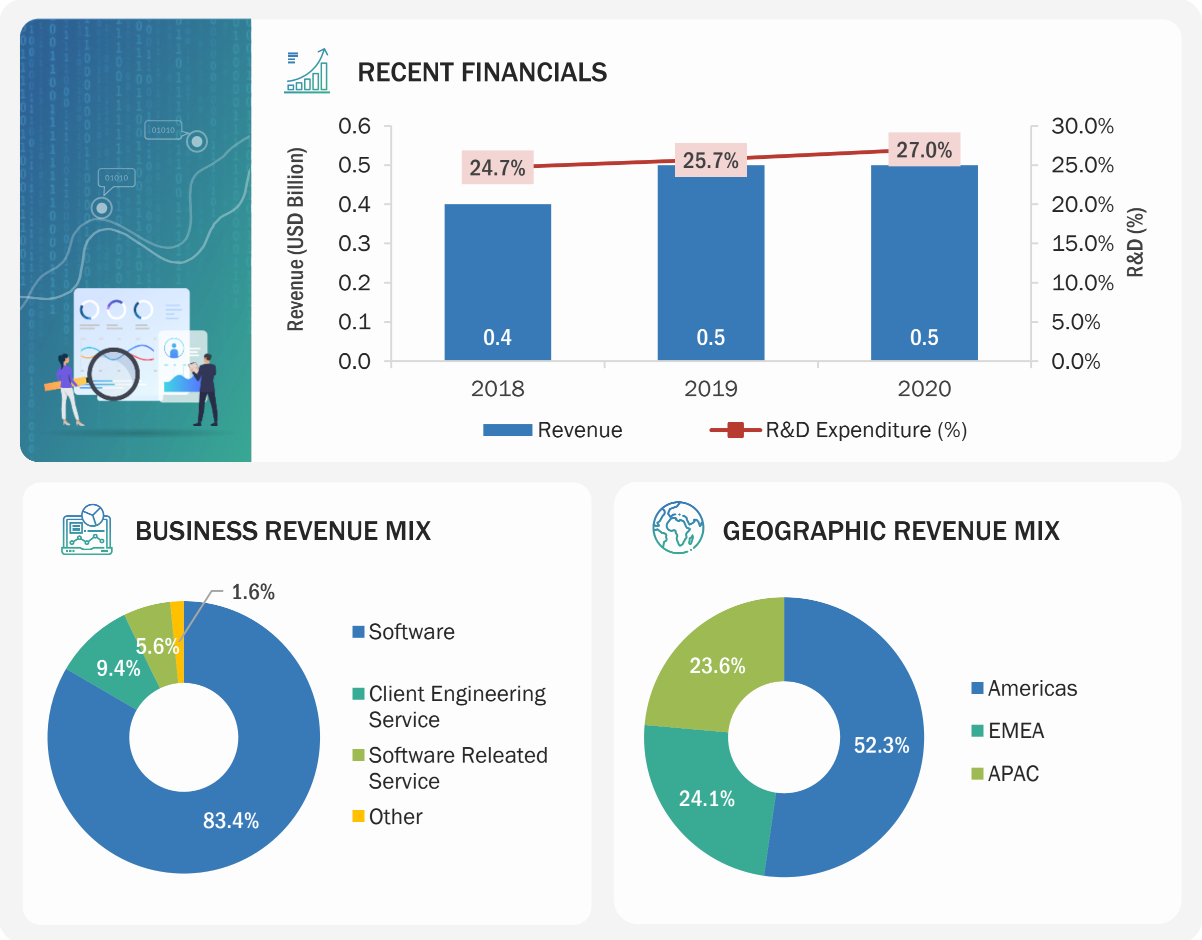
COMPANY AT A GLANCE – 2021	
Founded	1985
Country	US
City	Michigan
No. of Employees	3,300+
Overall Annual Revenue	USD 0.5 Billion
R&D Expenditure	USD 0.1 Billion
Ownership	Public (NYSE: ALTR)

Source: Company Website

Altair is a global technology company, which provides software and cloud solutions in the areas of simulation, High-Performance Computing (HPC), and AI. It enables organizations in nearly every industry to compete more effectively in a connected world while creating a more sustainable future. Its vision is to help customers drive decisions leveraging the convergence of simulation, HPC, and AI. It has helped companies in their digital transformation efforts to make smarter decisions, save costs, reduce waste, accelerate time-to-market, improve business performance, and evolve to a data-driven culture.

In the advanced analytics market, it offers engineering analytics solutions, data science, and data analytics. Altair Knowledge Works is a data automation and ML platform comprising Altair Monarch, Altair Knowledge Hub, and many others that can help users connect with data sources and create dashboards.

FIGURE 65 ALTAIR: COMPANY SNAPSHOT



Note: The recent financials and geographic revenue mix are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.9.2 SOLUTIONS OFFERED

TABLE 253 ALTAIR: SOLUTIONS OFFERED

PRODUCT TYPE	SOLUTION NAME	DESCRIPTION
Solution	Altair One	Altair One offers dynamic, collaborative access to simulation and data analytics technologies with scalable HPC and cloud resources on a single platform.
Solution	Altair Data Analytics	Altair Data Analytics enables organizations to operationalize data analytics and AI with secure, governed, and scalable strategies.
Solution	Altair SmartSight	Altair SmartSight is a self-service BI platform that helps unlock data insights. It is deployable on both on-premises and cloud.

Source: Company Website

TABLE 254 ALTAIR: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
Consulting Service	Professional service	It helps select, deploy, and manage solutions that fit into the organization.
Services	Industrial Design Services	It is used for design research, product, and brand strategy.

14.9.3 RECENT DEVELOPMENTS

TABLE 255 ALTAIR: SOLUTION LAUNCHES AND ENHANCEMENTS

DATE	NATURE OF DEVELOPMENT	SOLUTION NAME	DESCRIPTION
2021-Oct	Launched	Toggled iQ Smart	Toggled, a wholly owned subsidiary of Altair, focused on smart building management solutions and lighting technology. It announced the roll-out of its comprehensive connected building data and device management solution, Toggled iQ. Leveraging IoT, Toggled iQ helps businesses optimize their building environments by streamlining operations, improving energy efficiency, saving money, and reducing their carbon footprint, through the connection of smart connected lighting with wireless sensors, controls, and intuitive analytics.
2021-Oct	Enhancement	Integrated Simulation and Analysis Portfolio	Altair announced significant enhancements to its integrated portfolio of simulation and design tools. This update to Altair's simulation software suite is focused on accelerating simulation-driven design and unleashing the power of AI by embedding augmented intelligence in the broadest possible range of Computer-Aided Engineering (CAE) tools.

Source: Press Releases

TABLE 256 ALTAIR: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Sept	Partnership	Altair	TrueInsight	Altair partner with TrueInsight. It will exclusively offer Altair's simulation, data analytics, and AI software solutions. With deep roots in structural, manufacturing, and electromagnetics analysis, TrueInsight will manage middle-market customer relationships end-to-end, from strategic sales to robust technical support.
2021-Aug	Acquisition	Altair	S-FRAME Software	Altair acquired S-FRAME Software, powerful structural analysis, and design software to strengthen and accelerate global footprint in Architecture, Engineering, and Construction (AEC). It will utilizing S-FRAME Software's finite element structural analysis and code support. Moreover, the acquisition will allow architects and civil engineers to be more innovative and bring their visions to life while adhering to local code requirements.
2020-Dec	Collaborated	AWS	Novartis	AWS collaborated with Novartis to accelerate the digital transformation of its business operations. This collaboration enables Novartis to reimagine its core pharmaceutical manufacturing, supply chain, and delivery operations using AWS' portfolio of cloud services.

Source: Press Releases

14.10 RAPIDMINER

14.10.1 BUSINESS OVERVIEW

TABLE 257 RAPIDMINER: BUSINESS OVERVIEW

COMPANY AT A GLANCE – 2021	
Founded	2007
Country	Massachusetts, US
Ownership	Private

Source: Company Website

RapidMiner, which was formerly known yet Another Learning Environment (YALE), is an open data science platform for analytics. In 2013, it rebranded itself to RapidMiner. RapidMiner unites the life cycle of data science, from preparation of data to ML and the deployment of predictive models that offer an end-to-end collaboration platform. It works on all the stages of Automated Data Science, from exploring data to modeling data, using RapidMiner Auto Model that is integrated with Turbo Prep and Model Ops. It offers different products in the field of Automated Data Science and managed as well as on-demand AI cloud. It also offers various platforms, including Studio, Server, Real-Time Scoring, and Radoop. The customers of RapidMiner are BMW, CISCO, ezCater, Hewlett Packard Enterprise, Sanofi, and Transport for London. RapidMiner mission is to be reinventing in enterprise AI so that anyone has the power to positively shape the future. It is doing this by enabling ‘data loving’ people of all skill levels, across the enterprise, to rapidly create and operate AI solutions to drive immediate business impact. It offers an end-to-end platform that unifies data prep, ML, and model operations with a user experience that provides depth for data scientists and simplifies complex tasks for everyone else.

In the advanced analytics market, RapidMiner offers Data Mining, Text Mining, Predictive Analytics, Sentiment Analysis, Business Analytics, BI, ETL, Data Integration, Data Science, Customer Insight, Competitive Intelligence, and many other.

14.10.2 SOLUTIONS OFFERED

TABLE 258 RAPIDMINER: SOLUTIONS OFFERED

SOLUTION TYPE	SOLUTION NAME	DESCRIPTION
Solution	RapidMiner Turbo Prep	RapidMiner Turbo Prep helps prepare the data for predictive modeling. Predictive models are prepared using RapidMiner Studio and RapidMiner Auto Model. RapidMiner Auto Model uses automated ML and best practices to build predictive models in five mouse clicks.
Solution	RapidMiner AI Cloud	RapidMiner AI Cloud enables employees across the enterprise to rapidly create and operate AI solutions and drive business impact. It provides high processing power that leverages the predictive power of big data.
Solution	RapidMiner AI Hub	RapidMiner AI Hub connects AI to people, process, and value. It also connects enterprise-wide collaboration, decision automation, deployment, and control. It provides analytics for anyone as the RapidMiner Go is included in the AI Hub.
Solution	RapidMiner Radoop	RapidMiner Radoop enables building and running predictive models in Hadoop using the RapidMiner Studio visual workflow designer, without having to code in Spark.

Source: Company Website

14.10.3 RECENT DEVELOPMENTS

TABLE 259 RAPIDMINER: SOLUTION LAUNCHES AND ENHANCEMENTS

DATE	NATURE OF DEVELOPMENT	SOLUTION NAME	DESCRIPTION
2020-Jun	Product Launch	RapidMiner AI Hub	RapidMiner AI Hub it is built to connect AI to people, processes, systems, and value to tear down silos and make enterprise AI successful, wide-spread and sustainable. The RapidMiner AI Hub allows people with different tooling preferences and skillsets to collaborate on AI projects within any organization. Creating a shared infrastructure such as this improves costs, agility, data access, transparency, security, and compliance.
2020-Feb	Product Enhancement	RapidMiner 9.6	RapidMiner announced the release of its platform enhancement, RapidMiner 9.6. It provides new, unique experiences to empower users of varying backgrounds and abilities. The RapidMiner 9.6 platform will have a broader array of users, enabling collaboration between coders, non-coding data scientists, and business users on a platform seamlessly combining automated data science, visual workflows, and coding as needed or preferred.

TABLE 260 RAPIDMINER: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Oct	Partnership	RapidMiner	Braincube	RapidMiner and Braincube has a strategic partnership to improve the Overall Analytical Effectiveness (OAE) in manufacturing. It focuses on reducing data complexity through streamlined data centralization and structuration and helps in improving data literacy. It helps utilize advanced analytics techniques to identify anomalies, predict manufacturing outcomes, and optimize production through prescriptive modeling.
2021-Feb	Partnership	RapidMiner	Hivecell	RapidMiner announced a partnership with Hivecell the Edge-as-a-Service company, to allow users to rapidly create and operate models with streaming data from the edge. With this integration, the RapidMiner platform can be easily integrated into the Hivecell architecture, allowing Hivecell users to pool together data and build models centrally.

Source: Press Releases

14.11 AWS

14.11.1 BUSINESS OVERVIEW

TABLE 261 AWS: BUSINESS OVERVIEW

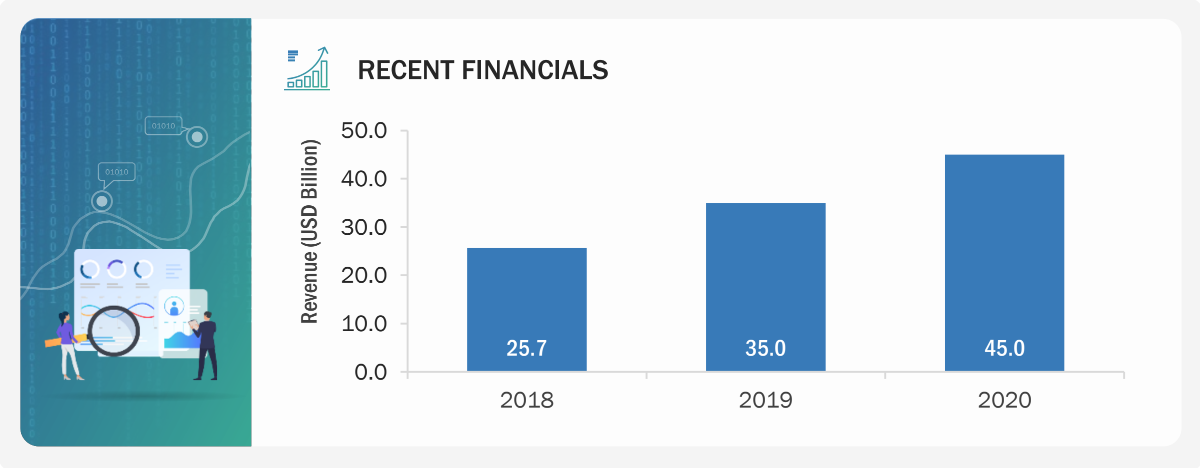
COMPANY AT A GLANCE – 2021	
Founded	2006
Country	US
City	Washington
No. of Employees	105,850
Overall Annual Revenue	USD 45 billion
R&D Expenditure	USD 12 billion
Ownership	Subsidiary (Amazon)

Source: Company Website

The company is a subsidiary of Amazon and primarily offers cloud computing services in the form of web services. It offers a wide range of products and services to customers present in 190 countries. The company offers a wide range of products in the fields of analytics, application integration, Augmented Reality (AR) and VR, IoT, blockchain, ML, robotics, satellite, quantum technologies, security, and many other fields. AWS caters to various solutions in applications, such as blockchain, cloud migration, analytics, and data lake, enterprise application, end-user computing, media services, and SaaS applications. It caters to various verticals, such as media and entertainment, healthcare, government, education, and utilities. With AWS, businesses can take advantage of Amazon's expertise and economies of scale to access resources when their business needs them and deliver the results faster at a lower cost. AWS provides a highly reliable, scalable, low-cost infrastructure platform in the cloud with data center locations in the US, Europe, Singapore, and Japan, customers across all industries are taking advantage of the low cost, elastic, open and flexible, secure platform.

In the advanced analytics market, AWS offers Amazon Athena, Amazon Kinesis, and many others. For instance, Amazon SageMaker is a cloud ML platform that enables developers to deploy ML models on embedded systems and edge devices. The platform removes heavy lifting from each step of the ML process to make it easier to develop high-quality models.

FIGURE 66 AWS: COMPANY SNAPSHOT



Note: The recent financials are given for the overall company

Source: Company Website, Annual Reports, and SEC Filings

14.11.2 SOLUTION OFFERED

TABLE 262 AWS: SOLUTION OFFERED

PRODUCT TYPE	SOLUTION NAME	DESCRIPTION
Solution	Amazon Athena	Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 using standard SQL. Athena is serverless so there is no infrastructure to manage and need to pay only for those queries which we run.
Solution	Amazon Quicksight	Amazon Quicksight allows users in the organization to understand the data by asking questions in natural language, exploring through interactive dashboards, or automatically looking for patterns and outliers powered by ML.
Solution	Amazon Redshift	Amazon redshift uses SQL to analyze structured and semi-structured data across data warehouses, operational databases, and data lakes, using AWS-designed hardware and ML to deliver the best price performance at any scale.
Solution	Amazon Kinesis Data Analytics	Amazon kinesis helps collect, process, and analyze real-time streaming data so that users can get timely insights and react quickly to new information. Amazon kinesis data analytics reduces the complexity of building, managing, and integrating streaming applications with other AWS services.
Solution	Amazon EMR	Amazon EMR is a cloud big data platform for running large-scale distributed data processing jobs, interactive sql queries, and ML applications using open-source analytics frameworks, such as Apache Spark.
Solution	Amazon OpenSearch	Amazon opensearch service makes it easy to perform interactive log analytics, real-time application monitoring, website search, and more. Opensearch is an open source, distributed search and analytics suite derived from Elastic Search.
Solution	Amazon SageMaker	It enables data lake environment for building, training, and deploying ML models using amazon Sage Maker on AWS cloud. It uses AWS services, such as amazon simple storage service (Amazon S3), amazon application programming interface (API) gateway, AWS Lambda, Amazon Kinesis Data Streams, and Amazon Kinesis Data Firehose.

14.11.3 SERVICES OFFERED

TABLE 263 AWS: SERVICES OFFERED

SERVICE TYPE	SERVICE NAME	DESCRIPTION
Professional Services	▪ Support and Maintenance ▪ Consulting and Training	AWS provide these professional services in the advanced analytics market.

Source: Company Website

14.11.4 RECENT DEVELOPMENTS

TABLE 264 AWS: SOLUTION LAUNCHES AND ENHANCEMENTS

DATE	NATURE OF DEVELOPMENT	SOLUTION NAME	DESCRIPTION
2021-Nov	Product Launch	Amazon Kinesis Data Analytics	Amazon Kinesis Data Streams added On-demand, a new capacity mode, which eliminates capacity provisioning and management for streaming workloads.
2021-Aug	Enhancement	Call Transcription Analytics Tool	AWS enhanced its AWS Contact Center Intelligence offering with a new Call transcription analytics tool that it said can generate more insights from customer conversations.
2021-May	Product Enhancement	Amazon Kinesis Data Analytics	Amazon Kinesis Data Analytics has a new version that deals with streaming data using SQL, Python, or Scala. It selects a Kinesis data stream in the Amazon Kinesis UI and creates a Kinesis data analytics studio notebook powered by Apache Zeppelin and Apache Flink with a single click to interactively analyze data in the stream.
2020-May	Product Launch	Amazon Kendra	AWS introduced the general availability of Amazon Kendra. Amazon Kendra is a highly accurate and easy-to-use enterprise search service powered by ML. With just a few clicks, Amazon Kendra uses ML to enable organizations to index all their internal data sources, make data searchable, and allow users to get precise answers to natural language queries.

TABLE 265 AWS: DEALS

YEAR-MONTH	DEAL TYPE	COMPANY 1 NAME	COMPANY 2 NAME	DESCRIPTION
2021-Sept	Partnership	AWS	UCI Track Champions League	AWS will provide Discovery Sports Events with cloud, AI, ML, and deep learning technology for its coverage of the UCI Track Champions League. The two companies plan to work together to deliver new fan engagement experiences around the UCI cycling competition, including a new UCI Track Champions League app, competition insights, and in-race statistics to help fans dive deeper into the race strategy.
2020-Dec	Collaborated	AWS	Novartis	AWS collaborated with Novartis to accelerate the digital transformation of its business operations. This collaboration enables Novartis to reimagine its core pharmaceutical manufacturing, supply chain, and delivery operations using AWS' portfolio of cloud services.

Source: Press Releases

14.12 SALESFORCE

Founded	1999
Headquarters	California, US
Ownership	Public (NYSE: CRM)
Business Overview	Salesforce is a Customer Relationship Management (CRM) solutions and services provider company. Salesforce acquired Tableau, a leading provider of an analytics platform. This acquisition would strengthen Salesforce's position in the digital transformation space. The company's integrated platform provides a single shared view of every customer across departments, such as marketing, sales, commerce, and services. The company offers a wide range of products and services across segments, such as sales, services, marketing, application, analytics, employee experience, trailblazers and reskilling, and enablement and collaboration. The organization caters to various industries, including BFSI, healthcare and life sciences, government, communications, retail, consumer goods, media, manufacturing, transportation, and hospitality, automotive, and education.
Products	Tableau Analytics and Salesforce Einstein Analytics
Services	Assurance Service, Transformation Services, and Implementation Delivery
Geographic Presence	North America, Europe, APAC, and MEA

Source: Company Website

14.13 TIBCO SOFTWARE

Founded	1997
Headquarters	California, US
Ownership	Private
Business Overview	TIBCO Software unlocks the potential of real-time data for making faster, smarter decisions. Its Connected Intelligence platform seamlessly connects any application or data source; intelligently it unifies data for greater access, trust, and control and it confidently helps predict outcomes in real time and at scale. It captures data in real-time and augments the intelligence of businesses through analytical insights, ranging from APIs and systems to devices and people. It provides products for business optimization, integration and core infrastructure, and process automation and collaboration. It offers consulting, education, and support services. With API-led microservices, serverless functions, and integration flows, the company helps businesses of any size and users of any technical level to interconnect resources to accelerate digital transformation.
Solutions Offered	TIBCO Spotfire and TIBCO Data Science
Services	Professional Services
Geographic Presence	North America

Source: Company Website

14.14 ALTERYX

Founded	2010
Headquarters	California, US
Ownership	Public (NYSE: AYX)
Business Overview	Alteryx is one of the leading data blending and advanced analytics software companies. It provides an end-to-end data science and analytics platform for the enterprise. The company has been expanding its statistical and analytics capabilities, increasing its big data and SaaS connectivity, and delivering a cloud platform for analytics applications called Alteryx Analytics Gallery. Alteryx provides solutions and services to a wide range of industry verticals, including BFSI, healthcare, retail, transportation and logistics, oil and gas, real estate, communication, energy and utilities, education, manufacturing, media and entertainment, travel and hospitality, and public sector.
Solutions Offered	Alteryx APA Platform, Alteryx Intelligence Suite, Alteryx Predictive Analytics, Alteryx Analytic Process Automation (APA) Platform
Geographic Presence	North America, Europe, APAC, MENA, and Latin America

Source: Company Website

14.15 TERADATA

Founded	1979
Headquarters	California, US
Ownership	Public
Business Overview	Teradata is the connected multi-cloud data platform for enterprise analytics, solving data challenges from start to scale. It gives the flexibility to handle the massive and mixed data workload. is a pervasive data intelligence company, catering to several industries, including communications, media and entertainment, financial services, life sciences, healthcare, retail, energy, travel and transportation, and manufacturing. Its broad customer base comprises O2 Czech Republic, BMW, Siemens Healthineers, Vodafone, American Red Cross, and U.S. Bank. It has partnered with various companies, including Accenture, Alation, AlphaZetta, Alteryx, General Dynamics Information technology, Erwin, HCL Technologies, and Informatica, to deliver end-to-end analytics solutions. Its vast product portfolio consists of various categories, including Software, Cloud, Ecosystem Management, Hardware, and Applications.
Solution Offered	Teradata Vantage
Services	Teradata Managed Services
Geographic Presence	North America and Europe

Source: Company Website

14.16 ADOBE

Founded	1982
Headquarters	California, US
Business Overview	Adobe is the global leader in digital media and digital marketing solutions. Its creative, marketing, and document solutions empower everyone to bring digital creations to life and deliver immersive, compelling experiences to the right person at the right moment for the best results. Its innovations have redefined the possibilities of digital experiences. We connect content and data and introduce new technologies that democratize creativity, shape the next generation of storytelling, and inspire new categories of business. Adobe has helped businesses meet the challenges of digital transformation. For instance, Adobe Experience Cloud has harnessed the data of business and delivered personalized experiences that drive business growth and customer loyalty.
Solutions Offered	Adobe Analytics
Geographic Presence	North America, APAC, and Europe

Source: Company Website

14.17 ABSOLUTDATA ANALYTICS

Founded	2001
Headquarters	California, US
Ownership	Private
Business Overview	Absolutdata is an Infogain company. Its products and services deliver scalable business impact across the enterprise by combining cutting-edge AI and ML with its heritage in analytical frameworks, business understanding, and technology. The NAVIK AI Platform for the enterprise blends AI, technology, data, and analytics to serve as the intelligence layer for forward-thinking companies. It has a suite of pre-built AI solutions, or it can be licensed and customized using the deep bench of data scientists and engineers. Its mission is to transform how the world makes decisions. Since 2001, it has connected data to business results for global brands. It has a powerful suite of AI-based products and advanced analytics services that give its clients a competitive edge.
Solutions Offered	NAVIK AI Platform
Services	Customer Analytics and Marketing Analytics
Geographic Presence	North America, Europe, APAC, and MEA

Source: Company Website

14.18 MOODY'S ANALYTICS

Founded	2008
Headquarters	California, US
Ownership	Private
Business Overview	Moody's Analytics provides financial intelligence and analytical tools to help business leaders make better, faster decisions. Its deep risk expertise, expansive information resources, and innovative application of technology help the clients confidently navigate through an evolving marketplace. It is known for its industry-leading and award-winning solutions, made up of research, data, software, and professional services, assembled to deliver a seamless customer experience. Its mission is to empower the customers to make better and faster decisions.
Solutions Offered	▪ Financial Analytics Software ▪ Economic Analysis and Forecasting
Geographic Presence	North America, EMEA, and Europe

Source: Company Website

14.19 QLIK

Founded	1993
Headquarters	Pennsylvania, US
Ownership	Private
Business Overview	Qlik offers a platform with an end-to-end solution to get value out of data. It enables open-ended, curiosity-driven exploration for transformative changes. It caters to various industries, including healthcare, financial services, retail, manufacturing, high tech, public sector, energy and utilities, communications, life sciences, and consumer products, with more than 50,000 customers across the globe, including Bridgestone, Cisco, Lloyd's, Samsung, Deloitte, and Paccar. Qlik has over 1,700 partners across the globe. These partners include solution providers, consulting and service partners, various companies, managed service providers, technology partners, and OEM partners.
Solution Offered	Qlik Sense is a BI and analytics platform that enables users to tackle every BI use case, from interactive dashboards and KPI reports to embedded, mobile, and conversational analytics. Qlik Sense combines data governance with AI-assisted data visualization and data exploration, advanced analytics, and intuitive, self-service data preparation capabilities to enable everyone in organization to make impactful discoveries from data. It provides access to automated ML modeling and on-demand predictions directly from within Qlik Sense that enables to generate predictive insights whenever and wherever needed. QlikView is a data analytics solution that provides all BI capabilities, dashboards, KPIs, balanced scorecards, and predictive analysis for enterprises to gain better data insights to make strategic business decisions. Qlik Data Catalyst builds a secure, enterprise-scale catalog that helps businesses understand data preparation and gain insights for business decisions. Qlik Analytics Platform helps build, embed, and integrate engaging and beautiful visual analytics for any device or app on a single platform.
Geographic Presence	North America, EMEA, and APAC

Source: Company Website

14.20 DATABRICKS

Founded	2013
Headquarters	California, US
Ownership	Private
Business Overview	Databricks is a data and AI company. It is the world's first and only lake house platform in the cloud. It combines the best of data warehouses and data lakes to offer an open and unified platform for data and AI. Its mission is to help data teams solve the world's toughest problems. The Databricks Leak Platform unifies all the data, analytics, and AI workload.
Solution Offered	▪ SQL Analytics ▪ Data Science ▪ ML ▪ Data Bricks Lakehouse Platform
Services	Professional Services
Geographic Presence	North America, Europe, and APAC

Source: Company Website

14.21 DATAIKU

Founded	2013
Headquarters	New York, US
Business Overview	Dataiku was formed to help companies in using data science, ML, and AI for continuous innovations. It works by enabling organizations to embed data systems and competencies at each stage. The company has started an initiative, 'AI for Good', to work with non-profit organizations. It has launched an academic program that provides free software licenses to academicians, researchers, and students. Dataiku's partners are analytics consultants, Original Equipment Manufacturer (OEM) partners, service partners, technology partners, and value-added resellers, such as Micropole, InformationsFabrik, SVA, Ondata, Integra, Microsoft, and 4C. The valuable customers of Dataiku are GE, Sephora, Unilever, BNP Paribas, Ubisoft, Capgemini, and Loreal.
Solutions and Services Offered	Dataiku DSS is a data science software platform that provides visual interfaces at any step of the predictive dataflow prototyping process, from wrangling to analysis and modeling. It helps bundle the entire workflow as a single deployable package for real-time predictions with REST API and enables monitoring live data in-production with dashboard and data validation policies. The platform helps create powerful predictive analytics solutions using the latest data analysis and ML technologies.
Geographic Presence	North America, Europe, and APAC

Source: Company Website

14.22 KINETICA

Founded	2009
Headquarters	California, US
Ownership	Private
Business Overview	Kinetica is an analytics database for fusing data across streams and data lakes to unlock value from spatial and temporal data at scale and speed. Kinetica helps many of the world's largest companies drive outcomes from machine data that includes time series and spatial data, including Citibank, GSK, OVO, Softbank, NORAD, and Telkomsel. It has a rich partner ecosystem, including Dell, HP, IBM, NVIDIA, and Oracle, and is privately held, backed by leading global venture capital firms such as Canvas Ventures, Citi Ventures, GreatPoint Ventures, and Meritech Capital Partners.
Solutions Offered	Real-time Analytics and IoT Analytics
Services	Kinetica Enterprise Support and Kinetica Managed Services
Geographic Presence	North America, Europe, and APAC

Source: Company Website

14.23 SMES/STARTUP COMPANIES

14.23.1 H2O.AI

Founded	2012
Headquarters	California, US
Ownership	Private
Business Overview	H2O.ai is the leading AI cloud company, its mission is to democratize AI for everyone. It is the trusted AI partner to more than 20,000 global organizations, including AT&T, Aegon/Transamerica, and Allergan. Customers use the H2O.ai Cloud platform to rapidly make, operate, and innovate to solve complex business problems and accelerate the discovery of new ideas. It is recognized as a global visionary and thought leader in automated ML (autoML), time series forecasting, and responsible AI. The platform, H2O AI Cloud, enables businesses, government entities, nonprofits, and academic institutions to make, operate, and innovate with AI to accelerate responsible innovation and push the boundaries of what is possible with AI.
Product Description	H2O AI Cloud
Geographic Presence	North America and Europe

Source: Company Website

14.23.2 DOMINO DATA LAB

Founded	2013
Headquarters	California, US
Ownership	Private
Business Overview	Domino Data lab is a leading provider of advanced analytics solution which powers the model-driven business for the world's most advanced enterprises. It has MLOps platform, which speeds up the development and deployment of data science work while increasing collaboration and governance, to scale data science into a competitive advantage. Its platform enables thousands of data scientists to develop better medicines, grow more productive crops, adapt risk models to major economic shifts, build better cars, improve customer support, or simply the best purchase to make at the right time. It is backed by leading venture capital firms such as Sequoia Capital, Bloomberg Beta, Coatue Management, Dell Technologies Capital, Highland Capital Partners, In-Q-Tel, and Zetta Venture Partners. Its mission is to unleash the power of data science to address the world's most important challenges.
Solutions Offered	Domino Data Center and Domino Data Science
Geographic Presence	North America

Source: Company Website

14.23.3 DATAROBOT

Founded	2012
Headquarters	Massachusetts, US
Ownership	Private
Business Overview	DataRobot is the AI cloud leader, delivering a unified platform for all users, all data types, and all environments to accelerate the delivery of AI to production. It is trusted by global customers across industries and verticals. It helps in delivering over a trillion predictions for leading companies globally.
Solutions Offered	Automated ML and Predictive Analytics Platform
Geographic Presence	APAC and North America

Source: Company Website

14.23.4 DATACHAT

Founded	2017
Headquarters	Wisconsin, US
Ownership	Private
Business Overview	DataChat has the vision of democratizing the power of data science. Developing a programming language that has the naturalness of English and the precision/accuracy of data programming languages, it had made analytics accessible and approachable to everyone. It is pushing data analytics into the future with conversational intelligence as it empowers individuals to get more out of their data with patented conversational intelligence, revolutionizing how users, non-programmers and coders gain insights from their data. It enables users to complete sophisticated analytics, uncovering patterns in their data that they would have not unlocked before.
Solutions Offered	Data Analytics Platform
Geographic Presence	North America

Source: Company Website

14.23.5 IMPLY

Founded	2015
Headquarters	California, US
Ownership	Private
Business Overview	Imply, founded by the original creators of Apache Druid, develops an innovative database purpose-built for modern analytics applications. It is driving a new era in data analytics, called Analytics in Motion, where interactive queries, real-time and historical data at unlimited scale, combine with the best price/performance, to realize the full potential of data. It is driving a new era in data analytics, called Analytics in Motion, where interactive queries, real-time and historical data at unlimited scale, combine with the best price and performance, to realize the full potential of data with modern analytics applications.
Solutions Offered	Imply Enterprises
Services	Professional Services and Technical Support Services
Geographic Presence	North America

Source: Company Website

14.23.6 PROMETHIUM

Founded	2018
Headquarters	California, US
Ownership	Private
Business Overview	Promethium solves the problem of organizations by providing them with what questions will they need to answer for tomorrow's business by providing real-time access to enterprise data and answers without moving data. As it is not possible to move data into one location at the speed of business. Its mission is to help every business be data-driven by enabling every employee to make data-driven decisions in real time. It is focused on helping organizations break free from the cycle that is holding them back. It has invented the future of data analytics with a solution that prioritizes fast access to trusted answers in minutes instead of months. It has helped customers in reducing project backlogs, the number of tools they use from many to one, access and use fresh and current data without moving and allow them to meet business needs by reducing delays in answering important business questions.
Software Offered	Promethium's Augmented Analytics
Geographic Presence	North America

Source: Company Website

14.23.7 SIREN

Founded	2014
Headquarters	Galway, Ireland
Ownership	Private
Business Overview	Siren provides the leading Investigative Intelligence platform to the world's largest and most complex organizations for Investigative Intelligence on their data. It is rooted in academic R&D in information retrieval, distributed computing, and knowledge representation. Also, the Siren platform provides integrated investigative intelligence combining the previously disconnected capabilities of search, BI, link analysis, big data operational logging, and alerting. Its vision is to build a Safer World enabled by unlocking insights from vast amounts of data.
Solutions Offered	Siren Platform and OSINT solution
Geographic Presence	North America and Europe

Source: Company Website

14.23.8 TELLIUS

Founded	2016
Headquarters	Virginia, US
Ownership	Private
Business Overview	Tellius is leading the era of decision intelligence with a guided insights platform powered by ML so that anyone can uncover the 'what' and the 'why' in the data. Tellius accelerates data-driven insights and decision-making for various industries such as pharmaceuticals, financial services, consumer goods, and high technology, and across many departments such as sales, marketing, and operations. It is an AI-driven decision intelligence platform that helps get relevant insights from all the data faster and easier than ever. It has made easy for business teams to ask questions while providing the power and flexibility for data experts. It can finally accelerate the path from data to decision and break down the silos between BI and AI. Its mission is to accelerate the journey from data to decisions, by augmenting human expertise and curiosity with intelligent automation.
Software Offered	Tellius (AI-driven Decision Intelligence Platform)
Geographic Presence	North America and APAC

Source: Company Website

14.23.9 SOTA SOLUTIONS

Founded	2012
Headquarters	Berlin, Germany
Ownership	Private
Business Overview	It is an innovative software company based in the Adlershof Technology Park located in Berlin, Germany. It supplies the customers with cutting-edge technology for optimizing their production processes. Its strategy focuses on long-term solutions that contribute to the energy transition. Optimization and increased efficiency are the core principles of Its work. The SOTA technology uses AI to automatically generate specialized customer applications and automatically adapts to changes in the environment. Its products are developed under strict consideration of quality requirements and customer needs. It uses platform technology to implement AI and predictive analytics in the field of big data.
Software Offered	SOTA IQ Platform
Geographic Presence	Europe

Source: Company Website

14.23.10 VANTI ANALYTICS

Founded	2019
Headquarters	Tel Aviv, Israel
Ownership	Private
Business Overview	Vanti Analytics integrates all the production lines with AI-based algorithms to detect and predict faulty units at an early stage. It uses advanced data-analytics technologies such as AI and ML, to optimize manufacturing processes. Its benefits are compelling such as reduced costs, less wastage, faster time-to-innovation, and enhanced customer satisfaction. It introduces the only scalable AI for the manufacturing sector.
Software Offered	Early Fault Prediction, Process Tuning Time Optimization, and Defect Detection
Geographic Presence	MEA

Source: Company Website

15 APPENDIX

15.1 INDUSTRY EXPERTS

"An advanced analytics platform is an essential tool to simplify complex data into easy-to-use dashboards and uses to connect disparate data to create an accurate canvas of the company's operations."

– Advanced Analytics Solution Vendor

"Advanced analytics is an umbrella term for a wide variety of analytics methods and solutions that work together mostly in a predictive way."

- Advanced Analytics Consultant

Source: Industry Experts

15.2 DISCUSSION GUIDE

Discovering a market is all about identifying opportunities in unrecognized and unexploited gaps that are worth pursuing. In most cases, these markets are not apparent or conventional. Discovering a product, unlike the markets, is quite different and significantly depends on its value and usability. We want to understand and discuss how your company is investing in discovering smart markets and smarter products to stay ahead of the curve.

Currently, we are conducting a global study on the advanced analytics market, with a focus on current trends and technology adoption. The report would provide a comprehensive market and forecast analysis of the overall market and its submarkets. The report also features a detailed analysis of opportunities, market drivers, restraints, roadmaps, competitive landscape, and profiles of key industry players.

Name	
Designation	
Company Name	
Email ID	
Mobile No.	
Phone No	
Fax No.	

Q. 1. According to you, would the COVID-19 pandemic have an impact on the advanced analytics market in the future? Please select from the options given below.

SR. NO.	YES	NO
1		

Primary's Viewpoint: ____

Q. 2. According to you, how would the COVID-19 pandemic impact the advanced analytics market? Please select from the options provided below.

SR. NO.	POSITIVE IMPACT	NO IMPACT	NEGATIVE IMPACT
1			

Primary's Viewpoint: ____

Q. 3. **Please brief us about the solution and service offerings of your company in the advanced analytics market.**

SR. NO.	SOLUTION/ SERVICES	DESCRIPTION
1		
2		
3		
4		
5		

Primary's Viewpoint: ____

Q. 4. **According to you, how does the advanced analytics market look like today in terms of opportunities, competitive landscape, and challenges, and where is it heading, i.e., how will the market shape in the next five years from now?**

SR. NO.	TODAY	AFTER FIVE YEARS
1		
2		
3		

Primary's Viewpoint: ____

Q. 5. **What could be major drivers, restraints, opportunities, and challenges in the advanced analytics market?**

SR. NO.	DRIVERS	RESTRAINTS	OPPORTUNITIES	CHALLENGES
1				
2				
3				

Primary's Viewpoint: ____

Q. 6. We have segmented the advanced analytics market based on components, so according to you,

- Which component is the highest revenue contributor for 2021? Please rank on a scale of 1 – 2 (1 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 2 (1 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET- BY COMPONENT	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2026
1	Solution		
2	Services (Professional and Managed Services)		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ____

Q. 7. We have segmented the advanced analytics market based on business function, so according to you,

- Which component is the highest revenue contributor for 2021? Please rank on a scale of 1 – 5 (5 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 5 (5 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET- BY BUSINESS FUNCTION	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 5, 5 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 5, 5 BEING THE HIGHEST) – 2026
1	Sales and Marketing		
2	Finance		
3	Human Resources (HR)		
4	Operations		
5	Supply Chain		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ____

Q. 8. We have segmented the advanced analytics market based on type, so according to you,

- Which component is the highest revenue contributor for 2021? Please rank on a scale of 1 – 7 (7 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 7 (7 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET-BY TYPE	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 7, 7 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 7, 7 BEING THE HIGHEST) – 2026
1	Customer Analytics		
2	Risk Analytics		
3	Predictive Analytics		
4	Big Data Analytics		
5	Statistical Analytics		
6	Prescriptive Analytics		
7	Others (voice and social analytics, augmented analytics, and multimedia analytics)		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ______________________________

Q. 9. We have segmented the advanced analytics market based on organization size, so according to you

- Which organization size is the highest revenue contributor for 2021? Please rank on a scale of 1 – 2 (1 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 2 (1 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET-BY ORGANIZATION SIZE	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2026
1	Large Enterprises		
2	SMEs		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ______________________________

Q. 10. We have segmented the advanced analytics market based on deployment mode, so according to you,

- Which deployment mode is the highest revenue contributor for 2021? Please rank on a scale of 1 – 2 (1 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 2 (1 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET-BY DEPLOYMENT MODE	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 2, 1 BEING THE HIGHEST) – 2026
1	On-premises		
2	Cloud		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ____

Q. 11. We have segmented the advanced analytics market based on verticals, so according to you,

- Which vertical is the highest revenue contributor for 2021? Please rank on a scale of 1 – 9 (1 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 9 (1 being the highest).

SR. NO.	CONVERSATIONAL AI MARKET-BY VERTICAL	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 9, 1 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 9, 1 BEING THE HIGHEST) – 2026
1	BFSI		
2	Retail and Consumer Goods		
3	IT and Telecom		
4	Transportation and Logistics		
5	Government and Defense		
6	Healthcare and Life Sciences		
7	Manufacturing		
8	Media and Entertainment		
9	Others (education, energy and utilities, and travel and hospitality)		

Note - Please mention in case of any overlapping segments or technologies that you perceive is not relevant for the given market

Primary's Viewpoint: ____

Q. 12. **Which region, according to you, is the largest in terms of opportunities in the advanced analytics market and is expected to be the largest in the future? Kindly give points on a scale of 1-5, with 1 being the largest market.**

- Which region is the highest revenue contributor for 2021? Please rank on a scale of 1 – 5 (1 being the highest).
- How do you think it will grow in the next five years (2026)? Please rank on a scale of 1 – 5 (1 being the highest).

SR. NO.	ADVANCED ANALYTICS MARKET- BY REGION	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 5, 1 BEING THE HIGHEST) – 2021	ATTRACTIVENESS OF OPPORTUNITY (SCALE OF 1 – 5, 1 BEING THE HIGHEST) – 2026
1	North America		
2	APAC		
3	Europe		
4	Latin America		
5	MEA		

Note - Please mention any overlapping segments or technologies that you perceive are not relevant for the given market

Primary's Viewpoint: ______________________________

Q. 13. **Which five companies do you see as fierce competitors with respect to your company in the advanced analytics market?**

INSTRUCTIONS:

- STEP 1: Please list down the top five competitors in this market in the column named 'Competitors.'
- STEP 2: Please assign the rank (scores from 1 to 5) relatively in the column named 'Market Impact' to each of these competitors such that no two competitors have the same score.

SR. NO.	COMPETITORS	MARKET IMPACT	OFFERINGS
1			
2			
3			
4			

Q. 14. **According to our pre-analysis, the market size during the forecast period (2021-2026) is given below. Please provide comments, if any, on the data given below?**

- Global Advanced Analytics Market Size (2021) = USD 30.0 – 40.0 billion
- CAGR (2021–2026) = 20% to 25%

Primary's Viewpoint: ______________________________

Q. 15. **Do you think the segmentation is comprehensive and covers major business segments of the market? Any other sections you would like to see in the report?**

AMENDMENTS/FEEDBACK

Q. 16. **What do you think about the advanced analytics market? Please give your opinion in a 2-3 liner quote about the growth of the Advanced Analytics market in the next five years (excluding any company or product name).**

Primary's Viewpoint: ____

We would like to discuss your company profile, understand the product/service portfolio and key strategies, and analyze your competition in the market over a short telephonic call. Please let us know a convenient date and time for the discussion. We can set up the bridge and conference call.

Date	
Time	
Number of Attendees	

ABOUT MARKETSANDMARKETS:

MarketsandMarkets (MnM) is a global market research and consulting company. We publish strategically analyzed market research reports and serve as business intelligence and consulting partner to Fortune 500 companies across the world. MarketsandMarkets covers seventeen industry verticals, including advanced materials, aerospace and defense, agriculture, automotive and transportation, biotechnology, building and construction, chemicals, energy and power, food and beverages, industrial automation, medical devices, mining, minerals and metals, packaging, pharmaceuticals, semiconductor and electronics, and telecommunications and IT.

Please feel free to download our Telecom and IT Research and Analysis Brochure from our group page for a comprehensive understanding of our offerings for your company.

Web Link: - http://www.marketsandmarkets.com

15.3 KNOWLEDGE STORE: MARKETSANDMARKETS' SUBSCRIPTION PORTAL

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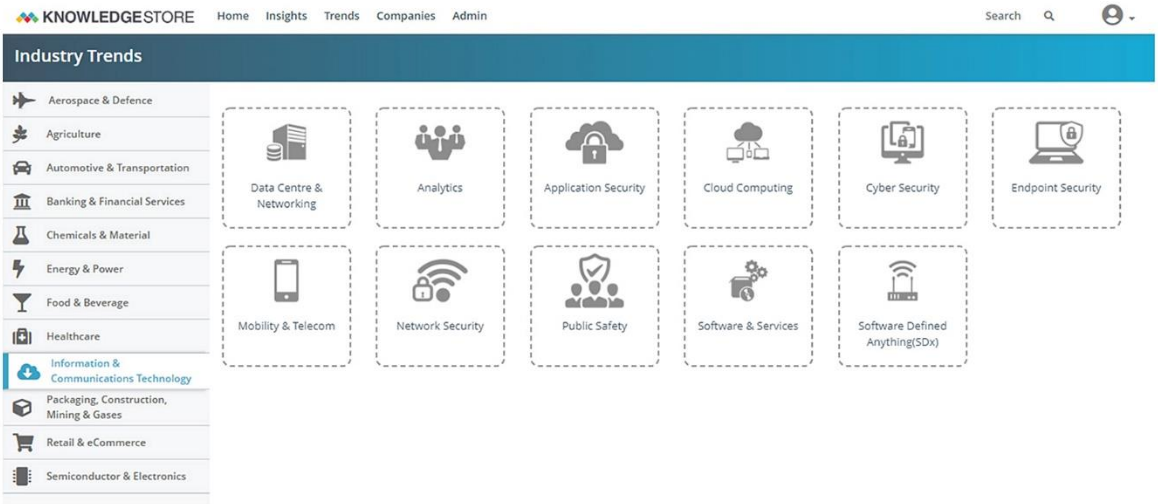

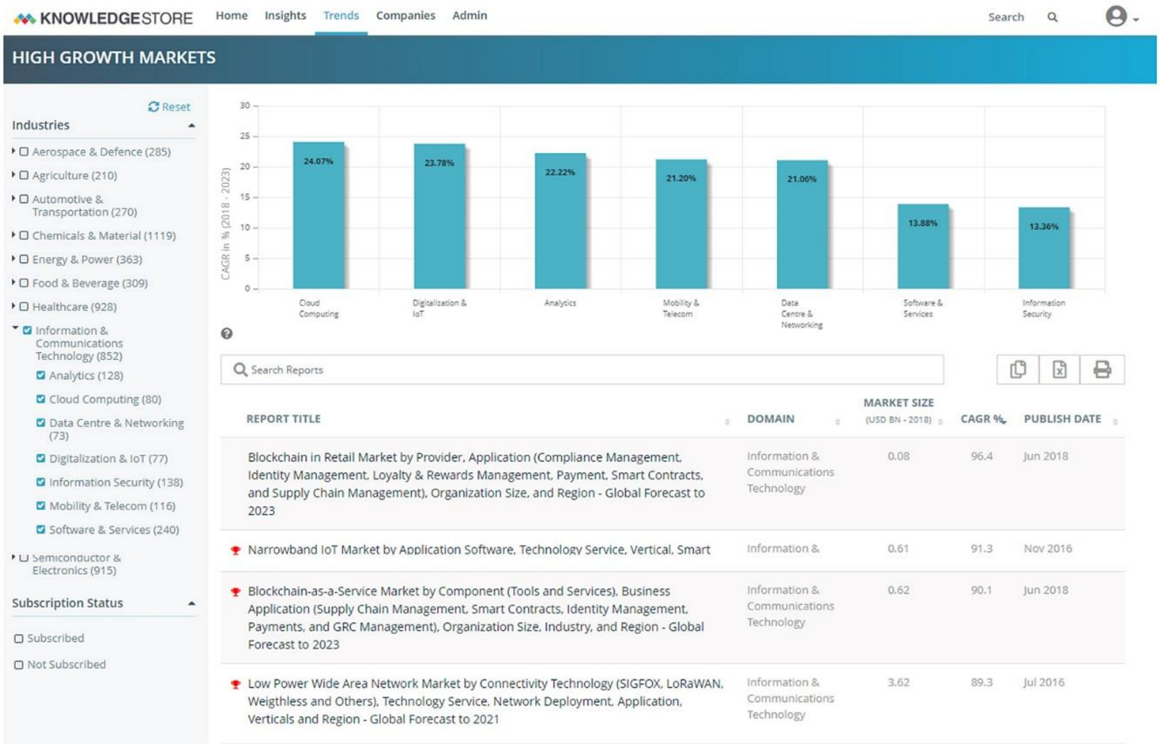

15.4 AVAILABLE CUSTOMIZATIONS

With the given market data, MarketsandMarkets offers customizations as per the company's specific needs. The following customization options are available for the report:

PRODUCT ANALYSIS

- Product quadrant which gives a detailed comparison of the product portfolio of each company

GEOGRAPHIC ANALYSIS

- Further breakup of the North American Advanced Analytics market
- Further breakup of the European Advanced Analytics market
- Further breakup of the APAC Advanced Analytics market
- Further breakup of the Latin American Advanced Analytics market
- Further breakup of the MEA Advanced Analytics market

COMPANY INFORMATION

- Detailed analysis and profiling of additional market players (up to 10)

15.5 RELATED REPORTS

SR. NO	REPORT TITLE	PUBLISHED DATE
1	**SPEECH ANALYTICS MARKET - GLOBAL FORECAST TO 2026** By Component (Solutions and Services), Application (Risk & Compliance Management, Sentiment Analysis), Organization Size, Deployment Mode (Cloud, On-premises), Vertical and Region https://www.marketsandmarkets.com/Market-Reports/speech-analytics-market-17297779.html	January 2022
2	**PREDICTIVE ANALYTICS MARKET - GLOBAL FORECAST TO 2026** By Solution (Financial Analytics, Risk Analytics, Marketing Analytics, Web & Social Media Analytics), Service, Deployment Mode, Organization Size, Vertical, and Region https://www.marketsandmarkets.com/Market-Reports/predictive-analytics-market-1181.html	October 2021

15.6 AUTHOR DETAILS

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With experience of over 14 years, he has been associated with ICT research for more than 10 years now. Prior to joining Markets and Markets, he has worked with firms like Frost and Sullivan, AMI-Partners, TNS Market Research and global IT major, IBM, where he worked for more than 5 years and his last role saw him as Market Segment Manager. He holds rich experience in the technology sector working on several customs and syndicated research assignments with large Multinational Organizations.

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The role encompasses handling syndicated and custom research projects in the ICT domain. Having more than 14 years of experience with 10+ years in research, Sameer joined MarketsandMarkets from Forrester Research where he was working as a Consultant, serving the Business Technology (BT) Professionals role. The role saw him working on varied consulting/custom-research projects and communicating directly with clients, internal resources (i.e., sales, research), and third-party vendors to address queries and manage expectations to create quality deliverables. Sameer has singlehandedly driven projects that include deliverables in the shape of data modeling, trend analysis, end-user requirements, competitive profiling, etc.

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